Supplementary Materials for Finite Classification of Transversal Diagonal Gates Beyond Stabilizer Codes

The 66 Orbits, Exact Enumerator Images, and No-Go Certificates

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Abstract

This supplement presents the exact enumeration, classification protocol, and orbit-by-orbit proofs for the lift-aware SSLP classification of exact $((6, 2, 3))$ codes with a nontrivial diagonal transversal logical element. It derives the sixty-six rank-compatible moment orbits, gives the reproducible protocol, and then lists a representative and the status of every orbit. Orbit 36 is proved rigid at the full local-monomial level. Ten further nonempty orbits—$7, 8, 9, 10, 11, 16, 21, 22, 23, 45$—have exact weight-enumerator images. Fifty-one orbits are excluded by exact Knill–Laflamme no-go certificates. All fifteen remaining orbits have exact realizations. The additional orbit-2 and orbit-3 converse proofs below determine two further images, leaving precisely orbits $1, 4$ unresolved. Complete enumeration tables are included at the end.

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Overview and organization

The supplement develops the finite, exact arguments supporting the six-qubit classification in the main text. The controlled-lift reduction first bounds the selected symmetry data, and the diagonal moment equations reduce them to sixty-six rank-compatible candidate orbits. These orbits classify labeled fiber and symmetry data: equivalent code projectors can occur in different orbits. Full Knill–Laflamme equations then decide whether the candidate data admit exact code completions, including their amplitude phases and boundary supports. Exact constructions realize fifteen orbits, and exact no-go certificates exclude the other fifty-one. Within the realizable orbits, converse bounds and constructions determine thirteen complete weight-enumerator images; the image questions for orbits 1 and 4 remain open. Orbit 36 also admits the stronger conclusion of code-level rigidity.

Part I fixes the common conventions, proves the finite enumeration, and lists all orbit representatives. Part II develops the exact realizations, rigidity theorem, and enumerator-image proofs, then assembles the main-text results and states the remaining questions. Part III supplies the orbit-by-orbit exclusion proofs, organized by targeted projective logical order. Part IV provides the complete enumeration tables and a code and data availability statement. Together, these parts distinguish realizability, complete enumerator images, and classification of code families up to equivalence.

For the assembled statements and their proof dependencies, start with Section 14. For a specific symmetry record, use Table 1 to locate its positive or no-go proof.

Part I
Finite enumeration and orbit representatives

1 Controlled lifts and enumeration conventions

Section 14 restates Results I–IV of the main text and proves them by assembling the exact results that follow.

We consider exact $((6,2,3))$ codes with a selected nontrivial projective diagonal logical action. The physical-order bound in Section III.C of the main text gives an exhaustive search over

$$m \mid q, \qquad 2 \le m, \qquad 2 \le q \le \beta_6 = 9, \tag{1}$$

where m is the targeted projective logical order and q is the primitive physical lift order. Here β_ν is the largest final Smith invariant factor of a nonsingular binary matrix of size at most ν. We write $M_3(6)$ for the maximum projective logical order of a diagonal transversal unitary preserving an exact $((6,2,3))$ code, maximized over all such codes and diagonal transversal unitaries in the chosen product basis. This is a maximum of logical orders m, not physical lift orders q. Write

$$D(a;q) = \bigotimes_{j=1}^{6} \mathrm{diag}(1, e^{2\pi i a_j/q}), \qquad \delta = \frac{q}{m}.$$

The physical order is exactly q when

$$\gcd(q, a_1, \ldots, a_6) = 1. \tag{2}$$

After the normal-form equivalences, the two logical eigenstates are supported in the full residue fibers

$$S_0(a;q), \quad S_\delta(a;q), \qquad S_b(a;q) = \{x \in \{0,1\}^6 : A_a(x) \equiv b \pmod q\}, \qquad A_a(x) = \sum_{j=1}^{6} a_j x_j,$$

with sorted angles

$$0 \le a_1 \le \cdots \le a_6 < q. \tag{3}$$

Zero amplitudes are allowed, so actual supports may be proper subsets of these fibers. The outer enumeration retains the labels (q,m), even when distinct labels describe the same exact code. The diagonal Knill–Laflamme equations require probability distributions on the two fibers with matching first and second moments. We retain a labeled pair only if these distributions can have common support affine dimension r satisfying $q \le \beta_r$. Quotienting these rank-compatible pairs by Boolean-cube isometries and logical-color interchange gives the SSLP moment orbits. An orbit is an SSLP realization orbit precisely when it admits complex amplitudes satisfying the full Knill–Laflamme equations.

2 Common moment and Knill–Laflamme conventions

2.1 Physical coordinates and mask encoding

Physical coordinates are numbered $1, \ldots, 6$. Orbit-specific relabelings are stated in the corresponding proofs. We label each computational-basis string $x = (x_1, \ldots, x_6)$ by the integer

$$k = \sum_{j=1}^{6} x_j 2^{j-1},$$

with x_1 as the least significant bit.

2.2 Diagonal moments

For a labeled fiber pair S_0, S_δ, introduce the degree-at-most-two feature map

$$\phi(x) := \big(x_1, \ldots, x_6, x_1x_2, x_1x_3, \ldots, x_5x_6\big) \in \mathbb{Z}^{21}. \tag{4}$$

The exact diagonal distance-three moment system used throughout this file is

$$\sum_{x\in S_0} p_0(x) = 1, \qquad \sum_{y\in S_\delta} p_1(y) = 1, \tag{5}$$

$$\sum_{x\in S_0} p_0(x)\phi(x) = \sum_{y\in S_\delta} p_1(y)\phi(y), \tag{6}$$

$$p_0(x) \geq 0, \qquad p_1(y) \geq 0. \tag{7}$$

These are the two normalization equations together with equality of the six one-bit and fifteen two-bit moments. The enumeration filters and the orbit-level certificates below use this common system.

2.3 Full Knill–Laflamme equations

For bit masks $u, v \in \mathbb{F}_2^6$, write $E_{u,v} = X^u Z^v$. With coefficient functions extended by zero outside the actual supports,

$$\langle k_L | E_{u,v} | \ell_L \rangle = \sum_{x\in\{0,1\}^6} (-1)^{v\cdot x} \overline{c^{(k)}_{x\oplus u}}\, c^{(\ell)}_x. \tag{8}$$

The moment LP solves the $u = 0$ diagonal part for every Pauli error of weight at most two. The remaining equations are, for $|\operatorname{supp}(u) \cup \operatorname{supp}(v)| \leq 2$ and $u \neq 0$,

$$\langle 0_L | E_{u,v} | 1_L \rangle = 0, \tag{9}$$

$$\langle 0_L | E_{u,v} | 0_L \rangle = \langle 1_L | E_{u,v} | 1_L \rangle. \tag{10}$$

These equations retain the amplitude phases and are polynomial in the real and imaginary parts of the coefficients.

3 The 66 SSLP moment orbits

Theorem 3.1 (The 66 SSLP moment orbits)**.** *After exact fiber hashing, quotienting the rank-compatible labeled fiber pairs by physical-qubit permutations, Boolean-cube translations by local X's, and logical-color interchange, while retaining the labels (q, m), gives exactly*

$$\boxed{66} \tag{11}$$

SSLP moment orbits. Their distribution by targeted logical order is

m	2	3	4	5	6	7	8	9
# *orbits*	25	11	10	8	4	5	2	1.

This is the complete intermediate enumeration of the controlled full-fiber moment skeletons used to construct the SSLP realization orbits; it is not a separate classification level or a classification of exact code projectors.

We prove the theorem by isolating every exact counting step in a lemma. The final proof is obtained by chaining those lemmas after the orbit quotient.

SS stage: outer angle records ⟶ labeled full-fiber pairs

$$13{,}877 \xrightarrow{\text{primitivity filter}} 13{,}449 \xrightarrow{\text{nonempty-fiber filter}} 12{,}768 \xrightarrow{\text{exact fiber hashing}} 11{,}422$$

LP stage: rank prefilter, moment feasibility, and active-face rank

$$11{,}422 \xrightarrow{\text{affine–Smith prefilter}} 508 \xrightarrow{\text{moment LP}} 291 \xrightarrow{\text{active-rank LP}} 215$$

Orbit level: SSLP moment-orbit formation

$$215 \xrightarrow{\text{Boolean-cube and color quotient}} \boxed{66}.$$

Figure 1: Overview of the reduction proved in this section. At the outer-record level, the subset-sum (SS) stage takes the 13,877 sorted angle records to 11,422 distinct labeled full-fiber pairs. At the fiber-pair level, the affine–Smith prefilter and the two LP tests leave 215 rank-compatible pairs. Only after the final symmetry quotient is the object an SSLP moment orbit, giving 66 orbits. These are computational object levels within the moment-orbit enumeration, not additional classification levels.

3.1 Raw outer count

Lemma 3.2 (Raw outer count)**.** *The exhaustive sorted outer list for the controlled data Eqs.* (1)–(3) *contains exactly* 13,877 *records.*

Proof. For fixed q, a sorted angle vector is a multiset of six elements chosen from the q residues $0, \dots, q-1$. Equivalently, let n_r be the multiplicity of residue r. Then

$$n_0 + \cdots + n_{q-1} = 6, \qquad n_r \geq 0.$$

By stars and bars, the number of sorted vectors is

$$\#\{0 \leq a_1 \leq \cdots \leq a_6 < q\} = \binom{q+5}{6}. \tag{12}$$

Logical-order choices. For a fixed physical lift order q, the canonical residue difference $\delta = q/m$ must be included for every nontrivial divisor $m \mid q$. There are $\tau(q) - 1$ such divisors, where τ is the divisor-counting function. Therefore

$$N_6(9) = \sum_{q=2}^{9} \big(\tau(q) - 1\big) \binom{q+5}{6}. \tag{13}$$

The complete breakdown is

q	nontrivial $m \mid q$	$\tau(q) - 1$	$\binom{q+5}{6}$	contribution
2	$\{2\}$	1	7	7
3	$\{3\}$	1	28	28
4	$\{2, 4\}$	2	84	168
5	$\{5\}$	1	210	210
6	$\{2, 3, 6\}$	3	462	1386
7	$\{7\}$	1	924	924
8	$\{2, 4, 8\}$	3	1716	5148
9	$\{3, 9\}$	2	3003	6006

Thus

$$\begin{aligned} N_6(9) &= 7 + 28 + 168 + 210 + 1386 + 924 + 5148 + 6006 \\ &= 13{,}877. \end{aligned} \tag{14}$$

The older specialization $q = m$ keeps only one logical-order choice for each q, and therefore contains

$$\sum_{q=2}^{9} \binom{q+5}{6} = 6{,}434 \tag{15}$$

records. The remaining

$$13{,}877 - 6{,}434 = 7{,}443$$

records are the genuinely lift-aware outer layer. They include, for example, $(q, m) = (4, 2)$, $(6, 2)$, $(6, 3)$, $(8, 2)$, $(8, 4)$, and $(9, 3)$. Even if some of these are later eliminated or reduce to smaller controlled lifts, they must be present initially for the normal form to be exhaustive. □

Remark 3.3 (What "before any filtering" means)**.** The number 13,877 still includes the all-zero angle vector, nonprimitive vectors, empty second fibers, vectors producing the same fiber pair, low-affine-rank fibers, moment-LP-infeasible pairs, and cases related by code equivalences. It is an upper count for an inexpensive discrete outer loop, not the number of nonlinear Knill–Laflamme solves.

3.2 Primitivity reduction

A nonprimitive vector does not have exact physical order q. It is therefore removed by Eq. (2). The number of sorted primitive vectors has a closed form.

Proposition 3.4 (Primitive sorted-vector count)**.** *Let $P_6(q)$ denote the number of sorted $a \in \{0, \dots, q-1\}^6$ satisfying $\gcd(q, a_1, \dots, a_6) = 1$. Then*

$$P_6(q) = \sum_{d \mid q} \mu(d) \binom{q/d + 5}{6}, \tag{16}$$

where μ is the Möbius function.

Proof. For $d \mid q$, vectors whose six entries are all divisible by d are multisets of size six drawn from the q/d residues $0, d, \dots, q-d$, and hence are counted by $\binom{q/d+5}{6}$. Möbius inversion imposes that no nontrivial divisor of q divide all six entries. □

For $q = 2, \dots, 9$, the primitive counts are

$$\big(P_6(2), \dots, P_6(9)\big) = (6, 27, 77, 209, 428, 923, 1632, 2975). \tag{17}$$

Lemma 3.5 (Primitivity reduction)**.** *Imposing Eq. (2) reduces the* 13,877 *raw records to*

$$13{,}449 \tag{18}$$

primitive records.

Proof. Multiply the primitive counts in Eq. (17) by the appropriate numbers of logical-order choices $m \mid q$ and sum over $q = 2, \dots, 9$. □

3.3 Nonempty fibers and exact hashing

Lemma 3.6 (Nonempty-fiber reduction)**.** *Discarding records with $S_\delta = \varnothing$ reduces the* 13,449 *primitive records to* 12,768 *records.*

Proof. For each primitive record, compute all 64 values $A_a(x) \bmod q$. The first fiber S_0 is never empty because $0^6 \in S_0$. The second fiber S_δ can be empty; such a record cannot support the second logical eigenvector and is discarded. This leaves

$$12{,}768 \tag{19}$$

angle records. □

Lemma 3.7 (Exact fiber hashing)**.** *Exact hashing of the labeled full-fiber pairs reduces the* 12,768 *nonempty records to* 11,422 *distinct labeled fiber pairs.*

Proof. For the diagonal moment problem, the angle vector matters only through the two full fibers and the labels (q, m). Store each fiber as a 64-bit mask and hash

$$(q, m; \operatorname{mask}(S_0), \operatorname{mask}(S_\delta)).$$

This identifies different angle vectors that induce the same labeled fiber pair. The 12,768 nonempty records collapse to

$$11{,}422 \tag{20}$$

distinct labeled fiber pairs. □

Hashing at this stage is lossless for the diagonal SSLP problem. It is not yet a quotient by all final code equivalences, because proper actual supports and accidental larger logical groups are not known.

3.4 The affine–Smith filter

Lemma 3.8 (Affine–Smith reduction)**.** *The exact coarse affine–Smith condition reduces the* 11,422 *labeled fiber pairs to* 508 *pairs.*

Proof. Let $\mathcal{T}_0 \subseteq S_0$ and $\mathcal{T}_1 \subseteq S_\delta$ be the actual nonzero supports of a hypothetical code, with positive probability distributions p_0, p_1. The Z_i and Z_iZ_j Knill–Laflamme equations give a common mean and common covariance:

$$\mu = \sum_{x \in \mathcal{T}_0} p_0(x)x = \sum_{y \in \mathcal{T}_1} p_1(y)y, \tag{21}$$

$$\Sigma = \sum_{x \in \mathcal{T}_0} p_0(x)(x-\mu)(x-\mu)^T = \sum_{y \in \mathcal{T}_1} p_1(y)(y-\mu)(y-\mu)^T. \tag{22}$$

Since all probabilities on the actual supports are positive,

$$\operatorname{range}\Sigma = \operatorname{span}_{\mathbb{R}}(\mathcal{T}_k - \mathcal{T}_k), \qquad k = 0, 1. \tag{23}$$

Thus

$$r := \operatorname{affdim}\mathcal{T}_0 = \operatorname{affdim}\mathcal{T}_1 = \operatorname{rank}\Sigma. \tag{24}$$

For a torsion-controlled primitive lift, the binary-Smith theorem gives

$$q \le \beta_r. \tag{25}$$

The exact small values needed here are

$$(\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6) = (1, 1, 2, 3, 5, 9). \tag{26}$$

The first four values follow by a direct enumeration of binary matrices of size at most four; $\beta_5 = 5$ and $\beta_6 = 9$ agree with the exhaustive tables of Ref. [2].

Define the smallest possible affine rank compatible with q by

$$r_{\min}(q) := \min\{r : \beta_r \geq q\}.$$

Then

q	2	3	4, 5	6, 7, 8, 9
$r_{\min}(q)$	3	4	5	6.

(27)

Because $\mathcal{T}_k \subseteq S_k$, a necessary condition is

$$\min\{\operatorname{affdim} S_0, \operatorname{affdim} S_\delta\} \geq r_{\min}(q). \tag{28}$$

This is an extremely inexpensive exact filter: compute the rank over $\mathbb{Q}$ of the difference vectors in each full fiber. It reduces the 11,422 labeled pairs to

$$508 \tag{29}$$

labeled pairs, represented by 510 sorted primitive angle records. □

This filter is only coarse. A full fiber can have large affine dimension while every feasible moment distribution is forced onto a lower-dimensional face. Section 3.6 removes precisely that possibility.

3.5 Moment-LP reduction

Lemma 3.9 (Moment-LP reduction)**.** *Exact degree-two moment feasibility reduces the* 508 *coarse affine–Smith survivors to* 291 *labeled fiber pairs.*

Proof. The common moment equations (5)–(7) are equivalent to

$$\operatorname{conv} \phi(S_0) \cap \operatorname{conv} \phi(S_\delta) \neq \varnothing. \tag{30}$$

If the variables are concatenated into $p = (p_0, p_1)$, the system has the form

$$Ap = b, \qquad p \geq 0, \tag{31}$$

where A has 23 rows: two normalization rows and 21 moment rows. Because the two residue fibers are disjoint, there are at most 64 variables in total.

Exact certificates. Feasibility is certified by rational moment points and infeasibility by integer separating vectors.

For a feasible case, a basic solution proposed by the LP solver is rationally reconstructed and checked exactly:

$$Ap = b \quad \text{in } \mathbb{Q}^{23}, \qquad p \geq 0. \tag{32}$$

Such a rational point must exist whenever the rational polytope is nonempty.

For an infeasible case, the two rational moment polytopes are strictly separated. This separation is certified by an integer vector $h \in \mathbb{Z}^{21}$ satisfying

$$\max_{x \in S_0} h \cdot \phi(x) < \min_{y \in S_\delta} h \cdot \phi(y), \tag{33}$$

or the same inequality with the two sides reversed. Equation (33) is an exact Farkas/separation certificate that the moment LP is infeasible.

Among the 508 coarse affine-Smith survivors, exact moment testing leaves

$$291 \tag{34}$$

distinct labeled fiber pairs, represented by 292 angle records. The single duplicate among these angle records is

$$(q, m) = (4, 2), \qquad a = (1, 1, 1, 1, 1, 1) \quad \text{and} \quad a = (3, 3, 3, 3, 3, 3),$$

which produce the same pair of fibers and are related by inversion of the physical generator. □

3.6 Active-rank reduction

Moment feasibility by itself is not enough to justify the lift order q. A feasible polytope may lie entirely on a face where the actual support rank is smaller than $r_{\min}(q)$. This can be decided exactly without adding a nonconvex search.

Let

$$\mathcal{P}(S_0, S_\delta)$$

denote the feasible polytope of Eqs. (5)–(7). A vertex $x \in S_0$ is called *active* if some $(p_0, p_1) \in \mathcal{P}$ has $p_0(x) > 0$; define active vertices in S_δ similarly. Let U_0, U_1 be the two active sets.

Lemma 3.10 (Maximum support rank from the active face)**.** *Assume $\mathcal{P}(S_0, S_\delta) \neq \varnothing$. Then*

$$r_* := \operatorname{affdim} U_0 = \operatorname{affdim} U_1 \tag{35}$$

is the largest affine rank attainable by a feasible moment distribution. In particular, a rank-$r_{\min}(q)$ moment realization exists if and only if

$$r_* \geq r_{\min}(q). \tag{36}$$

Proof. For every active variable choose a feasible point at which that variable is positive. The average of all these finitely many feasible points is feasible and is positive on every active variable simultaneously. Its two supports are exactly U_0, U_1. Equality of first and second moments gives equal covariance matrices, and positivity on the full active supports implies that the covariance range equals the real span of the within-support differences. Hence $\operatorname{affdim} U_0 = \operatorname{affdim} U_1$. Every other feasible support is contained in U_0, U_1, so no larger affine rank is possible; the averaged point attains r_*. □

Active variables are also decided with exact certificates. To test a variable p_i, maximize p_i over $Ap = b,\ p \geq 0$. A positive optimum is certified by an exact rational feasible point with $p_i > 0$. A zero optimum is certified by a dual vector $y \in \mathbb{Q}^{23}$ such that

$$A^T y \leq -e_i, \qquad b^T y = 0. \tag{37}$$

For any feasible p,

$$0 = b^T y = p^T A^T y \leq -p_i,$$

so $p_i = 0$ exactly.

Lemma 3.11 (Active-rank reduction)**.** *The exact active-face rank test reduces the* 291 *moment-feasible labeled pairs to* 215 *rank-compatible pairs, represented by* 216 *primitive angle records.*

Proof. Applying Eq. (36) removes 76 of the 291 moment-feasible labeled pairs and leaves

$$215 \tag{38}$$

rank-compatible labeled pairs, represented by 216 angle records.

The main losses occur in lift-inflated low-logical-order sectors:

(q, m)	$(6, 2)$	$(6, 3)$	$(8, 2)$	$(9, 3)$
moment-feasible pairs	52	33	38	4
rank-compatible pairs	30	21	0	0.

Every $(8, 2)$ moment solution has active rank 4, and every $(9, 3)$ moment solution has active rank 5; neither can support a torsion-controlled lift of order 8 or 9, respectively. This does not exclude order-two or order-three codes. It says only that these particular high-q controlled presentations are redundant or impossible and that a complete code, if one exists, must appear at a smaller compatible lift order. □

For later use, the relative dimension of the feasible moment polytope is

$$d_{\mathrm{LP}} = |U_0| + |U_1| - \operatorname{rank} A_U, \tag{39}$$

where A_U is the moment matrix restricted to active variables. Table 1 records d_{LP} for every surviving orbit.

3.7 Orbit quotient

Lemma 3.12 (Orbit reduction)**.** *Quotienting the* 215 *rank-compatible labeled pairs by Boolean-cube isometries and logical-color interchange gives exactly* 66 *SSLP moment orbits, with targeted-order distribution*

m	2	3	4	5	6	7	8	9
# *orbits*	25	11	10	8	4	5	2	1.

Proof. At the level of two colored subsets of the Boolean cube, the following operations preserve all Knill–Laflamme equations.

(i) A qubit permutation $\pi \in S_6$ permutes the six coordinates.

(ii) A local X operator X^t, $t \in \mathbb{F}_2^6$, translates every computational string by $x \mapsto x \oplus t$.

(iii) Interchanging the two logical eigenvectors swaps the two colors; at the cyclic-group level this is accompanied by inversion of the generator.

The Boolean-cube isometry group generated by the first two operations has order

$$2^6\, 6! = 46{,}080. \tag{40}$$

The enumeration acts with all 46,080 isometries on the two 64-bit fiber masks, treats the two colors as unordered, and chooses the lexicographically least image as the exact orbit key. Exact fiber hashing already removes angle-vector changes that leave the two fibers unchanged, including the residual unit action $a \mapsto ua$ with $u \equiv 1 \pmod m$.

The 215 rank-compatible labeled pairs collapse to

$$66 \tag{41}$$

fiber-pair orbits. The distribution by targeted logical order is

m	surviving controlled lift orders q	labeled pairs	fiber-pair orbits
2	$\{2,4,6\}$	73	25
3	$\{3,6\}$	36	11
4	$\{4,8\}$	31	10
5	$\{5\}$	30	8
6	$\{6\}$	13	4
7	$\{7\}$	18	5
8	$\{8\}$	9	2
9	$\{9\}$	5	1

□

Proof of Theorem 3.1. Lemma 3.2 gives the exhaustive outer list. Lemmas 3.5, 3.6, 3.7, 3.8, 3.9, and 3.11 give

$$13{,}877 \longrightarrow 13{,}449 \longrightarrow 12{,}768 \longrightarrow 11{,}422 \longrightarrow 508 \longrightarrow 291 \longrightarrow 215.$$

Lemma 3.12 then gives the 66 canonical orbit keys and their targeted-order distribution. Every reduction is exact, so the resulting list is exhaustive. The full representative list appears in Table 1; Section 25 gives the complete count tables. □

Thus every targeted logical order

$$m = 2, 3, 4, 5, 6, 7, 8, 9 \tag{42}$$

survives the exact diagonal SSLP layer. None of the orders $4, 6, 7, 8, 9$ can therefore be ruled out by subset sums, diagonal moments, or affine-Smith rank alone. The later non-diagonal arguments eliminate all twelve moment orbits at $m = 6, 7, 8, 9$ and all ten moment orbits at $m = 4$. Thus the full Knill–Laflamme analysis, rather than the moment polytope by itself, is what yields the exact order set $\{1, 2, 3, 5\}$.

Remark 3.13 (Meaning of the 66)**.** The 66 classes are moment-skeleton classes with the labels (q, m) retained. They are the input to the full Knill–Laflamme stage, which leaves the 15 SSLP realization orbits recorded in Table 1. Even those 15 are not inequivalent exact quantum codes: proper support choices can identify different full-fiber classes, and one exact code can appear under several targeted elements of a larger cyclic group. Final code-level deduplication requires the actual nonzero amplitudes and the intrinsic projector lattices.

The endpoint $m = 9$ consists of a single moment orbit. It survives every diagonal moment test but is eliminated by the off-diagonal Knill–Laflamme equations. A representative and its unique moment distribution are given in Section 17.1; Section 17 supplies the exact phase-inconsistency certificate.

4 Reproducible exact classification protocol

The complete diagonal reduction, followed by the transition to the non-diagonal Knill–Laflamme equations, can be organized as follows.

Step 1. Enumerate lift labels. Loop over $q = 2, \ldots, 9$. For every nontrivial divisor $m \mid q$, set $\delta = q/m$.

Step 2. Enumerate sorted angle vectors. Generate every weakly increasing $a \in \{0, \dots, q-1\}^6$. There are $\binom{q+5}{6}$ vectors for each (q, m).

Step 3. Enforce primitivity. Keep only vectors satisfying $\gcd(q, a_1, \dots, a_6) = 1$.

Step 4. Build the two fibers. Evaluate $A_a(x) \bmod q$ for all 64 strings and construct S_0, S_δ. Discard $S_\delta = \varnothing$.

Step 5. Hash labeled pairs. Store $(q, m; \operatorname{mask} S_0, \operatorname{mask} S_\delta)$ and retain the list of all angle vectors producing each pair.

Step 6. Apply the coarse affine-Smith test. Compute $\operatorname{affdim} S_0$ and $\operatorname{affdim} S_\delta$ over $\mathbb{Q}$. Discard the pair unless both are at least $r_{\min}(q)$.

Step 7. Decide the moment LP. Form the $23 \times (|S_0| + |S_\delta|)$ integer matrix A. For a feasible numerical proposal, reconstruct and verify a rational point. For an infeasible proposal, reconstruct and verify an integer separator as in Eq. (33).

Step 8. Determine the active face. Maximize every probability variable in turn. Certify active variables by rational witnesses and forced-zero variables by dual certificates (37).

Step 9. Enforce controlled-lift rank. Compute $r_* = \operatorname{affdim} U_0 = \operatorname{affdim} U_1$ and retain the pair only if $r_* \geq r_{\min}(q)$. Construct an exact feasible point positive on affine bases of both active sets as a rank witness.

Step 10. Compute exact orbits. Apply all $2^6 6!$ Boolean-cube isometries and logical interchange to the pair of masks. Use the least image as the canonical key.

Step 11. Record one representative and all multiplicities. For every orbit, save a representative (q, m, a), fiber sizes, active rank, moment-polytope dimension, and the number of surviving sorted angle records in the orbit.

Step 12. Proceed to the non-diagonal KL equations. Only the 66 orbit representatives, together with their exact feasible moment polytopes, need enter the next stage. The no-go proofs in Part III eliminate all twelve moment orbits of orders six through nine, all ten order-four moment orbits, seven of the eight order-five orbits, order-three orbits 5, 6, and 47–52, and the fourteen order-two orbits $12, 13, 14, 15, 17, 18, 19, 20, 24, 41, 42, 43, 44, 46$. Thus fifty-one orbits have self-contained exact infeasibility certificates. Exact constructions realize every one of the remaining fifteen orbits. Orbit 36, orbits 7–9, and order-two orbits 10, 11, 16, 21, 22, 23, and 45 have completely determined nonempty weight-enumerator images in Sections 6–13. The further proofs in Sections 11 and 12 determine the images of orbits 2 and 3. Orbits 1 and 4 are physically realized but remain unresolved at the weight-enumerator-image level.

The outer loop is tiny: only $13{,}877 \times 64$ modular subset sums. The exact moment LP is run only after hashing and affine screening, on 508 labeled pairs. The orbit reduction is performed only on the 215 rank-compatible pairs.

Numerical linear programming is used only to generate candidate primal solutions and dual certificates. All conclusions are established by exact verification over $\mathbb{Q}$ or $\mathbb{Z}$. Code and data availability is described in Section 26.

5 Representatives of all 66 rank-compatible moment orbits

These are the orbits that survive the diagonal moment and active-rank layers. The status map is

status	orbit IDs
open	$1, 4$
WE	$2, 3, 7, 8, 9, 10, 11, 16, 21, 22, 23, 45$
rigid	36
no-go	all remaining fifty-one IDs.

Here WE means that the exact weight-enumerator image is proved, while rigid means the stronger orbit-wide local-monomial rigidity theorem. Thus one orbit is settled beyond the enumerator level, twelve further nonempty orbits are settled at the enumerator level, fifty-one are excluded, and only two weight-enumerator images remain open. The status open concerns these images: exact realizations are known for all remaining orbits. The detailed proofs follow the list.

For each orbit, the table gives a representative sorted angle vector a, the full-fiber sizes $(|S_0|, |S_\delta|)$, the maximum attainable active affine rank r_*, the relative dimension d_{LP} of the exact moment polytope, the number ν of surviving sorted primitive angle records in the orbit, and its final status. The multiplicities sum to 216.

Table 1: One representative for every rank-compatible six-qubit diagonal moment orbit.

ID	q	m	representative a	$(\|S_0\|, \|S_\delta\|)$	r_*	d_{LP}	ν	status
1	2	2	$(0,0,0,1,1,1)$	$(32,32)$	6	41	1	open
2	2	2	$(0,0,1,1,1,1)$	$(32,32)$	6	41	1	WE
3	2	2	$(0,1,1,1,1,1)$	$(32,32)$	6	41	1	WE
4	2	2	$(1,1,1,1,1,1)$	$(32,32)$	6	41	1	open
5	3	3	$(0,0,1,1,1,1)$	$(20,20)$	6	19	2	no-go
6	3	3	$(0,1,1,1,1,1)$	$(22,20)$	6	19	4	no-go
7	3	3	$(0,1,1,1,2,2)$	$(22,22)$	6	21	2	WE
8	3	3	$(1,1,1,1,1,1)$	$(22,21)$	6	20	5	WE
9	3	3	$(1,1,1,1,1,2)$	$(21,21)$	6	19	2	WE
10	4	2	$(0,0,1,1,2,2)$	$(16,16)$	5	15	2	WE
11	4	2	$(0,0,1,2,2,2)$	$(16,16)$	5	15	2	WE
12	4	2	$(0,1,1,1,1,2)$	$(16,16)$	6	12	3	no-go
13	4	2	$(0,1,1,1,2,2)$	$(16,16)$	6	12	4	no-go
14	4	2	$(0,1,1,2,2,2)$	$(16,16)$	5	15	2	no-go
15	4	2	$(0,1,2,2,2,2)$	$(16,16)$	5	15	2	no-go
16	4	2	$(0,1,2,2,2,3)$	$(16,16)$	5	15	1	WE
17	4	2	$(1,1,1,1,1,1)$	$(16,16)$	6	9	4	no-go
18	4	2	$(1,1,1,1,1,2)$	$(16,16)$	6	9	6	no-go
19	4	2	$(1,1,1,1,2,2)$	$(16,16)$	6	12	3	no-go
20	4	2	$(1,1,1,2,2,2)$	$(16,16)$	6	12	4	no-go
21	4	2	$(1,1,1,2,2,3)$	$(16,16)$	6	12	2	WE
22	4	2	$(1,1,2,2,2,2)$	$(16,16)$	5	15	2	WE
23	4	2	$(1,2,2,2,2,2)$	$(16,16)$	5	15	2	WE
24	4	2	$(1,2,2,2,2,3)$	$(16,16)$	5	15	1	no-go
25	4	4	$(0,1,1,1,1,1)$	$(12,12)$	6	6	2	no-go

continued on next page

Table 1 (continued)

ID	q	m	representative a	$(\lvert S_0\rvert, \lvert S_\delta\rvert)$	r_*	d_{LP}	ν	status
26	4	4	$(0,1,1,2,2,3)$	$(16,16)$	6	12	2	no-go
27	4	4	$(1,1,1,1,1,1)$	$(16,12)$	6	5	4	no-go
28	4	4	$(1,1,1,1,1,2)$	$(16,16)$	5	0	3	no-go
29	4	4	$(1,1,1,1,2,2)$	$(16,16)$	6	8	5	no-go
30	4	4	$(1,1,1,1,2,3)$	$(16,16)$	6	9	3	no-go
31	4	4	$(1,1,1,2,2,2)$	$(16,16)$	6	9	2	no-go
32	4	4	$(1,1,2,2,2,3)$	$(16,16)$	6	9	2	no-go
33	5	5	$(0,1,1,3,3,3)$	$(12,12)$	6	6	3	no-go
34	5	5	$(1,1,1,1,1,1)$	$(7,7)$	6	0	2	no-go
35	5	5	$(1,1,1,2,2,3)$	$(12,13)$	6	2	7	no-go
36	5	5	$(1,1,1,2,2,4)$	$(12,12)$	6	5	3	rigid
37	5	5	$(1,1,1,2,3,3)$	$(13,13)$	6	6	3	no-go
38	5	5	$(1,1,2,2,2,3)$	$(13,13)$	6	6	3	no-go
39	5	5	$(1,1,2,3,3,3)$	$(12,13)$	6	3	6	no-go
40	5	5	$(1,2,2,2,2,2)$	$(12,12)$	6	5	3	no-go
41	6	2	$(1,1,1,1,2,3)$	$(10,10)$	6	3	4	no-go
42	6	2	$(1,1,1,1,3,3)$	$(10,10)$	6	3	4	no-go
43	6	2	$(1,1,1,2,3,3)$	$(10,10)$	6	3	6	no-go
44	6	2	$(1,1,2,3,3,4)$	$(10,10)$	6	3	6	no-go
45	6	2	$(1,1,2,3,4,4)$	$(10,10)$	6	3	4	WE
46	6	2	$(1,2,2,2,3,3)$	$(10,10)$	6	3	6	no-go
47	6	3	$(1,1,1,1,2,2)$	$(9,9)$	6	2	3	no-go
48	6	3	$(1,1,1,2,2,2)$	$(11,9)$	6	2	6	no-go
49	6	3	$(1,1,1,3,3,5)$	$(10,10)$	6	3	2	no-go
50	6	3	$(1,1,1,3,4,4)$	$(11,11)$	6	4	4	no-go
51	6	3	$(1,1,2,3,3,4)$	$(10,10)$	6	3	3	no-go
52	6	3	$(1,2,3,3,4,4)$	$(10,10)$	6	2	3	no-go
53	6	6	$(1,1,1,3,3,4)$	$(10,10)$	6	3	3	no-go
54	6	6	$(1,1,2,2,2,5)$	$(9,9)$	6	2	3	no-go
55	6	6	$(1,1,2,2,3,4)$	$(11,11)$	6	4	4	no-go
56	6	6	$(1,2,2,2,3,3)$	$(10,10)$	6	3	3	no-go
57	7	7	$(1,1,1,4,4,4)$	$(7,7)$	6	0	3	no-go
58	7	7	$(1,1,2,3,3,5)$	$(9,9)$	6	2	4	no-go
59	7	7	$(1,1,5,5,5,5)$	$(7,7)$	6	0	3	no-go
60	7	7	$(1,2,2,2,4,4)$	$(9,9)$	6	2	4	no-go
61	7	7	$(1,2,2,3,3,4)$	$(9,9)$	6	2	4	no-go
62	8	4	$(1,1,1,5,5,5)$	$(7,7)$	6	0	3	no-go
63	8	4	$(1,1,2,4,5,5)$	$(8,8)$	6	1	5	no-go
64	8	8	$(1,1,3,3,3,6)$	$(7,7)$	6	0	4	no-go
65	8	8	$(1,2,2,3,4,5)$	$(8,8)$	6	1	5	no-go
66	9	9	$(1,2,2,4,4,6)$	$(7,7)$	6	0	5	no-go

Part II
Exact realizations, rigidity, and weight-enumerator images

We first prove code-level rigidity for orbit 36 and determine the enumerator images of the three nonempty order-three orbits. The order-two results begin with common exact constructions in the four primitive records, followed by the complete converses for orbits 2 and 3 and the images of the seven other realizable order-two records. Section 14 assembles the main-text classification results; Section 15 identifies the remaining enumerator and code-family questions.

6 Rigidity of the order-five orbit 36

There are eight rank-compatible moment orbits with target order five, namely orbits 33–40 in Table 1. Orbit 36, represented by

$$(q, m, a) = \big(5, 5, (1, 1, 1, 2, 2, 4)\big), \tag{43}$$

contains the exact C_{10} code. Its explicit witness and isolated weight enumerator were reported in Ref. [1]. That construction fixed a twelve-term ansatz and then solved its phase equations. The purpose of this section is to prove the missing orbit-wide statement: every exact solution in orbit 36, including every possible boundary face of its moment polytope, is forced onto that same twelve-term ansatz and belongs to one local-monomial equivalence class. In particular, the Shor–Laflamme weight enumerator is unique on orbit 36.

This is an orbit-level theorem. Combined with the seven no-go theorems in Section 21, it gives the global uniqueness theorem for projective order five.

6.1 A symmetric presentation of the fiber pair

Conjugating a physical diagonal lift by a local X changes the corresponding angle coefficient from a_j to $-a_j$ and translates both residue fibers by that cube coordinate. Apply this operation to the coordinate carrying coefficient 4 and to one coordinate carrying coefficient 2. The sum of the flipped coefficients is $4 + 2 \equiv 1 \pmod 5$, and the new multiset of coefficients is

$$\{1, 1, 1, 1, 2, 3\}.$$

After a coordinate permutation and interchange of the two logical labels, orbit 36 is therefore represented by

$$a^\star = (1, 1, 1, 1, 2, 3), \qquad (S_0, S_4). \tag{44}$$

Indeed, if t is the two-bit translation used above, then

$$A_{a^\star}(x \oplus t) \equiv A_a(x) - 1 \pmod 5,$$

so the original pair (S_0, S_1) is taken to (S_4, S_0).

Let $V = \{1, 2, 3, 4\}$ index the first four coordinates, and separate the last two bits by a semicolon. Define

$$u = (0000; 00), \quad v = (0000; 11), \ s_i = (\mathbf{1}_{V\setminus\{i\}}; 10), \quad i \in V, \ e_{ij} = (\mathbf{1}_{\{i,j\}}; 01), \quad 1 \le i < j \le 4. \tag{45}$$

A direct residue check gives

$$S_0 = \{u, v\} \cup \{s_i : i \in V\} \cup \{e_{ij} : 1 \le i < j \le 4\}. \tag{46}$$

The coefficient sum is $1+1+1+1+2+3 \equiv 4 \pmod 5$, so

$$S_4 = \overline{S_0} := \{\bar{x} : x \in S_0\}. \tag{47}$$

Both fibers contain twelve points.

Write a general pair of logical vectors on these full fibers as

$$|0_L\rangle = a|u\rangle + b|v\rangle + \sum_{i=1}^{4} r_i|s_i\rangle + \sum_{1\le i<j\le 4} c_{ij}|e_{ij}\rangle, \tag{48}$$

$$|1_L\rangle = A|\bar{u}\rangle + B|\bar{v}\rangle + \sum_{i=1}^{4} R_i|\bar{s}_i\rangle + \sum_{1\le i<j\le 4} C_{ij}|\bar{e}_{ij}\rangle. \tag{49}$$

No coefficient is assumed nonzero at this stage, and coefficients outside the displayed fibers are zero.

6.2 Complete diagonal-moment normal form

For $x \in S_0$, put

$$p_x = |\langle x \mid 0_L\rangle|^2, \qquad \widetilde{p}_x = |\langle \bar{x} \mid 1_L\rangle|^2.$$

Let $\chi_{ij}(x) = (-1)^{x_i + x_j}$. The two normalizations and the Z_iZ_j Knill–Laflamme equalities imply that $p - \widetilde{p}$ is orthogonal on S_0 to

$$1, \qquad \chi_{ij} \quad (1 \le i < j \le 6).$$

These restricted characters span all functions on the twelve-point set S_0. More explicitly, in the column order

$$u, v, s_1, s_2, s_3, s_4, e_{12}, e_{13}, e_{14}, e_{23}, e_{24}, e_{34},$$

the square minor formed by

$$1, \chi_{12}, \chi_{13}, \chi_{14}, \chi_{15}, \chi_{16}, \chi_{23}, \chi_{24}, \chi_{25}, \chi_{34}, \chi_{35}, \chi_{45} \tag{50}$$

has determinant

$$-2^{19} \neq 0. \tag{51}$$

Consequently,

$$\boxed{\widetilde{p}_x = p_x \quad \text{for every } x \in S_0.} \tag{52}$$

For a single Z_i, complementation changes the character sign. Using Eq. (52), the single-Z_i equalities therefore say that the common probability distribution is balanced in each physical coordinate:

$$\sum_{x\in S_0} p_x(-1)^{x_i} = 0, \qquad i = 1, \ldots, 6. \tag{53}$$

Set

$$p_i = |r_i|^2, \qquad w_{ij} = |c_{ij}|^2, \qquad D_i = \sum_{j\neq i} w_{ij}, \tag{54}$$

and write

$$P = \sum_i p_i, \qquad W = \sum_{i<j} w_{ij}.$$

The balance equations for qubits five and six, the first four qubits, and normalization are respectively

$$|b|^2 + P = \frac{1}{2}, \qquad |b|^2 + W = \frac{1}{2}, \tag{55}$$

$$P - p_i + D_i = \frac{1}{2}, \qquad i = 1, \ldots, 4, \tag{56}$$

$$|a|^2 + |b|^2 + P + W = 1. \tag{57}$$

Thus $P = W$. Summing Eq. (56) over i gives $3P + 2W = 2$, and hence

$$P = W = \frac{2}{5}. \tag{58}$$

Substitution back into Eqs. (55)– (57) yields the complete moment normal form

$$\boxed{|a|^2 = |b|^2 = \frac{1}{10},} \qquad \boxed{\sum_i p_i = \sum_{i<j} w_{ij} = \frac{2}{5},} \tag{59}$$

and

$$\boxed{p_i = D_i - \frac{1}{10}, \qquad i = 1, \ldots, 4.} \tag{60}$$

The top amplitudes have the same individual magnitudes as the corresponding bottom amplitudes by Eq. (52).

6.3 The cross-KL equations and a symmetric K_4 kernel

Fix $i \in V$, and for $j \neq i$ define

$$U_j = \overline{c_{ij}} R_j, \qquad V_j = \overline{r_j} C_{ij}. \tag{61}$$

The X_i-incidences between the two fibers are precisely

$$(e_{ij}, \bar{s}_j), \qquad (s_j, \bar{e}_{ij}), \qquad j \neq i.$$

The equation for X_i is

$$\sum_{j \neq i} (U_j + V_j) = 0. \tag{62}$$

Let j_1, j_2, j_3 be the three elements of $V \setminus \{i\}$. The three weight-two equations for $X_i Z_{j_1}, X_i Z_{j_2}, X_i Z_{j_3}$ are

$$\begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} U_{j_1} - V_{j_1} \\ U_{j_2} - V_{j_2} \\ U_{j_3} - V_{j_3} \end{pmatrix} = 0. \tag{63}$$

The displayed matrix has determinant 4. Therefore

$$\boxed{\overline{c_{ij}} R_j = \overline{r_j} C_{ij} \quad (j \neq i),} \tag{64}$$

and Eq. (62) reduces to

$$\boxed{\sum_{j \neq i} \overline{c_{ij}} R_j = 0.} \tag{65}$$

The two weight-two flips $X_i X_5$ and $X_i X_6$ give

$$\bar{b} R_i + \overline{r_i} B = 0, \qquad \bar{a} R_i + \overline{r_i} A = 0. \tag{66}$$

At least one r_i is nonzero because $\sum_i |r_i|^2 = 2/5$. Since $|B| = |b|$, choose the unit phase

$$\tau = \frac{B}{\overline{b}}. \tag{67}$$

Equations (66), together with the equality of corresponding magnitudes, then imply

$$A = \tau\overline{a}, \qquad B = \tau\overline{b}, \qquad R_i = -\tau\overline{r_i} \quad (i = 1, \ldots, 4), \tag{68}$$

where the last identity is automatic when $r_i = 0$. Substituting into Eq. (65) gives

$$\sum_{j \neq i} c_{ij} r_j = 0, \qquad i = 1, \ldots, 4. \tag{69}$$

Equivalently,

$$\mathsf{C}r = 0, \qquad \mathsf{C} = \begin{pmatrix} 0 & c_{12} & c_{13} & c_{14} \\ c_{12} & 0 & c_{23} & c_{24} \\ c_{13} & c_{23} & 0 & c_{34} \\ c_{14} & c_{24} & c_{34} & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{pmatrix}. \tag{70}$$

6.4 No boundary support survives

Lemma 6.1 (Every vertex amplitude is active)**.** *Every exact orbit-36 solution satisfies*

$$r_1 r_2 r_3 r_4 \neq 0.$$

Proof. The case of no active r_i is excluded by Eq. (58). Suppose first that exactly one, say r_i, is active. The rows of Eq. (70) indexed by $j \neq i$ force $c_{ij} = 0$ for every $j \neq i$, so $D_i = 0$. Then Eq. (60) gives $p_i = -1/10$, impossible.

Suppose that exactly two, say r_i, r_j, are active. Row i of Eq. (70) gives $c_{ij} = 0$. Hence $D_i + D_j$ counts only edges joining the active pair to the inactive pair and is at most the total edge weight $W = 2/5$. On the other hand, Eq. (60) gives

$$D_i + D_j = p_i + p_j + \frac{1}{5} = \frac{3}{5},$$

a contradiction.

Finally, suppose exactly three vertex amplitudes are active. Restrict Eq. (70) to those three indices. The resulting zero-diagonal 3×3 matrix is singular, while its determinant is twice the product of its three edge amplitudes. Thus one active–active edge vanishes. The two row equations incident on that missing edge then force the other two active–active edges to vanish as well. All edge weight is therefore incident on the unique inactive vertex. Its weighted degree is $D_i = W = 2/5$, whereas Eq. (60) with $p_i = 0$ requires $D_i = 1/10$. This is again impossible. □

Because all r_j are nonzero, Eq. (64) and Eq. (68) now give

$$C_{ij} = -\tau\overline{c_{ij}} \qquad (1 \leq i < j \leq 4). \tag{71}$$

The cross-KL equation for X_5X_6 pairs each edge of K_4 with its opposite edge. Using Eq. (71), it becomes

$$\boxed{c_{12}c_{34} + c_{13}c_{24} + c_{14}c_{23} = 0.} \tag{72}$$

Let

$$G = \{ij : c_{ij} \neq 0\} \subseteq K_4$$

be the edge-support graph. Equation (70) and Lemma 6.1 imply that every vertex of G has degree at least two: a row containing only one active edge cannot sum to zero. If $G \neq K_4$, then up to relabeling there are only two possibilities, $G = C_4$ and $G = K_4 \setminus \{34\}$.

For $K_4 \setminus \{34\}$, rows three and four of Eq. (70) give

$$c_{23} = -c_{13}\frac{r_1}{r_2}, \qquad c_{24} = -c_{14}\frac{r_1}{r_2}.$$

Equation (72) then reduces to

$$-2c_{13}c_{14}\frac{r_1}{r_2} = 0,$$

contradicting the support assumption. For a four-cycle, take the active edges to be $12, 13, 24, 34$. Rows two and three give

$$c_{24} = -c_{12}\frac{r_1}{r_4}, \qquad c_{34} = -c_{13}\frac{r_1}{r_4},$$

and Eq. (72) becomes

$$-2c_{12}c_{13}\frac{r_1}{r_4} = 0,$$

again impossible. Therefore

$$\boxed{c_{ij} \neq 0 \quad \text{for every } 1 \leq i < j \leq 4.} \tag{73}$$

Thus every exact orbit-36 solution has the full twelve-term support in each logical word.

6.5 The magnitudes are forced

Define the three opposite-edge products

$$x = c_{12}c_{34}, \qquad y = c_{13}c_{24}, \qquad z = c_{14}c_{23}. \tag{74}$$

Equation (72) says $x + y + z = 0$. Since $\mathsf{C}r = 0$ with $r \neq 0$, one also has $\det \mathsf{C} = 0$. A direct determinant expansion gives

$$\det \mathsf{C} = x^2 + y^2 + z^2 - 2xy - 2xz - 2yz. \tag{75}$$

Combining Eqs. (72) and (75) yields

$$x^2 + y^2 + z^2 = 0. \tag{76}$$

All three products are nonzero by Eq. (73). Hence, up to a common nonzero factor and permutation,

$$(x, y, z) \propto (1, \omega, \omega^2), \qquad \omega = e^{2\pi i/3}. \tag{77}$$

In particular,

$$w_{12}w_{34} = w_{13}w_{24} = w_{14}w_{23}. \tag{78}$$

The positive relations (78) are equivalent to a vertex factorization

$$w_{ij} = s_i s_j, \qquad s_i > 0. \tag{79}$$

For example, one may set

$$s_1 = \sqrt{\frac{w_{12}w_{13}}{w_{23}}}, \qquad s_j = \frac{w_{1j}}{s_1} \quad (j = 2, 3, 4),$$

and use Eq. (78) to check the remaining edges. Let

$$S = s_1 + s_2 + s_3 + s_4. \tag{80}$$

Then

$$D_i = s_i(S - s_i). \tag{81}$$

Every principal 3×3 minor of C is a nonzero product of three active edge amplitudes, so $\operatorname{rank}\mathsf{C} = 3$. Since $\ker\mathsf{C} = \operatorname{span}\{r\}$, there is a nonzero scalar κ such that

$$\operatorname{adj}(\mathsf{C}) = \kappa r r^T. \tag{82}$$

The absolute value of the i-th diagonal cofactor is

$$2\prod_{j\neq i} s_j.$$

Taking absolute values in Eq. (82) therefore gives a constant $\gamma > 0$ such that

$$p_i s_i = \gamma \qquad (i = 1, \ldots, 4). \tag{83}$$

Combining Eqs. (60), (81), and (83), every s_i is a positive root of the same cubic

$$t^3 - St^2 + \frac{1}{10}t + \gamma = 0. \tag{84}$$

Moreover,

$$\sum_{i<j} s_i s_j = W = \frac{2}{5}. \tag{85}$$

We now show that the four positive numbers s_i are equal. They cannot take three distinct values: if α, β, δ are the three roots of Eq. (84), Vieta gives $\alpha + \beta + \delta = S$, whereas the sum of the four s_i's repeats one positive root and is strictly larger.

Suppose that exactly two values $\alpha \neq \beta$ occur, with α of multiplicity $m \in \{1, 2, 3\}$. Let δ be the third root of the cubic. Vieta's first two relations, together with Eq. (85), give the following three cases.

- If $m = 1$, then $\delta = 2\beta$,
$$3\alpha\beta + 2\beta^2 = \frac{1}{10}, \qquad 3\alpha\beta + 3\beta^2 = \frac{2}{5}.$$
Hence $\beta^2 = 3/10$, contradicting the first positive equation.
- If $m = 2$, then $\delta = \alpha + \beta$,
$$\alpha^2 + 3\alpha\beta + \beta^2 = \frac{1}{10}, \qquad \alpha^2 + 4\alpha\beta + \beta^2 = \frac{2}{5}.$$
Hence $\alpha\beta = 3/10$, again contradicting the first equation.
- If $m = 3$, the argument is symmetric to $m = 1$ and gives $\alpha^2 = 3/10$, with the same contradiction.

Thus all four s_i's are equal. Equation (85) gives

$$6s^2 = \frac{2}{5}, \qquad s^2 = \frac{1}{15}. \tag{86}$$

Consequently,

$$\boxed{w_{ij} = \frac{1}{15} \quad (i < j),} \qquad \boxed{p_i = \frac{1}{10} \quad (i = 1, \ldots, 4).} \tag{87}$$

Together with Eq. (59), each logical word has exactly six amplitudes of squared magnitude $1/10$ and six of squared magnitude $1/15$.

6.6 Phase rigidity and a canonical representative

Normalize the nonzero phases by

$$u_i = \sqrt{10}\, r_i, \qquad v_{ij} = \sqrt{15}\, c_{ij}, \qquad |u_i| = |v_{ij}| = 1, \tag{88}$$

and define a symmetric zero-diagonal matrix of edge phases by

$$M_{ij} = u_i u_j v_{ij}. \tag{89}$$

Multiplying the i-th equation of $\mathsf{C}r = 0$ by u_i shows that every row of M sums to zero:

$$\sum_{j \neq i} M_{ij} = 0. \tag{90}$$

Three unit complex numbers sum to zero only when they are a common phase times $\{1, \omega, \omega^2\}$. After permuting the first four qubits, we may therefore write

$$M_{12} = \mu, \qquad M_{13} = \mu\omega, \qquad M_{14} = \mu\omega^2, \tag{91}$$

for a unit phase μ. Row two must contain the other two colors $\mu\omega, \mu\omega^2$. Consistency with rows three and four forces

$$\boxed{M_{12} = M_{34} = \mu, \quad M_{13} = M_{24} = \mu\omega, \quad M_{14} = M_{23} = \mu\omega^2.} \tag{92}$$

Thus the discrete phase data are the unique proper three-edge-coloring of K_4. Interchanging ω and ω^2 is implemented by a permutation of the first four vertices.

It remains to remove the continuous phases. Let δ rephase $|0_L\rangle$, and let $\bigotimes_{j=1}^{6} \mathrm{diag}(1, e^{i\theta_j})$ be a local diagonal unitary. On the seven phases

$$\arg a, \arg b, \arg u_1, \arg u_2, \arg u_3, \arg u_4, \arg \mu$$

the phase increments are represented, in the column order $(\delta, \theta_1, \ldots, \theta_6)$, by

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 3 & 2 & 2 & 2 & 2 & 2 & 1 \end{pmatrix}. \tag{93}$$

Its determinant is

$$\det T = -5 \neq 0. \tag{94}$$

Hence the seven real phase parameters can all be fixed by local diagonal unitaries and a logical basis phase. Rephasing $|1_L\rangle$ also sets $\tau = 1$ in Eqs. (68) and (71).

We arrive at the canonical bottom word

$$\begin{aligned} |\psi_0\rangle = \frac{1}{\sqrt{10}} \left(|u\rangle + |v\rangle + \sum_{i=1}^{4} |s_i\rangle \right) \\ + \frac{1}{\sqrt{15}} \Big(|e_{12}\rangle + |e_{34}\rangle + \omega(|e_{13}\rangle + |e_{24}\rangle) + \omega^2(|e_{14}\rangle + |e_{23}\rangle) \Big), \end{aligned} \tag{95}$$

and

$$|\psi_1\rangle = Z_5 Z_6 X^{\otimes 6} K |\psi_0\rangle, \tag{96}$$

where K denotes complex conjugation in the computational basis. In coefficient form, Eq. (96) says

$$A = \overline{a}, \quad B = \overline{b}, \quad R_i = -\overline{r_i}, \quad C_{ij} = -\overline{c_{ij}}.$$

The inverse cube isometry from Eq. (44) returns this pair to the original representative (43).

For completeness, the full weight-one and weight-two KL check can be stated compactly. For a Pauli $E = i^\kappa X^f Z^g$, direct substitution into

$$\langle \psi_\alpha | E | \psi_\beta \rangle = i^\kappa \sum_{x \in \mathbb{F}_2^6} (-1)^{g \cdot x} \overline{c_\alpha(x \oplus f)} c_\beta(x) \tag{97}$$

shows that every off-diagonal matrix element vanishes for $1 \le \mathrm{wt}(E) \le 2$, and the common diagonal scalar λ_E is nonzero only in the following cases:

Pauli operator	λ_E
X_iX_j, Y_iY_j, Z_iZ_j, $1 \le i < j \le 4$	$1/15$
Z_iZ_5, $1 \le i \le 4$	$1/5$
Z_iZ_6, $1 \le i \le 4$	$-1/5$
X_5X_6, Y_5Y_6, Z_5Z_6	$1/5, -1/5, -3/5$

All other nonidentity Pauli operators of weight at most two have $\lambda_E = 0$. This verifies directly that the canonical pair is an exact $((6, 2, 3))$ code.

Theorem 6.2 (Orbit-36 rigidity)**.** *Every exact* $((6, 2, d \ge 3))$ *solution in orbit 36 has full twelve-term support in each logical word, has six probabilities* $1/10$ *and six probabilities* $1/15$ *in each word, and is equivalent to Eqs.* (95)–(96) *under physical-qubit permutations, local monomial unitaries, logical basis phases, and a possible interchange of the two logical basis vectors. Thus orbit 36 contains one essential exact code class.*

Proof. The complete moment equations give Eqs. (52)–(60). The cross-KL equations give the kernel system Eq. (70) and the matching equation Eq. (72). The boundary analysis proves Lemma 6.1 and Eq. (73). The adjugate and common-cubic argument gives the unique magnitudes Eq. (87). Finally, Eqs. (90)–(94) reduce all phase data to the canonical representative. The final overlap table verifies existence and the complete KL equations. □

6.7 The unique weight enumerator on orbit 36

For a rank-two projector Π, define the Shor–Laflamme enumerators by

$$A_j = \frac{1}{4} \sum_{\mathrm{wt}(E)=j} |\operatorname{Tr}(E\Pi)|^2, \tag{98}$$

$$B_j = \frac{1}{2} \sum_{\mathrm{wt}(E)=j} \operatorname{Tr}(E\Pi E^\dagger \Pi). \tag{99}$$

The weight-two scalar table above gives

$$\begin{aligned} A_1 &= 0, \\ A_2 &= 18\left(\frac{1}{15}\right)^2 + 8\left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(-\frac{1}{5}\right)^2 + \left(-\frac{3}{5}\right)^2 = \frac{21}{25}. \end{aligned} \tag{100}$$

It remains to record why $A_3 = 0$. Equation (96) defines an antiunitary

$$\Theta = Z_5 Z_6 X^{\otimes 6} K$$

with $|\psi_1\rangle = \Theta|\psi_0\rangle$. For a Pauli E, let

$$\varepsilon(E) = (-1)^{N_Z^{(1:4)}(E) + N_{\mathrm{nontriv}}^{(5:6)}(E)}, \tag{101}$$

where the first exponent counts Z's on qubits one through four and the second counts nonidentity factors on qubits five and six. Antiunitary conjugation gives

$$\langle\psi_1|E|\psi_1\rangle = \varepsilon(E)\langle\psi_0|E|\psi_0\rangle. \tag{102}$$

Thus the projector trace cancels whenever $\varepsilon(E) = -1$.

For a weight-three Pauli with $\varepsilon(E) = +1$, let f be its bit flip mask. Apart from

$$f = \{i, 5, 6\}, \qquad i \in \{1, 2, 3, 4\}, \tag{103}$$

the residue equation for $a^\star = (1, 1, 1, 1, 2, 3)$ forbids two points of S_0 from differing by f, so the bottom expectation is zero. For the exceptional mask (103), the three incidences are between s_j and the edge complementary to $\{i, j\}$, for $j \neq i$. For every X/Y dressing, Eq. (97) is a common real or imaginary linear combination of the three edge phases in the triangle opposite i. Those phases are $1, \omega, \omega^2$, and hence their sum and the sum of their conjugates vanish. Therefore

$$A_3 = 0. \tag{104}$$

For any $((6, 2, 3))$ code, the quantum MacWilliams relation together with $A_j = B_j$ for $j = 0, 1, 2$ determines the remaining A-coefficients from $x = A_1$, $y = A_2$, and $z = A_3$:

$$(A_0, \ldots, A_6) = \big(1, x, y, z, 15 - 4y - 3z, 12 + 3x + 4y + 3z, 4 - 4x - y - z\big). \tag{105}$$

Substituting Eqs. (100)–(104) gives the unique orbit-36 enumerator

$$\boxed{(A_0, \ldots, A_6) = \left(1, 0, \frac{21}{25}, 0, \frac{291}{25}, \frac{384}{25}, \frac{79}{25}\right).} \tag{106}$$

Equivalently,

$$A(z) = 1 + \frac{21}{25}z^2 + \frac{291}{25}z^4 + \frac{384}{25}z^5 + \frac{79}{25}z^6. \tag{107}$$

The corresponding B-enumerator is

$$(B_0, \ldots, B_6) = \left(1, 0, \frac{21}{25}, \frac{584}{25}, \frac{183}{5}, \frac{984}{25}, \frac{671}{25}\right). \tag{108}$$

This is the exact rational form of the isolated order-five enumerator reported in Ref. [1].

7 Exact weight-enumerator image of order-three orbit 7

Orbit 7 is represented by

$$(q, m, a) = \big(3, 3, (0, 1, 1, 1, 2, 2)\big). \tag{109}$$

Conjugating the two coefficient- 2 coordinates by physical X's changes their local eigenvalue ratios from ω^2 to ω, where $\omega = e^{2\pi i/3}$. After the resulting cube translation and an interchange of the two logical eigenvectors, the same orbit has the uniform active-coordinate presentation

$$a^\star = (0, 1, 1, 1, 1, 1). \tag{110}$$

Let A denote the zero-angle physical qubit and let $V = \{0, 1, 2, 3, 4\}$ denote the other five qubits. The logical fibers are then the active-coordinate Hamming-residue sectors

$$\mathrm{wt}_V(x) \equiv 0 \pmod 3, \qquad \mathrm{wt}_V(x) \equiv 2 \pmod 3. \tag{111}$$

Let $\mathcal{W}_7$ be the set of Shor–Laflamme A-enumerators of all exact $((6, 2, 3))$ solutions in this orbit, including every boundary-support solution.

The numerical search of Ref. [1] placed the known order-three family on the affine line in its Eq. (10), and Appendix B.1 of that reference gives an exact five-qubit-extension construction covering the whole parameter interval. The theorem below proves the converse for the complete orbit-7 KL variety.

Theorem 7.1 (Exact weight-enumerator image of orbit 7)**.** *One has*

$$\boxed{\mathcal{W}_7 = \{(1, \alpha, 1-\alpha, 0, 11+4\alpha, 16-\alpha, 3-3\alpha) : 0 \le \alpha \le 1\}.} \tag{112}$$

The corresponding B-enumerators are

$$\boxed{(1, \alpha, 1-\alpha, 24+6\alpha, 35+10\alpha, 40-7\alpha, 27-9\alpha), \qquad 0 \le \alpha \le 1.} \tag{113}$$

Moreover, every exact orbit-7 solution has an active qubit $s \in V$ for which the normalized code-projector marginal

$$\rho_{As} := \mathrm{Tr}_{V\setminus\{s\}}(P/2) \tag{114}$$

has rank two.

The rank-two-pair assertion is the structural core of the proof. The remainder of this section establishes it and then derives the enumerator image.

7.1 The K_5-edge normal form and scalar moments

For an edge $e \in \binom{V}{2}$, write $|e\rangle$ for the active five-qubit basis vector supported on e, and write $|V \setminus e\rangle$ for its three-bit complement. An arbitrary orbit-7 solution has the form

$$|0_L\rangle = u \otimes |\varnothing\rangle + \sum_{e\in\binom{V}{2}} a_e \otimes |V \setminus e\rangle, \tag{115}$$

$$|1_L\rangle = \sum_{e\in\binom{V}{2}} b_e \otimes |e\rangle + v \otimes |V\rangle, \tag{116}$$

where

$$u, v, a_e, b_e \in \mathbb{C}^2 \tag{117}$$

are amplitude spinors on the zero-angle qubit A. The ten labels are the edges of K_5.

Lemma 7.2 (Complete scalar moment normal form)**.** *Every exact orbit-7 solution satisfies*

$$\|u\|^2 = \|v\|^2 = \frac{1}{6}, \tag{118}$$

and, for every edge e,

$$\|a_e\|^2 = \|b_e\|^2 =: w_e. \tag{119}$$

The edge weights form a fractional perfect matching of K_5:

$$\sum_{e \ni i} w_e = \frac{1}{3} \quad (i \in V), \qquad \sum_e w_e = \frac{5}{6}. \tag{120}$$

Consequently every active qubit has maximally mixed one-qubit marginal.

Proof. Let μ_0, μ_3 be the total probabilities of $|0_L\rangle$ on active weights zero and three, and let ν_2, ν_5 be those of $|1_L\rangle$ on active weights two and five. Normalization and the sums of the five one-coordinate and ten two-coordinate diagonal moment equalities give

$$\begin{aligned} \mu_0 + \mu_3 &= 1, & \nu_2 + \nu_5 &= 1, \\ 3\mu_3 &= 2\nu_2 + 5\nu_5, & 3\mu_3 &= \nu_2 + 10\nu_5. \end{aligned} \tag{121}$$

Thus

$$\mu_0 = \nu_5 = \frac{1}{6}, \qquad \mu_3 = \nu_2 = \frac{5}{6}, \tag{122}$$

which proves Eq. (118) and the total edge masses.

Put

$$\alpha_e = \|a_e\|^2, \qquad \beta_e = \|b_e\|^2,$$

and let

$$d_i^\alpha = \sum_{e \ni i} \alpha_e, \qquad d_i^\beta = \sum_{e \ni i} \beta_e.$$

Equality of the one-coordinate occupation moments at vertex i gives

$$\frac{5}{6} - d_i^\alpha = d_i^\beta + \frac{1}{6}, \qquad \text{hence} \qquad d_i^\alpha + d_i^\beta = \frac{2}{3}. \tag{123}$$

For $i \neq j$, equality of the two-coordinate occupation moments gives

$$\sum_{e \subseteq V \setminus \{i,j\}} \alpha_e = \beta_{ij} + \frac{1}{6}. \tag{124}$$

Sum Eq. (124) over the four choices of $j \neq i$. Every edge not incident on i occurs twice on the left, so

$$2\left(\frac{5}{6} - d_i^\alpha\right) = d_i^\beta + \frac{2}{3}.$$

Together with Eq. (123), this yields

$$d_i^\alpha = d_i^\beta = \frac{1}{3}. \tag{125}$$

Now the left side of Eq. (124) equals

$$\frac{5}{6} - d_i^\alpha - d_j^\alpha + \alpha_{ij} = \frac{1}{6} + \alpha_{ij},$$

so $\alpha_{ij} = \beta_{ij}$. This proves Eqs. (119)–(120).

For an active qubit i, the occupation probability in either logical state is

$$\frac{5}{6} - \sum_{e \ni i} w_e = \frac{1}{2}$$

in $|0_L\rangle$, and

$$\sum_{e \ni i} w_e + \frac{1}{6} = \frac{1}{2}$$

in $|1_L\rangle$. Hence its Z_i scalar vanishes. The global order-three symmetry makes the X_i and Y_i scalars vanish as well, so the active one-qubit marginal is $I/2$. □

7.2 Matrix-valued moment identities and the rank-two-pair target

Define the rank-one positive matrices

$$U = uu^\dagger, \qquad W = vv^\dagger, \qquad A_e = a_e a_e^\dagger, \qquad B_e = b_e b_e^\dagger. \tag{126}$$

Let

$$\rho_A = U + \sum_e A_e = \sum_e B_e + W \tag{127}$$

be the common reduced state on the zero-angle qubit. Because an arbitrary operator on A, with or without one active Z_i dressing, has physical weight at most two, the corresponding KL equalities hold as 2×2 matrix identities.

For each active vertex put

$$\mathsf{A}_i = \sum_{e \ni i} A_e, \qquad \mathsf{B}_i = \sum_{e \ni i} B_e. \tag{128}$$

The Z_i-dressed matrix equation is equivalent to

$$\mathsf{A}_i + \mathsf{B}_i = \rho_A - U - W. \tag{129}$$

Summing over i, using that every edge is counted twice, and using Eq. (127), gives

$$\boxed{\rho_A = 3(U + W).} \tag{130}$$

Consequently

$$\sum_e A_e = 2U + 3W, \qquad \sum_e B_e = 3U + 2W, \tag{131}$$

$$\mathsf{A}_i + \mathsf{B}_i = 2(U + W) \qquad (i \in V). \tag{132}$$

The common reduced state on A and active qubit i is block diagonal:

$$\rho_{Ai} = \big(U + \mathsf{A}_i\big) \otimes |0\rangle\langle 0|_i + \big(W + \mathsf{B}_i\big) \otimes |1\rangle\langle 1|_i. \tag{133}$$

Each block has trace $1/2$. Write

$$P_u = 6U, \qquad P_v = 6W. \tag{134}$$

Thus the rank-two-pair assertion follows once one finds a vertex s with

$$\mathsf{A}_s = \frac{1}{3} P_u, \qquad \mathsf{B}_s = \frac{1}{3} P_v. \tag{135}$$

Indeed, Eq. (133) then becomes

$$\rho_{As} = \frac{1}{2} P_u \otimes |0\rangle\langle 0|_s + \frac{1}{2} P_v \otimes |1\rangle\langle 1|_s, \tag{136}$$

which has rank two.

7.3 Spinor normalization and the cross-KL hafnians

Choose a basis and phases on A so that

$$u = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\0\end{pmatrix}, \qquad v = \frac{1}{\sqrt{6}}\begin{pmatrix}c\\d\end{pmatrix}, \qquad 0 \le c \le 1, \qquad d = \sqrt{1-c^2}. \tag{137}$$

Put

$$n = \begin{pmatrix}c\\d\end{pmatrix}, \qquad m = \begin{pmatrix}-d\\c\end{pmatrix}, \tag{138}$$

so n, m are an orthonormal basis. Write

$$a_e = x_e n + y_e m. \tag{139}$$

The weight-two cross-KL equation obtained by flipping the active pair e contains exactly the incidences $u \leftrightarrow b_e$ and $a_e \leftrightarrow v$; hence

$$u^\dagger b_e + a_e^\dagger v = 0. \tag{140}$$

Together with $\|a_e\| = \|b_e\|$, this gives the exact parametrization

$$\boxed{b_e = \begin{pmatrix}-\overline{x_e}\\ \overline{z_e}\end{pmatrix}, \qquad |z_e| = |y_e|.} \tag{141}$$

For a vertex i, let $\mathcal{M}_i$ be the three perfect matchings of the four-set $V \setminus \{i\}$. For edge arrays r, s, define

$$H_i(r) = \sum_{\{e,f\}\in\mathcal{M}_i} r_e r_f, \tag{142}$$

$$\mathcal{B}_i(r,s) = \sum_{\{e,f\}\in\mathcal{M}_i} (r_e s_f + r_f s_e). \tag{143}$$

The weight-one flip on active vertex i, accompanied by an arbitrary operator on A, gives the matrix identity

$$\sum_{\{e,f\}\in\mathcal{M}_i} \left(b_f a_e^\dagger + b_e a_f^\dagger\right) = 0. \tag{144}$$

Comparing its four scalar matrix entries after Eqs. (139) and (141) yields

$$\boxed{H_i(x) = 0, \quad \mathcal{B}_i(x,y) = 0, \quad \mathcal{B}_i(x,z) = 0, \quad \mathcal{B}_i(y,z) = 0} \qquad (i \in V). \tag{145}$$

The one-Z dressings of the same flip systems also give

$$\boxed{a_e^\dagger b_f = a_f^\dagger b_e} \qquad \text{whenever } e \cap f = \varnothing. \tag{146}$$

These relations control both boundary and full-support branches.

7.4 Boundary support reduction

Let $G_x \subseteq K_5$ be the graph whose edge set is

$$E(G_x) = \{e : x_e \neq 0\}. \tag{147}$$

Lemma 7.3 (Boundary reduction)**.** *Either Eq.* (135) *already holds for some vertex, or*

$$x_e \neq 0 \qquad \text{for every } e \in \binom{V}{2}. \tag{148}$$

Proof. First suppose that a vertex s is isolated in G_x. For every edge $e \ni s$, Eq. (139) gives $a_e = y_e m$. The star has total weight $1/3$, so

$$\mathsf{A}_s = \frac{1}{3} P_m, \qquad P_m = mm^\dagger. \tag{149}$$

If $c > 0$, then the positive semidefinite matrix formed by the nonincident edges would be

$$\sum_{e \not\ni s} A_e = \frac{1}{3} P_u + \frac{1}{2} P_v - \frac{1}{3} P_m. \tag{150}$$

Its expectation in m equals

$$\frac{1}{3} |\langle m, (1,0)^T \rangle|^2 - \frac{1}{3} = -\frac{c^2}{3} < 0,$$

a contradiction. If $c = 0$, then $P_m = P_u$, so Eq. (132) gives $\mathsf{B}_s = P_v/3$, and Eq. (135) holds.

It remains to assume that G_x has no isolated vertex. Since $H_i(x) = 0$, the graph induced on any four vertices cannot contain exactly one active perfect matching: the corresponding equation would consist of one nonzero monomial. An elementary enumeration of the graphs on five vertices with minimum degree at least one and with this property leaves, up to isomorphism, exactly

$$K_{1,4}, \quad K_{2,3}, \quad K_5 \setminus K_3, \quad K_5 \setminus 2K_2, \quad K_5 \setminus K_2, \quad K_5. \tag{151}$$

Here $K_5 \setminus K_3$ is $K_{2,3}$ with the edge joining the two vertices in its two-vertex part.

For $K_{2,3}$, and also for $K_5 \setminus K_3$, relabel the bipartition as $\{0,1\} \sqcup \{2,3,4\}$. Three of the hafnian equations are

$$\begin{aligned} x_{03}x_{14} + x_{04}x_{13} &= 0, \\ x_{02}x_{14} + x_{04}x_{12} &= 0, \\ x_{02}x_{13} + x_{03}x_{12} &= 0. \end{aligned} \tag{152}$$

All displayed factors are nonzero. The first two equations imply $x_{03}/x_{02} = x_{13}/x_{12}$, whereas the third gives the negative identity.

For $K_5 \setminus 2K_2$, and also for $K_5 \setminus K_2$, relabel so that the missing edge common to the following reductions is 34. One obtains

$$\begin{aligned} x_{13}x_{24} + x_{14}x_{23} &= 0, \\ x_{03}x_{24} + x_{04}x_{23} &= 0, \\ x_{03}x_{14} + x_{04}x_{13} &= 0. \end{aligned} \tag{153}$$

The last two equations give $x_{24}/x_{23} = x_{14}/x_{13}$, while the first gives the negative identity.

Finally, if $G_x = K_{1,4}$, all nonzero x_e's lie in one star. Taking the expectation of Eq. (131) in n gives

$$\sum_e |x_e|^2 = n^\dagger (2U + 3W) n = \frac{3 + 2c^2}{6} \geq \frac{1}{2}. \tag{154}$$

But the full physical weight of any star is only $1/3$, while $w_e = \|a_e\|^2 \geq |x_e|^2$. This is impossible. Hence every proper support graph is excluded unless the isolated-vertex branch has already proved Eq. (135). □

7.5 The full-support hafnian variety

Assume henceforth that Eq. (148) holds. The equations $H_i(x) = 0$ are invariant under algebraic vertex scalings

$$x_{ij} \longmapsto d_i d_j x_{ij}, \qquad d_i \in \mathbb{C}^\times. \tag{155}$$

This is used only to solve the homogeneous equations; it is not asserted to be a physical code equivalence. Normalize $x_{01} = x_{02} = x_{03} = x_{04} = 1$. The five equations then give

$$\begin{aligned} &x_{12} = x_{34} = \tau, \\ &x_{13} = x_{24} = \tau\omega, \qquad \tau \neq 0, \qquad 1 + \omega + \omega^2 = 0. \\ &x_{14} = x_{23} = \tau\omega^2, \end{aligned} \tag{156}$$

The conjugate branch interchanges ω and ω^2.

The Jacobian of $(H_0, \dots, H_4)$ has rank five at Eq. (156); for example, it contains the nonzero minor

$$-4\tau^4(\omega + 2). \tag{157}$$

The five infinitesimal vertex-scaling vectors

$$(L_x r)_{ij} = (r_i + r_j)x_{ij} \tag{158}$$

lie in the tangent kernel. Since K_5 is nonbipartite and x has full support, L_x is injective. Dimension counting therefore shows that the tangent kernel is exactly its image. The equations $\mathcal{B}_i(x, y) = \mathcal{B}_i(x, z) = 0$ imply that there are vertex parameters $r_i, q_i \in \mathbb{C}$ such that

$$\boxed{y_{ij} = (r_i + r_j)x_{ij}, \qquad z_{ij} = (q_i + q_j)x_{ij}.} \tag{159}$$

Put

$$t_{ij} = r_i + r_j, \qquad p_{ij} = q_i + q_j. \tag{160}$$

Then

$$a_{ij} = x_{ij}(n + t_{ij}m), \qquad b_{ij} = \overline{x_{ij}}\binom{-1}{\overline{p_{ij}}}, \qquad |p_{ij}| = |t_{ij}|. \tag{161}$$

7.6 Disjoint edges and affine vertex parameters

Assume first that $c < 1$, so $d > 0$. Substitution of Eq. (161) into the disjoint-edge relation (146) gives, for disjoint edges e, f,

$$d(t_e - t_f + p_f - p_e) + c(t_e p_f - t_f p_e) = 0. \tag{162}$$

Put $k = c/d$. This can be written as

$$(t_e - p_e)(1 + kt_f) = (t_f - p_f)(1 + kt_e). \tag{163}$$

Define

$$h_i = \frac{1}{2} + kr_i, \qquad \delta_i = q_i - r_i. \tag{164}$$

For four distinct vertices i, j, k, ℓ, Eq. (163) becomes

$$(h_i + h_j)(\delta_k + \delta_\ell) = (h_k + h_\ell)(\delta_i + \delta_j). \tag{165}$$

Lemma 7.4 (Pair-sum collinearity)**.** *If five vectors $w_i \in \mathbb{C}^2$ satisfy*

$$\det(w_i + w_j, w_k + w_\ell) = 0 \tag{166}$$

whenever i, j, k, ℓ are distinct, then all five vectors are collinear.

Proof. For a four-set Q, put $W_Q = \sum_{i \in Q} w_i$. If $W_Q \neq 0$, Eq. (166) says that every pair sum in Q is collinear with W_Q. Since

$$2w_i = (w_i + w_j) + (w_i + w_k) - (w_j + w_k),$$

all four vectors in Q lie on that line, and an overlapping four-set places the fifth vector on it as well. If every W_Q vanishes, subtracting two such equations shows that all five w_i's are equal, and then $4w_i = 0$; hence all are zero. □

Apply Lemma 7.4 to $w_i = (h_i, \delta_i)$. The alternative $h_i = 0$ for every i would make every t_{ij} equal and hence all ten a_e's collinear. This contradicts the rank-two frame

$$\sum_e A_e = \frac{1}{3} P_u + \frac{1}{2} P_v \tag{167}$$

when $c < 1$. Thus there is one scalar λ with $\delta_i = \lambda h_i$, and consequently

$$\boxed{p_e = \mu t_e + \nu,} \qquad \mu = 1 + \lambda k, \qquad \nu = \lambda. \tag{168}$$

The slope μ is nonzero. Indeed, if $\mu = 0$, then $c > 0$ and $p_e = -d/c$ for every edge. From Eq. (131),

$$\sum_e |x_e|^2 = \frac{3 + 2c^2}{6}, \qquad \sum_e \overline{x_e} z_e = -\frac{cd}{3}. \tag{169}$$

Since $z_e = p_e x_e$, these equations would imply

$$-\frac{d}{c} \frac{3 + 2c^2}{6} = -\frac{cd}{3},$$

which reduces to $3 = 0$.

Insert Eq. (168) into $\mathcal{B}_i(y, z) = 0$. For a perfect matching $\{e, f\}$ of $V \setminus \{i\}$, the sum $t_e + t_f$ is independent of the matching. Since $H_i(x) = 0$, one obtains

$$\mathcal{B}_i(y, z) = 2\mu H_i(y), \qquad \text{and hence} \qquad H_i(y) = 0 \quad (i \in V). \tag{170}$$

7.7 One exceptional vertex

Because $H_i(x) = \mathcal{B}_i(x, y) = 0$, adding one constant to all five r_i's does not change the equations $H_i(y) = 0$. Set $r_4 = 0$ and write $\xi_i = r_i - r_4$ for $i = 0, 1, 2, 3$. In the normal form (156), the five equations are linear in the six monomials

$$(\xi_0\xi_1, \xi_0\xi_2, \xi_0\xi_3, \xi_1\xi_2, \xi_1\xi_3, \xi_2\xi_3). \tag{171}$$

After harmless nonzero row scalings, their coefficient matrix is

$$\begin{pmatrix} 0 & 0 & 0 & -1 & \omega+1 & -\omega \\ 0 & -1 & -\omega & 0 & 0 & \omega+1 \\ -1 & 0 & \omega+1 & 0 & -\omega & 0 \\ -\omega & \omega+1 & 0 & -1 & 0 & 0 \\ \omega+1 & -\omega & -1 & -1 & -\omega & \omega+1 \end{pmatrix}. \tag{172}$$

It has rank five and kernel spanned by $(1,1,1,1,1,1)^T$. Therefore

$$\xi_0\xi_1 = \xi_0\xi_2 = \xi_0\xi_3 = \xi_1\xi_2 = \xi_1\xi_3 = \xi_2\xi_3. \tag{173}$$

If the common value is zero, at most one ξ_i is nonzero; if it is nonzero, all four ξ_i's are equal. In either case, four of the five numbers r_i coincide.

Let s be the possible exceptional vertex and write

$$r_i = R \quad (i \neq s), \qquad r_s = S. \tag{174}$$

Then every edge incident on s has

$$t_e = T := R + S, \tag{175}$$

while every edge not incident on s has

$$t_e = N := 2R. \tag{176}$$

By Eq. (168), the p_e's have the same star/nonstar partition; denote their two values by p_T, p_N.

Define the rank-one projectors

$$P_a(t) = \frac{(n+tm)(n+tm)^\dagger}{1+|t|^2}, \tag{177}$$

$$P_b(p) = \frac{\binom{-1}{\bar{p}}(-1,p)}{1+|p|^2}. \tag{178}$$

Equation (161) gives $|p_T| = |T|$ and $|p_N| = |N|$. The exceptional star has total physical weight $1/3$, so Eq. (132) becomes

$$P_a(T) + P_b(p_T) = P_u + P_v. \tag{179}$$

At a nonexceptional vertex, one incident edge is a star edge and the other three are nonstar edges. Full support makes the total nonstar weight there strictly positive. Subtracting the star contribution and using Eq. (179) therefore gives

$$P_a(N) + P_b(p_N) = P_u + P_v. \tag{180}$$

Assume now $0 < c < 1$. For a pair (t,p) with $|p| = |t|$, comparison of the $(0,0)$ and $(0,1)$ entries in

$$P_a(t) + P_b(p) = P_u + P_v \tag{181}$$

gives

$$c|t|^2 + d\,\mathrm{Re}\,t = 0, \qquad p = \bar{t}. \tag{182}$$

The first equation is the circle

$$t = \frac{d}{2c}(\zeta - 1), \qquad |\zeta| = 1. \tag{183}$$

Apply the disjoint-edge equation (162) to one star edge and one disjoint nonstar edge. Substitution of Eq. (183) factors the result as

$$(\zeta_T + 1)(\zeta_T - \zeta_N)(\zeta_N + 1) = 0. \tag{184}$$

Hence

$$T = N, \qquad \text{or} \qquad T = -\frac{d}{c}, \qquad \text{or} \qquad N = -\frac{d}{c}. \tag{185}$$

The case $T = N$ makes all ten a_e's collinear, contradicting the rank-two frame (167). If $N = -d/c$, then

$$n - \frac{d}{c}m = \frac{1}{c}\begin{pmatrix}1\\0\end{pmatrix}, \tag{186}$$

so all six nonstar a_e's lie along u. Their total physical weight is $5/6 - 1/3 = 1/2$. The remaining star sum would therefore be

$$\frac{1}{3}P_u + \frac{1}{2}P_v - \frac{1}{2}P_u = \frac{1}{2}P_v - \frac{1}{6}P_u, \tag{187}$$

whose expectation in $m = v^\perp$ is $-d^2/6$, impossible. Thus the only surviving case is

$$T = -\frac{d}{c}. \tag{188}$$

Equation (186) now says that every a_e on the exceptional star is parallel to u. Its total weight is $1/3$, so

$$\mathsf{A}_s = \frac{1}{3}P_u. \tag{189}$$

Equation (132) then gives

$$\mathsf{B}_s = \frac{1}{3}P_v. \tag{190}$$

Thus Eq. (135) holds for $0 < c < 1$.

It remains to treat the endpoints. If $c = 1$, then u and v are collinear, and Eq. (130) makes ρ_A pure. The code therefore factorizes a fixed pure state on A, and every ρ_{Ai} has rank at most two. If $c = 0$, the isolated-vertex branch of Lemma 7.3 already proves the claim. In the full-support branch, Eq. (168) becomes $p_e = t_e + \nu$. Since $P_u + P_v = I$, the projector equations (179)–(180) are equivalent to

$$p_T = -T, \qquad p_N = -N. \tag{191}$$

They imply $T = N$, again contradicting the rank-two frame. Hence only the isolated-vertex branch survives when $c = 0$.

We have proved the structural statement

$$\boxed{\text{for every exact orbit-7 solution, some } s \in V \text{ satisfies } \rho_{As} = \frac{1}{2}P_u \otimes |0\rangle\langle 0|_s + \frac{1}{2}P_v \otimes |1\rangle\langle 1|_s.} \tag{192}$$

7.8 Enumerator consequence and exact attainability

Since ρ_{As} has trace one and rank two,

$$\operatorname{Tr}(\rho_{As}^2) \geq \frac{1}{2}. \tag{193}$$

The two-qubit Pauli expansion gives

$$4\operatorname{Tr}(\rho_{As}^2) = 1 + \sum_{\substack{E \neq I \\ \operatorname{supp}(E) \subseteq \{A,s\}}} \left|\frac{1}{2}\operatorname{Tr}(PE)\right|^2. \tag{194}$$

Thus the weight-one and weight-two Pauli mass supported on the pair $\{A, s\}$ is at least one, and therefore

$$A_1 + A_2 \geq 1. \tag{195}$$

For every $((6,2,3))$ code, the zeroth Rains shadow coefficient is

$$S_0 = \frac{1}{4}(1 - A_1 - A_2 - A_3) \geq 0. \tag{196}$$

Combining Eqs. (195) and (196) gives

$$\boxed{A_1 + A_2 = 1, \qquad A_3 = 0.} \tag{197}$$

By Lemma 7.2, all active one-body Pauli scalars vanish. Hence A_1 is the squared Bloch-vector length of ρ_A. From Eq. (130),

$$\rho_A = \frac{1}{2}(P_u + P_v), \tag{198}$$

and therefore

$$A_1 = 2\,\mathrm{Tr}(\rho_A^2) - 1 = |\langle (1,0)^T, n\rangle|^2 = c^2. \tag{199}$$

Put $\alpha = c^2 \in [0,1]$. Equation (197) gives $A_2 = 1 - \alpha$ and $A_3 = 0$. Substitution in the six-qubit MacWilliams normal form (105) yields

$$(A_0, \ldots, A_6) = (1, \alpha, 1-\alpha, 0, 11+4\alpha, 16-\alpha, 3-3\alpha)\,. \tag{200}$$

The quantum MacWilliams transform gives Eq. (113). This proves the inclusion of $\mathcal{W}_7$ in the right side of Eq. (112).

For the reverse inclusion, use the exact construction in Appendix B.1 of Ref. [1]: append one ancilla to the five-qubit perfect code and apply its displayed two-qubit unitary. In its binary-tetrahedral specialization, the code has the order-three transversal operator

$$(YHS^\dagger)^{\otimes 4} \otimes Z(2\pi/3) \otimes I, \tag{201}$$

where $S = \mathrm{diag}(1, i)$ and $Z(2\pi/3) = \mathrm{diag}(e^{-i\pi/3}, e^{i\pi/3})$. The adjoint on S corrects the cited gate convention. In the logical basis of the five-qubit construction, the induced matrix is

$$M = \frac{1}{2}\begin{pmatrix} -1+i & 1+i \\ -1+i & -1-i \end{pmatrix}.$$

It is unitary, satisfies $M^3 = I$, and neither M nor M^2 is scalar. The corrected fifth factor is diagonal and commutes with the controlled-Z coupling to the ancilla, so the same action holds for every ancilla-rotation parameter θ, not just at an endpoint. After diagonalizing the local factors and applying the eigenvector interchanges already included in the lift-aware orbit equivalence, this is the uniform-active presentation (110). The same exact construction has enumerator parameter

$$\alpha = \cos^2\theta. \tag{202}$$

As θ varies, every $\alpha \in [0,1]$ occurs. This proves the reverse inclusion and completes the proof of Theorem 7.1.

Remark 7.5 (What is and is not classified)**.** Theorem 7.1 determines the complete image of the exact orbit-7 KL variety under the Shor–Laflamme enumerator map. It does not by itself prove that two orbit-7 projectors with the same value of α are locally unitarily equivalent.

8 Exact weight-enumerator image of order-three orbit 8

Orbit 8 is represented by

$$(q, m, a) = \big(3, 3, (1, 1, 1, 1, 1, 1)\big). \tag{203}$$

Put $\omega = e^{2\pi i/3}$ and $D = \mathrm{diag}(1, \omega)$. The two logical fibers are

$$S_0 = \{x \in \mathbb{F}_2^6 : \mathrm{wt}(x) \equiv 0 \pmod 3\}, \qquad S_1 = \{x \in \mathbb{F}_2^6 : \mathrm{wt}(x) \equiv 1 \pmod 3\}. \tag{204}$$

Thus S_0 consists of the shells of weights $0, 3, 6$, while S_1 consists of the shells of weights $1, 4$. Let $\mathcal{W}_8$ denote the set of Shor–Laflamme A-enumerators of all exact $((6, 2, 3))$ codes in this lift-aware moment orbit, including every boundary-support solution.

The numerical search in Ref. [1] placed the relevant codes on the affine enumerator line of its Eq. (10). The result below is an exact orbit-wide theorem. It classifies the weight-enumerator image of orbit 8; it does not assert that the weight enumerator is a complete local-unitary invariant of the code projector.

Theorem 8.1 (Exact weight-enumerator image of orbit 8)**.** *One has*

$$\boxed{\mathcal{W}_8 = \left\{(1, \alpha, 1-\alpha, 0, 11+4\alpha, 16-\alpha, 3-3\alpha) : \frac{1}{2} \le \alpha \le 1\right\}.} \tag{205}$$

The corresponding B-enumerators are

$$\boxed{(1, \alpha, 1-\alpha, 24+6\alpha, 35+10\alpha, 40-7\alpha, 27-9\alpha)\,, \qquad \frac{1}{2} \le \alpha \le 1.} \tag{206}$$

The proof occupies the remainder of this section.

8.1 The exact shell normal form

Write $[6] = \{1, \ldots, 6\}$. For a subset $T \subseteq [6]$, the same symbol $|T\rangle$ denotes the computational basis state whose support is T, and $\bar{T} = [6] \setminus T$. Let

$$|0_L\rangle = \sum_{x \in S_0} c_x |x\rangle, \qquad |1_L\rangle = \sum_{y \in S_1} d_y |y\rangle.$$

Denote by μ_0, μ_3, μ_6 the total probabilities of $|0_L\rangle$ on the three shells of S_0, and by ν_1, ν_4 the corresponding probabilities of $|1_L\rangle$.

Lemma 8.2 (Shell rigidity)**.** *Every exact orbit-8 solution has*

$$(\mu_0, \mu_3, \mu_6) = \left(0, \frac{5}{6}, \frac{1}{6}\right), \qquad (\nu_1, \nu_4) = \left(\frac{1}{6}, \frac{5}{6}\right). \tag{207}$$

Proof. Put $t = \nu_4$, so $\nu_1 = 1 - t$. Summing the six one-coordinate moment equalities and the fifteen two-coordinate moment equalities gives, respectively, equality of the first two binomial moments of the Hamming weight. Together with normalization,

$$\begin{aligned} \mu_0 + \mu_3 + \mu_6 &= 1, \\ 3\mu_3 + 6\mu_6 &= \nu_1 + 4\nu_4 = 1 + 3t, \\ 3\mu_3 + 15\mu_6 &= 6\nu_4 = 6t. \end{aligned} \tag{208}$$

Solving gives

$$\mu_0 = \frac{5}{9} - \frac{2t}{3}, \qquad \mu_3 = \frac{5}{9} + \frac{t}{3}, \qquad \mu_6 = \frac{3t-1}{9}. \tag{209}$$

In particular $\mu_0 \geq 0$ implies $t \leq 5/6$, and hence $\nu_1 = 1 - t \geq 1/6$.

It remains to eliminate the vacuum amplitude. Fix $i \in [6]$. In the cross-KL equation for X_i, write

$$m_0 = \overline{c_\varnothing} d_{\{i\}},$$

and index the other ten reciprocal pairs by $R \in \binom{[6]\setminus\{i\}}{2}$:

$$m_R = \overline{c_{[6]\setminus(R\cup\{i\})}} d_{\bar{R}}.$$

The equations for X_i and the five dressings $X_i Z_j$, $j \neq i$, imply

$$m_0 + \sum_R m_R = 0, \qquad \sum_{R:\, j\notin R} m_R = 0 \quad (j \neq i). \tag{210}$$

Each two-set R is omitted by exactly three of the five coordinates $j \neq i$. Summing the second family in Eq. (210) therefore gives $3\sum_R m_R = 0$. The first equation then yields $m_0 = 0$. Thus

$$\overline{c_\varnothing} d_{\{i\}} = 0 \qquad (i = 1, \ldots, 6). \tag{211}$$

Since the total singleton probability is $\nu_1 \geq 1/6$, at least one $d_{\{i\}}$ is nonzero. Hence $c_\varnothing = 0$, so $\mu_0 = 0$. Equation (209) now gives $t = 5/6$ and Eq. (207) follows. □

Choose the logical phase of $|0_L\rangle$ so that its all-one amplitude is positive. Lemma 8.2 gives the exact form

$$|0_L\rangle = b|[6]\rangle + \sum_{\substack{T\subseteq[6]\\|T|=3}} a_T |T\rangle, \qquad b = \frac{1}{\sqrt{6}}, \tag{212}$$

$$|1_L\rangle = \sum_{i=1}^{6} u_i |\{i\}\rangle + \sum_{1\leq i<j\leq 6} v_{ij} |\overline{\{i,j\}}\rangle, \tag{213}$$

where

$$\sum_i |u_i|^2 = \frac{1}{6}, \qquad \sum_{i<j} |v_{ij}|^2 = \frac{5}{6}. \tag{214}$$

Put

$$p_i = |u_i|^2, \qquad r = \sum_i p_i = \frac{1}{6} = b^2. \tag{215}$$

8.2 An anti-linear ladder identity

Define

$$\mathcal{R} = \sum_{i=1}^{6} u_i \sigma_i^+, \qquad \sigma_i^+ = |1\rangle\langle 0|_i, \qquad J = Y^{\otimes 6} K, \tag{216}$$

where K is computational-basis complex conjugation. Since

$$J|T\rangle = (-1)^{|T|+1} |\bar{T}\rangle, \tag{217}$$

the weight-two cross-KL equations have a compact operator form.

For every $i < j$, the cross equation for X_iX_j is

$$b\, v_{ij} + \sum_{k \notin \{i,j\}} u_k \overline{a_{\{i,j,k\}}} = 0. \tag{218}$$

Indeed, the first term is the incidence between $|[6]\rangle$ and $|\overline{\{i,j\}}\rangle$, while the other four terms are the incidences between the weight-three and singleton shells. Equations (218) are exactly the weight-four coefficients of the following identity.

Lemma 8.3 (Anti-linear ladder)**.** *Let P be the code projector and $Q = \mathcal{R}\mathcal{R}^\dagger$. Then*

$$\mathcal{R}J|0_L\rangle = -b|1_L\rangle, \tag{219}$$
$$\mathcal{R}J|1_L\rangle = b|0_L\rangle, \tag{220}$$
$$QP = rP. \tag{221}$$

Proof. Equation (217) sends the all-one term of $|0_L\rangle$ to $-b|\varnothing\rangle$; applying $\mathcal{R}$ therefore produces the singleton part of $-b|1_L\rangle$. On the weight-three shell, the coefficient at $|\overline{\{i,j\}}\rangle$ is the sum in Eq. (218). Thus Eq. (219) follows.

The anti-linear form

$$\beta(\phi,\psi) = \langle\phi|\mathcal{R}J|\psi\rangle$$

is antisymmetric. To see this, put $S = Y^{\otimes 6}$, so $J = SK$. Because $S^T = S$ and

$$(\sigma_i^+ S)^T = S\sigma_i^- = -\sigma_i^+ S,$$

one has $(\mathcal{R}S)^T = -\mathcal{R}S$, which is equivalent to $\beta(\phi,\psi) = -\beta(\psi,\phi)$.

Moreover,

$$\|\mathcal{R}J|\psi\rangle\|^2 = \langle\psi|Q|\psi\rangle. \tag{222}$$

The operator Q has Pauli weight at most two, so the KL equations make its diagonal expectation the same on both logical states. By Eq. (219), that common norm is b. Antisymmetry gives

$$\langle 0_L|\mathcal{R}J|1_L\rangle = -\langle 1_L|\mathcal{R}J|0_L\rangle = b.$$

Cauchy–Schwarz is saturated, and hence Eq. (220) follows.

Finally, $J\mathcal{R}J = -\mathcal{R}^\dagger$, so $(\mathcal{R}J)^2 = -\mathcal{R}\mathcal{R}^\dagger = -Q$. Applying this to Eqs. (219)–(220) gives $Q|0_L\rangle = b^2|0_L\rangle$ and $Q|1_L\rangle = b^2|1_L\rangle$. Since $b^2 = r$, this is Eq. (221). □

8.3 Saturation of the low-weight enumerator

For a Pauli operator E, put

$$\lambda_E = \frac{1}{2}\operatorname{Tr}(PE). \tag{223}$$

For weights one and two these are the detectable KL scalars, and

$$A_1 + A_2 = \sum_{1 \le \mathrm{wt}(E) \le 2} |\lambda_E|^2. \tag{224}$$

Expand the Hermitian operator Q in the Pauli basis:

$$Q = q_0 I + \sum_{1 \le \mathrm{wt}(E) \le 2} q_E E. \tag{225}$$

From

$$Q = \sum_i p_i \frac{I - Z_i}{2} + \sum_{i \neq j} u_i \overline{u_j}\, \sigma_i^+ \sigma_j^-,$$

one obtains

$$q_0 = \frac{r}{2}, \qquad \sum_{1 \leq \mathrm{wt}(E) \leq 2} q_E^2 = \frac{1}{4} \sum_i p_i^2 + \frac{1}{2} \sum_{i<j} p_i p_j = \frac{r^2}{4}. \tag{226}$$

Taking a normalized code expectation in Eq. (221) gives

$$\frac{r}{2} = \sum_{1 \leq \mathrm{wt}(E) \leq 2} q_E \lambda_E. \tag{227}$$

Therefore Cauchy–Schwarz and Eq. (226) imply

$$A_1 + A_2 \geq 1. \tag{228}$$

For completeness, the reverse inequality needed here follows directly from the even-qubit spin flip, without invoking an external linear-programming bound. Let

$$\widetilde{P} = JPJ. \tag{229}$$

This is another positive semidefinite rank-two projector, so $\mathrm{Tr}(P\widetilde{P}) \geq 0$. On every nonidentity single-qubit Pauli factor, spin flip contributes a minus sign. Hence the Pauli expansion of P gives

$$\mathrm{Tr}(P\widetilde{P}) = \frac{1}{16} \sum_{j=0}^{6} (-1)^j A_j. \tag{230}$$

Using the six-qubit MacWilliams normal form Eq. (105), the alternating sum equals $8(1 - A_1 - A_2 - A_3)$. Thus

$$A_1 + A_2 + A_3 \leq 1. \tag{231}$$

Combining Eqs. (228) and (231) yields

$$\boxed{A_1 + A_2 = 1, \qquad A_3 = 0.} \tag{232}$$

Equality also holds in the Cauchy–Schwarz step. Consequently the full weight-one-and-two signature vector is parallel to the nonidentity Pauli coefficient vector of Q:

$$\lambda_E = \frac{2q_E}{r} \qquad (1 \leq \mathrm{wt}(E) \leq 2). \tag{233}$$

In particular,

$$\lambda_{Z_i} = -\frac{p_i}{r} = -6p_i, \qquad \lambda_{Z_i Z_j} = 0. \tag{234}$$

All other weight-one entries vanish as well, and hence

$$\boxed{A_1 = 36 \sum_{i=1}^{6} p_i^2.} \tag{235}$$

8.4 Weight-four magnitudes and complementary hafnians

Put

$$w_{ij} = |v_{ij}|^2, \qquad D_i = \sum_{j \neq i} w_{ij}. \tag{236}$$

In $|1_L\rangle$, the probability that qubit i is one is

$$n_i = p_i + \frac{5}{6} - D_i.$$

The first identity in Eq. (234) therefore gives

$$D_i = \frac{1}{3} - 2p_i. \tag{237}$$

Similarly, the probability that qubits i, j are both one is

$$n_{ij} = \frac{5}{6} - D_i - D_j + w_{ij}.$$

Using $\lambda_{Z_iZ_j} = 1 - 2n_i - 2n_j + 4n_{ij} = 0$ gives

$$\boxed{w_{ij} = \frac{1}{12} - \frac{p_i + p_j}{2} = \frac{r - p_i - p_j}{2} = \frac{1}{2} \sum_{k \notin \{i,j\}} p_k.} \tag{238}$$

In particular, if at least three of the p_i's are positive, then every v_{ij} is nonzero.

The operator identity (221) also reconstructs the weight-three amplitudes from u and v. Conjugating Eq. (218) gives the coefficient of $\mathcal{R}^\dagger|0_L\rangle$ at $|\{i,j\}\rangle$. Applying $\mathcal{R}$ once more and using $Q|0_L\rangle = r|0_L\rangle$ yields

$$\boxed{a_{\{i,j,k\}} = -\frac{1}{b}\left(u_i\overline{v_{jk}} + u_j\overline{v_{ik}} + u_k\overline{v_{ij}}\right).} \tag{239}$$

We next use the weight-one cross systems. For fixed i, the reciprocal pairs for X_i are indexed by $R \in \binom{[6]\setminus\{i\}}{2}$. Walsh reduction of the equations for X_i and X_iZ_j, $j \neq i$, gives the five star equations

$$\sum_{R:\, j \in R} \overline{v_R}\, a_{[6]\setminus(R\cup\{i\})} = 0 \qquad (j \neq i). \tag{240}$$

For distinct i, ℓ, define the complementary four-vertex hafnian

$$\mathcal{H}_{i\ell} = v_{ab}v_{cd} + v_{ac}v_{bd} + v_{ad}v_{bc}, \qquad \{a,b,c,d\} = [6] \setminus \{i,\ell\}. \tag{241}$$

Substituting Eq. (239) into Eq. (240) and collecting the coefficient of each u_ℓ gives

$$\sum_{\ell \notin \{i,j\}} u_\ell \overline{\mathcal{H}_{i\ell}} = 0 \qquad (i \neq j). \tag{242}$$

For fixed i, put $t_\ell = u_\ell\overline{\mathcal{H}_{i\ell}}$ for $\ell \neq i$ and $T = \sum_{\ell \neq i} t_\ell$. Equation (242) says $T - t_j = 0$ for every $j \neq i$. Thus every $t_j = T$, while summing gives $T = 5T$. Therefore

$$\boxed{u_\ell \mathcal{H}_{i\ell} = 0 \qquad (i \neq \ell).} \tag{243}$$

By symmetry, $\mathcal{H}_{i\ell} = 0$ whenever at least one of i, ℓ belongs to $\operatorname{supp} u$.

8.5 The singleton support has size at most two

Lemma 8.4 (Singleton-support bound)**.** *For every exact orbit-8 solution,*

$$|\operatorname{supp} u| \le 2. \tag{244}$$

Proof. Assume $s := |\operatorname{supp} u| \ge 3$. Equation (238) then makes all fifteen edge amplitudes v_{ij} nonzero. Choose an active vertex and call it 0; relabel the other vertices as $1, \dots, 5$. The algebraic vertex scaling

$$v_{ij} \longmapsto d_i d_j v_{ij}, \qquad d_i \in \mathbb{C}^\times, \tag{245}$$

multiplies the three monomials in each four-vertex hafnian by a common nonzero factor and hence preserves its vanishing. Choose the d_i's so that

$$v_{0j} = 1 \qquad (j = 1, \dots, 5), \tag{246}$$

and write $y_{ij} = v_{ij}$ for $1 \le i < j \le 5$.

The hafnian on $\{0, i, j, k\}$ is then

$$y_{ij} + y_{ik} + y_{jk}. \tag{247}$$

By Eq. (243), this triangle equation is required unless the complementary pair consists of two inactive vertices.

If $s = 5$ or 6, every one of the ten triangle equations is required. The 10×10 triangle-versus-edge incidence matrix of K_5 has determinant -48, so all y_{ij} vanish. This contradicts the nonzero-edge conclusion above.

Suppose $s = 4$. Relabel so that $0, 1, 2, 3$ are active and $4, 5$ are inactive. The only omitted triangle equation is the one on $\{1, 2, 3\}$. Solving the other nine equations gives, for one scalar t,

$$\begin{aligned} y_{12} = y_{13} = y_{23} = y_{45} = t, \\ y_{14} = y_{15} = y_{24} = y_{25} = y_{34} = y_{35} = -\frac{t}{2}. \end{aligned} \tag{248}$$

All edges are nonzero, so $t \neq 0$. The four-set $\{1, 2, 4, 5\}$ has complementary pair $\{0, 3\}$, which meets the active support, and hence its hafnian must vanish. Equation (248) gives instead

$$y_{12}y_{45} + y_{14}y_{25} + y_{15}y_{24} = \frac{3}{2}t^2 \neq 0,$$

a contradiction.

Finally suppose $s = 3$, with active set $\{0, 1, 2\}$ and inactive set $\{3, 4, 5\}$. The omitted triangle equations are those on $\{1, 2, 3\}$, $\{1, 2, 4\}$, and $\{1, 2, 5\}$. The other seven equations give scalars a, b, d such that

$$\begin{aligned} y_{13} = y_{23} = y_{45} = a, \\ y_{14} = y_{24} = y_{35} = b, \\ y_{15} = y_{25} = y_{34} = -(a + b), \\ y_{12} = d. \end{aligned} \tag{249}$$

All edges are nonzero, so $a, b \neq 0$. The hafnians on $\{1, 3, 4, 5\}$, $\{1, 2, 3, 5\}$, and $\{1, 2, 4, 5\}$ are all forced to vanish because their complementary pairs meet the active set. They give, respectively,

$$a^2 + ab + b^2 = 0, \tag{250}$$

$$db - 2a(a + b) = 0, \tag{251}$$

$$da - 2b(a + b) = 0. \tag{252}$$

Equation (250) implies $a(a + b) = -b^2$ and $b(a + b) = -a^2$. Hence Eqs. (251)–(252) give $d = -2b = -2a$, so $a = b$. Then Eq. (250) becomes $3a^2 = 0$, contradicting $a \neq 0$.

The cases $s = 3, 4, 5, 6$ are all impossible, proving Eq. (244). □

8.6 Enumerator range and exact attainability

Equations (215), (235), and (244) give

$$\frac{1}{2} = 36\frac{(1/6)^2}{2} \le A_1 \le 36\left(\frac{1}{6}\right)^2 = 1. \tag{253}$$

Put $\alpha = A_1$. Equation (232) gives $A_2 = 1-\alpha$ and $A_3 = 0$. Substituting into the MacWilliams normal form (105) gives

$$(A_0, \dots, A_6) = (1, \alpha, 1-\alpha, 0, 11+4\alpha, 16-\alpha, 3-3\alpha)\,. \tag{254}$$

The quantum MacWilliams transform gives Eq. (206). This proves the inclusion $\mathcal{W}_8 \subseteq$ the right-hand side of Eq. (205).

It remains to realize every point. Start from the five-qubit perfect code. As recalled in Appendix B.1 of Ref. [1], it admits a projective-order-three transversal element $F^{\otimes 5}$. Diagonalize the one-qubit factor, choose a logical eigenbasis, and append a sixth qubit in one eigenstate. Up to a global phase and generator inversion this gives an exact orbit-8 endpoint with physical action $D^{\otimes 6}$, one physical qubit r fixed to $|1\rangle$, and

$$A = (1, 1, 0, 0, 15, 15, 0). \tag{255}$$

Choose one of the five code qubits, denoted s, and apply on r, s the excitation-preserving unitary

$$\begin{aligned} &U_{rs}(\theta)|00\rangle = |00\rangle, && U_{rs}(\theta)|11\rangle = |11\rangle, \\ &U_{rs}(\theta)|01\rangle = \cos\theta|01\rangle + \sin\theta|10\rangle, \\ &U_{rs}(\theta)|10\rangle = -\sin\theta|01\rangle + \cos\theta|10\rangle. \end{aligned} \tag{256}$$

Because $U_{rs}(\theta)$ commutes with $D \otimes D$, the transformed code stays in the same lift-aware orbit.

The distance-three KL equations also remain exact. Indeed, for any six-qubit Pauli E of weight at most two, compression through the fixed ancilla gives

$$F_E = \langle 1|_r U_{rs}(\theta)^\dagger E U_{rs}(\theta)|1\rangle_r, \tag{257}$$

which is an operator of support at most two on the five code qubits. The five-qubit code detects every such operator, so the transformed six-qubit projector satisfies the complete weight-two KL system.

Before the rotation, the reduced state of the normalized code projector on r, s is $|1\rangle\langle 1| \otimes I/2$. After the rotation it is

$$\rho_{rs}(\theta) = \frac{1}{2}|11\rangle\langle 11| + \frac{1}{2}|\psi_\theta\rangle\langle\psi_\theta|, \qquad |\psi_\theta\rangle = \cos\theta|10\rangle - \sin\theta|01\rangle. \tag{258}$$

The only nonzero weight-one expectations are

$$\langle Z_r\rangle = -\cos^2\theta, \qquad \langle Z_s\rangle = -\sin^2\theta,$$

and the only nonzero weight-two expectations are

$$\langle X_r X_s\rangle = \langle Y_r Y_s\rangle = -\sin\theta\cos\theta.$$

All other weight-one and weight-two expectations vanish by the nondegeneracy of the five-qubit perfect code together with the compression argument in Eq. (257). Therefore

$$\alpha = A_1 = \cos^4\theta + \sin^4\theta = 1 - \frac{1}{2}\sin^2(2\theta). \tag{259}$$

As θ varies, Eq. (259) attains every value in $[1/2, 1]$. This proves the reverse inclusion and completes the proof of Theorem 8.1.

Corollary 8.5 (Raising-Gram spectrum)**.** *Let*

$$(G_+)_{ij} = \langle 0_L | \sigma_i^- \sigma_j^+ | 0_L \rangle.$$

Every exact orbit-8 solution satisfies

$$\operatorname{spec}(G_+) = \left\{0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right\}, \qquad G_+^2 = \frac{1}{2} G_+. \tag{260}$$

Proof. Equality in Eq. (227) gives

$$G_+ = \frac{1}{2} I - 6 \operatorname{diag}(p_1, \ldots, p_6) + 3uu^\dagger.$$

By Lemma 8.4, u has support at most two, and $\sum_i p_i = 1/6$. If the support has size one, its active diagonal entry is zero and the other five eigenvalues are $1/2$. If the support has size two, the active 2×2 block has trace $1/2$ and determinant zero, while the remaining four coordinate directions have eigenvalue $1/2$. Equation (260) follows. □

9 Unique weight enumerator of order-three orbit 9

Orbit 9 is represented by

$$(q, m, a) = \big(3, 3, (1, 1, 1, 1, 1, 2)\big). \tag{261}$$

Let the coefficient 2 occur on qubit 6. Conjugating that qubit by X_6 changes its local factor $\operatorname{diag}(1, \omega^2)$ into a scalar multiple of $\operatorname{diag}(1, \omega)$, where $\omega = e^{2\pi i/3}$. The same cube translation sends the two logical fibers to the uniform Hamming-residue sectors

$$\mathcal{H}_1 = \{x \in \mathbb{F}_2^6 : \operatorname{wt}(x) \equiv 1 \pmod 3\}, \qquad \mathcal{H}_2 = \{x \in \mathbb{F}_2^6 : \operatorname{wt}(x) \equiv 2 \pmod 3\}. \tag{262}$$

Since a Boolean-cube translation is a local Pauli transformation, it does not change the Shor–Laflamme weight enumerators. We therefore work in the uniform gauge (262).

Let $\mathcal{W}_9$ denote the set of A-enumerators of all exact $((6, 2, 3))$ solutions in orbit 9, including every boundary-support solution. In contrast with orbit 8, whose enumerator image is an interval, orbit 9 is rigid at the enumerator level.

Theorem 9.1 (Exact weight-enumerator image of orbit 9)**.** *One has*

$$\boxed{\mathcal{W}_9 = \{(1, 0, 1, 0, 11, 16, 3)\}.} \tag{263}$$

The corresponding B-enumerator is uniquely

$$\boxed{(B_0, \ldots, B_6) = (1, 0, 1, 24, 35, 40, 27).} \tag{264}$$

Moreover, in the uniform gauge every exact orbit-9 code has a weight-two Pauli scalar $Z_r Z_s P = P$ for a pair of physical qubits r, s.

The proof occupies the remainder of this section.

9.1 Uniform-shell form and diagonal moment reduction

Using the subset notation introduced in the orbit-8 analysis, an arbitrary orbit-9 solution in the uniform gauge has the form

$$|0_L\rangle = \sum_{i=1}^{6} x_i|\{i\}\rangle + \sum_{1\le i<j\le 6} d_{ij}|\overline{\{i,j\}}\rangle, \tag{265}$$

$$|1_L\rangle = \sum_{1\le i<j\le 6} c_{ij}|\{i,j\}\rangle + \sum_{i=1}^{6} y_i|\overline{\{i\}}\rangle. \tag{266}$$

Thus $|0_L\rangle$ occupies the shells of weights one and four, while $|1_L\rangle$ occupies the shells of weights two and five. Put

$$p_i = |x_i|^2, \qquad q_i = |y_i|^2, \qquad r_{ij} = |d_{ij}|^2, \qquad s_{ij} = |c_{ij}|^2, \tag{267}$$

and define the weighted edge degrees

$$R_i = \sum_{j\neq i} r_{ij}, \qquad S_i = \sum_{j\neq i} s_{ij}. \tag{268}$$

Lemma 9.2 (Diagonal moment normal form)**.** *Every exact orbit-9 solution satisfies*

$$\sum_i p_i = \sum_i q_i = \frac{1}{3}, \qquad \sum_{i<j} r_{ij} = \sum_{i<j} s_{ij} = \frac{2}{3}, \tag{269}$$

and, for every i,

$$R_i = \frac{1/3 - p_i + 3q_i}{2}, \qquad S_i = \frac{1/3 - q_i + 3p_i}{2}. \tag{270}$$

Proof. Let

$$P_1 = \sum_i p_i, \qquad Q_5 = \sum_i q_i$$

be the total weight-one probability of $|0_L\rangle$ and the total weight-five probability of $|1_L\rangle$. Equality of the sums of the six one-coordinate moments is equality of mean Hamming weight:

$$4 - 3P_1 = 2 + 3Q_5. \tag{271}$$

Equality of the sums of the fifteen two-coordinate moments is equality of $\binom{\mathrm{wt}}{2}$:

$$6(1 - P_1) = 1 + 9Q_5. \tag{272}$$

Solving gives $P_1 = Q_5 = 1/3$, and normalization then gives the two edge masses in Eq. (269).

Fix a coordinate i. Equality of its occupation probabilities in the two logical states reads

$$p_i + \frac{2}{3} - R_i = S_i + \frac{1}{3} - q_i. \tag{273}$$

Now sum the common two-coordinate occupation probabilities over all pairs containing i. In $|0_L\rangle$, every weight-four word containing i contributes three times, while the singleton $|i\rangle$ contributes zero; hence the sum is $2 - 3R_i$. In $|1_L\rangle$, the weight-two part contributes S_i, while every weight-five word containing i contributes four times; hence the sum is $S_i + 4(1/3 - q_i)$. Therefore

$$2 - 3R_i = S_i + \frac{4}{3} - 4q_i. \tag{274}$$

Solving Eqs. (273) and (274) gives Eq. (270). □

9.2 The single-flip Walsh equations

Fix $i \in [6]$. For $j \neq i$, the flip X_i has exactly two reciprocal incidences indexed by j:

$$|\{i,j\}\rangle \longleftrightarrow |\{j\}\rangle, \qquad |\overline{\{j\}}\rangle \longleftrightarrow |\overline{\{i,j\}}\rangle.$$

Set

$$A_j = \overline{x_j} c_{ij}, \qquad B_j = \overline{d_{ij}} y_j. \tag{275}$$

The cross-KL equation for X_i is

$$\sum_{j \neq i} (A_j + B_j) = 0. \tag{276}$$

For a fixed $j \neq i$, dressing by Z_j changes the sign of the A_j incidence and of every B_k incidence with $k \neq j$, while leaving the other incidences unchanged. Thus the equation for $X_i Z_j$ is

$$\sum_{k \neq i} A_k - 2A_j - \sum_{k \neq i} B_k + 2B_j = 0. \tag{277}$$

Writing $A = \sum_{k \neq i} A_k$ and $B = \sum_{k \neq i} B_k$, Eq. (276) gives $B = -A$, and Eq. (277) gives $A_j - B_j = A$ for all five values of j. Summing these five identities gives $2A = 5A$, so $A = 0$. We have proved the exact Walsh reduction

$$\boxed{\overline{x_j} c_{ij} = \overline{d_{ij}} y_j \quad (j \neq i), \qquad \sum_{j \neq i} \overline{x_j} c_{ij} = 0.} \tag{278}$$

Taking the conjugate of the second equation after using the first also gives

$$\sum_{j \neq i} d_{ij} \overline{y_j} = 0. \tag{279}$$

All operators used above have Pauli weight at most two.

Lemma 9.3 (Complementary singleton probabilities)**.** *For every i,*

$$p_i = q_i, \qquad R_i = S_i = \frac{1}{6} + p_i. \tag{280}$$

Consequently all one-body Pauli scalars vanish and

$$A_1 = 0. \tag{281}$$

Proof. Taking absolute values in the first identity of Eq. (278), first with flip center i and then with flip center j, gives

$$p_j s_{ij} = q_j r_{ij}, \qquad p_i s_{ij} = q_i r_{ij}. \tag{282}$$

Summing the second identity over $j \neq i$ yields $p_i S_i = q_i R_i$. Substituting Eq. (270) gives

$$(p_i - q_i)(1 + 9p_i + 9q_i) = 0. \tag{283}$$

The second factor is strictly positive, so $p_i = q_i$. Equation (280) follows from Eq. (270).

The common occupation probability of qubit i is now

$$p_i + \frac{2}{3} - R_i = \frac{1}{2},$$

so its detectable Z_i scalar is zero. The code projector commutes with $D^{\otimes 6}$, and conjugation by D rotates the span of X_i, Y_i without a nonzero fixed vector. Hence the X_i and Y_i scalars also vanish. Summing their squared moduli over all six qubits proves Eq. (281). □

9.3 Pair flips and the active support graph

Let

$$\Sigma = \{i : p_i > 0\}. \tag{284}$$

For $i \in \Sigma$, define the unit phase

$$\rho_i = \frac{y_i}{\overline{x_i}}. \tag{285}$$

Equation (278) implies

$$c_{ij} = \rho_i \overline{d_{ij}} = \rho_j \overline{d_{ij}} \tag{286}$$

whenever $i, j \in \Sigma$ and $d_{ij} \neq 0$. Thus the phases ρ_i are constant on each connected component of the graph on Σ whose edges are the nonzero d_{ij}'s.

For a four-set $Q = \{i, j, k, \ell\} \subseteq [6]$, flip the two coordinates in its complement. The six reciprocal incidences in the cross-KL equation give

$$\begin{aligned} 0 = {} & \overline{d_{ij}} c_{k\ell} + \overline{d_{k\ell}} c_{ij} + \overline{d_{ik}} c_{j\ell} + \overline{d_{j\ell}} c_{ik} \\ & + \overline{d_{i\ell}} c_{jk} + \overline{d_{jk}} c_{i\ell}. \end{aligned} \tag{287}$$

The allowed X/Y dressings on the two flipped coordinates have the same sign on all six incidences, so Eq. (287) is the complete pair-flip reduction.

Lemma 9.4 (Active-component lemma)**.** *Every nontrivial connected component of the active support graph has exactly four vertices and is a complete graph K_4.*

Proof. Let $K \subseteq \Sigma$ be a connected component. The common phase on K, Eq. (287), and Eq. (279) give, for distinct vertices in K,

$$\sum_{j \in K} d_{ij} x_j = 0, \tag{288}$$

$$d_{ij} d_{k\ell} + d_{ik} d_{j\ell} + d_{i\ell} d_{jk} = 0. \tag{289}$$

Put

$$m_{ij} = d_{ij} x_i x_j \qquad (i, j \in K). \tag{290}$$

Then the symmetric zero-diagonal matrix $M = (m_{ij})$ satisfies

$$\sum_{j \in K} m_{ij} = 0 \tag{291}$$

and, on every four vertices,

$$m_{ij} m_{k\ell} + m_{ik} m_{j\ell} + m_{i\ell} m_{jk} = 0. \tag{292}$$

Fix a triangle $\{i, j, k\} \subseteq K$ and sum Eq. (292) over the fourth vertex. Using the three row-sum identities gives

$$m_{ij} m_{ik} + m_{ij} m_{jk} + m_{ik} m_{jk} = 0. \tag{293}$$

If two edges of a triangle are nonzero, Eq. (293) forces the third to be nonzero. Along a path in a connected component this closes every triangle, so the component is a clique.

A clique of size two is impossible by the row-sum equation. A clique of size three is impossible because its zero-diagonal matrix has determinant $2 m_{12} m_{13} m_{23} \neq 0$, whereas Eq. (291) supplies a nonzero kernel vector.

A clique of size five is also impossible. Vertex scaling $m_{ij} \mapsto t_i t_j m_{ij}$, with all $t_i \neq 0$, preserves the hafnian equations and singularity. Normalize the four edges incident on a chosen vertex to one. The four hafnian equations through that vertex force, for nonzero scalars $E, F, E+F$, the matrix

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & F & E & -E-F \\ 1 & F & 0 & -E-F & E \\ 1 & E & -E-F & 0 & F \\ 1 & -E-F & E & F & 0 \end{pmatrix}. \tag{294}$$

Its determinant is

$$32EF(E+F) \neq 0, \tag{295}$$

contradicting singularity.

Finally, a component of size six is impossible. Normalize the five edges from one vertex to one. Each four-set containing that vertex gives a triangle equation

$$y_{ij} + y_{ik} + y_{jk} = 0$$

on the five remaining vertices. The resulting ten-by-ten triangle-versus-edge incidence matrix of K_5 has determinant

$$-48 \neq 0. \tag{296}$$

Hence all ten neighbor edges would vanish, contradicting that the component is a clique. The only possible nontrivial component size is therefore four. □

9.4 The singleton support has exactly four vertices

Lemma 9.5 (Four active singleton coordinates)**.** *Every exact orbit-9 solution satisfies*

$$|\Sigma| = 4. \tag{297}$$

The active support graph is the complete graph on those four vertices.

Proof. Put $s = |\Sigma|$. Equation (280) and $\sum_i p_i = 1/3$ give the active and inactive degree budgets

$$\sum_{i \in \Sigma} R_i = \frac{s}{6} + \frac{1}{3}, \qquad \sum_{i \notin \Sigma} R_i = \frac{6-s}{6}. \tag{298}$$

The case $s = 1$ is impossible: for every inactive vertex, the kernel Eq. (279) contains only the one active summand and therefore kills the corresponding edge, leaving the active vertex with zero degree. If $s = 3$, Lemma 9.4 makes all active vertices isolated, so the active-to-inactive edge weight is $5/6$, larger than the total inactive degree $1/2$. If $s = 5$, there is an isolated active vertex, whether or not the other four form a K_4; its degree is strictly larger than $1/6$, but the sole inactive vertex has total degree exactly $1/6$. If $s = 6$, Lemma 9.4 again leaves at least one isolated active vertex, now with no inactive vertex to which it can connect. Thus only $s = 2$ and $s = 4$ remain.

Assume $s = 2$, with active vertices a, b. The row-kernel equation at an active vertex gives $d_{ab} = 0$. For every inactive vertex t, the row-kernel equation is

$$d_{ta}\overline{y_a} + d_{tb}\overline{y_b} = 0. \tag{299}$$

Taking moduli, summing over the four inactive vertices, and using Eq. (280) gives

$$p_a\left(\frac{1}{6} + p_a\right) = p_b\left(\frac{1}{6} + p_b\right). \tag{300}$$

The function $t \mapsto t(1/6+t)$ is strictly increasing for $t \geq 0$, while $p_a + p_b = 1/3$. Hence

$$p_a = p_b = \frac{1}{6}. \tag{301}$$

Equation (299) now gives $r_{ta} = r_{tb}$. Since the active degree is $1/3$, the average value of the four r_{ta}'s is $1/12$. The inactive degree is only $1/6$, so every $r_{ta} \leq 1/12$. Therefore

$$r_{ta} = r_{tb} = \frac{1}{12} \quad \text{for every inactive } t, \tag{302}$$

and all inactive–inactive d-edges vanish.

Local diagonal phases commute with the uniform physical lift. Use them to set $x_a = x_b = 1/\sqrt{6}$, and use the logical phase of $|1_L\rangle$ to set $\rho_a = 1$. Write $\rho_b = \eta$, $|\eta| = 1$. Equations (278) and (299) give

$$d_{bt} = -\eta d_{at}, \qquad c_{at} = \overline{d_{at}}, \qquad c_{bt} = -\overline{d_{at}}. \tag{303}$$

For two distinct inactive vertices t, u, the pair-flip equation on $\{a, b, t, u\}$ reduces to

$$-2(1+\overline{\eta})\overline{d_{at}}\,\overline{d_{au}} = 0. \tag{304}$$

All displayed edge amplitudes are nonzero, so $\eta = -1$.

Now consider the detectable weight-two operator $L_{ab} = \sigma_a^+ \sigma_b^-$, which lies in the linear span of weight-two Paulis. Equations (301)–(303) give

$$\langle 0_L | L_{ab} | 0_L \rangle = \overline{x_a} x_b + \sum_t \overline{d_{bt}} d_{at} = \frac{1}{6} + 4\left(\frac{1}{12}\right) = \frac{1}{2}, \tag{305}$$

$$\langle 1_L | L_{ab} | 1_L \rangle = \overline{y_b} y_a + \sum_t \overline{c_{at}} c_{bt} = -\frac{1}{6} - 4\left(\frac{1}{12}\right) = -\frac{1}{2}. \tag{306}$$

This violates the diagonal KL equality. Hence $s \neq 2$, and therefore $s = 4$.

If the four active vertices were all isolated, their total degree would be one, exceeding the total degree $1/3$ of the two inactive vertices. Consequently Lemma 9.4 forces the active graph to be a single K_4. □

9.5 A forced weight-two Pauli stabilizer

Relabel the four active vertices as $1, 2, 3, 4$ and the two inactive vertices as r, s. On the active K_4, define

$$m_{ij} = d_{ij} x_i x_j. \tag{307}$$

The row-sum equations and the single four-vertex hafnian equation give nonzero scalars A, B, C such that

$$\begin{aligned} &m_{12} = m_{34} = A, \\ &m_{13} = m_{24} = B, \qquad A + B + C = 0, \qquad A^2 + B^2 + C^2 = 0. \\ &m_{14} = m_{23} = C, \end{aligned} \tag{308}$$

For the inactive vertex r, put

$$z_i = d_{ri} x_i, \qquad i = 1, 2, 3, 4. \tag{309}$$

The four pair-flip equations involving r and three active vertices are

$$\begin{pmatrix} C & B & A & 0 \\ B & C & 0 & A \\ A & 0 & C & B \\ 0 & A & B & C \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{pmatrix} = 0. \tag{310}$$

The all-one vector lies in the kernel because $A + B + C = 0$. The upper-left three-by-three minor is

$$-C(A^2 + B^2 - C^2) = 2C^3 \neq 0, \tag{311}$$

where the second equality uses $A^2 + B^2 + C^2 = 0$. Thus the matrix in Eq. (310) has rank three and its kernel is precisely the span of the all-one vector. On the other hand, the inactive-row kernel Eq. (279) gives

$$z_1 + z_2 + z_3 + z_4 = 0. \tag{312}$$

Therefore every $z_i = 0$, and there is no d-edge between r and the active K_4. The same argument applies to s. The single-flip identity then removes the corresponding c-edges as well.

It follows that every basis string in either logical word has equal bits on qubits r and s:

- the singleton terms and the active–active weight-two terms have $(x_r, x_s) = (0, 0)$;
- the active–active weight-four terms and the active weight-five terms have $(x_r, x_s) = (1, 1)$;
- the possible edge on $\{r, s\}$ has equal bits as well.

Consequently

$$\boxed{Z_r Z_s P = P.} \tag{313}$$

In the original orbit-9 gauge the scalar may be -1, depending on the chosen cube translation, but its modulus remains one and the enumerator is unchanged.

9.6 Enumerator rigidity and exact attainment

Equation (313) contributes one unit to A_2, so

$$A_2 \geq 1. \tag{314}$$

The even-qubit spin-flip inequality (231) gives $A_1 + A_2 + A_3 \leq 1$. Combining this with Eqs. (281) and (314) yields

$$\boxed{A_1 = 0, \qquad A_2 = 1, \qquad A_3 = 0.} \tag{315}$$

Substitution into the six-qubit MacWilliams normal form (105) gives

$$\boxed{(A_0, \ldots, A_6) = (1, 0, 1, 0, 11, 16, 3).} \tag{316}$$

The quantum MacWilliams transform gives Eq. (264).

For completeness, nonemptiness can be exhibited exactly. Put $\zeta_6 = e^{i\pi/3}$, and for a triple (x, k, d) write the coefficient $\zeta_6^k/\sqrt{d}$ on $|x\rangle$. In the lists immediately below, the displayed string is

$x_6x_5x_4x_3x_2x_1$: its rightmost bit is qubit 1, and its integer value is $\sum_{j=1}^{6} x_j 2^{j-1}$. The tensor basis vector is $|x_1\rangle \otimes \cdots \otimes |x_6\rangle$. In the original gauge (261), take

$$\begin{aligned} L_0 = \{&(000000, 5, 12), (000111, 2, 12), (001110, 3, 12), (010110, 3, 12),\\ &(011001, 2, 6), (100001, 0, 12), (101000, 3, 12), (101111, 2, 12),\\ &(110000, 5, 12), (110111, 0, 12), (111110, 5, 12)\},\\ L_1 = \{&(000001, 5, 12), (001000, 4, 12), (001111, 5, 12), (010000, 2, 12),\\ &(010111, 1, 12), (011110, 4, 12), (100110, 5, 6), (101001, 1, 12),\\ &(110001, 1, 12), (111000, 2, 12), (111111, 5, 12)\}. \end{aligned}$$

Define $|j_L\rangle$ by summing these coefficients over L_j. Each word has ten probabilities $1/12$ and one probability $1/6$. Direct substitution over $\mathbb{Q}(\sqrt{3}, i)$ verifies the Knill–Laflamme conditions for all 153 Pauli operators of weights one and two, establishing an exact $((6, 2, 3))$ code in orbit 9. Hence the singleton set in Eq. (263) is attained, completing the proof of Theorem 9.1.

Corollary 9.6 (Orbit-9 magnitude pattern)**.** *In the uniform gauge, after relabeling the active set as* $\{1, 2, 3, 4\}$ *and the inactive pair as* $\{r, s\}$*, every exact orbit-9 solution satisfies*

$$p_i = q_i = \frac{1}{12} \quad (i = 1, 2, 3, 4), \qquad r_{ij} = s_{ij} = \frac{1}{12} \quad (1 \le i < j \le 4), \tag{317}$$

and

$$r_{rs} = s_{rs} = \frac{1}{6}, \tag{318}$$

with all other displayed probabilities zero. Thus each logical word has ten coefficients of squared magnitude $1/12$ *and one of squared magnitude* $1/6$*.*

Proof. The stabilizer in Eq. (313) already contributes all of $A_2 = 1$, so every other weight-two Pauli scalar vanishes. In particular, for active $i \neq j$, $\lambda_{Z_iZ_j} = 0$, and the common pair-occupation probability is $1/4$. Evaluating it in $|0_L\rangle$ gives

$$r_{ij} = p_i + p_j - \frac{1}{12}. \tag{319}$$

Summing this over the three active neighbors of i and using $R_i = 1/6 + p_i$ and $\sum_{j\in\Sigma} p_j = 1/3$ yields $p_i = 1/12$. Equation (319) then gives $r_{ij} = 1/12$. The inactive degree is $R_r = R_s = 1/6$, and the cross edges have already vanished, so $r_{rs} = 1/6$. For active–active edges, Eq. (278) gives the same magnitudes for the c-amplitudes. The inactive degree $S_r = S_s = 1/6$, together with the absence of cross edges, gives $s_{rs} = 1/6$. This proves the claim. □

10 Exact families in primitive order-two orbits

We now give the constructions needed for the positive inclusions in orbits 1–4. In the notation of the main text, set

$$\mathcal{L}(\alpha) = (1, \alpha, 1-\alpha, 0, 11+4\alpha, 16-\alpha, 3-3\alpha), \tag{320}$$

$$\mathcal{M}(t) = (1, 0, t, 1-t, 12-t, 15+t, 3). \tag{321}$$

The four records have $q = m = 2$, with active-weight representatives

r	a	$\mathrm{wt}(D(a;2))$
1	$(0,0,0,1,1,1)$	3
2	$(0,0,1,1,1,1)$	4
3	$(0,1,1,1,1,1)$	5
4	$(1,1,1,1,1,1)$	6

(322)

A product Pauli of the indicated weight acting as a nontrivial logical involution therefore suffices: local Clifford conjugation and a qubit permutation put it in the displayed diagonal form, and its logical eigenvectors lie in the required opposite-parity fibers. These operations preserve distance and the weight enumerators.

Theorem 10.1 (Exact family inclusions for orbits 1–4)**.** *One has*

$$\{\mathcal{L}(\alpha) : 0 \le \alpha \le 1\} \subseteq \bigcap_{r=1}^{4} \mathcal{W}_r, \tag{323}$$

$$\{\mathcal{M}(t) : 2/3 \le t \le 1\} \subseteq \mathcal{W}_1 \cap \mathcal{W}_4. \tag{324}$$

Every point in these inclusions is attained by an exact $((6,2,3))$ *code, including the endpoints.*

10.1 The split-qubit construction of the $\mathcal{L}$-interval

Let P_5 be the standard five-qubit perfect-code projector, with stabilizer generators

$$XZZXI, \qquad IXZZX, \qquad XIXZZ, \qquad ZXIXZ. \tag{325}$$

Every nonidentity Pauli of weight at most two anticommutes with at least one of these generators. Thus

$$P_5 E P_5 = 0 \qquad (1 \le \mathrm{wt}(E) \le 2). \tag{326}$$

For any isometry $W : \mathbb{C}^2 \longrightarrow \mathbb{C}^2 \otimes \mathbb{C}^2$, replace the first physical qubit by its two outputs and write $V = W \otimes I^{\otimes 4}$, $P_6 = V P_5 V^\dagger$. The compression $V^\dagger E V$ of a six-qubit Pauli of weight at most two is supported on at most two original qubits. Expanding it in the original Pauli basis and using Eq. (326) gives $P_6 E P_6 = \lambda_E P_6$. This proves the full distance-three KL conditions for every such isometry.

More precisely, if a nonidentity E of weight at most two has any nonidentity factor on an unsplit qubit, all Pauli terms in $V^\dagger E V$ are nonidentity, and $\lambda_E = 0$. If it is supported entirely on the split pair, the surviving identity coefficient is

$$\lambda_E = \frac{1}{2}\,\mathrm{Tr}(W^\dagger E W) = \frac{1}{2}\,\mathrm{Tr}(RE), \qquad R = WW^\dagger. \tag{327}$$

For $0 \le \alpha \le 1$, choose $0 \le \theta \le \pi/2$ with $c = \cos\theta = \sqrt{\alpha}$, $s = \sin\theta = \sqrt{1-\alpha}$. Consider the two isometries

$$W_2|0\rangle = \cos(\theta/2)|00\rangle + \sin(\theta/2)|11\rangle, \qquad W_2|1\rangle = \cos(\theta/2)|01\rangle + \sin(\theta/2)|10\rangle, \tag{328}$$

$$W_3|0\rangle = |0\rangle\big(\cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle\big), \qquad W_3|1\rangle = |1\rangle\big(\cos(\theta/2)|0\rangle - \sin(\theta/2)|1\rangle\big). \tag{329}$$

Their image projectors and intertwining relations are

$$W_2 W_2^\dagger = \tfrac{1}{2}(I + c\,Z\otimes I + s\,X\otimes X), \qquad (Z\otimes Z)W_2 = W_2 Z, \tag{330}$$

$$W_3 W_3^\dagger = \tfrac{1}{2}(I + c\,I\otimes Z + s\,Z\otimes X), \qquad (Z\otimes I)W_3 = W_3 Z. \tag{331}$$

Equation (327) shows that the only possibly nonzero scalar coefficients at weights one and two are respectively (c, s) on $(Z \otimes I, X \otimes X)$ for W_2, or on $(I \otimes Z, Z \otimes X)$ for W_3. Hence

$$A_1 = \alpha, \qquad A_2 = 1 - \alpha, \qquad A_3 = 0. \tag{332}$$

Here $A_3 = 0$ follows from $A_3 \geq 0$ and Eq. (231). The full enumerator is $\mathcal{L}(\alpha)$ by Eq. (105).

It remains to identify the logical gate in each orbit. The five-qubit code has logical $Z^{\otimes 5}$. Multiplication by the first generator in Eq. (325) gives $-YIIYZ$, a weight-three representative. After local Clifford conjugation and relabeling, the latter can be taken to be $Z_1 Z_2 Z_3$. Split a qubit in the support of either representative. Equations (330)– (331) then give

original logical weight	isometry	new logical weight	orbit
3	W_3	3	1
3	W_2	4	2
5	W_3	5	3
5	W_2	6	4

(333)

This proves Eq. (323) for the entire closed interval. The choices of the original five-qubit code's local basis may differ between the rows of the table.

10.2 An explicit five-cat construction of the $\mathcal{M}$-interval

The following specialization of the five-cat construction of Ref. [3] gives Eq. (324). Its KL conditions and enumerators are verified explicitly below. Label the first physical qubit separately and the remaining five by $j = 1, \ldots, 5$, so tail label j means physical qubit $j + 1$. For the length-five unit vector u_j and its bitwise complement $\bar{u}_j$, put

$$|S_j\rangle = \frac{|u_j\rangle + |\bar{u}_j\rangle}{\sqrt{2}}. \tag{334}$$

Choose real a, b satisfying $2a^2 + b^2 = 1/4$, and define

$$\gamma = (0, 0, 0, 1/2, i/2), \tag{335}$$

$$\delta = \left(\frac{1}{2\sqrt{2}} + \frac{ib}{\sqrt{2}}, -\frac{1}{2\sqrt{2}} + \frac{ib}{\sqrt{2}}, -i\sqrt{2}a, 0, 0 \right), \qquad e = (a, a, b, 0, 0). \tag{336}$$

For each tail label define the first-qubit spinors

$$x_j = \gamma_j |0\rangle + \delta_j |1\rangle, \qquad y_j = \overline{\delta_j} |0\rangle - \overline{\gamma_j} |1\rangle, \tag{337}$$

and the two codewords

$$|0_L\rangle = \sum_{j=1}^{5} x_j \otimes |S_j\rangle, \qquad |1_L\rangle = \sum_{j=1}^{5} y_j \otimes |S_j\rangle. \tag{338}$$

The displayed coefficients obey

$$\|\gamma\|^2 = \|\delta\|^2 = \tfrac{1}{2}, \qquad \gamma^\dagger \delta = 0, \qquad \gamma^T \gamma = \delta^T \delta = \gamma^T \delta = 0, \tag{339}$$

$$\mathrm{Re}(x_j^\dagger x_k) = \frac{1}{4} \delta_{jk} - e_j e_k. \tag{340}$$

They follow by substitution using $2a^2 + b^2 = 1/4$; in particular, Eq. (338) is an orthonormal pair.

The full KL check. Let $C : \mathbb{C}^5 \longrightarrow (\mathbb{C}^2)^{\otimes 5}$ send the standard basis vector f_j to $|S_j\rangle$. Directly from Eq. (334), every nonidentity single-tail-qubit Pauli compresses to zero. For distinct tail labels j, k, the only nonzero two-qubit compressions are

$$C^\dagger Z_j Z_k C = I_5 - 2f_j f_j^\dagger - 2f_k f_k^\dagger, \tag{341}$$

$$C^\dagger X_j X_k C = C^\dagger Y_j Y_k C = f_j f_k^\dagger + f_k f_j^\dagger. \tag{342}$$

All mixed two-qubit Pauli types compress to zero: terms with exactly one X/Y flip leave the five-cat support, while X_jY_k and Y_jX_k cancel between complementary strings. Thus every Hermitian tail error of weight at most two compresses to a real symmetric matrix.

For any real symmetric matrix $H = (H_{jk})$, the relations

$$y_j^\dagger y_k = x_k^\dagger x_j, \qquad x_j^\dagger y_k = \overline{\gamma_j \delta_k - \delta_j \gamma_k} = -x_k^\dagger y_j \tag{343}$$

imply

$$\sum_{j,k} H_{jk} x_j^\dagger x_k = \sum_{j,k} H_{jk} y_j^\dagger y_k, \qquad \sum_{j,k} H_{jk} x_j^\dagger y_k = 0.$$

This proves the KL equations for errors entirely in the tail. For errors on the first physical qubit alone, Eq. (339) gives

$$\sum_j x_j x_j^\dagger = \sum_j y_j y_j^\dagger = \tfrac{1}{2} I_2, \qquad \sum_j y_j x_j^\dagger = 0. \tag{344}$$

Finally, an error with nonidentity factors on both the first qubit and one tail qubit has zero matrix elements, since the single-tail-qubit compression is zero. These cases exhaust all Pauli errors of weight at most two, proving the full KL conditions without divisions or nonzero-amplitude assumptions.

The enumerator and its parameter range. The weight-one scalar coefficients all vanish. Equations (340)–(342) give all nonzero weight-two coefficients, in tail notation:

$$\lambda_{Z_jZ_k} = 2(e_j^2 + e_k^2), \qquad \lambda_{X_jX_k} = \lambda_{Y_jY_k} = -2e_j e_k \quad (j < k). \tag{345}$$

Since $\sum_j e_j^2 = 1/4$, their squared sum is

$$A_1 = 0, \qquad A_2 = 4 \sum_{j<k} \big((e_j^2 + e_k^2)^2 + 2e_j^2 e_k^2\big) = \frac{1}{2} + 8 \sum_j e_j^4 = \frac{1}{2} + 8(2a^4 + b^4). \tag{346}$$

Five-qubit spin flip sends each symmetric cat $|S_j\rangle$ into its antisymmetric cat, orthogonal to the span of all the S_k. Hence the six-qubit code is orthogonal to its spin flip, and $\mathrm{Tr}(P\widetilde{P}) = 0$. The equality case of Eq. (230) therefore yields $A_1 + A_2 + A_3 = 1$. With $t = A_2$, Eq. (105) now gives precisely $\mathcal{M}(t)$.

For an explicit path covering the required closed interval, take

$$a = \sqrt{\frac{1-r}{12}}, \qquad b = \sqrt{\frac{1+2r}{12}}, \qquad 0 \le r \le 1. \tag{347}$$

Then $2a^2 + b^2 = 1/4$ and Eq. (346) becomes

$$t = \frac{2}{3} + \frac{r^2}{3}. \tag{348}$$

Every $t \in [2/3, 1]$ is attained, including the zero-coefficient endpoint $r = 1$, for which the same KL proof applies.

The weight-three and weight-six logical gates. In physical-qubit notation put

$$L_3 = Z_1 Z_5 Z_6, \qquad S_X = X_2 X_3 X_4 X_5 X_6. \tag{349}$$

For cat labels $1, 2, 3$, the tail factor $Z_5 Z_6$ has eigenvalue $+1$, while x_j is proportional to $|1\rangle$ and y_j to $|0\rangle$. For labels $4, 5$, the tail eigenvalue is -1, and the first-qubit roles are reversed. Therefore

$$L_3|0_L\rangle = -|0_L\rangle, \qquad L_3|1_L\rangle = |1_L\rangle. \tag{350}$$

Moreover, $S_X|S_j\rangle = |S_j\rangle$ for every cat, so S_X stabilizes both codewords. Their commuting product is

$$L_3 S_X = -Z_1 X_2 X_3 X_4 Y_5 Y_6, \tag{351}$$

a weight-six product Pauli with the same nontrivial logical action. The weight-three operator already gives orbit 1 after relabeling and logical-color interchange. Local Clifford conjugation makes the weight-six product diagonal, giving orbit 4 with the same enumerator. This proves Eq. (324).

Both constructions have distance at least three by the displayed KL arguments. The quantum Singleton bound excludes distance at least four for six physical qubits encoding one qubit, so their distance is exactly three. This completes the proof of Theorem 10.1. The theorem establishes the stated inclusions; it does not assert that they exhaust any of $\mathcal{W}_1, \ldots, \mathcal{W}_4$.

11 Exact weight-enumerator image of orbit 2

Orbit 2 is the primitive order-two record

$$(q, m, a) = \big(2, 2, (0, 0, 1, 1, 1, 1)\big).$$

For the proof, relabel the physical sites in the order $B_1, B_2, A_1, A_2, A_3, A_4$. Write $\mathsf{B} = \{B_1, B_2\}$ for the inactive pair and $\mathsf{A} = \{A_1, A_2, A_3, A_4\}$ for the active sites. Its chosen physical logical involution is

$$D_{\mathrm{phys}} = D(a; 2) = I_{\mathsf{B}} \otimes D, \qquad D = Z_{A_1} Z_{A_2} Z_{A_3} Z_{A_4}. \tag{352}$$

Let $V : \mathbb{C}^2 \to (\mathbb{C}^2)^{\otimes 6}$ be an isometric encoder, with code projector $P = VV^\dagger$, satisfying

$$D_{\mathrm{phys}} V = V Z_L, \qquad V^\dagger E V = \lambda_E I_2 \quad (\mathrm{wt}(E) \le 2), \tag{353}$$

where E ranges over Hermitian Pauli strings. For all Pauli weights put $\lambda_E = \mathrm{Tr}(PE)/2$; this agrees with the KL scalar at weights at most two but does not assert scalar compression at higher weights. We use the supplement's normalization

$$A_w = \frac{1}{4} \sum_{\mathrm{wt}(E)=w} |\,\mathrm{Tr}(PE)|^2 = \sum_{\mathrm{wt}(E)=w} \lambda_E^2.$$

Theorem 11.1 (Complete orbit-2 enumerator image)**.** *Every exact orbit-2 realization has $A_3 = 0$. Its complete Shor–Laflamme A-enumerator image is*

$$\boxed{\mathcal{W}_2 = \{\mathcal{L}(\alpha) : 0 \le \alpha \le 1\}, \qquad \mathcal{L}(\alpha) = (1, \alpha, 1-\alpha, 0, 11+4\alpha, 16-\alpha, 3-3\alpha).} \tag{354}$$

The converse includes every zero-support and singular stratum. No stabilizer assumption is made.

The proof has an analytic reduction and an exact finite certificate. First, KL and complementary purities force a flat inactive pair and a specific rank-eight active projector. Mixed KL supplies one common antiunitary, whose spectral saturation restricts an associated partial isometry to 32 complex Pauli coordinates. A homogeneous quartic certificate then forces its weight-three coordinates, and the code's entire A_3, to vanish. The certificate identity and positivity are verified with integer and rational arithmetic.

11.1 Flatness of the inactive pair and the active reflection

Put $\rho = P/2$. The universal six-qubit rank-two MacWilliams normal form, with $x = A_1, y = A_2, z = A_3$, is

$$(A_0, \dots, A_6) = (1, x, y, z, 15 - 4y - 3z, 12 + 3x + 4y + 3z, 4 - 4x - y - z). \tag{355}$$

It follows from the quantum MacWilliams transform with factor $1/32$ and $A_j = B_j$ for $j = 0, 1, 2$. With full six-qubit spin flip $\widetilde{P} = Y^{\otimes 6} P^T Y^{\otimes 6}$, positivity gives

$$0 \le \mathrm{Tr}(P\widetilde{P}) = \frac{1}{16} \sum_{w=0}^{6} (-1)^w A_w = \frac{1}{2}(1 - A_1 - A_2 - A_3). \tag{356}$$

Lemma 11.2 (Universal flatness and weight-three support)**.** *Every encoder in* (353) *satisfies*

$$\rho_{\mathsf{B}} = \frac{I_4}{4}, \qquad \rho_{\mathsf{B}A_i} = \frac{I_4}{4} \otimes \rho_{A_i} \quad (1 \le i \le 4), \qquad A_1 + A_2 + A_3 = 1. \tag{357}$$

The only possibly nonzero weight-three coefficients are $\lambda_{p_i q_j Z_k}$ on distinct active sites, with $p, q \in \{X, Y\}$. Moreover,

$$\rho_{\mathsf{A}} = \frac{\Pi}{8}, \qquad \Pi^\dagger = \Pi = \Pi^2, \qquad \mathrm{rank}\, \Pi = 8, \qquad [\Pi, D] = 0, \tag{358}$$

and each active parity block of Π has rank four.

Proof. Purify the maximally mixed code with a reference qubit R:

$$|\Psi\rangle = \frac{|0\rangle_R |0_L\rangle + |1\rangle_R |1_L\rangle}{\sqrt{2}}.$$

The operator $G = Z_R I_{\mathsf{B}} D$ fixes this state. KL says that every correlation involving a nonidentity reference Pauli and a physical Pauli of weight at most two vanishes. Multiplication by G preserves squared correlations.

Physical coefficients with an odd number of active X/Y factors vanish by D-parity. An active-only coefficient with at least two Z factors also vanishes: multiplying by G gives a reference-Z correlation with physical weight $4 - z \le 2$, where z is the original active Z-count. For this purity computation, denote the remaining nonidentity active-only squared masses by

mass	a	u	v	w
pattern	Z_i	$p_i q_j$	$p_i q_j Z_k$	$p_1 p_2 p_3 p_4$

$(p, q, p_i \in \{X, Y\})$.

All displayed sites in a pattern are distinct. Let $s = a + u + v + w$, and define the nonnegative masses

$$b = \sum_{\substack{F \in \{I,X,Y,Z\}^{\otimes \mathsf{B}} \\ F \ne I}} \lambda_F^2, \qquad m = \sum_{i=1}^{4} \sum_{F \ne I} \lambda_{F Z_i}^2.$$

Here F in the second sum ranges over all nonidentity spectator Paulis, including those of spectator weight two.

Purity of a marginal on r qubits is 2^{-r} times the sum of its squared Pauli correlations. KL gives $\rho_{R\mathsf{B}} = I_R/2 \otimes \rho_{\mathsf{B}}$. Equality of complementary purities for A and $R\mathsf{B}$ therefore gives

$$1 + s = 2(1 + b). \tag{359}$$

Let h be the squared correlation mass on $R\mathsf{A}$ with reference factor X or Y. Multiplication by G identifies the reference-Z mass with the physical active mass $1+s$. Comparing the complementary purities of B and $R\mathsf{A}$ gives

$$2(1+s)+h=8(1+b), \qquad h=4(1+b).$$

A surviving reference-X/Y correlation has odd active transverse count. If that count is one and its active Z-count is z, the term and its G-multiple have physical weights $1+z$ and $4-z$. At least one weight is at most two, so both correlations vanish. Thus the surviving transverse count is three. Multiplication by G interchanges its physical-weight-three and physical-weight-four terms, giving

$$h_3=h_4=\frac{h}{2}=2(1+b). \tag{360}$$

For each i, compare the complementary marginals $\mathsf{B}A_i$ and $R(\mathsf{A}\setminus\{A_i\})$, then sum over all four active sites. The first marginal's mass is

$$1+b+\lambda_{Z_i}^2+\sum_{F\neq I}\lambda_{FZ_i}^2.$$

On the complementary side, reference identity and reference Z together count active strings whose ith factor is I or Z. A string with transverse count t is therefore counted $4-t$ times. Reference-X/Y physical-weight-three terms appear once and weight-four terms never appear. Accounting also for the dimensions of the two marginals gives

$$2\{4(1+b)+a+m\}=4(1+s)-2u-2v-4w+h_3. \tag{361}$$

Substitution of (359) and (360) yields

$$b+m+w=0. \tag{362}$$

Every summand is a sum of real squares. Hence $b=m=w=0$, every coefficient they contain vanishes, and $a+u+v=1$. In particular $\rho_{\mathsf{B}}=I_4/4$. Parity kills spectator-dressed active X/Y coefficients, and $m=0$ kills all spectator-dressed Z_i coefficients. This proves the product marginal in (357).

The full weight-one and weight-two masses are now $A_1=a$ and $A_2=u$. Write $A_3=v+n$, where $n\geq 0$ collects the remaining terms supported on two active sites and one spectator site. All other additional low-weight patterns have vanished. Equation (356) gives $1+n\leq 1$, so $n=0$, $A_3=v$, and $A_1+A_2+A_3=1$.

Finally, KL for all errors on B gives the same marginal $I_4/4$ for each logical word. Each word therefore has four equal Schmidt coefficients $1/2$. Its active Schmidt space lies in the corresponding D-parity sector. Each logical word's active marginal is its rank-four projector divided by four. Averaging these two marginals gives $\Pi/8$, proving (358). □

Write the resulting flat parity frames as 16×4 matrices W_0, W_1:

$$|z_L\rangle=\frac{1}{2}\sum_{b=0}^{3}|b\rangle_{\mathsf{B}}\otimes W_z|b\rangle, \qquad W_z^\dagger W_z=I_4, \qquad DW_z=(-1)^z W_z. \tag{363}$$

Their ranges are orthogonal, and

$$\Pi=W_0W_0^\dagger+W_1W_1^\dagger=4\,\mathrm{Tr}_{\mathsf{B}}\,P.$$

Set $K = 2\Pi - I_{16}$. Then $K^\dagger = K$, $K^2 = I_{16}$, and $[K, D] = 0$. For every nonidentity active Pauli E,

$$\lambda_{I_{\mathsf{B}}E} = \frac{1}{8}\operatorname{Tr}(\Pi E) = \frac{1}{16}\operatorname{Tr}(KE). \tag{364}$$

The active support proved above thus gives

$$\begin{aligned} K &= K_1 + K_2 + K_3, \\ K_1 &= \sum_{i=1}^{4} a_i Z_i, \\ K_2 &= \sum_{i<j} \sum_{p,q\in\{X,Y\}} u_{ij}^{pq} p_i q_j, \\ K_3 &= \sum_{i<j} \sum_{k\notin\{i,j\}} \sum_{p,q\in\{X,Y\}} v_{ij;k}^{pq} p_i q_j Z_k. \end{aligned} \tag{365}$$

The three sets have $4, 24, 48$ real coefficients, respectively. Their squared masses are

$$\alpha = \frac{1}{16}\operatorname{Tr}(K_1^2) = A_1, \qquad \beta = \frac{1}{16}\operatorname{Tr}(K_2^2) = A_2, \qquad \nu = \frac{1}{16}\operatorname{Tr}(K_3^2) = A_3, \quad \alpha+\beta+\nu = 1. \tag{366}$$

Consequently (355) already reduces the enumerator to

$$(A_0, \dots, A_6) = (1, \alpha, 1-\alpha-\nu, \nu, 11+4\alpha+\nu, 16-\alpha-\nu, 3-3\alpha). \tag{367}$$

The remaining converse is $\nu = 0$. An arbitrary reflection having (365) need not be completable to a six-qubit code; the mixed KL constraints are essential in the next step.

11.2 One common antiunitary from mixed KL

On the spectator pair choose the fixed basis of $\Theta_{\mathsf{B}} = (Y \otimes Y)\mathcal{K}$, where $\mathcal{K}$ denotes computational complex conjugation. Explicitly, let M_{B} have columns

$$i\Phi_+, \quad \Phi_-, \quad \Psi_+, \quad i\Psi_-, \qquad \Phi_\pm = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad \Psi_\pm = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

Each column is fixed by Θ_{B}. This is a common spectator magic basis, not an independently chosen basis for each observable. In that basis put

$$Q_z = W_z \overline{M_{\mathsf{B}}}, \qquad Q = (Q_0\ Q_1).$$

The conjugate follows from the bipartite expansion (363). We have $Q^\dagger Q = I_8$ and $QQ^\dagger = \Pi$.

Lemma 11.3 (Common antiunitary equations)**.** *For every real single-body active Hamiltonian H, put*

$$h = Q^\dagger H Q, \qquad \mu_H = \frac{1}{8}\operatorname{Tr}(\Pi H).$$

Then the same skew-unitary matrix

$$J_0 = \begin{pmatrix} 0 & I_4 \\ -I_4 & 0 \end{pmatrix} \tag{368}$$

satisfies

$$hJ_0 + J_0 h^T = 2\mu_H J_0. \tag{369}$$

Proof. Each single-spectator Pauli F changes sign under Θ_{B}. Hence $M_{\mathsf{B}}^\dagger F M_{\mathsf{B}}$ is imaginary, antisymmetric, and Hermitian. The six such Paulis span the six-dimensional space of imaginary antisymmetric Hermitian 4×4 matrices.

For every single-active Pauli E, mixed KL gives, for all six F and all z, z',

$$\frac{1}{4}\operatorname{Tr}\Big[(M_{\mathsf{B}}^\dagger F M_{\mathsf{B}})^T Q_z^\dagger E Q_{z'}\Big] = 0.$$

Orthogonality to this antisymmetric space makes every block $Q_z^\dagger E Q_{z'}$ complex symmetric. A diagonal block is also Hermitian and is therefore real symmetric.

The stronger product-marginal identity in (357), not merely spectator flatness, gives

$$h_{00} + h_{11} = 2\mu_H I_4.$$

Indeed, tracing $(I_{\mathsf{B}} \otimes H)\rho$ over the active sites gives the spectator matrix $(h_{00}+h_{11})^T/8$ in this basis; the product marginal instead gives $\mu_H I_4/4$. Thus

$$h = \begin{pmatrix} R & C \\ \overline{C} & 2\mu_H I_4 - R \end{pmatrix}, \qquad R = R^T = \overline{R}, \qquad C = C^T.$$

Direct block multiplication proves (369). □

The antiunitary $J_0\mathcal{K}$ squares to $-I_8$ and reverses every centered compression $h - \mu_H I_8$. The same J_0 works for all twelve single-active Paulis. Separate spectral pairings, or a condition using only the four Z_i, would not provide the full hypotheses used in the final certificate.

11.3 A sparse partial isometry on four active qubits

Define

$$\Omega = Q J_0 Q^T = Q_0 Q_1^T - Q_1 Q_0^T, \qquad S = Y^{\otimes 4}, \qquad T = \Omega S. \tag{370}$$

Here S is real, symmetric, and unitary. Using $Q^T \overline{Q} = I_8$ gives

$$\Omega^T = -\Omega, \qquad \Omega\Omega^\dagger = \Pi, \qquad \Omega^\dagger\Omega = \Pi^T.$$

Therefore

$$TT^\dagger = \Pi, \qquad T^\dagger T = \widetilde{\Pi} := S\Pi^T S, \qquad ST^T S = -T, \qquad \operatorname{Tr}(TT^\dagger) = 8. \tag{371}$$

The Pauli identity $SE^T S = (-1)^{\mathrm{wt}(E)} E$ means that only odd Pauli weights can occur in T.

Lemma 11.4 (Exact edge sparsity)**.** *For computational active strings $x, y \in \{0,1\}^4$,*

$$d_H(x,y) = 1 \quad \Longrightarrow \quad \Omega_{xy} = 0.$$

In fact Ω is supported only at Hamming distance three, and T only at Hamming distance one.

Proof. Parity already makes Ω zero between equal-parity strings, since it pairs Q_0 and Q_1. Fix an even string x, set $s_i = (-1)^{x_i}$, $t = \sum_i a_i s_i$, and write $\bar{x}$ for its complement. Consider

$$H_x = \sum_i s_i Z_i, \qquad r_x = \langle x|Q_0, \qquad r_{\bar{x}} = \langle \bar{x}|Q_0.$$

The allowed reflection support (365) gives

$$\|r_x\|^2 = \Pi_{xx} = \frac{1+t}{2}, \qquad \|r_{\bar{x}}\|^2 = \frac{1-t}{2}, \qquad r_x r_{\bar{x}}^\dagger = \Pi_{x\bar{x}} = 0.$$

The last equality uses the absence of any term flipping all four active bits.

The eigenvalue of H_x on y is $4 - 2d_H(x, y)$. Thus its even-parity restriction is $4(|x\rangle\langle x| - |\bar{x}\rangle\langle\bar{x}|)$, and its odd-parity spectrum is $\{-2, +2\}$. With $h_z = Q_z^\dagger H_x Q_z$, this gives

$$h_0 = 4r_x^\dagger r_x - 4r_{\bar{x}}^\dagger r_{\bar{x}}, \qquad h_0^T r_x^T = 2(1+t)r_x^T.$$

Moreover $\mu_{H_x} = t$. Equation (369) implies

$$h_1 = 2tI_4 - h_0^T, \qquad h_1 r_x^T = -2r_x^T.$$

Set $w = Q_1 r_x^T$. Then

$$\langle w|(H_x + 2I)|w\rangle = 0.$$

On odd parity $H_x + 2I$ is positive semidefinite, with kernel consisting exactly of the strings at distance three from x. Consequently w has no distance-one components. For even x and odd y,

$$\Omega_{xy} = r_x(\langle y|Q_1)^T = \langle y|w,$$

so the desired entries vanish. If $r_x = 0$, that row already vanishes; no division is needed. Skew symmetry supplies the reverse ordering. Only odd distances are possible, so the remaining distance is three. Right multiplication by S complements the column string up to a phase, taking distance three to distance one. □

An operator supported only on distance-one matrix entries has exactly one X/Y factor in its Pauli expansion. Combining this with the odd Pauli weights in (371) gives

$$\begin{aligned} T &= T_1 + T_3, \\ T_1 &= \sum_i (t_i^X X_i + t_i^Y Y_i), \\ T_3 &= \sum_i \sum_{\substack{j<k \\ j,k\neq i}} (t_{i;jk}^X X_i + t_{i;jk}^Y Y_i) Z_j Z_k. \end{aligned} \tag{372}$$

There are $8 + 24 = 32$ complex coefficients.

For each single-active Pauli σ, multiply (369) by Q, Q^T to obtain

$$\Pi\sigma\Omega + \Omega\sigma^T\Pi^T = 2\mu_\sigma\Omega.$$

Since $S\sigma^T S = -\sigma$, multiplication on the right by S yields the physical common-antiunitary equation

$$\Pi\sigma T - T\sigma\widetilde{\Pi} = 2\mu_\sigma T, \qquad \mu_\sigma = \frac{1}{8}\operatorname{Tr}(\Pi\sigma). \tag{373}$$

All steps so far are valid for arbitrary computational zeros and singular overlap patterns.

11.4 A homogeneous rigidity theorem

The remaining finite statement can be formulated independently of encoders. All operators in this subsection act on the four active qubits; active identity factors are suppressed. Write

$$\tau(M) = \frac{1}{16}\operatorname{Tr}(M), \qquad \mathcal{S}_1 = \{X_i, Y_i : 1 \le i \le 4\},$$

$$\mathcal{S}_3 = \{p_i Z_j Z_k : p \in \{X, Y\},\ j < k,\ i \notin \{j, k\}\}.$$

For an arbitrary T in the complex span of $\mathcal{S}_1 \cup \mathcal{S}_3$, let T_1, T_3 denote these two coordinate parts and put

$$N = \tau(TT^\dagger), \qquad c_E = \tau(ETT^\dagger). \tag{374}$$

Each c_E is real. Define the homogeneous cubic defects

$$\begin{aligned} F(T) &= TT^\dagger T - 2NT, \\ R_\sigma(T) &= TT^\dagger \sigma T - T\sigma T^\dagger T - 4c_\sigma T. \end{aligned} \tag{375}$$

The target and forbidden Pauli sets are

$$\begin{aligned} \mathcal{V}_3 &= \{p_i q_j Z_k : i<j,\ k \notin \{i,j\},\ p,q \in \{X,Y\}\}, \\ \mathcal{D}_2 &= \{Z_i Z_j : i<j\}, \\ \mathcal{D}_3 &= \{Z_i Z_j Z_k : i<j<k\}, \\ \mathcal{F}_4 &= \{p_i q_j Z_k Z_\ell : i<j,\ \{k,\ell\} = \{1,2,3,4\} \setminus \{i,j\},\ p,q \in \{X,Y\}\}. \end{aligned} \tag{376}$$

Their sizes are $48, 6, 4, 24$, respectively.

Proposition 11.5 (Global common-antiunitary rigidity)**.** *Suppose T has the support (372) and*

$$F(T) = 0, \qquad R_\sigma(T) = 0 \quad \text{for all twelve single-active Paulis}, \qquad c_E = 0 \quad (E \in \mathcal{D}_2 \cup \mathcal{D}_3 \cup \mathcal{F}_4). \tag{377}$$

Then $T_3 = 0$ and $c_E = 0$ for every $E \in \mathcal{V}_3$.

For the matrix constructed from an orbit-2 code, $N = \tau(\Pi) = 1/2$, so $F(T) = 0$ follows from its partial-isometry identities. Also $c_\sigma = \mu_\sigma/2$, so (373) gives $R_\sigma(T) = 0$ with exactly the factor 4 in (375). Finally, the support of $K = 2TT^\dagger - I$ gives all the forbidden zeros in (377). Thus the proposition applies once its certificate is proved.

11.5 The exact positive quartic certificate

Define the scalar defect contractions

$$\begin{aligned} g &= \Re\,\tau(T_3^\dagger F(T)), \\ c_E^{13} &= \tau\Big(E(T_1 T_3^\dagger + T_3 T_1^\dagger)\Big), \qquad c_E^{33} = \tau(E T_3 T_3^\dagger), \\ f_{\mathcal{C}}^{13} &= \sum_{E \in \mathcal{C}} c_E c_E^{13}, \qquad f_{\mathcal{C}}^{33} = \sum_{E \in \mathcal{C}} c_E c_E^{33}. \end{aligned} \tag{378}$$

The last two definitions apply to any Pauli set $\mathcal{C}$. All displayed quantities are real; g, $f_{\mathcal{C}}^{13}$, and $f_{\mathcal{C}}^{33}$ are homogeneous of degree four.

For a set $\mathcal{I}$ of ordered pairs (σ, P'), where $P' \in \mathcal{S}_1 \cup \mathcal{S}_3$ is a Pauli label, put

$$J(\mathcal{I}) = \sum_{(\sigma,P') \in \mathcal{I}} \Re\big[\overline{t_{P'}}\,\tau\big(P'\sigma R_\sigma(T)\big)\big]. \tag{379}$$

Here $t_{P'}$ is the coefficient of P' in T; P' is not the code projector P. The five choices needed below are defined by the following table. In each row, sum over every active site i and every indicated Pauli pair; P_i' is the factor at that site.

Sum	σ and P'	Condition at site i
$J_{3,L}$	$\sigma \in \{X_i, Y_i\},\ P' \in \mathcal{S}_3$	$P'_i = Z$
$J_{3,Z}$	$\sigma = Z_i,\ P' \in \mathcal{S}_3$	$P'_i \in \{X, Y\}$
$J_{3,\perp}$	$\sigma \in \{X_i, Y_i\},\ P' \in \mathcal{S}_3$	$P'_i \in \{X, Y\},\ P'_i \neq \sigma_i$
$J_{1,\perp}$	$\sigma \in \{X_i, Y_i\},\ P' \in \mathcal{S}_1$	$P'_i \in \{X, Y\},\ P'_i \neq \sigma_i$
$J_{1,Z}$	$\sigma = Z_i,\ P' \in \mathcal{S}_1$	$P'_i \in \{X, Y\}$

All five sums vanish if the common-antiunitary defects vanish.

The certificate polynomial is

$$\begin{aligned}\mathcal{Q}(T) = &-\sum_{E\in\mathcal{V}_3} c_E^2 + \frac{5}{12}J_{3,L} + \frac{5}{6}J_{3,Z} + \frac{7}{12}J_{3,\perp} + \frac{1}{2}g \\ &+ \frac{3}{2}J_{1,\perp} - \frac{3}{2}J_{1,Z} + \frac{17}{6}f^{13}_{\mathcal{F}_4} + \frac{19}{6}f^{33}_{\mathcal{F}_4} + \frac{25}{6}f^{13}_{\mathcal{D}_3} + \frac{1}{3}f^{33}_{\mathcal{D}_2}.\end{aligned} \tag{380}$$

Lemma 11.6 (Exact coefficient matrix and positivity)**.** *Order the 32 Pauli labels in $\mathcal{S}_1 \cup \mathcal{S}_3$ lexicographically using $I < X < Y < Z$. Let z contain all 528 monomials $t_{P'}t_{Q'}$, $P' \leq Q'$, with no $\sqrt{2}$ factors on unequal pairs. There is an integer real symmetric 528×528 matrix M such that, identically for every complex T in this coordinate space,*

$$\mathcal{Q}(T) = \frac{1}{12} z^\dagger M z. \tag{381}$$

The 36 rows and columns indexed by two $\mathcal{S}_1$ coordinates are zero. On the remaining 492 coordinates, M has 111 blocks and satisfies

$$M_{\text{remaining}} - 4I_{492} > 0. \tag{382}$$

In particular,

$$\mathcal{Q}(T) \geq \frac{1}{3} \sum_{\substack{P'\leq Q' \\ P'\in\mathcal{S}_3 \text{ or } Q'\in\mathcal{S}_3}} |t_{P'}t_{Q'}|^2. \tag{383}$$

Exact finite verification. Treat the $t_{P'}$ and their formal conjugates as independent variables and expand $12\mathcal{Q}$ in bidegree $(2,2)$. Each monomial has the unique form $\overline{t_{P'}t_{Q'}}t_{R'}t_{S'}$, with $P' \leq Q'$ and $R' \leq S'$. Accumulating all ordered trace contributions into these unordered pair indices uniquely determines the coefficient matrix M; it is not a fitted Gram representation.

Multiplying Paulis with phases $1, -1, i, -i$ and expanding the normalized traces and real parts in (378)–(380) gives integer entries and $M = M^T$. Exact coefficient comparison establishes (381) independently of the encoder equations.

Conjugation by a four-qubit Pauli multiplies each $t_{P'}$ by its Pauli-conjugation sign and leaves $\mathcal{Q}$ invariant. Hence a matrix entry between the monomials $t_{P'}t_{Q'}$ and $t_{R'}t_{S'}$ can be nonzero only when the Pauli labels of $P'Q'$ and $R'S'$ agree modulo phase. This gives the character blocks. Exact coefficient comparison confirms every off-block zero entry, all 36 zero rows and columns, and every remaining block.

The exact characteristic polynomials of all blocks are summarized in Table 2. Every linear root is greater than four. Each quadratic factor $x^2 - bx + c$ has real roots because it is a factor of the characteristic polynomial of a real symmetric matrix. Its shifted roots $y = x - 4$ have positive sum $b - 8$ and positive product $c - 4b + 16$, as is checked from every listed factor. Thus both roots exceed four. This proves (382) once the exact block polynomials have been verified.

Positivity can also be established directly by exact rational LDL^T elimination. Subtracting $4I$ from each of the 111 integer blocks gives strictly positive pivots in every case. Explicitly, at pivot $d > 0$, the remaining Schur complement is updated by

$$A_{ij} \longmapsto A_{ij} - \frac{A_{ik}A_{kj}}{d}.$$

All entries and pivots are rational, and positive pivots give positive definiteness by congruence. Every block is checked individually, so this argument does not assume that one representative block covers its entire symmetry class.

Finally, the zero coordinates are precisely the $\binom{8+1}{2} = 36$ products of two weight-one coordinates. On the remaining coordinates (382) implies (383). The matrix is defined completely by (380) and the specified monomial ordering. □

Character pattern	Size	Count	Characteristic polynomial of a block of M
I	24	1	$(x-424)(x-192)^4(x-24)^{11}(x-16)^2(x^2-48x+192)^3$
Z	3	4	$(x-384)(x-48)^2$
pq	3	24	$(x-52)(x^2-460x+47872)$
pqZ	4	48	$(x-50)(x-46)(x^2-248x+10716)$
ZZ	8	6	$(x-32)(x-24)(x^2-408x+9856)(x^2-352x+9216)^2$
ZZZ	12	4	$(x-136)(x-48)(x^2-352x+9216)^3(x^2-64x+720)^2$
$pqZZ$	4	24	$(x-56)(x-44)(x^2-380x+23936)$

Table 2: All exact character blocks after removal of the 36 zero $\mathcal{S}_1\mathcal{S}_1$ coordinates. Here $p, q \in \{X, Y\}$; all distinct-site and transverse-axis choices are included. The counts sum to 111, and the sizes with multiplicity sum to 492. The displayed polynomials give supplementary spectral information; positive definiteness is verified for every block by exact rational LDL^T elimination.

Proof of Proposition 11.5. Under (377), g and all five J-sums vanish. Each forbidden-set contraction also vanishes because its factor c_E is zero. Hence (380) and (383) give

$$-\sum_{E\in\mathcal{V}_3} c_E^2 = \mathcal{Q}(T) \geq \frac{1}{3} \sum_{\substack{P'\leq Q' \\ P'\in\mathcal{S}_3 \text{ or } Q'\in\mathcal{S}_3}} |t_{P'}t_{Q'}|^2 \geq 0.$$

The left side is nonpositive, so every c_E, $E \in \mathcal{V}_3$, vanishes. The right side includes $|t_{P'}|^4$ for every $P' \in \mathcal{S}_3$, so all those coordinates vanish as well. Thus $T_3 = 0$. □

11.6 Completion of the converse and exact attainment

Proof of Theorem 11.1. For every exact orbit-2 encoder, the analytic construction gives (372), the partial-isometry identities, and all twelve physical equations (373). As noted after Proposition 11.5, these imply every hypothesis (377). The proposition therefore gives $c_E = 0$ for $E \in \mathcal{V}_3$. By (364), the coefficient of E in $K = 2TT^\dagger - I$ is $2c_E$, so

$$A_3 = \frac{1}{16}\operatorname{Tr}(K_3^2) = 4\sum_{E\in\mathcal{V}_3} c_E^2 = 0. \tag{384}$$

Equation (367) gives $\mathcal{L}(\alpha)$, with $0 \leq \alpha \leq 1$ by $A_1, A_2 \geq 0$.

For completeness, the reverse inclusion has the following explicit split-qubit construction. Start with the standard perfect five-qubit code with stabilizer generators

$$XZZXI, \quad IXZZX, \quad XIXZZ, \quad ZXIXZ.$$

It is pure: every nonidentity Pauli of weight at most two has zero compression. Its logical $Z^{\otimes 5}$, multiplied by the first generator, gives the weight-three logical representative $-YIIYZ$. Local Clifford rotations and a permutation therefore give a pure five-qubit code with a chosen logical $Z_1Z_2Z_3$.

For $0 \le \theta \le \pi/2$, put $a = \cos(\theta/2)$, $b = \sin(\theta/2)$, and split original site 1 by the isometry

$$W_\theta|0\rangle = a|00\rangle + b|11\rangle, \qquad W_\theta|1\rangle = a|01\rangle + b|10\rangle.$$

Its exact identities are

$$W_\theta W_\theta^\dagger = \frac{1}{2}(I + \cos\theta\, Z \otimes I + \sin\theta\, X \otimes X), \qquad (Z \otimes Z)W_\theta = W_\theta Z. \tag{385}$$

Take the two split outputs as A_1, A_2, original sites $2, 3$ as A_3, A_4, and original sites $4, 5$ as B_1, B_2. The original weight-three logical lifts to $I_{\mathsf{B}}D$, which is precisely (352).

Compression of any six-qubit error of weight at most two through $W_\theta \otimes I^{\otimes 4}$ is supported on at most two original sites. Purity of the five-qubit code therefore proves all KL equations. If the error has a nonidentity factor on an unsplit site, its scalar compression is zero. On the split pair its scalar is the identity coefficient $\mathrm{Tr}(W_\theta^\dagger E W_\theta)/2$. Consequently the only nonzero weight-one and weight-two coefficients are $\cos\theta$ on Z_{A_1} and $\sin\theta$ on $X_{A_1}X_{A_2}$. Thus

$$A_1 = \cos^2\theta = \alpha, \qquad A_2 = \sin^2\theta = 1 - \alpha.$$

The universal inequality (356) gives $A_3 = 0$, and (355) gives $\mathcal{L}(\alpha)$. Every $\alpha \in [0, 1]$, including both endpoints, occurs. The quantum Singleton bound excludes distance at least four at length six and code dimension two, so these are exact $((6, 2, 3))$ realizations. □

Remark 11.7 (Scope of the converse)**.** The proof uses no division by a computational coefficient, row norm, polarization, Schmidt eigenvalue, or Gram minor. It covers arbitrary interactions among different Z-pairs in T_3 and both regular and singular frame configurations. The necessary common-antiunitary equations already suffice for this enumerator conclusion; no sufficiency claim for the matrix completion problem is required. The theorem determines $\mathcal{W}_2$, not all code projectors up to local unitaries, and does not assert global connectedness or uniqueness of the split-code family. It also does not prove a stronger partial-isometry inequality with the common-antiunitary equations omitted.

12 Exact weight-enumerator image of primitive order-two orbit 3

This section proves the complete enumerator image of SSLP orbit 3:

$$\boxed{\mathcal{W}_3 = \{\mathcal{L}(\alpha) : 0 \le \alpha \le 1\}, \qquad \mathcal{L}(\alpha) = (1, \alpha, 1-\alpha, 0, 11+4\alpha, 16-\alpha, 3-3\alpha).} \tag{386}$$

The converse is a computer-assisted exact proof. Its two finite ingredients are rational scalar dual identities and a degree-six coefficient identity with an explicit positive square. We give the analytic reduction, the meaning of every certificate row, and an exact construction attaining the full interval.

12.1 The orbit, the encoder equations, and the enumerator question

The primitive SSLP record is

$$q = m = 2, \qquad a = (0, 1, 1, 1, 1, 1).$$

For the proof we relabel physical sites as $0, 1, \ldots, 5$, so that the supplement's $D(a; 2)$ becomes

$$D = I_0 Z_1 Z_2 Z_3 Z_4 Z_5.$$

Site 0 is inactive for this chosen involution; sites $1, \ldots, 5$ are active. Inactive does not mean that site 0 is a product ancilla. An orbit-3 realization has a logical eigenbasis with encoder columns $u, v \in (\mathbb{C}^2)^{\otimes 6}$ satisfying

$$Du = u, \quad Dv = -v, \quad u^\dagger u = v^\dagger v = N > 0, \quad u^\dagger E u = v^\dagger E v, \quad u^\dagger E v = 0 \quad (\mathrm{wt}(E) \le 2). \tag{387}$$

Here and below E ranges over the Hermitian representatives $\{I, X, Y, Z\}^{\otimes 6}$, and $\mathrm{wt}(E)$ is its Pauli weight. The two parity sectors are orthogonal, so $u^\dagger v = 0$. Thus

$$P = \frac{uu^\dagger + vv^\dagger}{N}$$

is a rank-two orthogonal projector. Allowing N to vary makes the certificate homogeneous; a normalized encoder has $N = 1$. Equation (387) is the full Knill–Laflamme (KL) system, including all logical off-diagonal conditions for the 153 nonidentity Paulis of weight at most two. It characterizes distance at least three. The quantum Singleton bound then makes the distance exactly three. No stabilizer or additivity assumption is made.

Use the supplement's enumerator normalization:

$$\gamma_E = \tfrac{1}{2}\,\mathrm{Tr}(PE), \qquad A_w = \sum_{\mathrm{wt}(E)=w} \gamma_E^2, \qquad B_w = \tfrac{1}{2} \sum_{\mathrm{wt}(E)=w} \mathrm{Tr}(PEPE). \tag{388}$$

In particular, $A_0 = B_0 = 1$, $\sum_w A_w = 32$, and $A_w = B_w$ for $w = 0, 1, 2$. The image $\mathcal{W}_3$ consists of all vectors $(A_0, \ldots, A_6)$ attained by encoders in (387). An SSLP orbit records a chosen symmetry, rather than a disjoint LU class of projectors. A single code can realize several different orbits.

Theorem 12.1 (Global orbit-3 enumerator theorem)**.** *Every exact orbit-3 realization satisfies $A_3 = 0$. Its complete enumerator image is (386), including both endpoints. The converse includes zero amplitudes and all spectator Schmidt-rank strata.*

12.2 Universal identities and the exact scalar reduction

The six-qubit MacWilliams transform in this normalization is

$$B(X, Y) = \frac{1}{32} A(X + 3Y, X - Y), \qquad A(X, Y) = \sum_{w=0}^{6} A_w X^{6-w} Y^w.$$

Solving $A_w = B_w$ for $w = 0, 1, 2$, with $A_0 = 1$, gives

$$(A_0, \ldots, A_6) = (1, x, y, z, 15 - 4y - 3z, 12 + 3x + 4y + 3z, 4 - 4x - y - z), \quad (x, y, z) = (A_1, A_2, A_3). \tag{389}$$

Put $S = Y^{\otimes 6}$ and $\widetilde{P} = SP^TS$. Since $P, \widetilde{P}$ are positive semidefinite,

$$0 \le \mathrm{Tr}(P\widetilde{P}) = \frac{1}{16} \sum_{w=0}^{6} (-1)^w A_w = \frac{1}{2}(1 - A_1 - A_2 - A_3). \tag{390}$$

For a Pauli string let $a \in \{0,1\}$ indicate a nonidentity factor at site 0, let j count active Z factors, and let k count active X/Y factors. Define the category masses and sizes by

$$A_{a,j,k} = \sum_{E \text{ of type } (a,j,k)} \gamma_E^2, \qquad N_{a,j,k} = 3^a \frac{5!}{j!k!(5-j-k)!}\, 2^k. \tag{391}$$

The ranges are $a = 0, 1$ and $j, k \geq 0$, $j + k \leq 5$. There are 42 categories, of which 24 have even k. Since $[P, D] = 0$, all coefficients in odd-k categories vanish. These are category sums; equal coefficients or permutation symmetry of the original code are not assumed.

Lemma 12.2 (Universal orbit-3 scalar restrictions)**.** *Every encoder in* (387) *satisfies*

$$A_{0,1,0} = A_{0,0,2} = A_{0,1,2} = A_{1,2,0} = A_{0,3,0} = 0, \tag{392}$$

$$A_1 + A_2 + A_3 = 1. \tag{393}$$

Consequently

$$A_1 = A_{1,0,0}, \qquad A_2 = A_{0,2,0} + A_{1,1,0}, \qquad A_3 = A_{1,0,2}. \tag{394}$$

Proof. For individual Paulis set

$$a_E = \gamma_E^2, \qquad b_E = \tfrac{1}{2}\operatorname{Tr}(PEPE), \qquad s_E = \tfrac{1}{2}\operatorname{Tr}(PE\widetilde{P}E).$$

Positivity and KL imply $b_E \geq a_E \geq 0$, $s_E \geq 0$, and $b_E = a_E$ when $\mathrm{wt}(E) \leq 2$. There are two additional label-sensitive relations:

$$b_E = a_E + a_{DE} \quad ([E, D] = 0), \qquad s_E = 0 \quad (\{E, D\} = 0,\ \mathrm{wt}(E) \text{ odd}). \tag{395}$$

An overall sign in the Hermitian representative of DE has no effect on a_{DE}. For the first relation, write the compression of E in the normalized logical D-eigenbasis $u/\sqrt{N}, v/\sqrt{N}$ as $\mathrm{diag}(\eta_+, \eta_-)$. Then $a_E = (\eta_+ + \eta_-)^2/4$, $a_{DE} = (\eta_+ - \eta_-)^2/4$, and $b_E = (\eta_+^2 + \eta_-^2)/2$. For the second relation the antiunitary SK, with K complex conjugation, anticommutes with D. If E also anticommutes with D, then ESK preserves its eigenspaces, so the cross-parity code matrix elements vanish. Its same-parity code matrix elements vanish because $(ES)^T = (-1)^{\mathrm{wt}(E)} ES$ is skew symmetric at odd weight. Equivalently, every overlap contributing to s_E is zero.

The individual complete MacWilliams and shadow transforms are

$$b_E = \frac{1}{32} \sum_F (-1)^{[E,F]} a_F, \qquad s_E = \frac{1}{32} \sum_F (-1)^{[E,F]+\mathrm{wt}(F)} a_F, \tag{396}$$

where $[E, F] \in \{0, 1\}$ records commutation. Average the rows over each category. For an error category $\ell = (a, j, k)$ and a coefficient category $c = (b, z, t)$, the resulting kernel acting on category masses is

$$K_{\ell c} = \frac{m_{a,b}}{32 N_{0,z,t}} \left[U^z V^t\right] (1+U+2V)^{5-j-k} (1+U-2V)^j (1-U)^k, \quad m_{a,b} = \begin{cases} -1/3, & a = b = 1, \\ 1, & \text{otherwise.} \end{cases} \tag{397}$$

Here $[U^z V^t]$ means coefficient extraction. Define

$$\bar{b}_\ell = \sum_c K_{\ell c} A_c, \qquad \bar{s}_\ell = \sum_c (-1)^{b+z+t} K_{\ell c} A_c.$$

Only even-t columns are needed. The exact finite system used by the dual certificates is

$$\begin{aligned}
&A_{0,0,0}=1, \quad A_c\ge 0, \quad \bar b_\ell\ge 0, \quad \bar s_\ell\ge 0,\\
&e^{\mathrm{diag}}_{a,j,k} := \bar b_{a,j,k} - \frac{A_{a,j,k}+A_{a,5-j-k,k}}{N_{a,j,k}} = 0 && (k \text{ even}),\\
&e^{\mathrm{KL}}_{a,j,k} := \bar b_{a,j,k} - \frac{A_{a,j,k}}{N_{a,j,k}} = 0 && (a+j+k\le 2),\\
&e^{\mathrm{shadow}}_{a,j,k} := \bar s_{a,j,k} = 0 && (k \text{ odd},\ a+j+k \text{ odd}).
\end{aligned} \tag{398}$$

An odd-k mass in this display is interpreted as zero. The complementary category in the diagonal equation has the same size. The averaged relations follow from (395) and (396); averaging equations does not require averaging the code projector into another rank-two projector.

The six rational dual certificates have objectives

$$-A_{0,1,0}, \quad -A_{0,0,2}, \quad -A_{0,1,2}, \quad -A_{1,2,0}, \quad -A_{0,3,0}, \quad A_1+A_2+A_3.$$

Their lower bounds are $0,0,0,0,0,1$, respectively. Each certificate expresses its objective minus the claimed bound as a rational linear combination of the vanishing equality residuals in (398) and $A_{0,0,0}-1$, plus a nonnegative linear combination of $\bar b_\ell, \bar s_\ell, A_c$. The coefficient identities and the signs of all inequality multipliers and variable remainders are verified exactly over $\mathbb{Q}$. When the inequality residuals are written as $-\bar b_\ell, -\bar s_\ell$, their multipliers are nonpositive, giving the nonnegative terms described above. For example the fifth certificate is the elementary exact identity

$$-A_{0,3,0} = 10e^{\mathrm{diag}}_{0,2,0} - 10e^{\mathrm{KL}}_{0,2,0}.$$

Thus the five nonnegative masses must vanish, and $A_1+A_2+A_3\ge 1$. Equation (390) supplies the opposite inequality. Enumerating the surviving categories of weights one, two, and three proves (394). □

Each zero mass is a sum of real squares, so all its individual γ_E's vanish. The five categories contain, respectively, $5, 40, 120, 30, 10$ Paulis, hence 205 individual zero coefficients. The remaining weight-three target is

$$\mathcal{T} = \{\sigma_0 p_j q_k : 1\le j<k\le 5,\ \sigma\in\{X,Y,Z\},\ p,q\in\{X,Y\}\}, \quad |\mathcal{T}|=120, \quad A_3=\sum_{E\in\mathcal{T}}\gamma_E^2. \tag{399}$$

Writing $\alpha=A_1$, $\nu=A_3$, the scalar result and (389) give

$$A=(1,\alpha,1-\alpha-\nu,\nu,11+4\alpha+\nu,16-\alpha-\nu,3-3\alpha), \qquad \alpha,\nu\ge 0, \quad \alpha+\nu\le 1. \tag{400}$$

The remaining converse is exactly $\nu=0$. The scalar system also admits a formal nonnegative mass array with $A=(1,0,0,1,12,15,3)$ satisfying the scalar system. That array is not a code projector. It explains why the scalar restrictions alone do not prove the desired converse.

12.3 A single amplitude vector and the quadratic zero equations

Set $R_\pm=(I\pm D)/2$ and

$$w=\frac{u+v}{\sqrt{2}}, \qquad u=\sqrt{2}R_+w, \qquad v=\sqrt{2}R_-w, \qquad r_E=w^\dagger E w. \tag{401}$$

This is an invertible linear parameterization by 64 complex amplitudes: there are 32 allowed entries in each logical column. For an actual encoder $N = w^\dagger w$. If $[E, D] = 0$, a direct expansion gives

$$u^\dagger E u - v^\dagger E v = 2 r_{DE}, \qquad g_E := \tfrac{1}{2}(u^\dagger E u + v^\dagger E v) = r_E.$$

If $\{E, D\} = 0$, the Hermitian representative $E' = iDE$ gives

$$u^\dagger E v = r_E - i r_{E'}.$$

Therefore KL becomes a collection of real quadratic zero equations. Let $\mathcal{C}$ denote the following set of Pauli labels modulo phase:

$$\begin{aligned}\mathcal{C} = &\{DE : \mathrm{wt}(E) \le 2,\ [E, D] = 0\} \\ &\cup \{E, DE : \mathrm{wt}(E) \le 2,\ \{E, D\} = 0\} \\ &\cup \{E : E \text{ is in one of the five zero categories in (392)}\}.\end{aligned} \tag{402}$$

The first set contains 74 labels (including D, from $E = I$); the second contains 160; the last contains 205. The overlap has 10 labels, so $|\mathcal{C}| = 429$. The label D enforces equality of the two column norms. Every exact encoder obeys

$$r_E = 0 \quad (E \in \mathcal{C}), \qquad s(w) := w^T S w = 0. \tag{403}$$

Indeed, (393) saturates (390). Because S is symmetric and anticommutes with D, the same-parity bilinear terms $u^T S u, v^T S v$ vanish and

$$\mathrm{Tr}(P\widetilde{P}) = \frac{2|u^T S v|^2}{N^2}, \qquad s(w) = u^T S v. \tag{404}$$

This proves the second equation in (403). For every target label, $[E, D] = 0$, and hence

$$N \sum_{E \in \mathcal{T}} r_E^2 = N^3 A_3. \tag{405}$$

No individual amplitude or Schmidt coefficient has been divided out.

12.4 A positive three-copy moment for every exact encoder

We use a finite average of transformations of the actual encoder. Choose independently a permutation π of the five active sites, an X-translation X^b, and one of the 24 single-qubit Clifford representatives C on spectator site 0. After applying $C_0 X^b \pi$, apply

$$L_r = \left(\bigotimes_{j=0}^{5} \mathrm{diag}(1, i^{r_j}) \right) (R_+ + i^{r_D} R_-), \qquad r_j, r_D \in \{0, 1, 2, 3\}.$$

Take the uniform average over all these choices and then over complex conjugation; write $\mathbb{E}_{\mathcal{G}}$ for this finite average and ξ for a resulting vector. The average of a monomial with its conjugate is its real part. The spectator Clifford representatives are closed under left multiplication by H_0 and $\mathrm{diag}(1, i)_0$, up to a global phase. Consequently phase-invariant cubic moments have the required spectator Hadamard identity. The fourth-root averages distinguish the occupations 0 and 2 in an XOR-zero triple and its two logical charges. They therefore impose exactly the support and charge blocks used below; continuous phase integration is unnecessary.

Every selected transformation maps an exact encoder to an exact encoder with the same chosen orbit record. Spectator local unitaries commute with D; active axial rotations and active permutations preserve D. A parity-changing Pauli interchanges the two logical columns. The unitary $e^{i\phi D}$ multiplies them by opposite phases, which preserves KL and the code projector. Conjugation changes only Pauli signs in the zero equations. These observations also show preservation of the scalar zeros and of spin-flip saturation. More explicitly, local unitaries preserve $s(w) = 0$ since for a one-qubit matrix L, $L^T Y L = (\det L)Y$, and $(e^{i\phi D})^T S e^{i\phi D} = S$. All transformed vectors therefore satisfy (403).

The norm N and $\sum_{E\in\mathcal{T}} r_E^2$ are invariant. For spectator rotations, active axial rotations, and active permutations, the latter is the squared norm of a complete Pauli tensor on its specified support patterns. Paulis and conjugation change signs. Logical phases commute with every target Pauli, since each has two active X/Y factors. Thus averaging does not alter (405).

Form the three-copy moment operator

$$M_3 = \mathbb{E}_{\mathcal{G}}[\,|\xi\rangle^{\otimes 3}\langle\xi|^{\otimes 3}\,]. \tag{406}$$

It is positive semidefinite, supported on the symmetric tensor cube, and has trace N^3. This positivity follows directly from averaging rank-one positive operators; it requires no numerical eigenvalue test. We use a block of M_3 in ordinary monomial coordinates.

12.4.1 The XOR-zero block and its exact coordinate reduction

Index computational basis vectors by integers $0, \dots, 63$, with site 0 the least significant bit. For a sorted triple $t = (a, b, c)$, put $\xi_t = \xi_a \xi_b \xi_c$, with repeated entries allowed. Let

$$\mathcal{B} = \{\mathrm{sort}(a, b, a \oplus b) : 0 \le a, b < 64\}. \tag{407}$$

There are $715 = 1 + 63 + 651$ triples: $(0, 0, 0)$, the 63 triples $(0, a, a)$ with $a \neq 0$, and 651 triples of distinct entries. Physical Z-Pauli averaging kills a moment entry unless its two triples have equal bitwise XOR. Physical X-Paulis translate all three entries simultaneously. Since the degree is odd, translating by the XOR of a triple takes that block to XOR zero; all 64 XOR blocks are consequently equivalent. Define

$$Y_{tu} = 64\,\mathbb{E}_{\mathcal{G}}[\,\xi_t \overline{\xi_u}\,], \qquad t, u \in \mathcal{B}. \tag{408}$$

Then Y is real symmetric and positive semidefinite. We use unnormalized monomials, so there are no square-root multinomial factors in (408).

For a triple $t = (a, b, c) \in \mathcal{B}$, define

$$L(t) = a \,\mathrm{OR}\, b \,\mathrm{OR}\, c, \qquad q(t) = \sum_{x\in t} (-1)^{\mathrm{popcount}(x\&62)}.$$

Here 62 is the bit mask for the five active sites, and popcount counts set bits. At a site, an XOR-zero triple has either zero or two occupied bits. Local axial averaging therefore forces $Y_{tu} = 0$ unless $L(t) = L(u)$. Logical-phase averaging also forces $q(t) = q(u)$. The charge takes only the values $3, -1$. Active permutations identify entries under simultaneous permutation of the five active bits in both triples. Together with matrix symmetry, these conditions leave 269 real entry orbits in 116 phase blocks. These counts follow by enumerating the triples and their simultaneous permutation orbits. Write $y \in \mathbb{R}^{269}$ for their coordinates. Spectator-Hadamard invariance will supply further explicit linear rows; it has not been used to obtain the count 269.

12.5 The degree-six constraint rows

A Hermitian Pauli label is a pair of six-bit integers $E = (x_E, z_E)$, encoded by the integer $x_E + 64z_E$, with representative

$$P_E = i^{\mathrm{popcount}(x_E \& z_E)} X^{x_E} Z^{z_E}. \tag{409}$$

For labels E, F, G whose bitwise XOR is zero and whose total number of Y factors is even, put

$$\mathcal{R}(E, F, G) = \mathbb{E}_{\mathcal{G}}[\, r_E(\xi) r_F(\xi) r_G(\xi)\,]. \tag{410}$$

It is an integer linear form in y. With $\eta = \mathrm{popcount}(x_E \& z_E) + \mathrm{popcount}(x_F \& z_F) + \mathrm{popcount}(x_G \& z_G)$, its exact row is

$$\mathcal{R}(E, F, G) = i^{\eta} \sum_{a,b=0}^{63} (-1)^{z_E \cdot a + z_F \cdot b + z_G \cdot c} Y_{\mathrm{sort}(a,b,c),\mathrm{sort}(a \oplus x_E, b \oplus x_F, c \oplus x_G)}, \quad c = a \oplus b. \tag{411}$$

Dot products in the sign are binary; entries forbidden by the phase conditions are interpreted as zero. To derive this formula, expand each r_E using (409). In the full sum over a, b, c, partition by the triple XOR $h = a \oplus b \oplus c$. Translation by the physical X^h identifies each part with the XOR-zero part. Its sign is unchanged since $z_E \oplus z_F \oplus z_G = 0$. The 64 equivalent parts give exactly the factor already absorbed in (408). The sum over ordered a, b retains all monomial multiplicities, including repeated entries.

There are three kinds of vanishing row:

(i) If one of E, F, G belongs to $\mathcal{C}$, then $\mathcal{R}(E, F, G) = 0$, since the corresponding quadratic r_E vanishes pointwise on all transformed encoders. These are degree-six multiples of KL and scalar-zero equations.

(ii) For $t \in \mathcal{B}$, spin-flip zero gives

$$\mathcal{S}_t := 64\,\mathrm{Re}\,\mathbb{E}_{\mathcal{G}}[\,\overline{\xi_t} s(\xi) \xi_{63}\,] = -\sum_{a=0}^{63} (-1)^{\mathrm{popcount}(a)} Y_{t,\mathrm{sort}(a, a \oplus 63, 63)} = 0. \tag{412}$$

Here $S_{a,a\oplus 63} = -(-1)^{\mathrm{popcount}(a)}$. The displayed orientation of Y uses its real symmetry.

(iii) Averaging with $H = H_0$ gives

$$\mathcal{H}(E, F, G) := \mathcal{R}(E, F, G) - \epsilon_H(E, F, G) \mathcal{R}(HEH, HFH, HGH) = 0. \tag{413}$$

The labels HEH, HFH, HGH on the right omit their overall signs; $\epsilon_H = (-1)^{n_0}$, where n_0 is the number of spectator Y's among the three Paulis. Hadamard exchanges spectator X, Z and negates spectator Y. This is an identity of the averaged moment, not an additional symmetry assumption on the unaveraged code.

12.6 The explicit positive square and the exact coefficient identity

Define the real-coefficient cubic

$$p(w) = \sum_{(a,b,c),\, v_t \text{ in Table 3}} v_t w_a w_b w_c. \tag{414}$$

All 36 triples have XOR zero, $L = 63$, and $q = -1$. Thus the coefficient vector $\mathbf{v}$, extended by zero, belongs to a single 92-dimensional block of Y. The target and positive-square rows are

$$T(y) = \sum_{E \in \mathcal{T}} \mathcal{R}(E, E, I) = N^3 A_3, \qquad V(y) = \mathbf{v}^T Y \mathbf{v} = 64\, \mathbb{E}_{\mathcal{G}} |p(\xi)|^2 \geq 0. \tag{415}$$

The target equality uses $r_I = N$, invariance of the target sum, and (405). For computing its integer row one can reduce the 120 terms to

$$T = 80\, \mathcal{R}(X_0X_1X_2, X_0X_1X_2, I) + 40\, \mathcal{R}(Z_0X_1X_2, Z_0X_1X_2, I). \tag{416}$$

The multiplicities distinguish spectator X/Y from spectator Z, using the axial and active-permutation averages. Summing all 120 rows gives the same reduction by exact integer coefficient comparison. Here $T(y)$ is a scalar target linear form; it is unrelated to the auxiliary matrix T in the orbit-2 proof.

Table 3: All 36 coefficients of p. The decimal indices use site 0 as the least significant bit; the monomials have no multinomial normalization.

(a, b, c)	v_t	(a, b, c)	v_t
$(2, 61, 63)$	1	$(20, 43, 63)$	-1
$(3, 60, 63)$	1	$(21, 43, 62)$	1
$(3, 61, 62)$	-2	$(22, 43, 61)$	1
$(4, 59, 63)$	-1	$(22, 45, 59)$	-1
$(5, 58, 63)$	-1	$(23, 42, 61)$	1
$(5, 59, 62)$	2	$(23, 43, 60)$	-2
$(10, 53, 63)$	1	$(23, 44, 59)$	-1
$(11, 53, 62)$	-1	$(23, 45, 58)$	2
$(12, 51, 63)$	-1	$(26, 39, 61)$	1
$(13, 51, 62)$	1	$(27, 36, 63)$	-1
$(14, 51, 61)$	1	$(27, 37, 62)$	1
$(14, 53, 59)$	-1	$(27, 38, 61)$	1
$(15, 50, 61)$	1	$(27, 39, 60)$	-2
$(15, 51, 60)$	-2	$(28, 39, 59)$	-1
$(15, 52, 59)$	-1	$(29, 34, 63)$	1
$(15, 53, 58)$	2	$(29, 35, 62)$	-1
$(18, 45, 63)$	1	$(29, 38, 59)$	-1
$(19, 45, 62)$	-1	$(29, 39, 58)$	2

Lemma 12.3 (Exact rational degree-six certificate)**.** *As linear forms in all* 269 *real moment coordinates,*

$$\boxed{3T(y) + 1280V(y) = 80 \sum_{j=1}^{211} \lambda_j \widehat{\mathcal{E}}_j(y).} \tag{417}$$

Each $\widehat{\mathcal{E}}_j$ is an integer row obtained from (411), (412), *or* (413) *by its listed nonzero signed integer divisor. The list comprises* 203 *KL/scalar rows, five spin-flip rows, and three spectator-Hadamard rows. Every λ_j is rational.*

Exact finite verification. The finite certificate specifies the 36 cubic terms in Table 3, the coefficients $3, 1280, 80$, and all 211 row labels, signed divisors, and rational multipliers. The 429-label set $\mathcal{C}$, the 715 triples, their 269 real entry orbits, the target row, the square row $\mathbf{v}^T Y \mathbf{v}$, and every constraint row are reconstructed from the displayed formulas. Each KL/scalar row has a label in $\mathcal{C}$, and division by each specified divisor gives an integer row. Exact comparison over $\mathbb{Q}$ verifies equality of all 269 coefficients in (417). The positive term has the explicit squared-modulus representation in (415). This establishes the stated linear-form identity by finite rational computation. □

Global converse in Theorem 12.1. Start with any exact encoder and construct its averaged moment (408). Every row on the right of (417) vanishes by the arguments in Subsection 12.5. Therefore

$$0 = 3N^3 A_3 + 1280\, \mathbf{v}^T Y \mathbf{v} = 3N^3 A_3 + 81920\, \mathbb{E}_{\mathcal{G}} |p(\xi)|^2. \tag{418}$$

Both terms are nonnegative, and $N > 0$; hence $A_3 = 0$. Equation (400) now gives $A = \mathcal{L}(\alpha)$, $0 \le \alpha \le 1$. The construction below supplies the reverse inclusion. All changes of variables and constraint multiples were polynomial and homogeneous. The sole physical normalization is $N > 0$; no coordinate chart, nonzero-amplitude hypothesis, spectator Schmidt division, or regular-support assumption is needed. □

12.7 Exact attainment of the entire interval

Let P_5 be the perfect five-qubit code with stabilizer generators

$$XZZXI, \quad IXZZX, \quad XIXZZ, \quad ZXIXZ$$

on old sites $1, \dots, 5$. For completeness, the single-Pauli syndromes, in this generator order, are

old site	X	Y	Z
1	0001	1011	1010
2	1000	1101	0101
3	1100	1110	0010
4	0110	1111	1001
5	0011	0111	0100

They exhaust the 15 nonzero four-bit syndromes. A weight-two Pauli on different sites has the XOR of two distinct such syndromes, which is nonzero. Thus every nonidentity Pauli supported on at most two old sites anticommutes with a generator and has zero code compression. The four independent commuting generators define dimension two. Both $Z^{\otimes 5}$ and $X^{\otimes 5}$ normalize the stabilizer and anticommute, so they act as nontrivial logical Paulis. Choose its logical eigenbasis for $Z_1 Z_2 Z_3 Z_4 Z_5$. Replace old site 1 by output sites $0, 1$, leaving the other four sites unchanged. For $0 \le \theta \le \pi/2$, set

$$a_\theta = \cos(\theta/2), \qquad b_\theta = \sin(\theta/2)$$

and use the isometry

$$\begin{aligned} W_\theta |0\rangle &= (a_\theta |0\rangle + b_\theta |1\rangle)_0 |0\rangle_1, \\ W_\theta |1\rangle &= (a_\theta |0\rangle - b_\theta |1\rangle)_0 |1\rangle_1. \end{aligned} \tag{419}$$

One has $W_\theta^\dagger W_\theta = I$ and

$$(I_0 Z_1) W_\theta = W_\theta Z_1, \qquad W_\theta W_\theta^\dagger = \tfrac{1}{2}(I + \cos\theta\, Z_0 + \sin\theta\, X_0 Z_1).$$

Thus the new encoder realizes $D = I_0 Z_1 Z_2 Z_3 Z_4 Z_5$, as required.

If a six-qubit Pauli error has weight at most two, its pullback through $W_\theta \otimes I^{\otimes 4}$ is an operator supported on at most two old sites. Purity of P_5 makes its compression a scalar multiple of the logical identity, proving the full KL system. For its scalar expectation, a nonidentity Pauli on any of the four unsplit sites gives zero: after expanding the old-site-1 factor in Paulis, all terms are nonidentity operators on at most two old sites. If the error is supported only on the pair $0, 1$, its scalar is the identity coefficient of $W_\theta^\dagger E W_\theta$, namely

$$\gamma_E = \tfrac{1}{2} \operatorname{Tr}(E W_\theta W_\theta^\dagger).$$

Apart from $E = I$, the only nonzero scalars of weight at most two are

$$\gamma_{Z_0} = \cos\theta, \qquad \gamma_{X_0 Z_1} = \sin\theta.$$

Consequently $A_1 = \cos^2\theta$ and $A_2 = \sin^2\theta$. Equation (390), together with $A_3 \geq 0$, already gives $A_3 = 0$ for this construction. Substitution in (389) gives

$$A = \mathcal{L}(\cos^2\theta).$$

As θ runs from 0 to $\pi/2$, the parameter covers all $\alpha \in [0, 1]$. This completes the proof of Theorem 12.1.

12.8 Relation to orbit 2 and the scope of the result

Orbit 2 selects a weight-four product-Pauli logical involution, whereas orbit 3 selects the weight-five involution used here. The two global enumerator theorems have the same answer,

$$\mathcal{W}_2 = \mathcal{W}_3 = \{\mathcal{L}(\alpha) : 0 \leq \alpha \leq 1\}.$$

This equality of images does not identify the underlying sets of projectors. A code realizing both records belongs to their intersection, but the orbit-3 converse above neither assumes nor proves the existence of a weight-four product logical on every orbit-3 code. The degree-six certificate is proved directly in orbit-3 encoder amplitudes; it does not invoke the orbit-2 auxiliary partial-isometry classification or a universal shortening theorem.

The enumerator problem is therefore closed for orbit 3. Complete LU classification, the possibility of different code families sharing these enumerators, and the universal existence of additional product logical operators are separate questions. Together with the orbit-2 theorem, this leaves the enumerator converses for primitive order-two orbits 1 and 4 as the unresolved cases among the four records.

13 Further projective-order-two enumerator images

The projective-order-two sector contains twenty-five rank-compatible moment orbits. Fourteen are impossible by Section 24. Among the remaining eleven, the seven images treated in this section and the two images proved in Sections 11 and 12 leave only orbits 1 and 4 unresolved.

It is convenient to name the two rigid enumerator pairs

$$\mathcal{E}_1^A = (1, 1, 0, 0, 15, 15, 0), \qquad \mathcal{E}_1^B = (1, 1, 0, 30, 45, 33, 18), \tag{420}$$

$$\mathcal{E}_2^A = (1, 0, 1, 0, 11, 16, 3), \qquad \mathcal{E}_2^B = (1, 0, 1, 24, 35, 40, 27). \tag{421}$$

13.1 The five weight-one or weight-two Pauli-stabilizer orbits

Theorem 13.1 (Orbits 10, 11, 16, 22, and 23)**.** *The exact images are*

$$\mathcal{W}_{10} = \mathcal{W}_{16} = \mathcal{W}_{22} = \{\mathcal{E}_2^A\}, \tag{422}$$

$$\mathcal{W}_{11} = \mathcal{W}_{23} = \{\mathcal{E}_1^A\}. \tag{423}$$

The corresponding B-enumerators are respectively $\mathcal{E}_2^B$ and $\mathcal{E}_1^B$.

Proof. For every one of these representatives, the logical action of $D(a;4)$ is $\mathrm{diag}(1,-1)$. Hence $D(a;4)^2$ is scalar on the code. The odd entries of a give the following forced Pauli stabilizers:

orbit	representative a	forced $G = D(a;4)^2$	$\mathrm{wt}(G)$
10	$(0,0,1,1,2,2)$	Z_3Z_4	2
11	$(0,0,1,2,2,2)$	Z_3	1
16	$(0,1,2,2,2,3)$	Z_2Z_6	2
22	$(1,1,2,2,2,2)$	Z_1Z_2	2
23	$(1,2,2,2,2,2)$	Z_1	1

In each case $GP = P$, so the complete coefficient of G is one and therefore $A_{\mathrm{wt}(G)} \geq 1$. The zeroth shadow inequality

$$A_1 + A_2 + A_3 \leq 1 \tag{424}$$

then forces

$$(A_1, A_2, A_3) = \begin{cases} (1,0,0), & \mathrm{wt}(G) = 1, \\ (0,1,0), & \mathrm{wt}(G) = 2. \end{cases}$$

The universal six-qubit MacWilliams normal form (105) gives $\mathcal{E}_1^A$ or $\mathcal{E}_2^A$, and the quantum MacWilliams transform gives the displayed B-enumerators.

It remains to record exact attainment. Start with the five-qubit perfect $[[5,1,3]]$ code. It has a weight-five logical Pauli and, after multiplying by a stabilizer, a weight-three logical Pauli. Physical permutations and single-qubit Clifford conjugations may place either representative on any chosen support and turn all of its nonidentity factors into Z's.

For orbits 11 and 23, append a fixed $|0\rangle$ on the qubit supporting the forced weight-one stabilizer. The effective diagonal operator on the remaining five code qubits is, respectively, a weight-three or weight-five logical Z. For orbits 10, 16, and 22, replace one perfect-code qubit by the repetition isometry

$$V|0\rangle = |00\rangle, \qquad V|1\rangle = |11\rangle.$$

The split pairs and the resulting effective exponent vectors are

orbit	split pair	effective five-qubit exponents modulo four
10	$(3,4)$	$(0,0,2,2,2)$
16	$(2,6)$	$(0,0,2,2,2)$
22	$(1,2)$	$(2,2,2,2,2)$

Thus the effective diagonal is again a weight-three or weight-five logical Pauli. Appending a fixed qubit and applying the repetition split both preserve distance at least three: errors that change the fixed or repetition parity have zero compression, while every commuting physical error of weight at most two reduces to a perfect-code error of weight at most two. Choosing the logical eigenbasis of the effective Pauli gives logical action $\mathrm{diag}(1,-1)$, proving nonemptiness in all five orbits. □

13.2 Orbit 21: a complete label-sensitive certificate

Orbit 21 is represented by

$$(q, m, a) = \big(4, 2, (1, 1, 1, 2, 2, 3)\big), \qquad \delta = 2. \tag{425}$$

We prove

$$\boxed{\mathcal{W}_{21} = \{\mathcal{E}_2^A\}}, \qquad \boxed{B = \mathcal{E}_2^B}. \tag{426}$$

Unlike the five orbits above, the immediately forced Pauli stabilizer has weight four, so the scalar shadow bound alone does not determine the first three enumerator coefficients. Nevertheless, the exact result is the same enumerator as for orbits 10, 16, and 22.

13.2.1 Forced symmetries

Let

$$D = S_1 S_2 S_3 Z_4 Z_5 S_6^\dagger, \qquad S = \mathrm{diag}(1, i). \tag{427}$$

Since the logical action of D is $\mathrm{diag}(1, -1)$,

$$DPD^\dagger = P.$$

Squaring gives the scalar Pauli stabilizer

$$D^2 = Z_1 Z_2 Z_3 Z_6 =: G, \qquad GP = P. \tag{428}$$

Thus every computational-basis string in either logical support satisfies

$$x_1 \oplus x_2 \oplus x_3 \oplus x_6 = 0. \tag{429}$$

The weight-four stabilizer gives only $A_4 \geq 1$, which does not combine directly with $A_1 + A_2 + A_3 \leq 1$. A label-sensitive complete-enumerator argument is required.

13.2.2 Complete Pauli and shadow coefficients

For every Hermitian six-qubit Pauli E, define

$$a_E = \frac{1}{4} \left|\mathrm{Tr}(EP)\right|^2, \tag{430}$$

$$b_E = \frac{1}{2} \mathrm{Tr}(PEPE), \tag{431}$$

and, with

$$\widetilde{P} = Y^{\otimes 6} \overline{P}\, Y^{\otimes 6},$$

define the complete shadow coefficient

$$s_E = \frac{1}{2} \mathrm{Tr}(PE\widetilde{P}E). \tag{432}$$

Then

$$a_E \geq 0, \qquad b_E \geq 0, \qquad s_E \geq 0, \qquad A_j = \sum_{\mathrm{wt}(E)=j} a_E.$$

Writing $\langle E, F\rangle \in \mathbb{F}_2$ for the Pauli commutation pairing, the complete MacWilliams and shadow transforms are

$$b_E = \frac{1}{32}\sum_F (-1)^{\langle E,F\rangle} a_F, \tag{433}$$

$$s_E = \frac{1}{32}\sum_F (-1)^{\mathrm{wt}(F)} (-1)^{\langle E,F\rangle} a_F. \tag{434}$$

For every Pauli of weight at most two, the KL equations imply $b_E = a_E$. The forced symmetries imply further that

(i) if E anticommutes with G, then $a_E = 0$;

(ii) if E commutes with G, then $a_{GE} = a_E$, with the Pauli sign discarded;

(iii) $a_{DED^\dagger} = a_E$, again discarding Pauli signs.

Let $[E]$ denote the unsigned coefficient class generated by multiplication by G and conjugation by D, and let $a_{[E]}$ be its common coefficient.

13.2.3 Exact complete-MacWilliams/shadow certificate

For an orbit-21 completion, normalization and the six-qubit MacWilliams identity give

$$a_I = 1, \qquad A_6 + 4A_1 + A_2 + A_3 = 4,$$

The complete KL identities include

$$b_I = a_I, \qquad b_{Z_i} = a_{Z_i}, \quad i \in \{1,2,3,6\},$$

Expanding the complete transforms leaves a residual quadratic trace expression. Its Walsh transform vanishes on pairs of strings in the prescribed fibers, giving

$$\boxed{\begin{aligned} A_2 - 1 = 2\sum_{E\in\mathcal{B}_{21}} b_E + 2\sum_{E\in\mathcal{T}_{21}} s_E \\ + 2\sum_{F\in\mathcal{R}_{21,2}} a_{[F]} + 3\sum_{F\in\mathcal{R}_{21,3}} a_{[F]} + 4\sum_{F\in\mathcal{R}_{21,4}} a_{[F]}. \end{aligned}} \tag{435}$$

Here

$$\mathcal{B}_{21} = \{\mathrm{IIZIIZ}, \mathrm{IZIIIZ}, \mathrm{IZZIII}\} = \{Z_3Z_6, Z_2Z_6, Z_2Z_3\}. \tag{436}$$

The shadow labels are

$$\begin{aligned} \mathcal{T}_{21} = \{&\mathrm{IIIZZZ}, \\ &\mathrm{IIZIXZ}, \mathrm{IIZIYZ}, \mathrm{IIZIZZ}, \mathrm{IIZXIZ}, \mathrm{IIZYIZ}, \mathrm{IIZZIZ}, \mathrm{IIZZZI}, \\ &\mathrm{IZIIXZ}, \mathrm{IZIIYZ}, \mathrm{IZIIZZ}, \mathrm{IZIXIZ}, \mathrm{IZIYIZ}, \mathrm{IZIZIZ}, \mathrm{IZIZZI}, \\ &\mathrm{IZZIXI}, \mathrm{IZZIYI}, \mathrm{IZZIZI}, \mathrm{IZZXII}, \mathrm{IZZYII}, \mathrm{IZZZII}, \mathrm{IZZZZZ}\}. \end{aligned} \tag{437}$$

The residual complete-A classes are represented by

$$\begin{aligned} \mathcal{R}_{21,2} = \{&\mathrm{IIIXXI}, \mathrm{IIIXYI}, \mathrm{IIIYXI}, \mathrm{IIIYYI}, \mathrm{IIIZZI}, \\ &\mathrm{IIZIZZ}, \mathrm{IIZZIZ}, \mathrm{IZIIZZ}, \mathrm{IZIZIZ}, \mathrm{IZZIZI}, \mathrm{IZZZII}\}, \end{aligned} \tag{438}$$

$$\mathcal{R}_{21,3} = \{\mathrm{IIIIIZ}, \mathrm{IIIIZI}, \mathrm{IIIZII}, \mathrm{IIZIII}, \mathrm{IZIIII}, \mathrm{IZZIIZ}\}, \tag{439}$$

and

$$\begin{aligned}\mathcal{R}_{21,4} = \{&\mathrm{IIXIIX}, \mathrm{IIXIIY}, \mathrm{IXIIIX}, \mathrm{IXIIIY}, \mathrm{IXXIII}, \mathrm{IXYIII},\\ &\mathrm{XIIIIX}, \mathrm{XIIIIY}, \mathrm{XIXIII}, \mathrm{XIYIII}, \mathrm{XXIIII}, \mathrm{XYIIII}\}.\end{aligned} \tag{440}$$

Identity (435) is an exact rational certificate. The two forced symmetries reduce the 4096 Pauli labels to 640 unsigned classes after anticommuting labels are removed. To eliminate the residual, expand each a_E as a squared projector trace. Every resulting term contains two strings and their common X-mask translation in one of the two prescribed fibers. The Walsh coefficient of each such pair is zero. The remaining terms are exactly the five nonnegative sums in Eq. (435).

Every term on the right side of Eq. (435) is nonnegative; therefore

$$\boxed{A_2 \geq 1.} \tag{441}$$

The zeroth shadow inequality gives $A_1 + A_2 + A_3 \leq 1$. Hence

$$\boxed{A_1 = 0, \qquad A_2 = 1, \qquad A_3 = 0.} \tag{442}$$

Substitution into Eq. (105) and application of the quantum MacWilliams transform give

$$A = (1, 0, 1, 0, 11, 16, 3) = \mathcal{E}_2^A, \tag{443}$$

$$B = (1, 0, 1, 24, 35, 40, 27) = \mathcal{E}_2^B. \tag{444}$$

This proves uniqueness for every possible orbit-21 completion, including all boundary-support completions.

13.2.4 Exact realization

The following code realizes orbit 21:

$$\begin{aligned}\sqrt{8}\,|0_L\rangle = &|000000\rangle + |001001\rangle + |010111\rangle + |011100\rangle\\ &- |100111\rangle + |101100\rangle - |110010\rangle + |111011\rangle,\end{aligned} \tag{445}$$

and

$$\begin{aligned}\sqrt{8}\,|1_L\rangle = &|000100\rangle - |001101\rangle - |010011\rangle + |011000\rangle\\ &- |100011\rangle - |101000\rangle + |110110\rangle + |111111\rangle.\end{aligned} \tag{446}$$

Every basis vector in $|0_L\rangle$ satisfies $a \cdot x \equiv 0 \pmod 4$, while every basis vector in $|1_L\rangle$ satisfies $a \cdot x \equiv 2 \pmod 4$. Therefore

$$D(a;4)|0_L\rangle = |0_L\rangle, \qquad D(a;4)|1_L\rangle = -|1_L\rangle.$$

A direct exact evaluation gives

$$PEP = \lambda_E P \qquad \text{for every } \mathrm{wt}(E) \leq 2.$$

Apart from $\lambda_I = 1$, the only nonzero coefficients are

$$\lambda_{Z_5Z_6} = \frac{1}{2}, \qquad \lambda_{Z_3Z_5} = -\frac{1}{2}, \qquad \lambda_{Z_2Z_5} = \frac{1}{2}, \qquad \lambda_{Z_1Z_5} = \frac{1}{2}. \tag{447}$$

Thus the full KL system holds and

$$A_2 = 4\left(\frac{1}{2}\right)^2 = 1.$$

Since $B_3 > A_3$, the distance is exactly three.

13.2.5 Moment and phase geometry

The complete moment system for orbit 21 has rank 20, hence its affine moment polytope has dimension 12. Independent exact face exploration reveals twelve symmetry-related phase-feasible square faces. A representative bottom support is

$$T_0 = \{\texttt{000000}, \texttt{001001}, \texttt{010111}, \texttt{011100}, \\ \texttt{100111}, \texttt{101100}, \texttt{110010}, \texttt{111011}\}, \tag{448}$$

with the coordinate-four-matched top support

$$T_1 = \{\texttt{000100}, \texttt{001101}, \texttt{010011}, \texttt{011000}, \\ \texttt{100011}, \texttt{101000}, \texttt{110110}, \texttt{111111}\}. \tag{449}$$

Before imposing the half-angle phase equations, the two diagonal probability vectors on this face are

$$p = q = \left(\alpha, \frac{1}{4} - \alpha, \beta, \frac{1}{4} - \beta, \frac{1}{4} - \beta, \beta, \frac{1}{4} - \alpha, \alpha\right). \tag{450}$$

The restricted half-angle Walsh matrix has rank six and kernel

$$\operatorname{span}\{(0,0,1,-1,-1,1,0,0), (1,-1,0,0,0,0,-1,1)\}. \tag{451}$$

If $z_r = \bar{c}_r d_r$, then $|z_r| = p_r$. Comparing magnitudes in the two kernel blocks forces

$$\alpha = \beta = \frac{1}{8}. \tag{452}$$

Equations (445)–(446) give one exact sign solution. This face geometry is useful for a finer amplitude classification, but it is not used in the global enumerator theorem: the complete MacWilliams/shadow certificate proves $A_2 \geq 1$ without assuming that the twelve observed faces exhaust the phase variety.

Theorem 13.2 (Exact orbit-21 image)**.** *For every exact orbit-21 completion,*

$$A = \mathcal{E}_2^A, \qquad B = \mathcal{E}_2^B,$$

and the code in Eqs. (445)–(446) *proves attainment. Hence*

$$\boxed{\mathcal{W}_{21} = \{\mathcal{E}_2^A\}.}$$

Proof. Uniqueness follows from Eqs. (435)–(444); exact attainment and distance three follow from Eqs. (445)–(447). □

13.3 Orbit 45: a complete one-parameter enumerator interval

Orbit 45 is represented by

$$(q, m, a) = \big(6, 2, (1,1,2,3,4,4)\big), \qquad \delta = 3. \tag{453}$$

This orbit is exceptional among the $q = 6, m = 2$ records: the half-angle restricted Walsh matrix is singular, but its kernel is small enough that the remaining phase equations can be solved exactly.

Theorem 13.3 (Exact orbit-45 enumerator image)**.** *Orbit 45 admits exact* $((6,2,3))$ *completions. Its complete Shor–Laflamme A-enumerator image is*

$$\boxed{\mathcal{W}_{45} = \left\{ (1,0,t,1-t,12-t,15+t,3) : \frac{2}{3} \le t \le 1 \right\}.} \tag{454}$$

The corresponding B-enumerators are

$$\boxed{(B_0,\dots,B_6) = (1,0,t,22+2t,39-4t,42-2t,24+3t)\,, \qquad \frac{2}{3} \le t \le 1.} \tag{455}$$

More precisely, the exact moment/phase solutions are parametrized at the enumerator level by

$$\beta,\gamma,\sigma \ge 0, \qquad \beta+\gamma+\sigma = \frac{1}{4}, \qquad \max\{\beta,\gamma,\sigma\} \le \frac{1}{8}, \tag{456}$$

and

$$t = 32\left(\beta^2+\gamma^2+\sigma^2\right). \tag{457}$$

Proof. We divide the proof into the exact moment reduction, the half-angle kernel, the remaining phase equations, and the enumerator calculation.

The fibers and the diagonal moment polytope. Use coordinate 4, whose coefficient $a_4 = 3 = \delta$, as the half-angle coordinate. Order the bottom fiber as

$$\begin{aligned} S_0 = \{x_0,\dots,x_9\} = \{&000000,\ 001001,\ 001010,\ 010111,\ 011100,\\ &100111,\ 101100,\ 110001,\ 110010,\ 111011\}, \end{aligned} \tag{458}$$

and put

$$y_r = x_r \oplus e_4. \tag{459}$$

Then

$$\begin{aligned} S_3 = \{y_0,\dots,y_9\} = \{&000100,\ 001101,\ 001110,\ 010011,\ 011000,\\ &100011,\ 101000,\ 110101,\ 110110,\ 111111\}. \end{aligned} \tag{460}$$

Write

$$|0_L\rangle = \sum_{r=0}^{9} c_r |x_r\rangle, \qquad |1_L\rangle = \sum_{r=0}^{9} d_r |y_r\rangle, \tag{461}$$

and set

$$p_r = |c_r|^2, \qquad q_r = |d_r|^2. \tag{462}$$

Exact row reduction of the two normalizations and the degree-at-most-two diagonal KL moments gives

$$\begin{aligned} (p_0,\dots,p_9) &= (q_0,\dots,q_9) \\ &= \left(\gamma,\ \beta,\ \sigma,\ \alpha,\ \frac{1}{4}-\alpha,\ \frac{1}{4}-\alpha,\ \alpha,\ \sigma,\ \beta,\ \gamma\right). \end{aligned} \tag{463}$$

$$\sigma = \frac{1}{4} - \beta - \gamma. \tag{464}$$

The complete 23×20 moment matrix has rank 17; one 17×17 minor has determinant -108. Hence Eq. (463) is the complete three-dimensional diagonal moment polytope, including all of its boundary faces.

The half-angle Walsh kernel. Define the matched cross monomials

$$z_r = \overline{c_r} d_r. \tag{465}$$

For the half-angle coordinate 4, use all twelve character rows

$$1, \quad \chi_{\{4\}}, \quad \chi_{\{j\}}, \quad \chi_{\{4,j\}}, \qquad j \in \{1,2,3,5,6\}, \tag{466}$$

coming from

$$X_4,\ Y_4,\ X_4Z_j,\ Y_4Z_j.$$

Restricted to the ten matched incidences, this Walsh matrix has rank 7. For example, the minor using rows

$$1,\ \chi_{\{4\}},\ \chi_{\{1\}},\ \chi_{\{1,4\}},\ \chi_{\{3\}},\ \chi_{\{3,4\}},\ \chi_{\{5\}}$$

and columns $0,1,2,3,4,5,7$ has determinant of absolute value 256. Its exact kernel is

$$\boxed{(z_0,\dots,z_9) = (C,\ B,\ -B-C,\ A,\ -A,\ -A,\ A,\ -B-C,\ B,\ C)} \tag{467}$$

for some $A, B, C \in \mathbb{C}$.

Because the matched probabilities are equal,

$$|z_r| = \sqrt{p_r q_r} = p_r. \tag{468}$$

Comparing entries $3,4,5,6$ in Eqs. (463) and (467) gives

$$|A| = \alpha = \frac{1}{4} - \alpha, \tag{469}$$

and therefore

$$\boxed{\alpha = \frac{1}{8}, \qquad |A| = \frac{1}{8}.} \tag{470}$$

The other entries give

$$|B| = \beta, \qquad |C| = \gamma, \qquad |B+C| = \sigma. \tag{471}$$

Thus β, γ, σ are the side lengths of a possibly degenerate triangle. Since their sum is $1/4$, this is equivalent to Eq. (456).

After a relative logical phase, take

$$A = \frac{1}{8}. \tag{472}$$

Write

$$B = \beta\xi, \qquad C = \gamma\eta, \qquad -B-C = \sigma\zeta, \qquad \xi,\eta,\zeta \in \mathrm{U}(1). \tag{473}$$

The triangle equation is then

$$\beta\xi + \gamma\eta + \sigma\zeta = 0. \tag{474}$$

Shadow saturation and the identity $A_2 + A_3 = 1$. First note that every one-body scalar vanishes. Substitution of Eq. (463) with $\alpha = 1/8$ gives

$$\sum_{r=0}^{9} p_r (-1)^{(x_r)_i} = 0, \qquad i = 1, \ldots, 6. \tag{475}$$

Moreover, neither S_0 nor S_3 contains two strings at Hamming distance one. Hence every X_i- and Y_i-expectation also vanishes. Therefore

$$\boxed{A_1 = 0.} \tag{476}$$

Let

$$\Theta = Y^{\otimes 6} K, \qquad \widetilde{P} = \Theta P \Theta^{-1} \tag{477}$$

be the six-qubit spin flip. For a $((6, 2, 3))$ code, the MacWilliams normal form (105) gives

$$S_0 = \frac{1}{2} \operatorname{Tr}(P\widetilde{P}) = \frac{1}{4}(1 - A_1 - A_2 - A_3). \tag{478}$$

Up to a common phase,

$$\Theta |x\rangle = (-1)^{\mathrm{wt}(x)} |\overline{x}\rangle. \tag{479}$$

For the ordering above,

$$\overline{x_r} = y_{9-r}, \qquad p_r = p_{9-r}, \qquad z_r = z_{9-r}. \tag{480}$$

The first two equalities follow from Eqs. (458)–(463), and the third follows from Eq. (467).

The only potentially nonzero spin-flip overlap is, up to complex conjugation and a common phase,

$$T = \sum_{r=0}^{9} (-1)^{\mathrm{wt}(x_r)} c_r d_{9-r}. \tag{481}$$

Whenever $p_r > 0$,

$$d_r = \frac{z_r}{\overline{c_r}} = \frac{z_r c_r}{p_r}. \tag{482}$$

Consequently

$$c_r d_{9-r} = \frac{z_{9-r}}{p_{9-r}} c_r c_{9-r} = \frac{z_r}{p_r} c_r c_{9-r} = c_{9-r} d_r. \tag{483}$$

The same equality is trivial when the corresponding probability vanishes. On the other hand,

$$(-1)^{\mathrm{wt}(x_{9-r})} = -(-1)^{\mathrm{wt}(x_r)}, \tag{484}$$

because $x_{9-r} = \overline{x_r} \oplus e_4$. The terms r and $9 - r$ in Eq. (481) therefore cancel pairwise. Thus

$$T = 0, \qquad \operatorname{Tr}(P\widetilde{P}) = 0, \qquad S_0 = 0. \tag{485}$$

Combining Eqs. (476), (478), and (485) gives

$$\boxed{A_2 + A_3 = 1.} \tag{486}$$

Exact solution of the remaining phase equations. Assume first that

$$\beta\gamma\sigma > 0. \tag{487}$$

Put

$$h = \frac{1}{\sqrt{8}}. \tag{488}$$

A common phase of $|0_L\rangle$, together with six physical diagonal one-qubit phases, acts on the bottom amplitudes through the affine phase vectors $(1, x_r)$. The seven rows with

$$r = 0, 1, 2, 3, 4, 5, 7$$

have determinant 6. We may therefore fix those seven bottom amplitudes to be positive real numbers. Write the remaining three bottom phases as $U, V, W \in \mathrm{U}(1)$. Then

$$(c_0, \ldots, c_9) = \big(\sqrt{\gamma},\ \sqrt{\beta},\ \sqrt{\sigma},\ h,\ h,\ h,\ hU,\ \sqrt{\sigma},\ \sqrt{\beta}V,\ \sqrt{\gamma}W\big). \tag{489}$$

Equations (467), (473), and (482) then give

$$(d_0, \ldots, d_9) = \big(\sqrt{\gamma}\,\eta,\ \sqrt{\beta}\,\xi,\ \sqrt{\sigma}\,\zeta,\ h,\ -h,\ -h,\ hU,\ \sqrt{\sigma}\,\zeta,\ \sqrt{\beta}\,V\xi,\ \sqrt{\gamma}\,W\eta\big). \tag{490}$$

Besides the half-angle matching, the only weight-at-most-two flip masks connecting the two logical fibers are

$$\{2,6\},\quad \{2,5\},\quad \{2,3\},\quad \{1,6\},\quad \{1,5\},\quad \{1,3\}. \tag{491}$$

Substitution of Eqs. (489) and (490) into the four X/Y dressings of each mask gives the following exact factorization. Nonzero common factors such as $h\sqrt{\beta}$, $h\sqrt{\gamma}$, and $h\sqrt{\sigma}$ have been removed:

flip mask	independent cross-KL factors
$\{2,6\}$	$(V+1)(V\xi-1),\quad (V-1)(V\xi-1)$
$\{2,5\}$	$\zeta - 1$
$\{2,3\}$	$(W+1)(W\eta-1),\quad (W-1)(W\eta-1)$
$\{1,6\}$	$(U+V\xi)(UV+1),\quad (U+V\xi)(UV-1)$
$\{1,5\}$	$(U+1)(U+\zeta),\quad (U-1)(U+\zeta)$
$\{1,3\}$	$(U+W\eta)(UW+1),\quad (U+W\eta)(UW-1).$

(492)

Because all displayed variables have unit modulus, the two factors for $\{2,6\}$ force $V\xi = 1$, those for $\{2,3\}$ force $W\eta = 1$, and the $\{2,5\}$ row forces $\zeta = 1$. The $\{1,6\}$ pair then forces $U + V\xi = 0$. Therefore

$$\boxed{U = -1, \qquad V = \overline{\xi}, \qquad W = \overline{\eta}, \qquad \zeta = 1.} \tag{493}$$

The remaining factorized rows are then automatically satisfied. The triangle equation becomes

$$\boxed{\beta\xi + \gamma\eta + \sigma = 0.} \tag{494}$$

Thus every exact solution on the positive stratum is locally diagonally equivalent to

$$(c_0, \ldots, c_9) = \big(\sqrt{\gamma},\ \sqrt{\beta},\ \sqrt{\sigma},\ h,\ h,\ h,\ -h,\ \sqrt{\sigma},\ \sqrt{\beta}\,\overline{\xi},\ \sqrt{\gamma}\,\overline{\eta}\big), \tag{495}$$

$$(d_0, \ldots, d_9) = \big(\sqrt{\gamma}\,\eta,\ \sqrt{\beta}\,\xi,\ \sqrt{\sigma},\ h,\ -h,\ -h,\ -h,\ \sqrt{\sigma},\ \sqrt{\beta},\ \sqrt{\gamma}\big), \tag{496}$$

where $\xi, \eta \in \mathrm{U}(1)$ obey Eq. (494).

For completeness, the only two-bit flip masks occurring inside either logical fiber are

$$\{1,2\}, \quad \{3,5\}, \quad \{3,6\}, \quad \{5,6\}. \tag{497}$$

Extend each coefficient function by zero outside its logical fiber. For the canonical amplitudes, define the oriented incidence sum

$$F_{ij} := \sum_{\substack{x \in S_0 \\ x_i = 0}} c_x \overline{c_{x \oplus e_i \oplus e_j}}.$$

The corresponding sum in the other logical word has the same value. For each mask, at most one of the two channels $00 \leftrightarrow 11$ and $01 \leftrightarrow 10$ has a nonzero summed coherence. Their values are

mask	F_{ij}
$\{1,2\}$	0
$\{3,5\}$	$\sqrt{\gamma\sigma}\,(1+\eta)$
$\{3,6\}$	$\sqrt{\beta\gamma}\,\overline{\xi}\,(\xi+\eta)$
$\{5,6\}$	$\sqrt{\beta\sigma}\,(1+\xi)$.

(498)

Hence all diagonal KL equalities for the XX, XY, YX, YY dressings of these four masks hold. There are no other within-fiber or cross-fiber incidences of Hamming weight at most two. Together with the already solved diagonal moments and half-angle equations, this verifies the complete distance-at-least-three KL system.

Conversely, for every triple satisfying Eq. (456), elementary triangle geometry supplies $\xi, \eta \in \mathrm{U}(1)$ satisfying Eq. (494). Equations (495)–(496) then define normalized, orthogonal logical vectors satisfying all KL equations. Their supports lie in S_0 and S_3, respectively, so

$$D(a;6)|0_L\rangle = |0_L\rangle, \qquad D(a;6)|1_L\rangle = -|1_L\rangle. \tag{499}$$

Thus they are exact completions of orbit 45.

Calculation of A_2. Let

$$\lambda_{ij} = \langle 0_L|Z_iZ_j|0_L\rangle = \langle 1_L|Z_iZ_j|1_L\rangle. \tag{500}$$

Substitution of the moment normal form gives

$\{i,j\}$	λ_{ij}
$\{1,3\},\{2,3\}$	$4\gamma - \frac{1}{2}$
$\{1,5\},\{2,5\}$	$\frac{1}{2} - 4\sigma$
$\{1,6\},\{2,6\}$	$\frac{1}{2} - 4\beta$
$\{3,5\}$	-4β
$\{3,6\}$	-4σ
$\{5,6\}$	4γ
$\{1,2\},\{1,4\},\{2,4\},\{3,4\},\{4,5\},\{4,6\}$	0.

(501)

Put

$$R = \beta^2 + \gamma^2 + \sigma^2. \tag{502}$$

Squaring and summing Eq. (501) gives the purely diagonal contribution

$$A_2^{(Z)} = \sum_{i<j} |\lambda_{ij}|^2 = 48R - \frac{1}{2}. \tag{503}$$

For each two-bit flip mask in Eq. (497), the four Pauli dressings XX, XY, YX, YY are the four-point Walsh transform of the two oriented incidence sums. Parseval therefore gives a contribution $8|F_{ij}|^2$. Using Eq. (498),

$$A_2^{(\mathrm{off})} = 8\big[\beta\sigma|1+\xi|^2 + \beta\gamma|\xi+\eta|^2 + \gamma\sigma|1+\eta|^2\big]. \tag{504}$$

The vector-triangle identity $\beta\xi + \gamma\eta + \sigma = 0$ implies

$$\beta\sigma|1+\xi|^2 = \gamma^2 - (\beta-\sigma)^2, \tag{505}$$
$$\beta\gamma|\xi+\eta|^2 = \sigma^2 - (\beta-\gamma)^2, \tag{506}$$
$$\gamma\sigma|1+\eta|^2 = \beta^2 - (\gamma-\sigma)^2. \tag{507}$$

Adding these identities and using $\beta+\gamma+\sigma = 1/4$ gives

$$A_2^{(\mathrm{off})} = \frac{1}{2} - 16R. \tag{508}$$

Combining Eqs. (503) and (508) yields

$$\boxed{A_2 = 32R = 32\left(\beta^2+\gamma^2+\sigma^2\right).} \tag{509}$$

The only points of the triangle domain with $\beta\gamma\sigma = 0$ are the three permutations of

$$\left(0, \frac{1}{8}, \frac{1}{8}\right).$$

At each such point $R = 1/32$, and Eq. (503) already gives $A_2^{(Z)} = 1$. Since Eq. (486) gives $A_2 + A_3 = 1$, while both coefficients are nonnegative, one necessarily has

$$A_2 = 1, \qquad A_3 = 0.$$

Thus Eq. (509) also holds on every boundary-support solution.

Equation (486) now gives

$$A_3 = 1 - A_2. \tag{510}$$

The sharp interval and the full enumerators. By Cauchy–Schwarz,

$$R \geq \frac{(\beta+\gamma+\sigma)^2}{3} = \frac{1}{48}, \tag{511}$$

with equality exactly at

$$\beta = \gamma = \sigma = \frac{1}{12}.$$

On the other hand,

$$R \leq \max\{\beta,\gamma,\sigma\}(\beta+\gamma+\sigma) \leq \frac{1}{32}, \tag{512}$$

with equality at the three degenerate vertices. Hence

$$\frac{2}{3} \leq A_2 \leq 1. \tag{513}$$

Every intermediate value occurs. For example, the path

$$\beta_r = \frac{1-r}{12}, \qquad \gamma_r = \sigma_r = \frac{2+r}{24}, \qquad 0 \le r \le 1, \tag{514}$$

lies in the triangle domain and gives

$$32\left(\beta_r^2 + \gamma_r^2 + \sigma_r^2\right) = \frac{2}{3} + \frac{r^2}{3}. \tag{515}$$

Thus the complete A_2-image is $[2/3, 1]$.

Put $t = A_2$. Equations (476) and (510), followed by the universal six-qubit MacWilliams normal form (105), give

$$(A_0, \ldots, A_6) = (1, 0, t, 1-t, 12-t, 15+t, 3)\,. \tag{516}$$

The quantum MacWilliams transform gives Eq. (455). This proves Eq. (454) and completes the theorem. □

Remark 13.4 (Enumerator image versus code classification)**.** Theorem 13.3 determines the complete image of the orbit-45 exact KL variety under the Shor–Laflamme enumerator map. It also gives an explicit exact realization for every point of that image. It does not assert that two orbit-45 projectors with the same value of t are locally unitarily or locally monomially equivalent.

Corollary 13.5 (Current order-two status)**.** *Among the twenty-five projective-order-two moment orbits:*

(i) *fourteen have empty exact image;*

(ii) *the seven images for orbits 10, 11, 16, 21, 22, 23, and 45 are given above;*

(iii) *Sections 11 and 12 determine the images of orbits 2 and 3; only orbits 1 and 4 remain unresolved at the weight-enumerator-image level.*

Consequently the complete six-qubit classification now covers 64 *of the* 66 *moment-orbit images.*

14 Proofs of the main-text classification results

This section restates Results I–IV of Section IV of the main text and assembles their proofs from the exact enumeration, constructions, weight-enumerator converses, and no-go certificates in this supplement. The SSLP candidate classes of the main text are the moment orbits of Theorem 3.1; the index r refers to the common numbering in Table 1. These are classes of labeled symmetry data, and different classes can admit equivalent code projectors.

14.1 Result I: orders of the full diagonal logical groups

For a code projector P, let $\mathcal{G}_{\mathrm{diag}}(P)$ denote the projective logical image of all diagonal transversal unitaries preserving its range, in the chosen product basis. The maximum logical order $M_3(6)$ is defined in Section 1.

Theorem 14.1 (Main-text Result I: six-qubit projective diagonal groups)**.** *For an exact* $((6, 2, 3))$ *code whose projective diagonal logical group in the chosen product basis is nontrivial,*

$$\boxed{\mathcal{G}_{\mathrm{diag}}(P) \cong C_m, \qquad m \in \{2, 3, 5\}.} \tag{517}$$

The possible projective element orders are exactly $\{1, 2, 3, 5\}$*, including the identity, and* $M_3(6) = 5$*.*

Proof. Corollary 22.23 establishes the exact projective element-order set $\{1, 2, 3, 5\}$. Its proof combines the controlled-lift bound $m \mid q \leq 9$, the exact exclusions of orders $4, 6, 7, 8, 9$, and constructions realizing orders $2, 3, 5$. The rank-two SSLP normal-form corollary in the main text proves that the full projective diagonal logical group of a rank-two code of distance at least two is finite cyclic. The order of its generator therefore belongs to $\{2, 3, 5\}$ whenever the group is nontrivial. Conversely, a code realizing an element of order $m \in \{2, 3, 5\}$ must have full diagonal logical group C_m, since none of the other allowed group orders is divisible by m. Thus all three groups occur, and the maximum projective logical order is five. □

14.2 Result II: the realizable SSLP candidate classes

Theorem 14.2 (Main-text Result II: the fifteen realizable symmetry-data classes)**.** *Among the* 66 *SSLP candidate classes satisfying the support-dimension bound, the full weight-at-most-two Knill–Laflamme system admits an exact completion if and only if*

$$\boxed{r \in \{1, 2, 3, 4, 7, 8, 9, 10, 11, 16, 21, 22, 23, 36, 45\}.} \tag{518}$$

Their distribution by targeted projective logical order is

m	2	3	5
# *realizable classes*	11	3	1.

Each class contains an exact physical $((6, 2, 3))$ *code realization. These fifteen classes classify realizable symmetry data, not exact code projectors up to equivalence.*

Proof. Theorem 3.1 gives the exhaustive list of 66 rank-compatible candidate classes. The following exact no-go theorems exclude disjoint subsets, distinguished by their retained target-order labels:

Target order m	Number excluded	Exact no-go result
9	1	Theorem 17.2
8	2	Theorem 18.3
7	5	Theorem 19.6
6	4	Theorem 20.5
4	10	Theorem 22.22
5	7	Theorem 21.1
3	8	Theorem 23.10
2	14	Theorem 24.19
Total	51	

These exclusions include boundary solutions with zero amplitudes. They leave exactly the fifteen indices in Eq. (518). The converse direction follows from the exact constructions in the following results:

Realized classes	Exact construction result
$1, 2, 3, 4$	Theorem 10.1
7	Theorem 7.1
8	Theorem 8.1
9	Theorem 9.1
$10, 11, 16, 22, 23$	Theorem 13.1
21	Theorem 13.2
36	Theorem 6.2
45	Theorem 13.3

Each construction satisfies the complete weight-at-most-two Knill–Laflamme equations, giving distance at least three. The quantum Singleton bound excludes distance at least four for a six-qubit code encoding one qubit, so each has exact distance three. Finally, the retained labels give eleven realized classes with $m = 2$, three with $m = 3$, and one with $m = 5$. □

14.3 Result III: the proved weight-enumerator sets

Let $\mathcal{W}_r$ be the set of vectors $(A_0(P), \ldots, A_6(P))$ attained by exact code projectors in candidate class r, with the normalization in Eq. (98). We use $\mathcal{L}(\alpha)$ and $\mathcal{M}(t)$ from Eqs. (320)–(321), and $\mathcal{E}_1^A = \mathcal{L}(1)$, $\mathcal{E}_2^A = \mathcal{L}(0) = \mathcal{M}(1)$ from Eqs. (420)–(421). Write

$$\mathcal{R}_5^A := \left(1, 0, \frac{21}{25}, 0, \frac{291}{25}, \frac{384}{25}, \frac{79}{25}\right). \tag{519}$$

Theorem 14.3 (Main-text Result III: the proved weight-enumerator sets)**.** *For the fifteen realizable classes in Theorem 14.2, the following exact nonempty sets have been established:*

$$\mathcal{W}_{36} = \{\mathcal{R}_5^A\}, \tag{520}$$
$$\mathcal{W}_2 = \mathcal{W}_3 = \{\mathcal{L}(\alpha) : 0 \le \alpha \le 1\}, \tag{521}$$
$$\mathcal{W}_7 = \{\mathcal{L}(\alpha) : 0 \le \alpha \le 1\}, \tag{522}$$
$$\mathcal{W}_8 = \{\mathcal{L}(\alpha) : \tfrac{1}{2} \le \alpha \le 1\}, \tag{523}$$
$$\mathcal{W}_9 = \{\mathcal{E}_2^A\}, \tag{524}$$
$$\mathcal{W}_{10} = \mathcal{W}_{16} = \mathcal{W}_{21} = \mathcal{W}_{22} = \{\mathcal{E}_2^A\}, \tag{525}$$
$$\mathcal{W}_{11} = \mathcal{W}_{23} = \{\mathcal{E}_1^A\}, \tag{526}$$
$$\mathcal{W}_{45} = \{\mathcal{M}(t) : \tfrac{2}{3} \le t \le 1\}. \tag{527}$$

These formulas determine thirteen sets completely. The two nonempty sets $\mathcal{W}_1, \mathcal{W}_4$ remain unresolved. The union of the thirteen completely determined sets is exactly

$$\bigcup_{r \in \{2,3,7,8,9,10,11,16,21,22,23,36,45\}} \mathcal{W}_r = \{\mathcal{L}(\alpha) : 0 \le \alpha \le 1\} \cup \{\mathcal{M}(t) : \tfrac{2}{3} \le t \le 1\} \cup \{\mathcal{R}_5^A\}. \tag{528}$$

This determines neither all fifteen enumerator sets in the nontrivial diagonal sector nor the enumerator set of all exact $((6, 2, 3))$ codes. In particular, it does not prove the global weight-enumerator conjecture stated in Result III of the main text.

Proof. Theorem 6.2 reduces every exact class-36 completion to one code under local monomial unitaries and qubit permutations. The Shor–Laflamme enumerators are invariant under these operations, and Eq. (106) computes the enumerator of that code, giving $\mathcal{W}_{36} = \{\mathcal{R}_5^A\}$.

Theorems 7.1, 8.1, and 9.1 prove the complete images of classes 7, 8, and 9, respectively. Their converse arguments apply to every exact completion, and their constructions attain the full stated parameter intervals, including the endpoints. Theorem 13.1 settles classes 10, 11, 16, 22, and 23, and Theorem 13.2 settles class 21. In each of these six cases, the enumerator converse and an exact realization establish a singleton image. The flip reduction, Gram-block constraints, and phase family in Theorem 13.3 give both the bounds and every point of the interval $\mathcal{W}_{45}$. These results include all admissible zero-amplitude boundary cases.

Theorem 11.1 and Theorem 12.1 determine $\mathcal{W}_2$ and $\mathcal{W}_3$ on all rank and boundary strata.

For the four primitive order-two records, Theorem 10.1 proves

$$\{\mathcal{L}(\alpha) : 0 \le \alpha \le 1\} \subseteq \bigcap_{r=1}^{4} \mathcal{W}_r, \qquad \{\mathcal{M}(t) : \tfrac{2}{3} \le t \le 1\} \subseteq \mathcal{W}_1 \cap \mathcal{W}_4.$$

These constructions prove nonemptiness and the stated inclusions. The orbit-2/3 converses exclude additional enumerators in those two classes; the corresponding exclusion remains open for classes 1 and 4.

Finally, $\mathcal{W}_7$ contains the entire $\mathcal{L}$-interval, while $\mathcal{W}_8$ is a subinterval and the singleton vectors $\mathcal{E}_1^A, \mathcal{E}_2^A$ are the endpoints of $\mathcal{W}_7$. Together with $\mathcal{W}_{45}$ and $\mathcal{W}_{36}$, this gives Eq. (528). The global conjecture additionally concerns codes with no nontrivial projective diagonal action, which are outside the nontrivial-action enumeration. □

14.4 Result IV: the unique order-five code and its minimum length

Theorem 14.4 (Main-text Result IV: the unique projective-C_5 code)**.** *Every exact* $((6, 2, 3))$ *code admitting a projective diagonal logical element of order five belongs to class 36. All exact completions of class 36 are equivalent under physical-qubit permutations, single-qubit bit flips, and diagonal phase gates. Consequently the projective-*C_5 *sector consists of one local-unitary code class, with physical-qubit permutations allowed.*

Proof. For an element of projective logical order five, the controlled-lift bound forces $5 \mid q \le 9$, hence $q = 5$. The exhaustive enumeration in Theorem 3.1 places its symmetry data in one of classes 33–40. Theorem 21.1 excludes all of these except class 36.

Theorem 6.2 proves that every exact class-36 completion has full twelve-term support in each logical word, with six probabilities 1/10 and six probabilities 1/15. Its phase analysis reduces the discrete choices to the unique proper three-edge-coloring of K_4, up to the permitted equivalences, and removes the continuous phases by local diagonal unitaries and logical basis phases. The canonical code in Eqs. (95)–(96) satisfies every weight-at-most-two KL equation by the overlap check Eq. (97) and its accompanying table.

Logical basis phases and a logical-state interchange do not change the code subspace. The remaining equivalences are physical-qubit permutations and local monomial unitaries; each single-qubit monomial unitary is a product of a diagonal phase gate and a possible bit flip. Thus all exact class-36 codes are equivalent by the physical operations in the statement, and consequently belong to one LU class with permutations allowed. This is the uniqueness conclusion also recorded in Corollary 21.2. □

Corollary 14.5 (Main-text minimum-length corollary for projective order five)**.** *If an exact* $((n, 2, d))$ *code with* $d \ge 3$ *supports a single-block transversal unitary of projective logical order five, then* $n \ge 6$.

Equality is attained by the code in Eqs. (95)–(96), and every six-qubit code with this property is equivalent to it under local unitaries and qubit permutations.

Proof. A local change of basis diagonalizes each factor of the transversal unitary, preserving the code distance and projective logical order. Suppose $n < 6$, and let P be the resulting rank-two code projector. Append $6 - n$ fixed ancillas and extend the gate by the identity:

$$P' = P \otimes \big(|0\rangle\langle 0|\big)^{\otimes(6-n)}. \tag{529}$$

Every weight-at-most-two Pauli compression factors into a scalar compression on the original code and an ancilla expectation. Thus P' defines a six-qubit code of distance at least three, hence exactly three by the quantum Singleton bound, retaining the order-five logical action.

With the rank-two normalization in Eq. (98), each appended qubit contributes one to A_1 through its Pauli Z, while its X- and Y-contributions vanish. The weight-one contributions on the original qubits are unchanged because each ancilla projector has trace one. Consequently

$$A_1(P') = A_1(P) + (6 - n) > 0. \tag{530}$$

On the other hand, Theorem 14.4, LU invariance of the weight enumerator, and Eq. (106) imply $A_1(P') = 0$, a contradiction. The canonical class-36 code attains length six, and Theorem 14.4, after the initial local change of basis, gives uniqueness at that length. □

The same minimum length holds when the full projective transversal logical group cannot be made Clifford by any choice of logical basis. Let G be the full projective logical group induced by single-block tensor-product unitaries on an exact $((n, 2, d))$ code with $d \geq 3$ and $n \leq 5$. For each nonidentity element of G, a local change of physical basis diagonalizes its transversal representative while preserving the code distance and its projective logical order m. The physical-order bound in Section III.C of the main text and $\beta_5 = 5$ then give

$$m \leq q \leq \beta_n \leq \beta_5 = 5.$$

Corollary 14.5 excludes $m = 5$, so every element of G has order at most four. The local basis may depend on the chosen element; the uniform bound applies to each one.

By Eastin–Knill (Section II.B of the main text), G is a finite subgroup of $PU(2) \cong SO(3)$. The classification of finite rotation groups, equivalently the finite subgroups of $SU(2)$ reviewed in Section 2.3 of Ref. [1], now leaves only cyclic or dihedral groups whose principal rotation has order at most four, or the tetrahedral and octahedral rotation groups. All are conjugate to subgroups of the octahedral rotation group, which is the projective single-qubit Clifford group. Thus one logical basis makes every element of G Clifford. The six-qubit code above attains the minimum because its order-five logical element belongs to no conjugate of the Clifford group.

15 Remaining enumerator and code-family questions

15.1 Unresolved weight-enumerator images for orbits 1 and 4

Theorem 10.1 establishes exact families in each of orbits $1, 2, 3, 4$. Sections 11 and 12 now prove the complete images of orbits 2 and 3. Their common constructions and symmetry records are given in Section 10. The remaining partial converse analysis reduces orbit 4 to a parity-code dichotomy between the low-rank pair branch and the flat $SO(5)$-type branch. The latter analysis has been developed separately for its rank-three and rank-four pair strata, but the final dichotomy is not yet closed. Orbit 1 likewise has exact constructions and a stable numerical image, but no complete converse theorem. These partial reductions do not yet determine the full images of orbits 1 and 4.

Resolving these two images is the remaining finite task. Because their fibers are parity sectors of size 32, the useful next reduction should be label-sensitive rather than a further scalar MacWilliams argument: complete Pauli coefficients, marginal-rank strata, and exact phase-incidence factorizations are the natural tools.

15.2 Remaining classification at the code level

Even after those two images are decided, a complete family classification requires additional work:

(i) determine the full amplitude varieties and their connected components inside every nonempty order-two and order-three orbit, including the fibers over the enumerator parameters in orbits 7, 8, and 45;

(ii) decide whether equal enumerators in different moment orbits, or at different points of one interval, can nevertheless be locally equivalent;

(iii) reconstruct each exact projector and compute its intrinsic lattices L_P, M_P and quotient $H_P = M_P/L_P$, thereby identifying the maximal projective diagonal group rather than only the targeted subgroup;

(iv) quotient the exact solutions by the chosen code-equivalence relation, retaining genuine continuous moduli when they occur.

For six physical qubits encoding one qubit, the quantum Singleton bound rules out distance $d \geq 4$. Thus every solution of the distance-at-least-three KL equations has exact distance three.

Part III
Exact no-go proofs for fifty-one orbits

16 Guide to the exact no-go proofs

The preceding sections give the complete orbit list, the realized projective orders, and the nonempty weight-enumerator images. The sections in this part collect the exact nonexistence proofs in decreasing projective order: $9, 8, 7, 6, 5, 4, 3, 2$. The complete count tables and the code and data availability statement follow these proofs.

17 Exact nonexistence of projective order nine

The order-nine endpoint can be closed without a Gröbner-basis computation or a general real-algebraic decision procedure. The obstruction is a rank-two Walsh system attached to one weight-one flip.

17.1 The unique order-nine moment orbit

The most constrained endpoint is $m = 9$. There are five rank-compatible labeled $(q, m) = (9, 9)$ fiber pairs, but all five lie in one orbit. A convenient representative is

$$q = m = 9, \qquad \delta = 1, \qquad a = (1, 2, 2, 4, 4, 6). \tag{531}$$

The total angle sum is

$$\sum_{j=1}^{6} a_j = 19 \equiv 1 \pmod 9.$$

Therefore

$$A_a(\bar{x}) \equiv 1 - A_a(x) \pmod 9,$$

and hence

$$S_1 = \{\bar{x} : x \in S_0\}. \tag{532}$$

Both fibers have seven vertices. The exact moment polytope is zero-dimensional, so the probabilities are unique:

$x \in S_0$	$p_0(x)$	$\bar{x} \in S_1$	$p_1(\bar{x})$
000000	2/9	111111	2/9
111100	1/18	000011	1/18
111010	1/18	000101	1/18
100110	1/6	011001	1/6
110001	1/9	001110	1/9
101001	1/9	010110	1/9
011111	5/18	100000	5/18

The common mean is

$$\mu = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right), \tag{533}$$

and the common covariance is

$$\Sigma = \frac{1}{36}\begin{pmatrix} 9 & -1 & -1 & -1 & -1 & -1 \\ -1 & 9 & 5 & 3 & 3 & 5 \\ -1 & 5 & 9 & 3 & 3 & 5 \\ -1 & 3 & 3 & 9 & 7 & 1 \\ -1 & 3 & 3 & 7 & 9 & 1 \\ -1 & 5 & 5 & 1 & 1 & 9 \end{pmatrix}, \qquad \det \Sigma = \frac{5}{157464} > 0. \tag{534}$$

Thus the active affine rank is six, exactly what is required for a controlled lift of order nine.

Consequently, any exact code arising from this moment orbit must have all fourteen displayed basis strings in its actual support and must have magnitudes fixed by the table. The remaining freedom is phase:

$$c_x = \sqrt{p_0(x)}\, e^{i\alpha_x}, \qquad d_{\bar{x}} = \sqrt{p_0(x)}\, e^{i\beta_x}. \tag{535}$$

The order-nine existence problem has therefore been reduced, at this endpoint, to the exact solvability of a finite phase system. The two-edge obstruction below makes this phase system inconsistent using two off-diagonal KL equations.

17.2 The two-edge phase obstruction

Lemma 17.1 (Two-edge flip obstruction)**.** *Let*

$$|0_L\rangle = \sum_{x \in T_0} c_x |x\rangle, \qquad |1_L\rangle = \sum_{y \in T_1} d_y |y\rangle$$

be logical states of a binary code of distance at least three, where $T_0, T_1 \subseteq \{0,1\}^n$ are their actual supports. Suppose that for some coordinate i, exactly two top-support strings are joined to the bottom support by the flip e_i:

$$\{y \in T_1 : y \oplus e_i \in T_0\} = \{y, y'\}.$$

Suppose also that there is a coordinate $j \neq i$ with $y_j \neq y'_j$, and that all four endpoint amplitudes

$$c_{y \oplus e_i}, \quad c_{y' \oplus e_i}, \quad d_y, \quad d_{y'}$$

are nonzero. Then these two logical states cannot satisfy the distance-three Knill–Laflamme equations.

Proof. Set

$$A := \overline{c_{y \oplus e_i}} d_y, \qquad B := \overline{c_{y' \oplus e_i}} d_{y'}.$$

The off-diagonal Knill–Laflamme equation for X_i gives

$$A + B = 0.$$

The equation for the weight-two Pauli $X_i Z_j$ gives, up to multiplication of the whole equation by an irrelevant sign,

$$A - B = 0,$$

because the two contributing top strings have opposite j-th bits. Equivalently,

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = 0.$$

The Walsh matrix has determinant -2, so $A = B = 0$. This contradicts nonvanishing of all four endpoint amplitudes. □

Theorem 17.2 (Six-qubit order-nine no-go)**.** *No exact* $((6,2,d\geq 3))$ *qubit code admits a projective diagonal transversal logical element of order* 9*.*

Proof. Assume that such a code exists. For a controlled primitive physical lift, the distance-three Smith bound gives

$$9 = m \mid q, \qquad q \leq \beta_6 = 9.$$

Hence $q = 9$. The exhaustive lift-aware enumeration and the exact moment and affine-rank reductions of the preceding sections show that every such case is equivalent, under qubit permutations, local X conjugations, and logical-state interchange, to the unique orbit represented by Eq. (531). These operations preserve code existence, distance, and the projective order, so it suffices to rule out this representative.

For completeness, the uniqueness and positivity of the moment assignment in Section 17.1 have a compact exact certificate. Order the fourteen probability variables according to the seven rows of the displayed table, first for S_0 and then for the complementary strings in S_1. Take the following fourteen equations from the complete moment system:

(a) the two normalization equations;

(b) all six one-bit moment equalities;

(c) the five two-bit equalities for coordinates $(1, j)$, $j = 2, \ldots, 6$;

(d) the two-bit equality for coordinates $(2, 3)$.

The resulting 14×14 integer coefficient matrix has determinant

$$-36 \neq 0.$$

Its unique solution is exactly the probability table in Section 17.1; direct substitution verifies the remaining nine two-bit equations. In particular, every one of the fourteen probabilities is strictly positive, so every displayed basis string is in the actual support of any putative code.

Now consider a flip of the first qubit. A direct check of the two fibers gives

$$\{y \in S_1 : y \oplus e_1 \in S_0\} = \{100000, 111111\}. \tag{536}$$

The corresponding bottom endpoints are 000000 and 011111. Define

$$A := \overline{c_{000000}} d_{100000}, \qquad B := \overline{c_{011111}} d_{111111}. \tag{537}$$

The forced magnitudes are

$$|A| = \sqrt{\frac{2}{9}\frac{5}{18}} = \frac{\sqrt{5}}{9}, \qquad |B| = \sqrt{\frac{5}{18}\frac{2}{9}} = \frac{\sqrt{5}}{9}. \tag{538}$$

Thus A and B are both nonzero.

Only the two incidences in Eq. (536) contribute to the weight-one off-diagonal matrix element, so distance at least three requires

$$0 = \langle 0_L | X_1 | 1_L \rangle = A + B. \tag{539}$$

For $X_1 Z_2$, the top string 100000 has second bit zero while 111111 has second bit one. The same two contributions therefore acquire opposite signs, and the weight-two off-diagonal equation is

$$0 = \langle 0_L | X_1 Z_2 | 1_L \rangle = A - B. \tag{540}$$

Equations (539) and (540) force $A = B = 0$, contradicting Eq. (538). Hence the assumed code does not exist. □

Corollary 17.3 (Order range after the order-nine no-go)**.** *Every projective diagonal logical element of an exact* $((6,2,d\geq 3))$ *code has order at most* 8*. At this stage the known realized nontrivial orders are* $2,3,5$*, while* $m\in\{4,6,7,8\}$ *remain to be decided. The next section closes the order-eight endpoint.*

Remark 17.4 (Determinant-one convention)**.** In the determinant-one $SU(2)$ convention, projective order m corresponds to the cyclic subgroup C_{2m}. Thus Theorem 17.2 rules out a diagonal transversal C_{18} subgroup for a six-qubit distance-three code.

18 Exact nonexistence of projective order eight

We now close the next endpoint. The proof again uses only exact moment linear algebra and a small number of off-diagonal Knill–Laflamme equations, but the two surviving moment orbits exhibit two different obstruction mechanisms.

18.1 The two exhaustive order-eight moment orbits

Suppose that an exact $((6,2,d\geq 3))$ code admits a projective diagonal logical element of order $m=8$. For a controlled primitive physical lift, the physical-order bound in Section III.C of the main text gives

$$8=m\mid q,\qquad q\leq \beta_6=9. \tag{541}$$

Hence necessarily

$$q=8,\qquad \delta=q/m=1.$$

The exact enumeration in this note gives precisely two rank-compatible moment orbits:

orbit	representative $a\in\mathbb{Z}_8^6$	$\lvert S_0\rvert$	$\lvert S_1\rvert$	$\dim\mathcal{P}$	angle records
A	$(1,1,3,3,3,6)$	7	7	0	4
B	$(1,2,2,3,4,5)$	8	8	1	5

Here $\mathcal{P}$ is the exact diagonal-moment polytope. The four sorted angle records in orbit A are

$$\begin{aligned}&(1,1,3,3,3,6),\quad (1,2,5,5,5,7),\\ &(2,3,3,3,7,7),\quad (3,5,5,6,7,7),\end{aligned} \tag{542}$$

and the five records in orbit B are

$$\begin{aligned}&(1,2,2,3,4,5),\quad (1,3,4,5,6,6),\\ &(2,2,4,5,5,7),\quad (2,3,3,4,6,7),\\ &(4,5,5,6,6,7).\end{aligned} \tag{543}$$

The equivalence relation is generated by qubit permutations, local X conjugations, and logical-state interchange. These transformations preserve existence of an exact code and its projective logical order. It therefore suffices to rule out the two representatives displayed in the table.

For both representatives,

$$\sum_{j=1}^{6}a_j=17\equiv 1\pmod 8.$$

Consequently bitwise complementation exchanges the two fibers:

$$S_1=\{\bar{x}:x\in S_0\}. \tag{544}$$

The equality in Eq. (544) is a property of the fiber geometry; no binary-dihedral or transversal logical X symmetry is being assumed for the amplitudes.

18.2 Orbit A: a two-edge Walsh obstruction

Take

$$a_A = (1, 1, 3, 3, 3, 6). \tag{545}$$

The residue-zero fiber is

$$\begin{aligned} S_0^A = \{&000000, 111100, 111010, 110110, \\ &110001, 101111, 011111\}, \end{aligned} \tag{546}$$

and the residue-one fiber is

$$\begin{aligned} S_1^A = \{&100000, 010000, 001110, 001001, \\ &000101, 000011, 111111\} = \{\bar{x} : x \in S_0^A\}. \end{aligned} \tag{547}$$

Write

$$|0_L\rangle = \sum_{x \in S_0^A} c_x |x\rangle, \qquad |1_L\rangle = \sum_{y \in S_1^A} d_y |y\rangle,$$

and put $p_x = |c_x|^2$, $r_y = |d_y|^2$.

The two normalization equations and the twenty-one equal first- and second-moment equations have a unique solution:

$x \in S_0^A$	p_x	$\bar{x} \in S_1^A$	$r_{\bar{x}}$
000000	5/16	111111	5/16
111100	1/16	000011	1/16
111010	1/16	000101	1/16
110110	1/16	001001	1/16
110001	1/8	001110	1/8
101111	3/16	010000	3/16
011111	3/16	100000	3/16

An explicit exact uniqueness certificate is obtained by taking the two normalization equations, all six one-bit moment equations, the five two-bit equations for coordinate pairs $(1, j)$, $j = 2, \ldots, 6$, and the two-bit equation for $(2, 3)$. With the probability variables ordered according to the rows of the table, first for S_0^A and then for S_1^A, the resulting 14×14 integer matrix has determinant

$$-48 \neq 0. \tag{548}$$

The displayed vector solves these fourteen equations and, by direct substitution, all remaining two-bit equations. Every probability is strictly positive, so all fourteen strings are in the actual supports of a putative code.

Consider the first-qubit flip. Its complete incidence set is

$$\{y \in S_1^A : y \oplus e_1 \in S_0^A\} = \{100000, 111111\}, \tag{549}$$

with bottom endpoints 000000 and 011111. Define

$$U := \overline{c_{000000}} d_{100000}, \qquad V := \overline{c_{011111}} d_{111111}. \tag{550}$$

The forced magnitudes are

$$|U| = |V| = \sqrt{\frac{5}{16}\frac{3}{16}} = \frac{\sqrt{15}}{16} > 0. \tag{551}$$

The off-diagonal equations for X_1 and X_1Z_2 are respectively

$$U + V = 0, \tag{552}$$

$$U - V = 0. \tag{553}$$

Indeed, the two source strings in Eq. (549) have opposite second bits. The invertible Walsh matrix in Lemma 17.1 therefore forces $U = V = 0$, contradicting Eq. (551).

Proposition 18.1 (Orbit A is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-eight moment orbit represented by* $a_A = (1, 1, 3, 3, 3, 6)$.

Proof. The exact moment equations force all four endpoints in Eq. (550) to have nonzero amplitude, while Eqs. (552)–(553) force both associated edge products to vanish. This contradiction is independent of all amplitude phases. □

18.3 Orbit B: an odd cancellation cycle

Take

$$a_B = (1, 2, 2, 3, 4, 5). \tag{554}$$

The two fibers are

$$S_0^B = \{000000, 111100, 011010, 100110, 110001, 101001, 000101, 011111\}, \tag{555}$$

$$S_1^B = \{111111, 000011, 100101, 011001, 001110, 010110, 111010, 100000\} = \{\bar{x} : x \in S_0^B\}. \tag{556}$$

The moment matrix has rank 15 in the sixteen probability variables. A small exact rank certificate is obtained from the two normalizations, all six one-bit equations, the five two-bit equations $(1, j)$, $j = 2, \ldots, 6$, and the two equations $(2, 3)$ and $(2, 4)$. Delete the column for $p_0(000000)$; the resulting 15×15 minor has determinant

$$32 \neq 0. \tag{557}$$

Exact row reduction gives the complete one-parameter family

$x \in S_0^B$	$p_0(x)$	$\bar{x} \in S_1^B$	$p_1(\bar{x})$
000000	t	111111	t
111100	$1/16$	000011	$1/16$
011010	$1/4 - t$	100101	$1/4 - t$
100110	$3/16$	011001	$3/16$
110001	$1/8$	001110	$1/8$
101001	$1/8$	010110	$1/8$
000101	$3/16 - t$	111010	$3/16 - t$
011111	$t + 1/16$	100000	$t + 1/16$

Nonnegativity is equivalent to

$$0 \leq t \leq \frac{3}{16}. \tag{558}$$

We shall use only six amplitudes. Define

$$\begin{aligned} \alpha_0 &:= c_{100110}, \quad \alpha_1 := d_{011001}, \\ \beta_0 &:= c_{110001}, \quad \beta_1 := d_{001110}, \\ \gamma_0 &:= c_{011111}, \quad \gamma_1 := d_{100000}. \end{aligned} \tag{559}$$

Their squared magnitudes are

$$\begin{aligned} |\alpha_0|^2 = |\alpha_1|^2 &= \frac{3}{16}, \\ |\beta_0|^2 = |\beta_1|^2 &= \frac{1}{8}, \\ |\gamma_0|^2 = |\gamma_1|^2 &= t + \frac{1}{16}. \end{aligned} \tag{560}$$

Thus all six amplitudes are nonzero throughout the closed interval Eq. (558).

The mask $u = 101000$, corresponding to X_1X_3, has exactly the two incidences

$$011001 \longmapsto 110001, \qquad 001110 \longmapsto 100110.$$

Therefore

$$\overline{\beta_0}\alpha_1 + \overline{\alpha_0}\beta_1 = 0, \tag{561}$$

which, using nonvanishing, is equivalent to

$$\frac{\alpha_1}{\beta_1} = -\frac{\overline{\alpha_0}}{\overline{\beta_0}}. \tag{562}$$

The mask $u = 010001$, corresponding to X_2X_6, also has exactly two incidences:

$$001110 \longmapsto 011111, \qquad 100000 \longmapsto 110001.$$

Its off-diagonal equation is

$$\overline{\gamma_0}\beta_1 + \overline{\beta_0}\gamma_1 = 0, \tag{563}$$

so

$$\frac{\beta_1}{\gamma_1} = -\frac{\overline{\beta_0}}{\overline{\gamma_0}}. \tag{564}$$

Multiplying Eqs. (562) and (564) gives

$$\frac{\alpha_1}{\gamma_1} = +\frac{\overline{\alpha_0}}{\overline{\gamma_0}}. \tag{565}$$

The two antisymmetric cancellations have produced a positive sign.

It remains to derive the direct relation between the α and γ pairs. The mask $u = 000110$, corresponding to X_4X_5, has four incidences:

$$\begin{aligned} 000011 &\longmapsto 000101, \qquad 011001 \longmapsto 011111, \\ 111010 &\longmapsto 111100, \qquad 100000 \longmapsto 100110. \end{aligned} \tag{566}$$

Collect the first and third contributions into

$$R := \overline{c_{000101}} d_{000011} + \overline{c_{111100}} d_{111010}. \tag{567}$$

The equation for X_4X_5 is

$$R + \overline{\gamma_0}\alpha_1 + \overline{\alpha_0}\gamma_1 = 0. \tag{568}$$

Now multiply by Z_5. The operator $X_4X_5Z_5$ is, up to an overall phase, the ordinary weight-two Pauli X_4Y_5. The two source strings 000011 and 111010 have fifth bit one, while 011001 and 100000 have fifth bit zero. Hence the equation for $X_4X_5Z_5$ is

$$-R + \overline{\gamma_0}\alpha_1 + \overline{\alpha_0}\gamma_1 = 0. \tag{569}$$

Adding Eqs. (568) and (569) eliminates the nuisance term and yields

$$\overline{\gamma_0}\alpha_1 + \overline{\alpha_0}\gamma_1 = 0. \tag{570}$$

Therefore

$$\frac{\alpha_1}{\gamma_1} = -\frac{\overline{\alpha_0}}{\overline{\gamma_0}}, \tag{571}$$

which contradicts Eq. (565) because all four amplitudes in the ratio are nonzero.

Proposition 18.2 (Orbit B is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-eight moment orbit represented by* $a_B = (1, 2, 2, 3, 4, 5)$.

Proof. Equations (561), (563), and (570) form an odd cycle of antisymmetric two-edge cancellations. The first two force the α-to-γ ratio to have positive sign, whereas the third forces the same ratio to have negative sign. The exact moment family guarantees that no amplitude used in this cycle can vanish, even at an endpoint of the moment interval. □

Theorem 18.3 (Six-qubit order-eight no-go)**.** *No exact* $((6, 2, d \geq 3))$ *qubit code admits a projective diagonal transversal logical element of order* 8.

Proof. Equation (541) forces every order-eight code into the $(q, m) = (8, 8)$ search. The exact diagonal enumeration gives only the two moment orbits A and B. Proposition 18.1 excludes orbit A, and Proposition 18.2 excludes orbit B. Hence no order-eight code exists. □

Corollary 18.4 (Range after the order-eight no-go)**.** *Every projective diagonal logical element of an exact* $((6, 2, d \geq 3))$ *code has order at most* 7*. At this intermediate stage the orders* $2, 3, 5$ *are realized, while* $m \in \{4, 6, 7\}$ *have not yet been decided by the arguments above. In the notation of the order–length function of the parent draft,*

$$5 \leq M_3(6) \leq 7. \tag{572}$$

Remark 18.5 (Determinant-one convention)**.** Projective orders eight and nine correspond, respectively, to cyclic subgroups C_{16} and C_{18} in the determinant-one $SU(2)$ convention. Theorems 18.3 and 17.2 therefore upgrade both numerical "N/A" entries at the top of the six-qubit cyclic table to exact no-go theorems.

19 Exact nonexistence of projective order seven

We now close the next endpoint. Unlike orders eight and nine, projective order seven has five diagonal moment orbits. Four are removed by immediate two-edge or odd-cycle obstructions. The remaining orbit has a two-dimensional moment polytope; there the weight-one Walsh equations first force a unique strictly positive moment point, after which a finite phase contradiction completes the proof.

19.1 The five exhaustive order-seven moment orbits

Suppose that an exact $((6, 2, d \geq 3))$ code admits a projective diagonal logical element of order $m = 7$. For a controlled primitive physical lift, the distance-three Smith bound gives

$$7 = m \mid q, \qquad q \leq \beta_6 = 9. \tag{573}$$

Hence necessarily

$$q = 7, \qquad \delta = q/m = 1.$$

The exact enumeration gives precisely five rank-compatible moment orbits:

orbit	representative $a \in \mathbb{Z}_7^6$	$\lvert S_0\rvert$	$\lvert S_1\rvert$	$\dim \mathcal{P}$	angle records
57	$(1,1,1,4,4,4)$	7	7	0	3
58	$(1,1,2,3,3,5)$	9	9	2	4
59	$(1,1,5,5,5,5)$	7	7	0	3
60	$(1,2,2,2,4,4)$	9	9	2	4
61	$(1,2,2,3,3,4)$	9	9	2	4

Here $\mathcal{P}$ is the exact diagonal-moment polytope. For reference, the complete sets of surviving sorted angle records are

$$\mathcal{A}_{57} = \{(1,1,1,4,4,4), (1,3,3,3,6,6), (3,4,4,6,6,6)\}, \tag{574}$$
$$\mathcal{A}_{58} = \{(1,1,2,3,3,5), (1,2,4,4,5,6), (2,2,3,3,6,6), (3,4,5,5,6,6)\}, \tag{575}$$
$$\mathcal{A}_{59} = \{(1,1,5,5,5,5), (1,2,2,2,2,6), (2,5,5,5,6,6)\}, \tag{576}$$
$$\mathcal{A}_{60} = \{(1,2,2,2,4,4), (1,3,3,5,5,5), (2,2,3,4,5,6), (4,4,5,5,5,6)\}, \tag{577}$$
$$\mathcal{A}_{61} = \{(1,2,2,3,3,4), (1,3,4,4,5,5), (2,2,4,4,4,6), (2,3,3,3,5,6)\}. \tag{578}$$

The equivalence relation is generated by qubit permutations, local X conjugations, and logical-state interchange. These operations preserve code existence, distance, and projective order, so it suffices to analyze the five representatives in the table.

For every representative,

$$\sum_{j=1}^{6} a_j \equiv 1 \pmod 7.$$

Consequently bitwise complementation exchanges the two full fibers:

$$S_1 = \{\bar{x} : x \in S_0\}. \tag{579}$$

This is only a statement about the fiber geometry; no transversal logical bit flip and no equality of complex amplitudes is assumed.

For each orbit, order the bottom fiber as

$$S_0 = \{x_0, \ldots, x_{r-1}\}$$

and write

$$|0_L\rangle = \sum_i c_i |x_i\rangle, \qquad |1_L\rangle = \sum_i d_i |\bar{x}_i\rangle. \tag{580}$$

Exact row reduction of the diagonal moment system shows in every one of the five cases that

$$|c_i|^2 = |d_i|^2 =: p_i. \tag{581}$$

For an off-diagonal flip incidence define

$$z_{ij} := \overline{c_i}\, d_j. \tag{582}$$

The product z_{ij} occurs for a mask u exactly when

$$x_i = \bar{x}_j \oplus u.$$

Whenever $p_i > 0$, put

$$\rho_i := \frac{d_i}{\overline{c_i}}, \qquad |\rho_i| = 1. \tag{583}$$

Then

$$z_{ij} = \rho_j \, \overline{c_i c_j}. \tag{584}$$

In particular, for a symmetric two-edge incidence pair,

$$z_{ij} + z_{ji} = 0 \Longrightarrow \rho_j = -\rho_i, \tag{585}$$

$$z_{ij} - z_{ji} = 0 \Longrightarrow \rho_j = \rho_i, \tag{586}$$

provided the endpoint amplitudes are nonzero.

19.2 Orbit 57: a two-edge Walsh obstruction

Take

$$a_{57} = (1, 1, 1, 4, 4, 4). \tag{587}$$

Order the bottom fiber as

$$\begin{aligned} &x_0 = 000000, \quad x_1 = 111100, \quad x_2 = 111010, \\ &x_3 = 111001, \quad x_4 = 110111, \quad x_5 = 101111, \quad x_6 = 011111. \end{aligned} \tag{588}$$

The complete moment system has rank fourteen in the fourteen probability variables and has the unique solution

$$(p_0, \ldots, p_6) = \frac{1}{14}(5, 1, 1, 1, 2, 2, 2). \tag{589}$$

A compact exact uniqueness certificate is obtained from the two normalizations, all six one-bit moment equations, the five two-bit equations $(1, j)$, $j = 2, \ldots, 6$, and the equation $(2, 3)$. In the variable order

$$(p_0, \ldots, p_6; |d_0|^2, \ldots, |d_6|^2),$$

the resulting 14×14 integer matrix has determinant

$$-28 \neq 0. \tag{590}$$

Direct substitution verifies the remaining two-bit equations. In particular, every displayed amplitude is nonzero.

For the first-qubit flip, the complete incidence set is

$$\{y \in S_1 : y \oplus e_1 \in S_0\} = \{100000, 111111\}. \tag{591}$$

The two edge products are z_{06} and z_{60}. The equation for X_1 gives

$$z_{06} + z_{60} = 0, \tag{592}$$

whereas the equation for $X_1 Z_2$ gives

$$z_{06} - z_{60} = 0, \tag{593}$$

because the two source strings in Eq. (591) have opposite second bits. Hence both edge products vanish. This contradicts

$$|z_{06}| = |z_{60}| = \sqrt{p_0 p_6} = \sqrt{\frac{5}{14}\frac{1}{7}} > 0. \tag{594}$$

Proposition 19.1 (Orbit 57 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-seven moment orbit represented by* $(1, 1, 1, 4, 4, 4)$.

Proof. Equations (592)–(593) are the invertible two-edge Walsh system of Lemma 17.1, while Eq. (589) forces all four endpoints to have nonzero amplitude. □

19.3 Orbit 58: the weight-one equations fix the moment point

Take

$$a_{58} = (1, 1, 2, 3, 3, 5) \tag{595}$$

and order the bottom fiber as

$$\begin{aligned} x_0 &= 000000, \quad x_1 = 111100, \quad x_2 = 111010, \\ x_3 &= 100110, \quad x_4 = 010110, \quad x_5 = 110001, \\ x_6 &= 001001, \quad x_7 = 101111, \quad x_8 = 011111. \end{aligned} \tag{596}$$

The moment matrix has rank sixteen in the eighteen probability variables. One exact rank certificate uses the two normalizations, all six one-bit rows, and the eight two-bit rows

$$(1,2), (1,3), (1,4), (1,5), (1,6), (2,3), (2,4), (3,4).$$

Keeping the first sixteen columns in the variable order

$$(p_0, \ldots, p_8; |d_0|^2, \ldots, |d_8|^2)$$

produces a 16×16 minor of determinant

$$42 \neq 0. \tag{597}$$

Exact row reduction gives the complete two-parameter family

i	p_i	i	p_i
0	t	1	$1/14$
2	$1/14$	3	s
4	$5/14 - s - t$	5	$1/7$
6	$2/7 - t$	7	$3/14 - s$
8	$s + t - 1/7$		

(598)

with the same probabilities on the complementary top strings. The exact nonnegativity region is

$$\begin{aligned} s \geq 0, \qquad t \geq 0, \qquad \frac{1}{7} \leq s + t \leq \frac{5}{14}, \\ s \leq \frac{3}{14}, \qquad t \leq \frac{2}{7}. \end{aligned} \tag{599}$$

For the first-qubit flip, order the four incidences as

$$(z_{08}, z_{46}, z_{64}, z_{80}). \tag{600}$$

The three weight-at-most-two equations for X_1, X_1Z_2, X_1Z_3 are

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} z_{08} \\ z_{46} \\ z_{64} \\ z_{80} \end{pmatrix} = 0. \tag{601}$$

The matrix has rank three and kernel

$$\operatorname{span}\{(1, -1, -1, 1)^T\}.$$

Therefore

$$z_{08} = z_{80}, \qquad z_{46} = z_{64}, \qquad z_{46} = -z_{08}. \tag{602}$$

Taking absolute values gives

$$p_0 p_8 = p_4 p_6. \tag{603}$$

Substituting Eq. (598) reduces this equation to

$$28s + 49t = 10. \tag{604}$$

For the second-qubit flip, the four incidences are

$$(z_{07}, z_{36}, z_{63}, z_{70}). \tag{605}$$

The equations for X_2, X_2Z_1, X_2Z_3 have the same Walsh matrix and therefore imply

$$z_{07} = z_{70}, \qquad z_{36} = z_{63}, \qquad z_{36} = -z_{07}. \tag{606}$$

Their magnitude consequence is

$$p_0 p_7 = p_3 p_6, \tag{607}$$

which becomes

$$4s = 3t. \tag{608}$$

Equations (604) and (608) have the unique solution

$$s = \frac{3}{28}, \qquad t = \frac{1}{7}. \tag{609}$$

Thus the non-diagonal weight-one equations force the moment point to

$$(p_0, \ldots, p_8) = \left(\frac{1}{7}, \frac{1}{14}, \frac{1}{14}, \frac{3}{28}, \frac{3}{28}, \frac{1}{7}, \frac{1}{7}, \frac{3}{28}, \frac{3}{28}\right). \tag{610}$$

Every amplitude is now strictly nonzero, so the phase ratios Eq. (583) are defined for all indices.

Equations (602) and (606) first give

$$\rho_8 = \rho_0, \qquad \rho_6 = \rho_4, \qquad \rho_7 = \rho_0, \qquad \rho_6 = \rho_3. \tag{611}$$

The mask 101000, corresponding to X_1X_3, has exactly the two incidences z_{35}, z_{53}. Hence

$$z_{35} + z_{53} = 0 \quad \Longrightarrow \quad \rho_5 = -\rho_3.$$

The mask 100001, corresponding to X_1X_6, has exactly the two incidences z_{57}, z_{75}. Hence

$$z_{57} + z_{75} = 0 \quad \Longrightarrow \quad \rho_7 = -\rho_5 = \rho_3.$$

Combining these relations with Eq. (611) yields

$$\rho_0 = \rho_3 = \rho_4 = \rho_6 = \rho_7 = \rho_8 =: \rho. \tag{612}$$

The last relations in Eqs. (602) and (606) now become

$$c_4 c_6 = -c_0 c_8, \tag{613}$$

$$c_3 c_6 = -c_0 c_7. \tag{614}$$

Finally, the mask 000110, corresponding to X_4X_5, has exactly four incidences:

$$z_{38}, \qquad z_{47}, \qquad z_{74}, \qquad z_{83}. \tag{615}$$

Its off-diagonal equation is

$$z_{38} + z_{47} + z_{74} + z_{83} = 0. \tag{616}$$

Using Eq. (612), this reduces to

$$c_3c_8 + c_4c_7 = 0. \tag{617}$$

But Eqs. (613)–(614) give

$$\begin{aligned} c_3c_8 + c_4c_7 &= -\frac{c_3c_4c_6}{c_0} - \frac{c_4c_3c_6}{c_0} \\ &= -\frac{2c_3c_4c_6}{c_0} \neq 0, \end{aligned} \tag{618}$$

because Eq. (610) makes all four amplitudes in the last fraction nonzero. This contradicts Eq. (617).

Proposition 19.2 (Orbit 58 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-seven moment orbit represented by* $(1, 1, 2, 3, 3, 5)$.

Proof. The first- and second-qubit Walsh systems force the unique interior moment point in Eq. (610). At that point all phase ratios are well-defined, and the remaining weight-two flip equations imply both Eq. (617) and its nonzero value Eq. (618). □

19.4 Orbit 59: a second two-edge Walsh obstruction

Take

$$a_{59} = (1, 1, 5, 5, 5, 5) \tag{619}$$

and order the bottom fiber as

$$\begin{aligned} &x_0 = 000000, \quad x_1 = 111000, \quad x_2 = 110100, \\ &x_3 = 110010, \quad x_4 = 110001, \quad x_5 = 101111, \quad x_6 = 011111. \end{aligned} \tag{620}$$

The exact moment solution is unique:

$$(p_0, \ldots, p_6) = \frac{1}{14}(4, 1, 1, 1, 1, 3, 3). \tag{621}$$

Using the same fourteen moment rows as in Subsection 19.2, the full 14×14 coefficient matrix has determinant

$$-42 \neq 0. \tag{622}$$

Thus all amplitudes are forced to be nonzero.

Again the first-qubit flip has only the two incidences

$$z_{06}, \qquad z_{60}.$$

The equations for X_1 and X_1Z_2 give

$$z_{06} + z_{60} = 0, \qquad z_{06} - z_{60} = 0.$$

They force both edge products to vanish, contradicting

$$|z_{06}| = |z_{60}| = \sqrt{p_0p_6} = \sqrt{\frac{2}{7}\frac{3}{14}} > 0. \tag{623}$$

Proposition 19.3 (Orbit 59 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-seven moment orbit represented by* $(1, 1, 5, 5, 5, 5)$.

Proof. This is Lemma 17.1 applied to the first-qubit flip, together with the strictly positive unique moment assignment in Eq. (621). □

19.5 Orbit 60: a two-edge obstruction uniform over a moment polygon

Take

$$a_{60} = (1, 2, 2, 2, 4, 4) \tag{624}$$

and order the bottom fiber as

$$\begin{aligned} x_0 &= 000000, \quad x_1 = 111100, \quad x_2 = 110010, \\ x_3 &= 101010, \quad x_4 = 100110, \quad x_5 = 110001, \\ x_6 &= 101001, \quad x_7 = 100101, \quad x_8 = 011111. \end{aligned} \tag{625}$$

The moment matrix has rank sixteen. An exact rank certificate is obtained from the two normalizations, all six one-bit rows, and the eight two-bit rows

$$(1,2), (1,3), (1,4), (1,5), (1,6), (2,3), (2,5), (3,5).$$

Keeping the columns for

$$(p_0, \ldots, p_8; |d_0|^2, \ldots, |d_5|^2, |d_8|^2)$$

gives a 16×16 minor with determinant

$$-28 \neq 0. \tag{626}$$

The complete moment family is

$$\begin{aligned} (p_0, \ldots, p_8) = \Big(& \frac{3}{14}, \frac{1}{14}, t, s, \frac{3}{14} - s - t, \frac{1}{7} - t, \\ & \frac{1}{7} - s, s + t - \frac{1}{14}, \frac{2}{7} \Big), \end{aligned} \tag{627}$$

with the same probabilities on the complementary top strings, where

$$0 \leq s, t \leq \frac{1}{7}, \qquad \frac{1}{14} \leq s + t \leq \frac{3}{14}. \tag{628}$$

Although the moment polytope is two-dimensional, the endpoint probabilities relevant to the first-qubit flip are fixed:

$$p_0 = \frac{3}{14} > 0, \qquad p_8 = \frac{2}{7} > 0. \tag{629}$$

The first-qubit flip has exactly the two incidences z_{08}, z_{80}. The X_1 and $X_1 Z_2$ equations give

$$z_{08} + z_{80} = 0, \qquad z_{08} - z_{80} = 0.$$

Hence both products vanish, contradicting

$$|z_{08}| = |z_{80}| = \sqrt{p_0 p_8} = \sqrt{\frac{3}{14}\frac{2}{7}} > 0. \tag{630}$$

Proposition 19.4 (Orbit 60 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-seven moment orbit represented by* $(1, 2, 2, 2, 4, 4)$.

Proof. The two-edge Walsh obstruction uses only the fixed positive probabilities in Eq. (629); it therefore excludes every point of the exact moment polygon. □

19.6 Orbit 61: an odd antisymmetric cancellation cycle

Take

$$a_{61} = (1, 2, 2, 3, 3, 4) \tag{631}$$

and order the bottom fiber as

$$\begin{aligned} x_0 &= 000000, \quad x_1 = 011100, \quad x_2 = 011010, \\ x_3 &= 100110, \quad x_4 = 110001, \quad x_5 = 101001, \\ x_6 &= 000101, \quad x_7 = 000011, \quad x_8 = 011111. \end{aligned} \tag{632}$$

The moment matrix again has rank sixteen. Use the two normalizations, all six one-bit rows, and the eight two-bit rows

$$(1,2), (1,3), (1,4), (1,5), (1,6), (2,3), (2,4), (2,5).$$

Keeping the first sixteen probability columns gives a minor of determinant

$$-28 \neq 0. \tag{633}$$

Exact row reduction gives

$$\begin{aligned} (p_0, \ldots, p_8) = \Big(&t, s, \frac{2}{7} - s - t, \frac{3}{14}, \frac{1}{7}, \frac{1}{7}, \\ &\frac{3}{14} - s - t, s - \frac{1}{14}, t + \frac{1}{14}\Big), \end{aligned} \tag{634}$$

where

$$t \geq 0, \qquad s \geq \frac{1}{14}, \qquad s + t \leq \frac{3}{14}. \tag{635}$$

In particular,

$$p_3 = \frac{3}{14} > 0, \qquad p_4 = \frac{1}{7} > 0, \qquad p_8 = t + \frac{1}{14} > 0. \tag{636}$$

Thus ρ_3, ρ_4, ρ_8 are all well-defined.

The mask 101000, corresponding to X_1X_3, has exactly the two incidences z_{34}, z_{43}. Hence

$$\rho_4 = -\rho_3. \tag{637}$$

The mask 010001, corresponding to X_2X_6, has exactly the two incidences z_{48}, z_{84}, so

$$\rho_8 = -\rho_4. \tag{638}$$

Equations (637)–(638) imply

$$\rho_8 = \rho_3. \tag{639}$$

But the mask 000110, corresponding to X_4X_5, has exactly the two incidences z_{38}, z_{83}, and therefore requires

$$\rho_8 = -\rho_3. \tag{640}$$

This contradicts Eq. (639). The obstruction is an odd cycle of antisymmetric two-edge cancellations on the indices

$$3 \longleftrightarrow 4 \longleftrightarrow 8 \longleftrightarrow 3.$$

Proposition 19.5 (Orbit 61 is impossible)**.** *No exact* $((6,2,d\geq 3))$ *code can arise from the order-seven moment orbit represented by* $(1,2,2,3,3,4)$*.*

Proof. The exact moment family forces the three amplitudes used in the cancellation cycle to be nonzero throughout the entire feasible region. The three weight-two off-diagonal equations then impose an inconsistent odd sign cycle. □

Theorem 19.6 (Six-qubit order-seven no-go)**.** *No exact* $((6,2,d\geq 3))$ *qubit code admits a projective diagonal transversal logical element of order* 7*.*

Proof. Equation (573) forces every order-seven code into the $(q,m)=(7,7)$ search. The exact diagonal enumeration gives only the five moment orbits 57–61. Propositions 19.1, 19.2, 19.3, 19.4, and 19.5 exclude them one by one. Hence no order-seven code exists. □

Corollary 19.7 (Updated six-qubit order range)**.** *The orders* $2,3,5$ *are realized by exact six-qubit distance-three codes, while projective orders* $7,8,9$ *are impossible. At this stage of the argument, only* $m=4$ *and* $m=6$ *remain to be analyzed. Consequently,*

$$5\leq M_3(6)\leq 6. \tag{641}$$

Remark 19.8 (Determinant-one convention)**.** Projective order seven corresponds to the cyclic subgroup C_{14} in the determinant-one $SU(2)$ convention. Theorem 19.6 upgrades the corresponding numerical "N/A" entry to an exact no-go theorem.

20 Exact nonexistence of projective order six

We now close the final order above the known order-five example. Projective order six has four diagonal moment orbits. None is removed by the diagonal LP alone, but each has a short exact obstruction once the weight-one and weight-two flip equations are imposed. Three cases are forced to strictly positive moment points before the phase contradiction; the largest four-dimensional moment polytope requires a boundary-safe argument.

20.1 The four exhaustive order-six moment orbits

Suppose that an exact $((6,2,d\geq 3))$ code admits a projective diagonal logical element of order $m=6$. For a controlled primitive physical lift, the distance-three Smith bound gives

$$6=m\mid q,\qquad q\leq \beta_6=9. \tag{642}$$

Hence necessarily

$$q=6,\qquad \delta=q/m=1.$$

The exact enumeration gives precisely four rank-compatible moment orbits:

orbit	representative $a\in\mathbb{Z}_6^6$	$\lvert S_0\rvert$	$\lvert S_1\rvert$	$\dim\mathcal{P}$	angle records
53	$(1,1,1,3,3,4)$	10	10	3	3
54	$(1,1,2,2,2,5)$	9	9	2	3
55	$(1,1,2,2,3,4)$	11	11	4	4
56	$(1,2,2,2,3,3)$	10	10	3	3

Here $\mathcal{P}$ is the exact diagonal-moment polytope. The complete sets of surviving sorted angle records are

$$\mathcal{A}_{53} = \{(1,1,1,3,3,4),(1,2,3,3,5,5),(3,3,4,5,5,5)\}, \tag{643}$$
$$\mathcal{A}_{54} = \{(1,1,2,2,2,5),(1,1,4,4,4,5),(2,4,4,5,5,5)\}, \tag{644}$$
$$\mathcal{A}_{55} = \{(1,1,2,2,3,4),(1,2,3,4,4,5),(2,2,2,3,5,5),(3,4,4,4,5,5)\}, \tag{645}$$
$$\mathcal{A}_{56} = \{(1,2,2,2,3,3),(1,3,3,4,4,4),(2,2,3,3,4,5)\}. \tag{646}$$

As before, the equivalence relation is generated by qubit permutations, local X conjugations, and logical-state interchange, so it suffices to analyze the four displayed representatives.

For every representative,

$$\sum_{j=1}^{6} a_j = 13 \equiv 1 \pmod 6.$$

Thus bitwise complementation exchanges the two full fibers:

$$S_1 = \{\bar{x} : x \in S_0\}. \tag{647}$$

This is a statement only about the fiber geometry; no binary-dihedral symmetry and no equality of complex amplitudes is assumed.

For each orbit, order the bottom fiber as

$$S_0 = \{x_0, \ldots, x_{r-1}\}$$

and write

$$|0_L\rangle = \sum_i c_i |x_i\rangle, \qquad |1_L\rangle = \sum_i d_i |\bar{x}_i\rangle. \tag{648}$$

Exact row reduction of the diagonal moment system shows in all four cases that

$$|c_i|^2 = |d_i|^2 =: p_i. \tag{649}$$

For an off-diagonal flip incidence define

$$z_{ij} := \overline{c_i}\, d_j. \tag{650}$$

The product z_{ij} occurs for a flip mask u exactly when

$$x_i = \bar{x}_j \oplus u.$$

Whenever $p_i > 0$, put

$$\rho_i := \frac{d_i}{\overline{c_i}}, \qquad |\rho_i| = 1. \tag{651}$$

Then

$$z_{ij} = \rho_j\, \overline{c_i c_j}. \tag{652}$$

In particular, a symmetric two-edge equation $z_{ij} + z_{ji} = 0$ gives $\rho_j = -\rho_i$, while $z_{ij} - z_{ji} = 0$ gives $\rho_j = \rho_i$, provided the two endpoint amplitudes are nonzero.

We repeatedly use the following four-edge Walsh pattern. If four incidence products are ordered as (A, B, C, D), the three independent equations

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = 0 \tag{653}$$

force

$$(A, B, C, D) = \lambda(1, -1, -1, 1). \tag{654}$$

In each occurrence below, the three rows are supplied by an undressed weight-one flip and two allowed single-Z dressings, all of total Pauli weight at most two.

20.2 Orbit 53: moment pinning and a four-edge product contradiction

Take

$$a_{53} = (1, 1, 1, 3, 3, 4) \tag{655}$$

and order the bottom fiber as

$$\begin{aligned} x_0 &= 000000, \quad x_1 = 111100, \quad x_2 = 111010, \\ x_3 &= 000110, \quad x_4 = 110001, \quad x_5 = 101001, \\ x_6 &= 011001, \quad x_7 = 110111, \quad x_8 = 101111, \\ x_9 &= 011111. \end{aligned} \tag{656}$$

The moment matrix has rank seventeen in the twenty probability variables. One exact rank certificate uses the two normalizations, all six one-bit rows, and the nine two-bit rows

$$(1,2), (1,3), (1,4), (1,5), (1,6), (2,3), (2,4), (3,4), (4,6).$$

Keeping the columns

$$(p_0, \ldots, p_9; |d_0|^2, \ldots, |d_6|^2)$$

gives a 17×17 minor of determinant

$$24 \neq 0. \tag{657}$$

Exact row reduction gives the complete three-parameter family

$$\begin{aligned} (p_0, \ldots, p_9) = \Big(& t, \frac{1}{12}, \frac{1}{12}, \frac{1}{3} - t, s, r, \frac{5}{12} - r - s - t, \\ & \frac{1}{6} - s, \frac{1}{6} - r, r + s + t - \frac{1}{4}\Big), \end{aligned} \tag{658}$$

with the same probabilities on the complementary top strings. Its exact nonnegativity region is

$$0 \leq r, s \leq \frac{1}{6}, \qquad 0 \leq t \leq \frac{1}{3}, \qquad \frac{1}{4} \leq r + s + t \leq \frac{5}{12}. \tag{659}$$

For the flips of qubits one, two, and three, order the four incidences as

$$X_1 : \quad (z_{90}, z_{63}, z_{36}, z_{09}), \tag{660}$$

$$X_2 : \quad (z_{80}, z_{53}, z_{35}, z_{08}), \tag{661}$$

$$X_3 : \quad (z_{70}, z_{43}, z_{34}, z_{07}). \tag{662}$$

In each case the allowed dressings have the Walsh kernel Eq. (654). Taking absolute values therefore gives

$$p_0 p_9 = p_3 p_6, \tag{663}$$

$$p_0 p_8 = p_3 p_5, \tag{664}$$

$$p_0 p_7 = p_3 p_4. \tag{665}$$

Substitution of Eq. (658) into Eqs. (664)–(665) gives, without division,

$$r = \frac{t}{2}, \qquad s = \frac{t}{2}. \tag{666}$$

Equation (663) then becomes

$$t\left(2t - \frac{1}{4}\right) = \left(\frac{1}{3} - t\right)\left(\frac{5}{12} - 2t\right),$$

or equivalently

$$6t = 1. \tag{667}$$

Hence the weight-one equations force the unique moment point

$$p_0 = p_3 = \frac{1}{6}, \qquad p_i = \frac{1}{12} \quad (i \neq 0, 3). \tag{668}$$

Every amplitude is nonzero.

The equalities within the three Walsh kernels give

$$\begin{aligned} \rho_0 &= \rho_9, & \rho_3 &= \rho_6, \\ \rho_0 &= \rho_8, & \rho_3 &= \rho_5, \\ \rho_0 &= \rho_7, & \rho_3 &= \rho_4. \end{aligned} \tag{669}$$

Now use two weight-two flip masks. For $u = e_4 + e_6$, row reduction of the allowed Z-dressings gives

$$z_{20} + z_{02} = 0, \qquad z_{31} + z_{13} = 0, \tag{670}$$

so $\rho_2 = -\rho_0$ and $\rho_1 = -\rho_3$. For $u = e_5 + e_6$, it gives

$$z_{10} + z_{01} = 0, \qquad z_{32} + z_{23} = 0, \tag{671}$$

so $\rho_1 = -\rho_0$ and $\rho_2 = -\rho_3$. Consequently

$$\rho_0 = \rho_3 = \rho_4 = \rho_5 = \rho_6 = \rho_7 = \rho_8 = \rho_9 =: \rho. \tag{672}$$

The nontrivial relations in the X_2 and X_3 Walsh kernels now reduce to

$$c_3 c_5 = -c_0 c_8, \tag{673}$$

$$c_3 c_4 = -c_0 c_7. \tag{674}$$

Finally, the mask $u = e_1 + e_6$ has the four incidences

$$z_{84}, \qquad z_{75}, \qquad z_{57}, \qquad z_{48}. \tag{675}$$

Its only independent off-diagonal equation, together with Eq. (672), is

$$c_4 c_8 + c_5 c_7 = 0. \tag{676}$$

But Eqs. (673)–(674) give

$$\begin{aligned} c_4 c_8 + c_5 c_7 &= c_4\left(-\frac{c_3 c_5}{c_0}\right) + c_5\left(-\frac{c_3 c_4}{c_0}\right) \\ &= -\frac{2 c_3 c_4 c_5}{c_0} \neq 0, \end{aligned} \tag{677}$$

where nonvanishing follows from Eq. (668). This contradicts Eq. (676).

Proposition 20.1 (Orbit 53 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-six moment orbit represented by* $(1, 1, 1, 3, 3, 4)$.

Proof. The three weight-one Walsh systems force the unique positive point Eq. (668). At that point the weight-two systems synchronize the phase ratios and imply both Eqs. (676) and (677). □

20.3 Orbit 54: a three-product odd sign cycle

Take

$$a_{54} = (1, 1, 2, 2, 2, 5) \tag{678}$$

and order the bottom fiber as

$$\begin{aligned} x_0 &= 000000, \quad x_1 = 111100, \quad x_2 = 111010, \\ x_3 &= 110110, \quad x_4 = 001110, \quad x_5 = 100001, \\ x_6 &= 010001, \quad x_7 = 101111, \quad x_8 = 011111. \end{aligned} \tag{679}$$

The moment matrix has rank sixteen in the eighteen probability variables. Use the two normalizations, all six one-bit rows, and the eight two-bit rows

$$(1,2), (1,3), (1,4), (1,5), (1,6), (2,3), (2,6), (3,6).$$

Keeping the columns

$$(p_0, \ldots, p_8; |d_0|^2, \ldots, |d_6|^2)$$

gives a 16×16 minor of determinant

$$36 \neq 0. \tag{680}$$

The complete moment family is

$$(p_0, \ldots, p_8) = \left(t, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{4} - t, s, \frac{5}{12} - s - t, \frac{1}{4} - s, s + t - \frac{1}{6}\right), \tag{681}$$

where

$$0 \leq s, t \leq \frac{1}{4}, \qquad \frac{1}{6} \leq s + t \leq \frac{5}{12}. \tag{682}$$

For X_1, X_2, X_6, respectively, order the four incidences as

$$(z_{80}, z_{64}, z_{46}, z_{08}), \tag{683}$$

$$(z_{70}, z_{54}, z_{45}, z_{07}), \tag{684}$$

$$(z_{85}, z_{76}, z_{67}, z_{58}). \tag{685}$$

Each system has kernel Eq. (654). Its magnitude consequences are

$$p_0 p_8 = p_4 p_6, \qquad p_0 p_7 = p_4 p_5, \qquad p_5 p_8 = p_6 p_7. \tag{686}$$

The middle equality gives

$$t\left(\frac{1}{4} - s\right) = \left(\frac{1}{4} - t\right) s,$$

hence

$$t = s. \tag{687}$$

The first equality then becomes

$$t\left(2t - \frac{1}{6}\right) = \left(\frac{1}{4} - t\right)\left(\frac{5}{12} - 2t\right),$$

which is equivalent to

$$36t = 5. \tag{688}$$

The third equality is then automatically satisfied. Therefore

$$t = s = \frac{5}{36} \tag{689}$$

and

$$\begin{aligned} p_0 = p_5 = p_6 &= \frac{5}{36}, \\ p_4 = p_7 = p_8 &= \frac{1}{9}, \\ p_1 = p_2 = p_3 &= \frac{1}{12}. \end{aligned} \tag{690}$$

All amplitudes are nonzero.

The equalities inside the three Walsh kernels imply

$$\rho_0 = \rho_4 = \rho_5 = \rho_6 = \rho_7 = \rho_8 =: \rho. \tag{691}$$

The three nontrivial product relations become

$$c_4 c_6 = -c_0 c_8, \tag{692}$$

$$c_4 c_5 = -c_0 c_7, \tag{693}$$

$$c_6 c_7 = -c_5 c_8. \tag{694}$$

The first two equations give

$$c_6 = -\frac{c_0 c_8}{c_4}, \qquad c_5 = -\frac{c_0 c_7}{c_4}.$$

Substitution into Eq. (694) requires

$$-\frac{c_0 c_7 c_8}{c_4} = +\frac{c_0 c_7 c_8}{c_4}, \tag{695}$$

which is impossible because Eq. (690) makes every factor nonzero.

Proposition 20.2 (Orbit 54 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-six moment orbit represented by* $(1, 1, 2, 2, 2, 5)$.

Proof. The weight-one Walsh equations force the positive point Eq. (690); their three residual product relations form the inconsistent odd sign cycle in Eq. (695). □

20.4 Orbit 55: a boundary-safe binomial contradiction

Take

$$a_{55} = (1, 1, 2, 2, 3, 4) \tag{696}$$

and order the bottom fiber as

$$\begin{aligned} x_0 &= 000000, \quad x_1 = 111100, \quad x_2 = 101010, \\ x_3 &= 011010, \quad x_4 = 100110, \quad x_5 = 010110, \\ x_6 &= 110001, \quad x_7 = 001001, \quad x_8 = 000101, \\ x_9 &= 101111, \quad x_{10} = 011111. \end{aligned} \tag{697}$$

The moment matrix has rank eighteen in the twenty-two probability variables. Use the two normalizations, all six one-bit rows, and the ten two-bit rows

$$(1,2), (1,3), (1,4), (1,5), (1,6), (2,3), (2,4), (2,5), (3,4), (3,5).$$

Keeping the columns

$$(p_0, \dots, p_{10}; |d_0|^2, \dots, |d_4|^2, |d_6|^2, |d_7|^2)$$

gives an 18×18 minor of determinant

$$-36 \neq 0. \tag{698}$$

Put

$$\alpha = p_4, \qquad \beta = p_3, \qquad \gamma = p_2, \qquad \delta = p_0.$$

The complete moment family is

$$\begin{aligned} &p_0 = \delta, && p_1 = \frac{1}{12}, && p_2 = \gamma, \quad p_3 = \beta, \\ &p_4 = \alpha, && p_5 = \frac{5}{12} - \alpha - \beta - \gamma - \delta, && p_6 = \frac{1}{6}, \\ &p_7 = \frac{1}{3} - \beta - \gamma - \delta, && p_8 = \beta + \gamma - \frac{1}{12}, \\ &p_9 = \frac{1}{4} - \alpha - \gamma, && p_{10} = \alpha + \gamma + \delta - \frac{1}{6}. \end{aligned} \tag{699}$$

Its exact feasible region is specified by

$$\begin{aligned} \alpha, \beta, \gamma, \delta \geq 0, \qquad & \alpha + \beta + \gamma + \delta \leq \frac{5}{12}, \\ \beta + \gamma + \delta \leq \frac{1}{3}, \qquad & \beta + \gamma \geq \frac{1}{12}, \\ \alpha + \gamma \leq \frac{1}{4}, \qquad & \alpha + \gamma + \delta \geq \frac{1}{6}. \end{aligned} \tag{700}$$

Unlike the previous two cases, zeros can occur on boundary faces and must not be divided away prematurely.

The fixed probability $p_6 = 1/6$ makes ρ_6 well-defined. Six weight-two masks have exactly two incidences each:

mask	incidence pair	consequence when active
$e_1 + e_3$	z_{64}, z_{46}	$\rho_4 = -\rho_6$
$e_1 + e_4$	z_{62}, z_{26}	$\rho_2 = -\rho_6$
$e_1 + e_6$	z_{96}, z_{69}	$\rho_9 = -\rho_6$
$e_2 + e_3$	z_{65}, z_{56}	$\rho_5 = -\rho_6$
$e_2 + e_4$	z_{63}, z_{36}	$\rho_3 = -\rho_6$
$e_2 + e_6$	$z_{10,6}, z_{6,10}$	$\rho_{10} = -\rho_6$

Let $\rho := \rho_6$. If one of the six listed amplitudes vanishes, the corresponding top amplitude also vanishes by Eq. (649); therefore in every case, active or inactive, we may write

$$d_i = -\rho\, \overline{c_i}, \qquad i \in \{2, 3, 4, 5, 9, 10\}. \tag{701}$$

Three further flip masks give binomial equations. For $u = e_3 + e_5$, the six incidences are

$$z_{71}, z_{10,2}, z_{9,3}, z_{17}, z_{3,9}, z_{2,10}.$$

The allowed dressings split the equations into the isolated pair $z_{71} + z_{17} = 0$ and the remaining four-edge sum. Using Eq. (701), the latter is

$$c_2 c_{10} + c_3 c_9 = 0. \tag{702}$$

Likewise, the mask $u = e_4 + e_5$ has incidences

$$z_{81}, z_{10,4}, z_{9,5}, z_{18}, z_{5,9}, z_{4,10}$$

and gives

$$c_4 c_{10} + c_5 c_9 = 0. \tag{703}$$

Finally, the mask $u = e_5 + e_6$ has incidences

$$z_{1,0}, z_{0,1}, z_{5,2}, z_{4,3}, z_{3,4}, z_{2,5}$$

and gives

$$c_2 c_5 + c_3 c_4 = 0. \tag{704}$$

These identities hold on the entire closed moment polytope, including all boundary faces.

Taking absolute values yields

$$p_2 p_{10} = p_3 p_9, \tag{705}$$

$$p_4 p_{10} = p_5 p_9, \tag{706}$$

$$p_2 p_5 = p_3 p_4. \tag{707}$$

The moment family also gives the strictly positive pair sums

$$p_2 + p_3 = p_8 + \frac{1}{12} > 0, \tag{708}$$

$$p_4 + p_5 = p_7 + \frac{1}{12} > 0, \tag{709}$$

$$p_9 + p_{10} = p_0 + \frac{1}{12} > 0. \tag{710}$$

We now prove that all six amplitudes occurring in Eqs. (702)–(704) are nonzero.

If $p_2 = 0$, then Eq. (708) gives $p_3 > 0$, and Eqs. (705) and (707) force $p_9 = p_4 = 0$. But $p_2 = \gamma = 0$ and $p_4 = \alpha = 0$ make $p_9 = 1/4$ in Eq. (699), a contradiction. Thus $p_2 > 0$.

If $p_3 = 0$, then $p_2 > 0$, and Eqs. (705) and (707) force $p_{10} = p_5 = 0$. But $p_{10} = 0$ and $p_3 = \beta = 0$ imply $\alpha + \gamma + \delta = 1/6$, so Eq. (699) gives $p_5 = 1/4$, again a contradiction. Hence $p_3 > 0$.

If $p_4 = 0$, then Eq. (709) gives $p_5 > 0$, and Eq. (707) contradicts $p_2 > 0$. The same equation rules out $p_5 = 0$, because then $p_4 > 0$ and $p_3 > 0$. If $p_9 = 0$, then Eq. (710) gives $p_{10} > 0$, contradicting Eq. (705) and $p_2 > 0$. Similarly $p_{10} = 0$ contradicts $p_3 p_9 > 0$. Therefore

$$p_2, p_3, p_4, p_5, p_9, p_{10} > 0. \tag{711}$$

We may now divide the three binomials. Equations (702)–(703) give

$$\frac{c_2}{c_3} = -\frac{c_9}{c_{10}} = \frac{c_4}{c_5}, \tag{712}$$

whereas Eq. (704) gives

$$\frac{c_2}{c_3} = -\frac{c_4}{c_5}. \tag{713}$$

The common ratio is nonzero by Eq. (711), so these two requirements are incompatible.

Proposition 20.3 (Orbit 55 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-six moment orbit represented by* $(1, 1, 2, 2, 3, 4)$.

Proof. The two-edge masks synchronize all phase ratios needed for the three binomial equations. Their magnitude consequences first exclude every zero-amplitude boundary escape, after which Eqs. (712)–(713) give the phase contradiction. □

20.5 Orbit 56: a forced interior point and an odd product cycle

Take

$$a_{56} = (1, 2, 2, 2, 3, 3) \tag{714}$$

and order the bottom fiber as

$$\begin{aligned} &x_0 = 000000, \quad x_1 = 011100, \quad x_2 = 110010, \\ &x_3 = 101010, \quad x_4 = 100110, \quad x_5 = 110001, \\ &x_6 = 101001, \quad x_7 = 100101, \quad x_8 = 000011, \\ &x_9 = 011111. \end{aligned} \tag{715}$$

The moment matrix has rank seventeen. Use the two normalizations, all six one-bit rows, and the nine two-bit rows

$$(1,2), (1,3), (1,4), (1,5), (1,6), (2,3), (2,5), (2,6), (3,5).$$

Keeping the columns

$$(p_0, \ldots, p_9; |d_0|^2, \ldots, |d_5|^2, |d_8|^2)$$

gives a 17×17 minor of determinant

$$48 \neq 0. \tag{716}$$

The complete moment family is

$$\begin{aligned} (p_0, \ldots, p_9) = \Big(&t, \frac{1}{4} - t, s, r, \frac{1}{4} - r - s, \frac{1}{6} - s, \frac{1}{6} - r, \\ &r + s - \frac{1}{12}, \frac{1}{6} - t, t + \frac{1}{12}\Big), \end{aligned} \tag{717}$$

where

$$0 \le t \le \frac{1}{6}, \qquad 0 \le r, s \le \frac{1}{6}, \qquad \frac{1}{12} \le r + s \le \frac{1}{4}. \tag{718}$$

For the first-qubit flip, order the incidences as

$$(z_{90}, z_{81}, z_{18}, z_{09}). \tag{719}$$

The Walsh kernel gives

$$p_0 p_9 = p_1 p_8. \tag{720}$$

Substituting Eq. (717),

$$t\left(t + \frac{1}{12}\right) = \left(\frac{1}{4} - t\right)\left(\frac{1}{6} - t\right),$$

which is equivalent to

$$12t = 1. \tag{721}$$

Thus

$$p_0 = p_8 = \frac{1}{12}, \qquad p_1 = p_9 = \frac{1}{6}, \tag{722}$$

and the equalities inside the same Walsh kernel give

$$\rho_0 = \rho_9, \qquad \rho_1 = \rho_8. \tag{723}$$

Since $p_2 + p_5 = 1/6$, at least one of these two probabilities is positive. The masks $e_2 + e_5$ and $e_2 + e_6$ imply

$$p_2 > 0 \Longrightarrow \rho_2 = -\rho_9 = -\rho_1,$$
$$p_5 > 0 \Longrightarrow \rho_5 = -\rho_1 = -\rho_9.$$

In either case $\rho_1 = \rho_9$. Combining this with Eq. (723) gives

$$\rho_0 = \rho_1 = \rho_8 = \rho_9 =: \rho. \tag{724}$$

The same argument with the mask pairs

$$(e_3 + e_5, e_3 + e_6), \qquad (e_4 + e_5, e_4 + e_6)$$

shows that every active index in each of

$$\{2, 5\}, \qquad \{3, 6\}, \qquad \{4, 7\} \tag{725}$$

has phase ratio $-\rho$. As in orbit 55, zero amplitudes cause no exception, because $c_i = d_i = 0$. Thus for every $2 \le i \le 7$ we may write

$$d_i = -\rho\, \overline{c_i}. \tag{726}$$

The masks $e_1 + e_2$, $e_1 + e_3$, and $e_1 + e_4$ now reduce to

$$c_3 c_7 + c_4 c_6 = 0, \tag{727}$$
$$c_2 c_7 + c_4 c_5 = 0, \tag{728}$$
$$c_2 c_6 + c_3 c_5 = 0. \tag{729}$$

Their magnitude consequences are

$$r\left(r + s - \frac{1}{12}\right) = \left(\frac{1}{4} - r - s\right)\left(\frac{1}{6} - r\right), \tag{730}$$
$$s\left(r + s - \frac{1}{12}\right) = \left(\frac{1}{4} - r - s\right)\left(\frac{1}{6} - s\right), \tag{731}$$
$$s\left(\frac{1}{6} - r\right) = r\left(\frac{1}{6} - s\right). \tag{732}$$

The last equation simplifies without division to

$$r = s. \tag{733}$$

Substitution into Eq. (730) gives

$$r\left(2r - \frac{1}{12}\right) = \left(\frac{1}{4} - 2r\right)\left(\frac{1}{6} - r\right),$$

which is equivalent to

$$12r = 1. \tag{734}$$

Thus

$$r = s = \frac{1}{12}, \qquad p_2 = p_3 = p_4 = p_5 = p_6 = p_7 = \frac{1}{12}. \tag{735}$$

Every amplitude in Eqs. (727)–(729) is nonzero.

Dividing Eq. (728) by Eq. (729) gives

$$\frac{c_3}{c_4} = \frac{c_6}{c_7}, \tag{736}$$

whereas Eq. (727) requires

$$\frac{c_3}{c_4} = -\frac{c_6}{c_7}. \tag{737}$$

The ratios are nonzero, so this is impossible.

Proposition 20.4 (Orbit 56 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code can arise from the order-six moment orbit represented by* $(1, 2, 2, 2, 3, 3)$*.*

Proof. The first-qubit equations fix t, the remaining flip equations force $r = s = 1/12$, and the resulting strictly positive phase system contains the odd product cycle in Eqs. (736)–(737). □

Theorem 20.5 (Six-qubit order-six no-go)**.** *No exact* $((6, 2, d \geq 3))$ *qubit code admits a projective diagonal transversal logical element of order* 6*.*

Proof. Equation (642) forces every order-six code into the $(q, m) = (6, 6)$ search. The exact diagonal enumeration gives only the four moment orbits 53–56. Propositions 20.1, 20.2, 20.3, and 20.4 exclude them one by one. Hence no order-six code exists. □

Corollary 20.6 (Maximum six-qubit projective diagonal order)**.** *The orders* $2, 3, 5$ *are realized by exact six-qubit distance-three codes, while projective orders* $6, 7, 8, 9$ *are impossible. Therefore*

$$M_3(6) = 5. \tag{738}$$

At this stage of the argument, projective order $m = 4$ *is the only nontrivial order not yet analyzed; Section 22 excludes it completely. It cannot affect the maximum in any case.*

Remark 20.7 (Determinant-one convention)**.** Projective order six corresponds to the cyclic subgroup C_{12} in the determinant-one $SU(2)$ convention. Theorem 20.5 upgrades the corresponding numerical "N/A" entry to an exact no-go theorem. Together with Theorems 19.6, 18.3, and 17.2, this proves that every projective order strictly larger than five is impossible on six qubits.

21 No-go theorems for the other seven order-five orbits

The rank-compatible projective-order-five moment orbits are orbits 33–40. Section 6 proves that orbit 36 contains one rigid exact code class. We now prove that each of the other seven orbits is impossible. Consequently orbit 36 is the unique order-five orbit, without any remaining numerical assumption.

Throughout this section,

$$|0_L\rangle = \sum_{x \in S_0} c_x |x\rangle, \qquad |1_L\rangle = \sum_{y \in S_1} d_y |y\rangle,$$

and

$$p_x = |c_x|^2, \qquad q_y = |d_y|^2.$$

All row reductions below are over $\mathbb{Q}$. The only off-diagonal KL operators used are Pauli operators of weight at most two.

Theorem 21.1 (The other seven order-five orbits are impossible)**.** *None of the moment orbits*

$$33, 34, 35, 37, 38, 39, 40$$

admits an exact $((6, 2, 3))$ *completion.*

21.1 Orbit 34: a two-term Walsh separation

Orbit 34 is represented by

$$a = (1, 1, 1, 1, 1, 1).$$

Its fibers are

$$S_0 = \{0^6\} \cup \{\overline{e_i} : 1 \le i \le 6\}, \qquad S_1 = \{e_i : 1 \le i \le 6\} \cup \{1^6\}.$$

The diagonal moment system has a unique solution:

$$p_{0^6} = q_{1^6} = \frac{2}{5}, \qquad p_{\overline{e_i}} = q_{e_i} = \frac{1}{10}.$$

Thus every displayed amplitude is nonzero.

Fix distinct i, j. The cross-KL equations for X_i and $X_i Z_j$ are

$$\overline{c_{0^6}} d_{e_i} + \overline{c_{\overline{e_i}}} d_{1^6} = 0,$$

$$\overline{c_{0^6}} d_{e_i} - \overline{c_{\overline{e_i}}} d_{1^6} = 0.$$

Adding and subtracting force both nonzero monomials to vanish, a contradiction. Hence orbit 34 is impossible.

21.2 Orbit 40: the same obstruction on two forced points

Orbit 40 is represented by

$$a = (1, 2, 2, 2, 2, 2).$$

Exact moment reduction gives, among other relations,

$$p_{000000} = \frac{1}{5}, \qquad p_{011111} = \frac{3}{10},$$

$$q_{100000} = \frac{3}{10}, \qquad q_{111111} = \frac{1}{5}.$$

The flip X_1 has exactly the two incidences

$$100000 \longleftrightarrow 000000, \qquad 111111 \longleftrightarrow 011111.$$

For any $j \in \{2, \ldots, 6\}$, the equations for X_1 and $X_1 Z_j$ separate the corresponding two monomials. Since all four amplitudes have strictly positive squared magnitude, this is impossible. Therefore orbit 40 is excluded.

21.3 Orbit 39: one forced monomial in a full-rank flip system

Orbit 39 is represented by

$$a = (1, 1, 2, 3, 3, 3).$$

In lexicographic order on the two fibers, exact row reduction gives

$$p_{010111} = \frac{1}{5}, \qquad q_{110111} = \frac{3}{10}.$$

The X_1 cross system has the five incidences

$$\begin{aligned} 000000 &\longleftrightarrow 100000,\\ 001001 &\longleftrightarrow 101001,\\ 001010 &\longleftrightarrow 101010,\\ 001100 &\longleftrightarrow 101100,\\ 010111 &\longleftrightarrow 110111. \end{aligned}$$

On these five source strings, the character rows

$$1, Z_2, Z_3, Z_4, Z_5$$

form a 5×5 matrix of determinant 16. Hence the dressed equations isolate all five cross monomials. In particular they force

$$\overline{c_{010111}} d_{110111} = 0,$$

contradicting the two positive probabilities above. Thus orbit 39 is impossible.

21.4 Orbit 35: three full-rank flip systems

Orbit 35 is represented by

$$a = (1, 1, 1, 2, 2, 3).$$

Let the bottom and top fibers be ordered lexicographically as in Table 1, and write p_i, q_j for the corresponding probabilities. The complete moment normal form contains

$$p_5 = \frac{2}{5} - q_{10} - q_{11}, \qquad p_7 = q_{10}, \qquad p_8 = q_{11}, \qquad q_{12} = \frac{3}{10}. \tag{739}$$

Here

$$x_5 = 011001, \qquad x_7 = 101001, \qquad x_8 = 110001, \qquad y_{12} = 111001.$$

For each $i = 1, 2, 3$, use X_i and the five dressings $X_i Z_j$ with $j \neq i$. The resulting 6×6 character matrix has determinant -160. Hence every monomial in that flip system is isolated. The forced top point 111001 is paired respectively with

$$011001, \qquad 101001, \qquad 110001$$

under X_1, X_2, X_3. Since $q_{12} > 0$, the three systems force

$$p_5 = p_7 = p_8 = 0.$$

Equation (739) then gives $q_{10} = q_{11} = 0$ and simultaneously $p_5 = 2/5$, a contradiction. Therefore orbit 35 is impossible.

21.5 Orbit 38: a 2×4 permanent obstruction

Orbit 38 is represented by

$$a = (1, 1, 2, 2, 2, 3).$$

Its bottom fiber can be labeled as

$$\{A_1, A_2, A_3, A_4\} \cup \{B_{1j}, B_{2j} : 1 \leq j \leq 4\} \cup \{S\},$$

where

	1	2	3	4
A_j	001001	000101	000011	000000
B_{1j}	010110	011010	011100	011111
B_{2j}	100110	101010	101100	101111

$S = 110001.$

The top fiber is the bitwise complement of the bottom fiber. Denote the corresponding bottom amplitudes by a_j, b_{ij}, s. Exact moment reduction gives equality of complementary probabilities. With

$$w_{ij} = |b_{ij}|^2,$$

the complete moment normal form is

$$|s|^2 = \frac{1}{5}, \qquad \sum_{j=1}^{4} w_{1j} = \sum_{j=1}^{4} w_{2j} = \frac{3}{10}, \qquad |a_j|^2 = w_{1j} + w_{2j} - \frac{1}{10}. \tag{740}$$

In particular every column has positive total weight:

$$w_{1j} + w_{2j} \geq \frac{1}{10} > 0. \tag{741}$$

On every active point x, write

$$d_{\bar{x}} = \rho_x \overline{c_x}, \qquad |\rho_x| = 1, \qquad \sigma_x = \overline{\rho_x}.$$

For every grid point B_{ij}, one weight-two flip has exactly the reciprocal pair $B_{ij} \leftrightarrow S$. Its cross-KL equation is

$$(\sigma_{B_{ij}} + \sigma_S) b_{ij} s = 0.$$

Thus every active grid point has the same phase ratio,

$$\sigma_{B_{ij}} = -\sigma_S. \tag{742}$$

For every pair of distinct columns j, k, another weight-two flip gives

$$(\sigma_{B_{1j}} + \sigma_{B_{2k}}) b_{1j} b_{2k} + (\sigma_{B_{1k}} + \sigma_{B_{2j}}) b_{1k} b_{2j} = 0.$$

Using (742), this becomes the permanent equation

$$b_{1j} b_{2k} + b_{1k} b_{2j} = 0 \qquad (j \neq k). \tag{743}$$

Let

$$R_i = \{j : b_{ij} \neq 0\}.$$

Both sets are nonempty by the row sums in (740), and $R_1 \cup R_2 = \{1, 2, 3, 4\}$ by (741). If $R_1 \neq R_2$, choose a column occurring in one support but not the other and a column occurring in the other row. Equation (743) then reduces to a single nonzero monomial. This contradiction shows that $R_1 = R_2$. Their union condition therefore forces

$$R_1 = R_2 = \{1, 2, 3, 4\}.$$

Now put $r_j = b_{1j}/b_{2j}$. Equation (743) gives $r_j = -r_k$ for every $j \neq k$. Already three nonzero ratios cannot be pairwise negatives. Hence orbit 38 is impossible.

21.6 Orbit 37: a 3×3 star-and-permanent obstruction

Orbit 37 is represented by

$$a = (1, 1, 1, 2, 3, 3).$$

Label its bottom fiber by

$$\{A_1, A_2, A_3\} \cup \{B_{ij} : 1 \le i, j \le 3\} \cup \{S\},$$

with

	1	2	3
A_j	000110	000101	000000
B_{1j}	011001	011010	011111
B_{2j}	101001	101010	101111
B_{3j}	110001	110010	110111

$S = 111100.$

Again the top fiber is the bitwise complement. Denote the bottom amplitudes by a_j, b_{ij}, s. Exact moment reduction gives complementary equality and

$$|s|^2 = \frac{1}{10}, \qquad \sum_{j=1}^{3} w_{ij} = \frac{1}{5}, \qquad |a_j|^2 = \sum_{i=1}^{3} w_{ij} - \frac{1}{10}, \tag{744}$$

where $w_{ij} = |b_{ij}|^2$. Consequently

$$\sum_{j=1}^{3} |a_j|^2 = \frac{3}{10}.$$

Use the same phase-ratio notation as above. Walsh inversion of the weight-one system centered at row i gives

$$(\sigma_{A_j} - \sigma_{B_{ij}}) a_j b_{ij} = 0 \qquad (j = 1, 2, 3) \tag{745}$$

and one star equation

$$\sum_{j=1}^{3} \sigma_{B_{ij}} a_j b_{ij} = 0. \tag{746}$$

The three weight-two systems pairing A_j with S give

$$(\sigma_{A_j} + \sigma_S) a_j s = 0. \tag{747}$$

Finally, for every pair of rows $i < k$ and columns $j < \ell$, one of the nine remaining weight-two systems gives

$$\begin{aligned} &(\sigma_{B_{ij}} + \sigma_{B_{k\ell}}) b_{ij} b_{k\ell} \\ &\quad + (\sigma_{B_{i\ell}} + \sigma_{B_{kj}}) b_{i\ell} b_{kj} = 0. \end{aligned} \tag{748}$$

Let

$$J = \{j : a_j \neq 0\}, \qquad R_i = \{j \in J : b_{ij} \neq 0\}.$$

For $j \in J$, Eqs. (745) and (747) imply that every active B_{ij} in column j has

$$\sigma_{B_{ij}} = \sigma_{A_j} = -\sigma_S.$$

Hence on columns in J, Eq. (748) reduces to

$$b_{ij} b_{k\ell} + b_{i\ell} b_{kj} = 0. \tag{749}$$

Also, the star equation shows that $|R_i| \neq 1$ for every row.

The set J is nonempty because the A-probabilities sum to $3/10$. If $|J| = 1$, the unique active column has positive grid weight, so some row has exactly one active star pair, contradicting $|R_i| \neq 1$.

Suppose $|J| = 2$. The remaining inactive column has total grid weight $1/10$. A row with $R_i = \varnothing$ would place its full row weight $1/5$ in that one inactive column, impossible. Hence every row contains both columns of J. If $J = \{j, k\}$, define $r_i = b_{ij}/b_{ik}$. Equation (749) gives $r_i = -r_\ell$ for each pair of rows, which is impossible for three nonzero ratios.

It remains to take $|J| = 3$. Every R_i has size two or three. If two row supports differ, their intersection is nonempty; choose a common column and a column present in one row but absent in the other. The corresponding permanent equation has exactly one nonzero monomial, a contradiction. Thus all three R_i are equal. If their common size is two, the preceding three-row ratio contradiction applies. If their common size is three, already two rows give three nonzero ratios that would have to be pairwise negative. This is again impossible. Therefore orbit 37 is excluded.

21.7 Orbit 33: a six-block qubit reflection

Orbit 33 is represented by

$$a = (0, 1, 1, 3, 3, 3).$$

The first physical qubit carries zero angle. On the remaining five qubits, let

$$\begin{aligned} t_0 &= 00000, \quad t_1 = 01111, \quad t_2 = 10111, \\ t_3 &= 11001, \quad t_4 = 11010, \quad t_5 = 11100. \end{aligned}$$

Then

$$S_0 = \{(r, t_i) : r \in \mathbb{F}_2,\ 0 \le i \le 5\}, \qquad S_1 = \{(r, \bar{t}_i) : r \in \mathbb{F}_2,\ 0 \le i \le 5\}.$$

Write

$$|0_L\rangle = \sum_{i=0}^{5} \sum_{r=0}^{1} c_{ri} |r, t_i\rangle, \qquad |1_L\rangle = \sum_{i=0}^{5} \sum_{r=0}^{1} d_{ri} |r, \bar{t}_i\rangle,$$

and regard

$$c_i = (c_{0i}, c_{1i})^T, \qquad d_i = (d_{0i}, d_{1i})^T$$

as two-component vectors on the zero-angle qubit.

Exact diagonal moment reduction gives

$$\|c_i\|^2 = \|d_i\|^2 = R_i, \qquad R = \frac{1}{10}(3, 2, 2, 1, 1, 1). \tag{750}$$

In particular all twelve block vectors are nonzero.

There is a useful stronger matrix form of the KL equations involving the first qubit. Let

$$C = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & -1 & -1 & -1 & 1 & 1 \end{pmatrix}, \qquad \det C = 160,$$

and $J = \mathrm{diag}(1, -1, -1, -1, -1, -1)$. For any 2×2 matrix F, the KL equations for F on the first qubit and for FZ_j with j on one of the last five qubits imply

$$f = C^{-1} J C\, g,$$

where

$$f_i = c_i^\dagger F c_i, \qquad g_i = d_i^\dagger F d_i.$$

A direct calculation gives

$$C^{-1} J C = 2R\mathbf{1}^T - I_6. \tag{751}$$

Let

$$\rho = \sum_i c_i c_i^\dagger = \sum_i d_i d_i^\dagger$$

be the common reduced state on the first qubit. Since Eq. (751) holds for a basis of 2×2 matrices, it is equivalent to the six matrix identities

$$c_i c_i^\dagger + d_i d_i^\dagger = 2R_i \rho \qquad (0 \le i \le 5). \tag{752}$$

Now use cross-KL equations. Flipping the first and second coordinates of the five-bit block gives respectively the reciprocal pairs $\{t_0, t_1\}$ and $\{t_0, t_2\}$. Since an arbitrary Pauli may also act on the zero-angle qubit while keeping total weight at most two, the four cross equations are equivalent to

$$d_0 c_1^\dagger + d_1 c_0^\dagger = 0, \qquad d_0 c_2^\dagger + d_2 c_0^\dagger = 0. \tag{753}$$

A sum of two nonzero rank-one matrices can vanish only if their left vectors are collinear and their right vectors are collinear. Hence there exist unit vectors u, v and nonzero scalars α_i, β_i such that

$$c_i = \alpha_i u, \qquad d_i = \beta_i v, \qquad i = 0, 1, 2.$$

Put

$$\gamma_i = \frac{\beta_i}{\alpha_i}.$$

Equation (753) gives

$$\gamma_1 = \gamma_2 = -\gamma_0. \tag{754}$$

Every pair $\{t_i, t_j\}$ with

$$i \in \{0, 1, 2\}, \qquad j \in \{3, 4, 5\}$$

is the unique reciprocal pair of one weight-two flip on the last five qubits. The corresponding cross equation is

$$c_j^\dagger d_i + c_i^\dagger d_j = 0. \tag{755}$$

For fixed $j \ge 3$, define

$$A_j = c_j^\dagger v, \qquad B_j = u^\dagger d_j.$$

Using $i = 0$ and $i = 1$ in (755) yields

$$\gamma_0 A_j + B_j = 0, \qquad \gamma_1 A_j + B_j = 0.$$

Equation (754) therefore forces

$$A_j = B_j = 0.$$

Thus

$$c_j \perp v, \qquad d_j \perp u, \qquad j = 3, 4, 5. \tag{756}$$

Let $P_u = uu^\dagger$ and $P_v = vv^\dagger$. Dividing Eq. (752) by R_i for $i \le 2$ gives

$$P_u + P_v = 2\rho.$$

For $j \geq 3$, Eq. (756) gives instead

$$(I - P_v) + (I - P_u) = 2\rho.$$

Hence $P_u + P_v = I$ and $\rho = I/2$. In two dimensions this means $u \perp v$. Equation (756) then implies that every c_j is parallel to u and every d_j is parallel to v. Using (750),

$$\sum_i c_i c_i^\dagger = \left(\sum_i R_i\right) P_u = P_u,$$

contradicting $\rho = I/2$. Thus orbit 33 is impossible.

21.8 Consequences for projective order five

Corollary 21.2 (Unconditional uniqueness of the order-five sector)**.** *Orbit 36 is the only rank-compatible moment orbit admitting a six-qubit* $((6, 2, 3))$ *code with projective diagonal logical order five. Consequently every such code has the unique Shor–Laflamme enumerator*

$$A = \left(1, 0, \frac{21}{25}, 0, \frac{291}{25}, \frac{384}{25}, \frac{79}{25}\right)$$

and belongs, up to the equivalences of the lift-aware normal form, to the rigid orbit-36 class.

Proof. Orbits 33–40 exhaust the order-five moment skeletons. Theorem 21.1 excludes all except orbit 36, and Theorem 6.2 classifies every exact solution in orbit 36. □

22 Projective order four: complete ten-orbit no-go

After the high-order theorems, projective order four is the only remaining candidate among $2, \ldots, 9$ not already realized or excluded. This section closes it. For $m = 4$, the controlled-lift bound gives

$$4 \mid q, \qquad q \leq 9, \tag{757}$$

so that

$$q \in \{4, 8\}. \tag{758}$$

The exact diagonal enumeration contains eight $q = 4$ moment orbits and two $q = 8$ moment orbits. The ten cases in Table 4 are therefore exhaustive; no higher-denominator controlled lift of a projective order-four element is omitted.

Remark 22.1 (Finite computational step)**.** For orbit 31, the classification of support masks into sixteen symmetry types is computer-assisted and uses exact rational arithmetic and symbolic row reduction. Each type is then excluded by an explicit analytic contradiction. The remaining orbits in Table 4 are excluded by the analytic arguments below.

The remainder treats and excludes the ten orbits in numerical order.

orbit	q	representative a	d_{LP}	proof method
25	4	$(0,1,1,1,1,1)$	6	analytic exclusion
26	4	$(0,1,1,2,2,3)$	12	analytic exclusion, Theorem 22.5
27	4	$(1,1,1,1,1,1)$	5	analytic exclusion
28	4	$(1,1,1,1,1,2)$	0	analytic exclusion
29	4	$(1,1,1,1,2,2)$	8	analytic exclusion on fifteen faces
30	4	$(1,1,1,1,2,3)$	9	analytic exclusion, Theorem 22.12
31	4	$(1,1,1,2,2,2)$	9	exact support classification and analytic exclusion, Theorem 22.16
32	4	$(1,1,2,2,2,3)$	9	analytic exclusion, Theorem 22.18
62	8	$(1,1,1,5,5,5)$	0	analytic exclusion
63	8	$(1,1,2,4,5,5)$	1	analytic exclusion

Table 4: Exact no-go proofs for the ten projective-order-four moment orbits. Orbit 31 uses a finite rational support classification followed by an analytic contradiction for every type; the remaining orbits are excluded by the displayed analytic arguments.

22.1 Orbit 25: the zero-angle coherence contradiction

Take

$$a = (0,1,1,1,1,1) \in \mathbb{Z}_4^6. \tag{759}$$

Let $\Omega = 1^5$, let e_j be a unit vector in the last-five block, and put $\widehat{e}_j = \Omega - e_j$. The two fibers are

$$S_0 = \{(b,0^5), (b,\widehat{e}_j) : b \in \mathbb{F}_2,\ 1 \le j \le 5\}, \tag{760}$$

$$S_1 = \{(b,e_j), (b,\Omega) : b \in \mathbb{F}_2,\ 1 \le j \le 5\}. \tag{761}$$

Write

$$|0_L\rangle = \sum_{b=0}^{1} a_b |b,0^5\rangle + \sum_{b=0}^{1}\sum_{j=1}^{5} b_{b,j}|b,\widehat{e}_j\rangle, \tag{762}$$

$$|1_L\rangle = \sum_{b=0}^{1}\sum_{j=1}^{5} r_{b,j}|b,e_j\rangle + \sum_{b=0}^{1} s_b|b,\Omega\rangle. \tag{763}$$

Zero amplitudes are initially allowed.

Set

$$t_j := |r_{1,j}|^2, \qquad \sigma := |s_1|^2, \qquad T := \sigma + \sum_{j=1}^{5} t_j. \tag{764}$$

Exact row reduction of the normalization and degree-at-most-two diagonal moment equations gives the complete six-dimensional moment family

$$|r_{0,j}|^2 = \frac{1}{8} - t_j, \qquad |r_{1,j}|^2 = t_j, \tag{765}$$

$$|s_0|^2 = \frac{3}{8} - \sigma, \qquad |s_1|^2 = \sigma, \tag{766}$$

$$|a_0|^2 = \frac{3}{8} - \frac{3T}{4} + \sigma, \qquad |a_1|^2 = \frac{3T}{4} - \sigma, \tag{767}$$

$$|b_{0,j}|^2 = \frac{1}{8} - \frac{T}{4} + t_j, \qquad |b_{1,j}|^2 = \frac{T}{4} - t_j. \tag{768}$$

In particular,

$$|a_0|^2 + |a_1|^2 = \frac{3}{8}, \qquad |b_{0,j}|^2 + |b_{1,j}|^2 = \frac{1}{8}. \tag{769}$$

Fix a tail coordinate j. Flipping that qubit gives exactly four cross-logical incidences. With

$$z_j = \begin{pmatrix} \overline{a_0} r_{0,j} \\ \overline{a_1} r_{1,j} \\ \overline{b_{0,j}} s_0 \\ \overline{b_{1,j}} s_1 \end{pmatrix}, \tag{770}$$

the equations for X_{j+1}, $Z_1 X_{j+1}$, and one $Z_k X_{j+1}$ with $k \notin \{1, j+1\}$ are

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix} z_j = 0. \tag{771}$$

The kernel is generated by $(1, -1, -1, 1)^T$. Thus there is a scalar λ_j such that

$$\overline{a_0} r_{0,j} = \lambda_j, \quad \overline{a_1} r_{1,j} = -\lambda_j, \quad \overline{b_{0,j}} s_0 = -\lambda_j, \quad \overline{b_{1,j}} s_1 = \lambda_j. \tag{772}$$

The first two magnitudes and Eq. (769) give

$$|a_0|^2 \left(\frac{1}{8} - t_j \right) = |a_1|^2 t_j \quad \Longrightarrow \quad |a_0|^2 = 3t_j. \tag{773}$$

Hence all t_j are equal; write $t_j = t$. Comparing the first and third magnitudes in Eq. (772), and using $T = 5t + \sigma$, reduces exactly to

$$(T - 1)(2T - 1) = 0. \tag{774}$$

The branch $T = 1$ would give $t = 5/64$ and $\sigma = 39/64$, contradicting $|s_0|^2 = 3/8 - \sigma \geq 0$. Therefore $T = 1/2$, and

$$t = \frac{1}{16}, \qquad \sigma = \frac{3}{16}. \tag{775}$$

All twenty-four amplitudes are consequently nonzero, with

$$|a_0|^2 = |a_1|^2 = |s_0|^2 = |s_1|^2 = \frac{3}{16}, \qquad |b_{b,j}|^2 = |r_{b,j}|^2 = \frac{1}{16}. \tag{776}$$

Define the first-qubit anchor coherences

$$A := \overline{a_0} a_1, \qquad S := \overline{s_0} s_1. \tag{777}$$

The weight-two equations for flipping the first qubit together with the j-th tail qubit contain

$$\overline{a_1} r_{0,j} + \overline{b_{1,j}} s_0 = 0. \tag{778}$$

Using Eq. (772) and the equal nonzero anchor magnitudes in Eq. (776), this is equivalent to

$$S = -A. \tag{779}$$

The same kernel relations give

$$\overline{r_{0,j}} r_{1,j} = -\frac{A}{3}, \qquad \overline{b_{0,j}} b_{1,j} = -\frac{S}{3}. \tag{780}$$

Hence the complex first-qubit coherences of the two logical states are

$$K_0 = A - \frac{5}{3}S, \qquad K_1 = S - \frac{5}{3}A. \tag{781}$$

Detection of both X_1 and Y_1 is equivalent to $K_0 = K_1$, which forces $A = S$. Together with Eq. (779), this would give $A = 0$, contradicting $|A| = 3/16$.

Proposition 22.2 (Orbit 25 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code arises from the moment orbit represented by* $(q, m, a) = (4, 4, (0, 1, 1, 1, 1, 1))$.

22.2 Orbit 26: a matrix-valued Walsh obstruction

A representative and the other surviving sorted angle record in the same Boolean-cube orbit are

$$a = (0, 1, 1, 2, 2, 3), \qquad a' = (0, 2, 2, 3, 3, 3) \qquad (\mathrm{mod} 4). \tag{782}$$

It is enough to treat the first representative.

The proof uses a feature absent from the higher-order cases: the first physical angle is zero. The first qubit can therefore be retained as a 2-component amplitude index. Scalar diagonal moments give a low-dimensional polytope of total tail weights, while the Knill–Laflamme equations containing arbitrary operators on the zero-angle qubit become matrix-valued Walsh identities. These identities exclude every interior and boundary branch.

22.2.1 The two fibers and the virtual three-bit tail cube

Write a six-bit string as (b, t), where $b \in \mathbb{F}_2$ is the first bit and $t \in \mathbb{F}_2^5$ consists of the remaining bits. For the tail angle vector $(1, 1, 2, 2, 3)$, the residue-zero tail fiber has the parametrization

$$t(x, y, z) := \big(x, y, z, z \oplus xy, x \oplus y\big), \qquad (x, y, z) \in \mathbb{F}_2^3. \tag{783}$$

We label the eight points as follows:

label	(x, y, z)	$t(x, y, z)$	label	(x, y, z)	$t(x, y, z)$
A	000	00000	B	001	00110
C	100	10001	D	101	10111
E	010	01001	F	011	01111
G	110	11010	H	111	11100

(784)

Let $t_0, \ldots, t_7$ denote A, B, C, D, E, F, G, H in this order. Because

$$0 + 1 + 1 + 2 + 2 + 3 = 9 \equiv 1 \pmod 4, \tag{785}$$

bitwise complementation maps residue zero to residue one. The full fibers are therefore

$$S_0 = \{(b, t_i) : b \in \mathbb{F}_2,\ 0 \leq i \leq 7\}, \qquad S_1 = \{(b, \bar{t}_i) : b \in \mathbb{F}_2,\ 0 \leq i \leq 7\}. \tag{786}$$

Both have size sixteen.

Suppose that an exact code exists. Collect the amplitudes over the first qubit into spinors

$$C_i := \begin{pmatrix} c_{0,i} \\ c_{1,i} \end{pmatrix}, \qquad D_i := \begin{pmatrix} d_{0,i} \\ d_{1,i} \end{pmatrix}, \tag{787}$$

so that

$$|0_L\rangle = \sum_{i=0}^{7} \sum_{b=0}^{1} c_{b,i} |b, t_i\rangle, \tag{788}$$

$$|1_L\rangle = \sum_{i=0}^{7} \sum_{b=0}^{1} d_{b,i} |b, \bar{t}_i\rangle. \tag{789}$$

Zero spinors are allowed at this stage.

22.2.2 Exact aggregate moment polytope

Put

$$P_i := \|C_i\|^2, \qquad Q_i := \|D_i\|^2. \tag{790}$$

After summing over the zero-angle first qubit, normalization and equality of all first and second tail moments form a rank-fourteen rational system in the sixteen variables P_i, Q_i. Exact row reduction gives

$$P_i = Q_i \qquad (0 \le i \le 7) \tag{791}$$

and the two-parameter family

$$\begin{aligned} P_0 &= \alpha + \beta - \frac{1}{8}, & P_1 &= \frac{3}{8} - \alpha - \beta, \\ P_2 &= \frac{1}{4} - \alpha, & P_3 &= \alpha, \\ P_4 &= \frac{1}{4} - \beta, & P_5 &= \beta, \\ P_6 &= \frac{1}{8}, & P_7 &= \frac{1}{8}, \end{aligned} \tag{792}$$

where

$$0 \le \alpha, \beta \le \frac{1}{4}, \qquad \frac{1}{8} \le \alpha + \beta \le \frac{3}{8}. \tag{793}$$

Thus the aggregate moment region is a hexagon. The full moment polytope has additional degrees of freedom describing how each P_i is split between the two first-qubit components, but the identities below hold throughout that larger polytope.

Three consequences of Eq. (792) will be used repeatedly:

$$P_0 + P_1 = P_2 + P_3 = P_4 + P_5 = P_6 + P_7 = \frac{1}{4}, \tag{794}$$

$$P_0 + P_2 + P_4 = P_1 + P_3 + P_5 = \frac{3}{8}, \tag{795}$$

and no three of $P_0, \ldots, P_5$ can vanish simultaneously.

22.2.3 Matrix-valued moment and flip equations

Group the two values of the virtual bit z into four 2×2 matrices:

$$\mathsf{C}_g := \begin{pmatrix} C_{2g} & C_{2g+1} \end{pmatrix}, \qquad \mathsf{D}_g := \begin{pmatrix} D_{2g} & D_{2g+1} \end{pmatrix}, \qquad 0 \le g \le 3. \tag{796}$$

The group $g = 3$, containing G, H, will be called the *anchor group*. Every group has total bottom and top probability $1/4$:

$$\operatorname{tr}(\mathsf{C}_g \mathsf{C}_g^\dagger) = \operatorname{tr}(\mathsf{D}_g \mathsf{D}_g^\dagger) = \frac{1}{4}. \tag{797}$$

Let

$$\varrho := \sum_{g=0}^{3} \mathsf{C}_g \mathsf{C}_g^\dagger = \sum_{g=0}^{3} \mathsf{D}_g \mathsf{D}_g^\dagger \tag{798}$$

be the common reduced density matrix on the first qubit. The equality follows from detection of arbitrary one-qubit operators on that qubit.

The tail coordinates numbered $2, 3, 6$ in the six-qubit block are the virtual functions $x, y, x \oplus y$. For every first-qubit operator F, the KL equations for F, FZ_2, FZ_3, and FZ_6 give the four-point matrix Walsh transform on (x, y). Since all three tail bits are complemented in the top fiber, the three nontrivial Fourier coefficients have opposite signs for the two logical states. Inverting the transform gives

$$\boxed{\mathsf{C}_g \mathsf{C}_g^\dagger + \mathsf{D}_g \mathsf{D}_g^\dagger = \frac{1}{2}\varrho, \qquad 0 \le g \le 3.} \tag{799}$$

This matrix-valued identity is stronger than the scalar moment equations.

Next set

$$J := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{800}$$

The three single-tail-bit flip classes are

$$(A, D) \cup (B, C), \qquad (A, F) \cup (B, E), \qquad (C, F) \cup (D, E). \tag{801}$$

A tail flip has weight one, so it may be multiplied by an arbitrary operator on the first qubit while remaining of weight at most two. The resulting four scalar equations are equivalent to the matrix identities

$$\boxed{\overline{\mathsf{C}_g} J \mathsf{D}_h^T + \overline{\mathsf{C}_h} J \mathsf{D}_g^T = 0, \qquad 0 \le g < h \le 2.} \tag{802}$$

Finally, each nonanchor group is connected reciprocally to the anchor group by two tail masks of weight two. The available Z-dressings separate the four reciprocal pair sums. For example, the masks 00101 and 00011 connect (A, B) to (G, H). Collecting the four equations gives

$$\boxed{\mathsf{C}_3^\dagger \mathsf{D}_g + \left(\mathsf{C}_g^\dagger \mathsf{D}_3\right)^T = 0, \qquad 0 \le g \le 2.} \tag{803}$$

Equations (799)–(803) use only weight-one and weight-two Pauli detection.

We will use the following elementary bilinear fact.

Lemma 22.3 (Three-vector J-orthogonality)**.** *Let $r_0, r_1, r_2 \in \mathbb{C}^{1\times 2}$ be nonzero row vectors satisfying*

$$r_g J r_h^T = 0 \qquad (g \ne h). \tag{804}$$

Then all three vectors lie on the same coordinate axis.

Proof. If two of the vectors were linearly independent, the nondegeneracy of J would make their common J-orthogonal complement zero, contradicting the existence of the third nonzero vector. Hence all three are proportional to one row $r = (r_0, r_1)$. Equation (804) then gives $r J r^T = 2 r_0 r_1 = 0$, so one coordinate of r vanishes. □

22.2.4 The anchor matrices are singular

Lemma 22.4. *Both* C_3 *and* D_3 *are singular.*

Proof. Assume first that C_3 is invertible. Equation (803) gives

$$\mathsf{D}_g = -M\overline{\mathsf{C}_g}, \qquad M := (\mathsf{C}_3^\dagger)^{-1}\mathsf{D}_3^T, \qquad 0 \le g \le 2. \tag{805}$$

Substitution into Eq. (802) yields

$$F_{gh}M^T = 0, \qquad F_{gh} := \overline{\mathsf{C}_g}J\overline{\mathsf{C}_h}^T + \overline{\mathsf{C}_h}J\overline{\mathsf{C}_g}^T. \tag{806}$$

Suppose first that D_3 is invertible. Then M is invertible, so $F_{gh} = 0$. For a generic column $w \in \mathbb{C}^2$, put

$$r_g(w) := w^T\overline{\mathsf{C}_g}. \tag{807}$$

Equation (806) implies $r_g(w)Jr_h(w)^T = 0$. Every nonanchor group has trace $1/4$, so none of the three matrices C_g is zero. Moreover, Eq. (793) permits at most two zero columns among $C_0, \dots, C_5$, so at least one nonanchor group has both columns nonzero. Choose w outside the finitely many kernels of the nonzero columns. Then all three rows $r_g(w)$ are nonzero and one has both entries nonzero, in contradiction with Lemma 22.3.

It remains to consider $\operatorname{rank}\mathsf{D}_3 = 1$. Let d be the unit vector spanning its columns. Since the anchor group has total top probability $1/4$,

$$\mathsf{D}_3\mathsf{D}_3^\dagger = \frac{1}{4}|d\rangle\langle d|. \tag{808}$$

The matrix M^T has image spanned by d, so Eq. (806) gives $F_{gh}d = 0$. Define

$$r_g := d^T\overline{\mathsf{C}_g}. \tag{809}$$

Then the three nonzero rows are pairwise J-orthogonal. They are nonzero because, using Eq. (799) for the anchor group,

$$\begin{aligned}\sum_{g=0}^{2}\|r_g\|^2 &= d^\dagger\left(\sum_{g=0}^{2}\mathsf{C}_g\mathsf{C}_g^\dagger\right)d \\ &= d^\dagger\mathsf{C}_3\mathsf{C}_3^\dagger d + 2d^\dagger\mathsf{D}_3\mathsf{D}_3^\dagger d \\ &> \frac{1}{2}.\end{aligned} \tag{810}$$

Each group contributes at most $1/4$, so strict inequality forces all three rows to be nonzero. Lemma 22.3 places them on one common coordinate axis. If it is the first axis, then

$$\sum_{g=0}^{2}\|r_g\|^2 \le P_0 + P_2 + P_4 = \frac{3}{8}; \tag{811}$$

and if it is the second axis, the same bound follows from $P_1 + P_3 + P_5 = 3/8$. Both contradict Eq. (810). Thus C_3 cannot be invertible.

The equations are symmetric under interchanging the two logical states, so the same argument proves that D_3 is singular. □

22.2.5 The two dependent-anchor branches

The two anchor columns have fixed nonzero norm $1/\sqrt{8}$. By Lemma 22.4, there are vectors $c, d \in \mathbb{C}^2$ and phases $\lambda, \mu \in U(1)$ such that

$$\mathsf{C}_3 = \begin{pmatrix} c & \lambda c \end{pmatrix}, \qquad \mathsf{D}_3 = \begin{pmatrix} d & \mu d \end{pmatrix}, \qquad \|c\|^2 = \|d\|^2 = \frac{1}{8}. \tag{812}$$

For $0 \le g \le 2$, define the rows

$$u_g := c^\dagger \mathsf{D}_g, \qquad v_g := (\mathsf{C}_g^\dagger d)^T. \tag{813}$$

The two rows of Eq. (803) are

$$u_g + v_g = 0, \qquad \overline{\lambda}\, u_g + \mu v_g = 0. \tag{814}$$

Nonmatching anchor phases: $\mu \neq \overline{\lambda}$. Equation (814) gives $u_g = v_g = 0$. Hence every nonanchor bottom spinor is orthogonal to d, and every nonanchor top spinor is orthogonal to c. If Π_c, Π_d denote the projectors onto the directions of c, d, then for every $g = 0, 1, 2$,

$$\mathsf{C}_g \mathsf{C}_g^\dagger = \frac{1}{4}\Pi_{d^\perp}, \qquad \mathsf{D}_g \mathsf{D}_g^\dagger = \frac{1}{4}\Pi_{c^\perp}, \tag{815}$$

while

$$\mathsf{C}_3 \mathsf{C}_3^\dagger = \frac{1}{4}\Pi_c, \qquad \mathsf{D}_3 \mathsf{D}_3^\dagger = \frac{1}{4}\Pi_d. \tag{816}$$

Comparing Eq. (799) for a nonanchor group and the anchor group gives

$$\Pi_{d^\perp} + \Pi_{c^\perp} = \Pi_c + \Pi_d, \tag{817}$$

so $\Pi_c + \Pi_d = I$ and therefore $c \perp d$. Then every bottom spinor, including the anchors, lies along c, while every top spinor lies along d. The two first-qubit reduced density matrices are the distinct pure states Π_c and Π_d, contradicting Eq. (798). Thus the nonmatching branch is impossible.

Matching anchor phases: $\mu = \overline{\lambda}$. Let

$$\widehat{c} := \frac{c}{\|c\|}, \qquad \widehat{d} := \frac{d}{\|d\|}, \qquad \kappa := |\widehat{c}^\dagger \widehat{d}|. \tag{818}$$

A unitary change of basis on the zero-angle first qubit and harmless phase choices may be used to take

$$\widehat{c} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \widehat{d} = \begin{pmatrix} \kappa \\ \sqrt{1-\kappa^2} \end{pmatrix}, \qquad 0 \le \kappa \le 1. \tag{819}$$

Define

$$r_g := \widehat{d}^\dagger \mathsf{C}_g, \qquad 0 \le g \le 2. \tag{820}$$

The first equation in (814) says

$$\widehat{c}^\dagger \mathsf{D}_g = -\overline{r_g}. \tag{821}$$

Contracting Eq. (802) with $\widehat{d}^T$ on the left and $\overline{\widehat{c}}$ on the right now gives

$$r_g J r_h^T = 0, \qquad 0 \le g < h \le 2. \tag{822}$$

The anchor case of Eq. (799) gives

$$\varrho = \frac{1}{2}\big(\Pi_{\widehat{c}} + \Pi_{\widehat{d}}\big). \tag{823}$$

Subtracting the bottom anchor contribution and taking the expectation in $\widehat{d}$ yields

$$\sum_{g=0}^{2}\|r_g\|^2=\frac{1}{2}+\frac{\kappa^2}{4}. \tag{824}$$

If $\kappa>0$, the right-hand side is larger than $1/2$. Since each $\|r_g\|^2\le 1/4$, all three rows are nonzero. Lemma 22.3 puts them on one coordinate axis, and Eq. (795) again bounds their total squared norm by $3/8$, contradicting Eq. (824). Therefore

$$\kappa=0, \qquad \widehat{c}\perp\widehat{d}. \tag{825}$$

Now Eq. (824) equals $1/2$. If all three rows were nonzero, Lemma 22.3 would again give the impossible upper bound $3/8$. Hence exactly one row, say r_k, vanishes, and the other two, r_i, r_j, satisfy

$$\|r_i\|^2=\|r_j\|^2=\frac{1}{4}. \tag{826}$$

Because $\widehat{c}\perp\widehat{d}$, the condition $r_k=0$ forces every column of C_k to lie along $\widehat{c}$. Equality in Eq. (826) forces every column of $\mathsf{C}_i,\mathsf{C}_j$ to lie along $\widehat{d}$. Put

$$s_k:=\widehat{c}^{\dagger}\mathsf{C}_k. \tag{827}$$

The row s_k is nonzero. The outer matrix equations between group k and groups i,j, together with Eq. (821), give

$$s_kJr_i^T=0, \qquad s_kJr_j^T=0. \tag{828}$$

Together with $r_iJr_j^T=0$, Lemma 22.3 shows that s_k,r_i,r_j all lie on the same coordinate axis. Since the two columns inside each group correspond to $z=0,1$, either

$$P_1=P_3=P_5=0 \tag{829}$$

or

$$P_0=P_2=P_4=0. \tag{830}$$

The first alternative gives $\alpha=\beta=0$ from $P_3=P_5=0$, but then $P_1=3/8$, contradicting Eq. (829). The second gives $\alpha=\beta=1/4$ from $P_2=P_4=0$, but then $P_0=3/8$, contradicting Eq. (830). Thus the matching branch is also impossible.

Theorem 22.5 (Orbit 26 no-go)**.** *No exact* $((6,2,d\ge 3))$ *qubit code can arise from the projective order-four moment orbit represented by*

$$(q,m,a)=\big(4,4,(0,1,1,2,2,3)\big). \tag{831}$$

Equivalently, the entire orbit containing the two records in Eq. (782) *is impossible.*

Proof. Lemma 22.4 reduces every putative solution to the dependent-anchor form (812). The non-matching and matching anchor-phase branches are both contradictory, including every facet and vertex of the aggregate moment hexagon. Hence the complete weight-at-most-two Knill–Laflamme system has no solution. □

22.3 Orbit 27: a full-rank six-edge Walsh system

For

$$a = (1, 1, 1, 1, 1, 1) \in \mathbb{Z}_4^6, \tag{832}$$

S_0 consists of the weight-zero and weight-four strings, while S_1 consists of the weight-one and weight-five strings. The exact moment system fixes

$$p_0(000000) = \frac{3}{8} \tag{833}$$

and gives total top probability

$$\sum_{i=1}^{6} p_1(e_i) = \frac{5}{8} \tag{834}$$

on the six weight-one strings.

Fix $i = 1$; all other coordinates are equivalent. The X_1-incidence sources are e_1 and the five weight-five strings missing one of qubits $2, \ldots, 6$. Using the rows corresponding to no Z-dressing and to $Z_2, \ldots, Z_6$, the six edge products obey

$$W_6 z = 0, \qquad W_6 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & 1 & -1 & -1 \\ 1 & -1 & -1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 & -1 & 1 \end{pmatrix}. \tag{835}$$

A direct determinant calculation gives

$$\det W_6 = -128 \neq 0. \tag{836}$$

Therefore every edge product vanishes separately. In particular, the edge $e_1 \mapsto 000000$ gives

$$\overline{c_{000000}} d_{e_1} = 0. \tag{837}$$

Since $c_{000000} \neq 0$, one has $d_{e_1} = 0$. Repeating the argument for all six qubits forces every weight-one top amplitude to vanish, contradicting Eq. (834).

Proposition 22.6 (Orbit 27 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code arises from orbit 27.*

22.4 Orbit 28: a forced two-edge contradiction

For

$$a = (1, 1, 1, 1, 1, 2), \tag{838}$$

the moment polytope is zero-dimensional. Its actual supports are

$$T_0 = \{000000, 111100, 111010, 110110, 101110, 011110\}, \tag{839}$$

$$T_1 = \{100000, 010000, 001000, 000100, 000010, 111110\}, \tag{840}$$

with

$$p_0(000000) = \frac{3}{8}, \qquad p_0(x) = \frac{1}{8} \quad (x \neq 000000), \tag{841}$$

and

$$p_1(111110) = \frac{3}{8}, \qquad p_1(y) = \frac{1}{8} \quad (y \neq 111110). \tag{842}$$

The first-qubit flip has exactly the two incidences

$$100000 \longmapsto 000000, \qquad 111110 \longmapsto 011110. \tag{843}$$

The sources have opposite second bits, so the equations for X_1 and X_1Z_2 form the invertible Walsh system

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \overline{c_{000000}} d_{100000} \\ \overline{c_{011110}} d_{111110} \end{pmatrix} = 0. \tag{844}$$

Both products have magnitude $\sqrt{3}/8$, a contradiction.

Proposition 22.7 (Orbit 28 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code arises from orbit 28.*

22.5 Orbit 29: a fifteen-face pair-ratio audit

Take

$$a = (1, 1, 1, 1, 2, 2). \tag{845}$$

The diagonal moment polytope has dimension eight. Exact elimination of the modulus consequences of the non-diagonal flip equations reduces every putative code to amplitudes arranged in pairs differing by X_5X_6. The nonzero bottom pairs are

$$B_0 = (111100, 111111), \qquad B_1 = (110010, 110001), \qquad B_2 = (101010, 101001), \tag{846}$$

$$B_3 = (011010, 011001), \qquad B_4 = (100110, 100101), \qquad B_5 = (010110, 010101), \tag{847}$$

$$B_6 = (001110, 001101), \tag{848}$$

and the top pairs are

$$T_0 = (100000, 100011), \quad T_1 = (010000, 010011), \quad T_2 = (001000, 001011), \quad T_3 = (000100, 000111), \tag{849}$$

$$T_4 = (111010, 111001), \quad T_5 = (110110, 110101), \quad T_6 = (101110, 101101), \quad T_7 = (011110, 011101). \tag{850}$$

The omitted bottom pair $(000000, 000011)$ is forced to zero. The two strings in every displayed pair have equal probability. With $a, b, c \geq 0$ and $a + b + c \leq 1/8$, the probability of each string in a bottom pair is

	B_0	B_1	B_2	B_3	B_4	B_5	B_6
p	$\frac{1}{8}$	$\frac{1}{8} - b - c$	$\frac{1}{8} - a - c$	$a + b$	$\frac{1}{8} - a - b$	$a + c$	$b + c,$

(851)

and for the top pairs it is

	T_0	T_1	T_2	T_3	T_4	T_5	T_6	T_7
p	$\frac{1}{8} - a - b - c$	a	b	c	$\frac{1}{8} - c$	$\frac{1}{8} - b$	$\frac{1}{8} - a$	$a + b + c.$

(852)

Introduce barycentric coordinates

$$\lambda_0 = \frac{1}{8} - a - b - c, \quad \lambda_1 = a, \quad \lambda_2 = b, \quad \lambda_3 = c. \tag{853}$$

They are nonnegative and sum to 1/8. The reduced probability region is therefore a three-simplex and has exactly fifteen nonempty relative interiors. The weight-one pair-ratio graph has vertices

$$\{B_1, \ldots, B_6, T_4, \ldots, T_7\} \tag{854}$$

and edges

$$X_1: \quad B_3T_4,\ B_5T_5,\ B_6T_6, \tag{855}$$
$$X_2: \quad B_2T_4,\ B_4T_5,\ B_6T_7, \tag{856}$$
$$X_3: \quad B_1T_4,\ B_4T_6,\ B_5T_7, \tag{857}$$
$$X_4: \quad B_1T_5,\ B_2T_6,\ B_3T_7. \tag{858}$$

A vertex is active precisely when its displayed probability is positive. The following table is the complete face audit. In every row the induced active graph is connected.

face dimension	active λ's	zero graph vertices	$\lvert V\rvert$	$\lvert E\rvert$
0	$\{0\}$	B_3, B_5, B_6, T_7	6	6
0	$\{1\}$	B_2, B_4, B_6, T_6	6	6
0	$\{2\}$	B_1, B_4, B_5, T_5	6	6
0	$\{3\}$	B_1, B_2, B_3, T_4	6	6
1	$\{0,1\}$	B_6	9	10
1	$\{0,2\}$	B_5	9	10
1	$\{0,3\}$	B_3	9	10
1	$\{1,2\}$	B_4	9	10
1	$\{1,3\}$	B_2	9	10
1	$\{2,3\}$	B_1	9	10
2	$\{0,1,2\}$	none	10	12
2	$\{0,1,3\}$	none	10	12
2	$\{0,2,3\}$	none	10	12
2	$\{1,2,3\}$	none	10	12
3	$\{0,1,2,3\}$	none	10	12

Table 5: The fifteen relative interiors of the orbit-29 moment simplex. The edge set is Eq. (858); direct inspection shows that every listed induced graph is connected.

For every active graph edge, the two relevant Walsh rows relate the unit-modulus internal ratios of the two X_5X_6-pairs with opposite sign. Because every active graph in Table 5 is connected, all active pair ratios are fixed by one common unit phase. Absorbing that phase into the amplitude gauge—equivalently, using one common local diagonal rephasing on qubit five—sets every active bottom ratio $B_1, \ldots, B_6$ to $+1$ and every active high-top ratio $T_4, \ldots, T_7$ to -1.

The pair B_0 has fixed positive probability 1/8. Put

$$\eta = \frac{\overline{c_{111111}}}{\overline{c_{111100}}}. \tag{859}$$

The total probability in T_4, T_5, T_6, T_7 is 3/4, so at least one of these pairs is active. Choose an active T_k. The corresponding two flip masks are X_jX_5 and X_jX_6, where

$$(k, j) = (4, 4), (5, 3), (6, 2), (7, 1). \tag{860}$$

For the two masks X_jX_5 and X_jX_6, choose the two Walsh rows that separate the two strings of B_0. After the active pair ratios have been fixed as above, all incidences other than the B_0–T_k edge contribute identical remainders to the corresponding pair of equations. The resulting four Walsh equations have the form

$$(1-\eta)A + R = 0, \qquad -(1-\eta)A + R = 0, \tag{861}$$
$$(1+\eta)A + S = 0, \qquad -(1+\eta)A + S = 0, \tag{862}$$

where $A \neq 0$ is the edge product joining B_0 to the chosen active T_k, and R, S collect the other incidences. Subtracting the first pair gives $\eta = 1$; subtracting the second gives $\eta = -1$. This is impossible. Table 5 shows that the argument covers every interior, facet, edge, and vertex of the exact moment simplex.

Proposition 22.8 (Orbit 29 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code arises from orbit 29.*

22.6 Orbit 30: a K_5 support-graph obstruction

For

$$(q, m, a) = \big(4, 4, (1, 1, 1, 1, 2, 3)\big), \tag{863}$$

the two fibers have size sixteen and are complementary. There is a useful coordinate system in which both the exact moment equations and the non-diagonal incidence equations become transparent. For $u \in \mathbb{F}_2^4$, put

$$x(u) := \big(u, t(\mathrm{wt}(u))\big), \qquad t(0) = t(4) = 00, \quad t(1) = 01, \quad t(2) = 10, \quad t(3) = 11. \tag{864}$$

Then

$$S_0 = \{x(u) : u \in \mathbb{F}_2^4\}, \qquad S_1 = \{\overline{x(u)} : u \in \mathbb{F}_2^4\}. \tag{865}$$

Label the sixteen points of S_0 by one special symbol $\star$, five vertices v_i, and the ten edges $e_{ij} = e_{ji}$ of a complete graph on $V = \{0, 1, 2, 3, 4\}$, as follows:

label	u	label	u
$\star$	1111	v_0	0000
v_1	0001	v_2	0010
v_3	0100	v_4	1000
e_{01}	1110	e_{02}	1101
e_{03}	1011	e_{04}	0111
e_{12}	1100	e_{13}	1010
e_{14}	0110	e_{23}	1001
e_{24}	0101	e_{34}	0011

(866)

The physical string attached to each entry is obtained from Eq. (864).

22.6.1 Exact moment normal form

Let p_x be the bottom-fiber probability at $x \in S_0$, and index the top-fiber probabilities by $q_x := p_1(\overline{x})$. Substitution of Eq. (866) into the two normalization equations and the twenty-one degree-at-most-two moment equations gives a rational matrix of rank twenty-three. Exact row reduction gives the equivalent normal form

$$q_x = p_x \quad (x \in S_0), \qquad \sum_{x \in S_0} p_x = 1, \qquad \sum_{x \in S_0} p_x x_r = \frac{1}{2} \quad (r = 1, \ldots, 6). \tag{867}$$

Thus the moment polytope has dimension $16 - 7 = 9$, in agreement with Table 4; the second moments are automatic once Eq. (867) holds.

Write

$$p_\star := p(\star), \qquad r_i := p(v_i), \qquad w_{ij} := p(e_{ij}), \qquad D_i := \sum_{j \neq i} w_{ij}. \tag{868}$$

The balanced physical marginals have a particularly symmetric translation into these variables. On the sixteen points of S_0, the following are pointwise identities:

$$\mathbf{1}_\star = \frac{1}{4}(x_1 + x_2 + x_3 + x_4) - \frac{1}{2}x_5 - \frac{1}{4}x_6, \tag{869}$$

$$\sum_{j\neq 0} \mathbf{1}_{e_{0j}} - \mathbf{1}_{v_0} = -1 + \frac{1}{4}\sum_{r=1}^{4} x_r + \frac{1}{2}x_5 + \frac{3}{4}x_6, \tag{870}$$

$$\sum_{j\neq i} \mathbf{1}_{e_{ij}} - \mathbf{1}_{v_i} = \frac{1}{4}\sum_{r=1}^{4} x_r - x_{5-i} + \frac{1}{2}x_5 - \frac{1}{4}x_6, \qquad i = 1,2,3,4. \tag{871}$$

Taking expectations and using Eq. (867) gives

$$p_\star = \frac{1}{8}, \qquad D_i - r_i = \frac{1}{8} \quad (i \in V). \tag{872}$$

If

$$R := \sum_i r_i, \qquad W := \sum_{i<j} w_{ij}, \tag{873}$$

then summing Eq. (872) and using normalization yields

$$R = \frac{3}{8}, \qquad W = \frac{1}{2}. \tag{874}$$

In particular, $\star$ is always active and every vertex has positive weighted edge degree $D_i = r_i + 1/8$.

22.6.2 The two canonical cross-KL systems

Denote the bottom amplitudes at $\star, v_i, e_{ij}$ by

$$s, \qquad b_i, \qquad c_{ij} = c_{ji}, \tag{875}$$

and denote the top amplitude at the complement of a bottom point α by $d_{\overline{\alpha}}$. By Eq. (867), whenever the bottom amplitude C_α is nonzero we may define

$$\rho_\alpha := \frac{d_{\overline{\alpha}}}{\overline{C_\alpha}}, \qquad |\rho_\alpha| = 1. \tag{876}$$

For a reciprocal incidence pair α, β, put

$$z_{\alpha\beta} := \overline{C_\alpha}\, d_{\overline{\beta}} = \rho_\beta\, \overline{C_\alpha C_\beta}. \tag{877}$$

The five nonempty weight-one cross-incidence systems are naturally indexed by $i \in V$. The corresponding physical qubits are

$$\mu(0) = 6, \qquad \mu(1) = 4, \qquad \mu(2) = 3, \qquad \mu(3) = 2, \qquad \mu(4) = 1. \tag{878}$$

For the flip $X_{\mu(i)}$, the four reciprocal pairs are

$$v_j \longleftrightarrow e_{ij}, \qquad j \in V \setminus \{i\}. \tag{879}$$

The allowed weight-at-most-two Z-dressings have rank five and row-reduce to

$$z_{v_j,e_{ij}} - z_{e_{ij},v_j} = 0, \qquad j \neq i, \tag{880}$$

$$\sum_{j\neq i} \left(z_{v_j,e_{ij}} + z_{e_{ij},v_j}\right) = 0. \tag{881}$$

The remaining physical weight-one flip, X_5, has no cross incidence.

For the weight-two flip $X_{\mu(i)}X_5$, the four reciprocal pairs are

$$\star \longleftrightarrow v_i \quad \text{and} \quad e_{jk} \longleftrightarrow e_{\ell m} \quad \big(\{j,k,\ell,m\} = V \setminus \{i\}\big), \tag{882}$$

where the last three pairs run over the three perfect matchings of $V \setminus \{i\}$. Its rank-two system is

$$z_{\star,v_i} + z_{v_i,\star} = 0, \tag{883}$$

$$\sum_{\{jk|\ell m\}\in\mathcal{M}_i} \left(z_{e_{jk},e_{\ell m}} + z_{e_{\ell m},e_{jk}}\right) = 0, \tag{884}$$

where $\mathcal{M}_i$ is the set of the three perfect matchings of $V \setminus \{i\}$. Equations (880)–(884) are obtained by an elementary Walsh row reduction of the explicitly listed fiber; no support assumption has been made.

22.6.3 Support reduction

Let

$$A := \{i \in V : r_i > 0\} \tag{885}$$

be the active point set, and let $G \subseteq K_5$ be the graph whose edge ij is present exactly when $w_{ij} > 0$, equivalently $c_{ij} \neq 0$.

Lemma 22.9 (At least three point amplitudes are active)**.** *Every putative solution has* $|A| \geq 3$.

Proof. Suppose first that $A = \{a\}$. Since $D_a = r_a + 1/8 > 0$, some edge ak is active. The weight-one sum Eq. (881) centered at k then has exactly one nonzero term, which is impossible.

Now suppose $A = \{a, b\}$. The edge ab cannot be active, because the weight-one sum centered at a or b would again have exactly one nonzero term. For every inactive k, the two edges ak, bk are therefore either both active or both inactive. When both are active, Eqs. (880) and (883) give a common phase ratio on the two terms in the sum centered at k. Hence that two-term sum implies

$$r_a w_{ak} = r_b w_{bk}. \tag{886}$$

The same equality is trivial when both edges vanish. Summing over the three inactive vertices gives

$$r_a\left(r_a + \frac{1}{8}\right) = r_b\left(r_b + \frac{1}{8}\right). \tag{887}$$

The function $t \mapsto t(t + 1/8)$ is strictly increasing for $t \geq 0$, so $r_a = r_b$. Equation (874) gives $r_a = r_b = 3/16$, and hence $D_a = D_b = 5/16$. Because ab is absent, the two weighted stars at a and b use disjoint edge sets, so

$$W \geq D_a + D_b = \frac{5}{8}, \tag{888}$$

contradicting $W = 1/2$. □

There is one possible obstruction to immediate global synchronization of the phase ratios: when $|A| = 3$, the edge joining the two inactive vertices might be active. This branch is itself impossible.

Lemma 22.10 (The inactive–inactive edge branch is impossible)**.** *If $|A| = 3$, the edge joining the two vertices in $V \setminus A$ cannot be active.*

Proof. Let r, s be the two inactive vertices and suppose $w_{rs} > 0$. Write W_{AA} for the total weight of edges internal to A, and W_{AI} for the total weight between A and $\{r, s\}$. Summing Eq. (872) over the three active vertices and comparing with Eq. (874) gives

$$2W_{AA} + W_{AI} = \frac{3}{4}, \tag{889}$$

$$W_{AA} + W_{AI} + w_{rs} = \frac{1}{2}, \tag{890}$$

so

$$W_{AA} - w_{rs} = \frac{1}{4}. \tag{891}$$

Thus the graph induced on A is nonempty. At an active vertex, the number of nonzero terms in Eq. (881) is precisely its degree in this three-vertex induced graph, and that degree cannot be one. Consequently the induced graph is the full triangle.

Relabel its vertices as $0, 1, 2$. Equations (880) and (883) synchronize all three point ratios and all three triangle-edge ratios. The three weight-one sums therefore reduce, after removing a common nonzero factor and complex conjugating, to

$$b_1 c_{01} + b_2 c_{02} = 0, \tag{892}$$

$$b_0 c_{01} + b_2 c_{12} = 0, \tag{893}$$

$$b_0 c_{02} + b_1 c_{12} = 0. \tag{894}$$

All six factors are nonzero. The first two equations give $c_{02}/c_{12} = b_1/b_0$, while the third gives $c_{02}/c_{12} = -b_1/b_0$, a contradiction. □

We may therefore assume the *regular branch*:

$$ij \in E(G) \quad \Longrightarrow \quad i \in A \text{ or } j \in A. \tag{895}$$

Every active edge is incident to an active point. Equations (880) and (883) then imply the global synchronization

$$\rho_{v_i} = \rho_{e_{ij}} = -\rho_\star \tag{896}$$

for every active point and every active edge.

Lemma 22.11 (Minimum support degree)**.** *On the regular branch, the support graph G has minimum degree at least two.*

Proof. Equation (872) rules out degree zero. Suppose vertex i has the unique neighbor j. If $j \in A$, then the weight-one sum centered at i has exactly one nonzero term. If $j \notin A$, regularity forces $i \in A$, and

$$w_{ij} = D_i = r_i + \frac{1}{8} > \frac{1}{8}. \tag{897}$$

But $r_j = 0$, so $D_j = 1/8$, while $D_j \geq w_{ij} > 1/8$, again a contradiction. □

Using Eq. (896) in the matching equation (884) gives, for every $i \in V$,

$$H_i := \sum_{\{jk|\ell m\} \in \mathcal{M}_i} c_{jk} c_{\ell m} = 0. \tag{898}$$

Thus H_i is the all-plus sum over the three perfect matchings of the four vertices in $V \setminus \{i\}$.

22.6.4 Every proper support graph is impossible

Assume first that $G \neq K_5$. By Lemma 22.11, the complement $\overline{G}$ has maximum degree at most two. Up to graph isomorphism, a nonempty graph on five vertices of maximum degree at most two has exactly the ten types

$$K_2,\ 2K_2,\ P_3,\ K_3,\ P_3 \sqcup K_2,\ P_4,\ K_3 \sqcup K_2,\ P_5,\ C_4,\ C_5. \tag{899}$$

Six types are excluded because one of the equations $H_i = 0$ contains exactly one nonzero monomial:

$\overline{G}$	$E(\overline{G})$	one-term equation
C_5	$03, 04, 12, 14, 23$	$H_0 = c_{13}c_{24}$
C_4	$13, 14, 23, 24$	$H_0 = c_{12}c_{34}$
P_5	$04, 13, 23, 24$	$H_0 = c_{12}c_{34}$
P_4	$14, 23, 34$	$H_0 = c_{13}c_{24}$
$P_3 \sqcup K_2$	$04, 14, 23$	$H_2 = c_{01}c_{34}$
P_3	$24, 34$	$H_0 = c_{14}c_{23}$

(900)

Here, for example, "03" denotes the missing edge $\{0, 3\}$. Every factor displayed in the last column is nonzero by definition of the support graph, contradicting Eq. (898).

The remaining four complement types have an odd binomial cycle. The table below lists three equations in the form

$$A_r + B_r = 0, \qquad r = 1, 2, 3. \tag{901}$$

In every row, direct multiplication gives $A_1A_2A_3 = B_1B_2B_3 \neq 0$, whereas Eq. (901) gives $A_1A_2A_3 = -B_1B_2B_3$, a contradiction.

$\overline{G}$	$E(\overline{G})$	(A_1, A_2, A_3)	(B_1, B_2, B_3)
$K_3 \sqcup K_2$	$04, 12, 13, 23$	$(c_{02}c_{34},\ c_{03}c_{14},\ c_{01}c_{24})$	$(c_{03}c_{24},\ c_{01}c_{34},\ c_{02}c_{14})$
K_3	$23, 24, 34$	$(c_{03}c_{14},\ c_{04}c_{12},\ c_{02}c_{13})$	$(c_{04}c_{13},\ c_{02}c_{14},\ c_{03}c_{12})$
$2K_2$	$14, 23$	$(c_{12}c_{34},\ c_{03}c_{24},\ c_{02}c_{13})$	$(c_{13}c_{24},\ c_{02}c_{34},\ c_{03}c_{12})$
K_2	34	$(c_{13}c_{24},\ c_{04}c_{23},\ c_{03}c_{14})$	$(c_{14}c_{23},\ c_{03}c_{24},\ c_{04}c_{13})$

(902)

For the four rows, the three equations come respectively from (H_1, H_2, H_3), (H_2, H_3, H_4), (H_0, H_1, H_4), and (H_0, H_1, H_2), with the order of the two monomials reversed where needed. This exhausts Eq. (899). Hence no proper support graph survives.

22.6.5 The full support graph is also impossible

It remains to consider $G = K_5$, so all ten edge amplitudes are nonzero. Let

$$C = (c_{ij})_{i,j \in V}, \qquad c_{ii} = 0, \tag{903}$$

be the symmetric zero-diagonal edge-amplitude matrix, and let $b = (b_0, \ldots, b_4)^T$, with inactive point amplitudes entered as zero. The synchronized weight-one sums Eq. (881) give

$$Cb = 0. \tag{904}$$

But Eq. (874) gives $b \neq 0$. We now show that the five matching equations force C to be invertible.

The vertex scaling

$$c_{ij} \longmapsto s_i s_j c_{ij}, \qquad s_i \in \mathbb{C}^\times, \tag{905}$$

preserves the equations $H_i = 0$ and preserves singularity of C. Because every edge is nonzero, choose the scaling so that

$$c_{01} = c_{02} = c_{12} = c_{34} = 1. \tag{906}$$

Write

$$c_{03} = c, \quad c_{04} = d, \quad c_{13} = f, \quad c_{14} = g, \quad c_{23} = h, \quad c_{24} = k. \tag{907}$$

The five equations $H_i = 0$ become

$$1 + fk + gh = 0, \tag{908}$$
$$1 + ck + dh = 0, \tag{909}$$
$$1 + cg + df = 0, \tag{910}$$
$$k + g + d = 0, \tag{911}$$
$$h + f + c = 0. \tag{912}$$

Equations (911)–(912) give $k = -(g + d)$ and $h = -(f + c)$. Substituting these into Eqs. (908)–(910) gives

$$cd = 1, \qquad fg = 1. \tag{913}$$

With $t := cg$, Eq. (910) is therefore

$$1 + t + t^{-1} = 0, \qquad \text{or equivalently} \qquad t^2 + t + 1 = 0. \tag{914}$$

A direct determinant expansion, using Eqs. (911)–(913), gives

$$\det C = \frac{2(t^2 - 3t + 1)(3t^2 + 7t + 3)}{t^2}. \tag{915}$$

Modulo Eq. (914), the two factors in the numerator are $-4t$ and $4t$, respectively. Hence

$$\det C = -32 \neq 0. \tag{916}$$

This contradicts Eq. (904) with $b \neq 0$.

Theorem 22.12 (Orbit 30 no-go)**.** *No exact* $((6, 2, d \geq 3))$ *qubit code can arise from the projective order-four moment orbit represented by*

$$(q, m, a) = \big(4, 4, (1, 1, 1, 1, 2, 3)\big). \tag{917}$$

Proof. Lemma 22.9 leaves at least three active point amplitudes. Lemma 22.10 removes the only branch in which an active edge is disconnected from all active points. On the regular branch, Lemma 22.11 and Eqs. (898)–(902) exclude every proper support graph. For the full graph, Eqs. (904)–(916) give the final contradiction. □

Remark 22.13 (Coverage of support cases)**.** The support-graph argument covers one exceptional three-point branch, ten proper-graph complement types, and the full K_5 branch. Every branch is excluded by the equations above.

22.7 Orbit 31: a complete exact support audit and no-go theorem

For

$$(q, m, a) = \big(4, 4, (1, 1, 1, 2, 2, 2)\big), \tag{918}$$

the two residue fibers are complementary. We classify all support faces by finite exact computation using rational arithmetic and symbolic row reduction. Every one of the sixteen resulting symmetry types is then excluded by a displayed analytic certificate.

22.7.1 A four-by-four model of the fiber pair

Put

$$\mathcal{E} = \{e_0, e_1, e_2, e_3\} := \{000, 011, 101, 110\}, \qquad \overline{e} := e \oplus 111, \tag{919}$$

and

$$u_1 = 011, \qquad u_2 = 101, \qquad u_3 = 110. \tag{920}$$

Index the sixteen points of the residue-zero fiber by the four-by-four table

$$A_e := (000, e), \qquad B_{i,e} := (u_i, \overline{e}), \qquad e \in \mathcal{E}, \quad 1 \le i \le 3. \tag{921}$$

Here the comma separates the first and last triples of physical qubits. A direct residue check gives

$$S_0 = \{A_e : e \in \mathcal{E}\} \cup \{B_{i,e} : 1 \le i \le 3,\ e \in \mathcal{E}\}, \qquad S_1 = \overline{S_0}. \tag{922}$$

Thus a bottom support can be written as four binary words

$$A \mid B_1 \mid B_2 \mid B_3, \tag{923}$$

where the four positions in each word are ordered as e_0, e_1, e_2, e_3.

Let

$$\alpha_e = p_0(A_e), \qquad \beta_{i,e} = p_0(B_{i,e}). \tag{924}$$

The complete degree-at-most-two moment system has the following particularly simple normal form.

Proposition 22.14 (Complete orbit-31 moment normal form)**.** *The two normalization equations and the twenty-one diagonal moment equations have rank twenty-three. They are equivalent to*

$$p_1(\overline{A_e}) = \alpha_e, \qquad p_1(\overline{B_{i,e}}) = \beta_{i,e}, \tag{925}$$

$$\sum_{e \in \mathcal{E}} \beta_{i,e} = \frac{1}{4}, \qquad 1 \le i \le 3, \tag{926}$$

$$\alpha_e = \sum_{i=1}^{3} \beta_{i,e} - \frac{1}{8}, \qquad e \in \mathcal{E}, \tag{927}$$

together with nonnegativity. In particular, the moment polytope has dimension nine.

Proof. Order both fibers according to Eq. (921), put the sixteen bottom columns first, and form the usual 23×32 rational moment matrix. The minor consisting of all sixteen bottom columns and the seven top columns

$$\overline{A_{e_0}},\ \overline{A_{e_1}},\ \overline{A_{e_2}},\ \overline{A_{e_3}},\ \overline{B_{1,e_0}},\ \overline{B_{2,e_0}},\ \overline{B_{3,e_0}} \tag{928}$$

has determinant of absolute value 128. Hence the rank is twenty-three. Exact row reduction gives Eqs. (925)–(927); direct substitution checks all twenty-three equations. The twelve variables $\beta_{i,e}$ subject to the three independent row sums in Eq. (926) leave nine free parameters. □

A useful interpretation is that each B_i-row carries total probability $1/4$, while the excess of the e-th B-column above $1/8$ is exactly α_e. Equation (925) also shows that the top support is exactly the complement of the bottom support, including on every boundary face.

22.7.2 Canonical cross-KL equations

Write the bottom amplitudes at A_e and $B_{i,e}$ as

$$a_e, \qquad b_{i,e}, \tag{929}$$

respectively. On every active table position v, Eq. (925) permits the parametrization

$$d_{\overline{v}} = \rho_v \, \overline{c_v}, \qquad |\rho_v| = 1, \qquad \sigma_v := \overline{\rho_v}. \tag{930}$$

Here c_v denotes the corresponding bottom amplitude. Conjugating the cross-KL equations makes them bilinear in the bottom amplitudes.

For a weight-one flip on the i-th qubit of the first triple, the four reciprocal incidences are

$$A_e \longleftrightarrow B_{i,e}, \qquad e \in \mathcal{E}. \tag{931}$$

The allowed Z-dressings give an eight-column Walsh system of rank five. Its row space is equivalently

$$\sigma_{A_e} = \sigma_{B_{i,e}} \quad \text{on every active reciprocal pair,} \tag{932}$$

and

$$\sum_{e:\, A_e, B_{i,e} \text{ active}} \sigma_{B_{i,e}} a_e b_{i,e} = 0. \tag{933}$$

In particular, exactly one active reciprocal pair is impossible.

The nonempty weight-two cross-flip systems reduce to eighteen rectangle equations, one for every pair of distinct B-rows and every pair of distinct columns:

$$\boxed{(\sigma_{B_{i,e}} + \sigma_{B_{j,f}}) b_{i,e} b_{j,f} + (\sigma_{B_{i,f}} + \sigma_{B_{j,e}}) b_{i,f} b_{j,e} = 0,} \tag{934}$$

where $1 \le i < j \le 3$ and $0 \le e < f \le 3$; a term is omitted when one of its endpoint amplitudes vanishes. These eighteen equations arise from nine physical flip masks, each of whose Pauli row space has rank two.

If exactly one of the two rectangle diagonals in Eq. (934) is active, its two endpoint phase ratios must be opposite. We therefore form a signed graph on the active table positions: an active vertical incidence in Eq. (932) is a positive edge, and a forced opposition from a one-diagonal rectangle is a negative edge. An odd signed cycle is an immediate contradiction.

22.7.3 Exact support enumeration and the sixteen residual types

The moment normal form makes the support audit finite and transparent. Let A_{mom} be the 7×16 matrix whose first row is all ones and whose remaining six rows are the coordinate functions on S_0. Exact enumeration of its nonsingular seven-column bases gives 204 distinct rational vertices of the moment polytope: 108 have support six and 96 have support seven.

For a mask $M \subseteq S_0$, the face obtained by setting all variables outside M to zero is the convex hull of the full-polytope vertices whose supports lie in M. Consequently, M occurs as the exact support of a relative-interior point if and only if the union of those vertex supports is exactly M. Applying this criterion to all 2^{16} masks gives the exact chain

$$15{,}603 \xrightarrow{\text{no singleton in Eq. (933)}} 2{,}903 \xrightarrow{\text{no odd signed cycle}} 311. \tag{935}$$

The enumeration includes every moment face, including all high-support faces.

The physical symmetry group used in the quotient is

$$S_3 \times \mathrm{AGL}(2,2) \cong S_3 \times S_4. \tag{936}$$

The first factor permutes the first three equal-angle qubits. On the last triple, coordinate permutations together with even-weight local X translations induce every permutation of the four points of $\mathcal{E}$. These operations preserve the fiber pair and the complete weight-at-most-two KL system. The 311 signed-cycle-compatible masks split into exactly the sixteen support orbits in Table 6.

Type	$A \mid B_1 \mid B_2 \mid B_3$	Orbit size	Support size	Certificate
1	0011|1100|0011|0011	18	8	mixed odd triangle
2	1111|1111|1100|0011	18	12	final class
3	0011|1111|1111|0011	18	12	direct P^{23}-triangle
4	0110|1111|0111|0111	36	12	direct P^{12}-triangle
5	0111|1111|0111|0111	12	13	direct P^{12}-triangle
6	1011|1111|1111|0011	36	13	direct P^{23}-triangle
7	1110|1111|0111|0111	36	13	direct P^{12}-triangle
8	0110|1111|1111|0111	36	13	direct P^{12}-triangle
9	1100|1111|1111|1111	6	14	propagated triangle
10	1111|1111|0111|0111	12	14	direct P^{12}-triangle
11	0111|1111|1111|0111	12	14	direct P^{12}-triangle
12	1111|1111|1111|0011	18	14	direct P^{23}-triangle
13	1110|1111|1111|0111	36	14	direct P^{12}-triangle
14	1110|1111|1111|1111	4	15	column-phase triangle
15	1111|1111|1111|0111	12	15	direct P^{12}-triangle
16	1111|1111|1111|1111	1	16	column-phase triangle

Table 6: The complete $S_3 \times S_4$ quotient of the 311 support masks surviving the signed-cycle test. The orbit sizes sum to 311.

22.7.4 Permanent-triangle certificates

For two B-rows and two columns, define the two-by-two permanent

$$P_{ij}^{ef} := b_{i,e}b_{j,f} + b_{i,f}b_{j,e}. \tag{937}$$

We use the following elementary obstruction repeatedly.

Lemma 22.15 (Odd permanent triangle)**.** *Let $(r_i, s_i) \in (\mathbb{C}^\times)^2$, $1 \le i \le 3$. The three equations*

$$r_i s_j + s_i r_j = 0, \qquad 1 \le i < j \le 3, \tag{938}$$

cannot hold simultaneously.

Proof. Put $z_i = r_i/s_i$. Equation (938) says $z_i = -z_j$ for every pair. The first two relations give $z_2 = z_3$, while the third gives $z_2 = -z_3$, contradicting $z_2 \neq 0$. □

For types

$$3, 6, 12, \tag{939}$$

substitution of the forced signed-phase pattern into Eq. (934) gives

$$P_{12}^{23} = P_{13}^{23} = P_{23}^{23} = 0, \tag{940}$$

up to a common nonzero factor $\pm 2\tau$, $\tau \in U(1)$. For types

$$4, 5, 7, 8, 10, 11, 13, 15, \tag{941}$$

the same calculation gives

$$P^{12}_{12} = P^{12}_{13} = P^{12}_{23} = 0 \tag{942}$$

with the same kind of nonzero phase factor. Lemma 22.15 excludes all eleven types.

Type 1 has a mixed version of the same obstruction. Its active amplitudes obey

$$a_2 b_{2,2} + a_3 b_{2,3} = 0, \tag{943}$$
$$a_2 b_{3,2} + a_3 b_{3,3} = 0, \tag{944}$$
$$b_{2,2} b_{3,3} + b_{2,3} b_{3,2} = 0. \tag{945}$$

The first two equations make the two row ratios equal, whereas the third makes them opposite. Since every displayed amplitude is nonzero, type 1 is impossible.

It remains in this subsection to remove types 9, 14, and 16. In type 9 the A-row is active only in columns $0, 1$, while all twelve B-entries are active. Let τ_0, τ_1 be the synchronized phase ratios in those two columns. For each pair of B-rows, Eq. (934) on columns $0, 1$ is

$$(\tau_0 + \tau_1) P^{01}_{ij} = 0. \tag{946}$$

Lemma 22.15 shows that the three permanents cannot all vanish, so

$$\tau_1 = -\tau_0. \tag{947}$$

The three weight-one equations then give a common nonzero ratio r such that

$$b_{i,0} = r b_{i,1}, \qquad 1 \le i \le 3. \tag{948}$$

For any row pair $i < j$, divide the column-pair $(0, 2)$ rectangle equation by r and subtract the $(1, 2)$ rectangle equation. All unknown phase ratios in column 2 cancel, leaving

$$(\tau_0 - \tau_1) P^{12}_{ij} = 0. \tag{949}$$

By Eq. (947), the prefactor is $2\tau_0 \neq 0$. Thus all three P^{12}_{ij} vanish, contradicting the permanent-triangle lemma.

In types 14 and 16, all twelve B-entries are active and at least three A-columns are active. Let τ_e be the common phase ratio in an active column e, as forced by Eq. (932). For every active column pair $e < f$, the three row-pair equations become

$$(\tau_e + \tau_f) P^{ef}_{ij} = 0, \qquad 1 \le i < j \le 3. \tag{950}$$

The permanent-triangle lemma forces $\tau_f = -\tau_e$. Applying this to the three pairs among any three active columns requires three nonzero phases to be pairwise opposite, which is impossible. Hence types 14 and 16 are also excluded.

We have now excluded every support type except type 2.

22.7.5 The final twelve-string class

For type 2, write the amplitudes in the canonical table

	e_0	e_1	e_2	e_3
A	a_0	a_1	a_2	a_3
B_1	x_0	x_1	x_2	x_3
B_2	y_0	y_1	0	0
B_3	0	0	z_2	z_3

(951)

with all twelve displayed amplitudes nonzero. This is equivalent under a first-triple permutation to the support

$$\{000000, 101100, 011100, 110010, 101010, 000110, \\ 110001, 101001, 000101, 000011, 101111, 011111\}. \tag{952}$$

Its support orbit has size eighteen.

The signed graph is connected. After a global phase gauge, its signs are constant on columns $0, 1$ and opposite on columns $2, 3$. The cross-KL system contains the five equations

$$a_0y_0 + a_1y_1 = 0, \tag{953}$$
$$x_0y_1 + x_1y_0 = 0, \tag{954}$$
$$a_2z_2 + a_3z_3 = 0, \tag{955}$$
$$x_2z_3 + x_3z_2 = 0, \tag{956}$$
$$a_0x_0 + a_1x_1 - a_2x_2 - a_3x_3 = 0. \tag{957}$$

Equations (953)–(954) imply

$$a_0x_0 = a_1x_1, \tag{958}$$

and Eqs. (955)–(956) imply

$$a_2x_2 = a_3x_3. \tag{959}$$

Equation (957) therefore gives one common nonzero number

$$K := a_0x_0 = a_1x_1 = a_2x_2 = a_3x_3 \neq 0. \tag{960}$$

Now use the same-state KL equations for the flip X_4X_5 and its three Z_4, Z_5 dressings. Exact row reduction, after substituting the phase signs fixed above, isolates

$$\overline{a_0}a_3 + \overline{x_3}x_0 = 0. \tag{961}$$

By Eq. (960),

$$x_0 = \frac{K}{a_0}, \qquad x_3 = \frac{K}{a_3}. \tag{962}$$

Multiplying Eq. (961) by $a_0\overline{a_3}$ gives the division-free identity

$$|a_0a_3|^2 + |K|^2 = 0, \tag{963}$$

which is impossible because $a_0, a_3, K \neq 0$. Thus the final support type is excluded.

Theorem 22.16 (Orbit 31 no-go)**.** *No exact* $((6, 2, d \geq 3))$ *qubit code can arise from the projective order-four moment orbit represented by*

$$(q, m, a) = \big(4, 4, (1, 1, 1, 2, 2, 2)\big). \tag{964}$$

Proof. Proposition 22.14 reduces every solution, including all boundary faces, to a support of the sixteen-cell table. The exact support enumeration in Eq. (935) starts from all 2^{16} masks, leaves 311 after the necessary cross-KL signed-cycle tests, and quotients them into the sixteen physical-symmetry types in Table 6. The permanent-triangle arguments exclude types 1 and 3–16. Equations (953)–(963) exclude type 2. Hence no support face admits the complete weight-at-most-two Knill–Laflamme system. □

Remark 22.17 (Scope of the finite computation)**.** The computer-assisted step in Theorem 22.16 consists of an exact support classification: construction of the fibers and rank-23 moment normal form, enumeration of the 204 rational vertices and 15,603 exact support masks, the reductions to 2903 and then 311 masks, and the quotient into sixteen $S_3 \times S_4$ support orbits. The Pauli row spaces and the displayed certificate for every representative are verified by exact rational arithmetic and symbolic row reduction.

22.8 Orbit 32: a local moment-reflection and Walsh-triangle obstruction

For

$$(q, m, a) = \big(4, 4, (1, 1, 2, 2, 2, 3)\big), \tag{965}$$

the no-go follows from a moment-reflection argument. Write

$$u = (x_3, x_4, x_5), \qquad o = (x_1, x_2, x_6), \tag{966}$$

and put

$$\mathcal{E} = \{000, 011, 101, 110\}, \qquad \mathcal{O} = \{001, 010, 100, 111\}. \tag{967}$$

Thus $\mathcal{E}$ and $\mathcal{O}$ are the even- and odd-parity three-bit strings. Introduce the outer labels

	A	B	C	D
o	000	011	101	110

	E	F	G	H
o	010	100	111	001.

(968)

Because

$$x_1 + x_2 + 2(x_3 + x_4 + x_5) + 3x_6 \equiv x_1 + x_2 - x_6 + 2\,\mathrm{wt}(u) \pmod 4, \tag{969}$$

the two fibers are

$$S_0 = \{(A, u), (B, u), (C, u) : u \in \mathcal{E}\} \cup \{(D, v) : v \in \mathcal{O}\}, \tag{970}$$

$$S_1 = \{(E, u), (F, u), (G, u) : u \in \mathcal{E}\} \cup \{(H, v) : v \in \mathcal{O}\}. \tag{971}$$

Here (o, u) means the physical string $(o_1, o_2, u_1, u_2, u_3, o_3)$. Both fibers have sixteen points.

For each $u \in \mathcal{E}$, denote the bottom amplitudes at $(A, u), (B, u), (C, u)$ by a_u, b_u, c_u, and the top amplitudes at $(E, u), (F, u), (G, u)$ by e_u, f_u, g_u. Write

$$A_u = |a_u|^2, \quad B_u = |b_u|^2, \quad C_u = |c_u|^2, \qquad E_u = |e_u|^2, \quad F_u = |f_u|^2, \quad G_u = |g_u|^2. \tag{972}$$

For $v \in \mathcal{O}$, let R_v and S_v be the bottom and top probabilities at (D, v) and (H, v), respectively.

22.8.1 Complete exact moment normal form

Substitution of the two displayed fibers into the two normalization equations and the twenty-one degree-at-most-two diagonal moment equations give a 23×32 rational system of rank twenty-three. One explicit rank certificate is as follows. Order each fiber lexicographically, place the sixteen S_0 columns first, and then the sixteen S_1 columns. The 23×23 minor consisting of all sixteen S_0 columns and the seven S_1 columns

$$000011,\ 000101,\ 001001,\ 001111,\ 010000,\ 100000,\ 110001 \tag{973}$$

has determinant 3456.

Exact row reduction is equivalent to

$$\boxed{\begin{pmatrix} A_u \\ B_u \\ C_u \end{pmatrix} = T \begin{pmatrix} E_u \\ F_u \\ G_u \end{pmatrix}, \qquad T := \frac{1}{3}\begin{pmatrix} 2 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{pmatrix}, \qquad u \in \mathcal{E},} \tag{974}$$

$$\boxed{\sum_{u\in\mathcal{E}} E_u = \sum_{u\in\mathcal{E}} F_u = \sum_{u\in\mathcal{E}} G_u = \frac{1}{4},} \tag{975}$$

and

$$\boxed{R_{\bar{u}} = S_{\bar{u}} = \frac{1}{8} - \frac{E_u + F_u + G_u}{3}, \qquad u \in \mathcal{E},} \tag{976}$$

where $\bar{u} = u \oplus 111$. Conversely, direct substitution shows that Eqs. (974)–(976) satisfy every diagonal moment equation. They contain twelve displayed variables subject to the three independent row sums in Eq. (975), and the variables E_u, F_u, G_u occur explicitly in the probability vector. The parametrization is therefore injective and nine-dimensional. Since the exact moment system has nullity $32 - 23 = 9$, this is its complete affine normal form. Notice also that $T^2 = I_3$.

22.8.2 Walsh inversion of three weight-one flip systems

Order the four elements of $\mathcal{E}$ as $000, 011, 101, 110$. The restrictions to $\mathcal{E}$ of the four characters corresponding to no Z-dressing and to Z_3, Z_4, Z_5 form the Walsh matrix

$$W = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad \det W = -16. \tag{977}$$

For the flip X_1, the only cross-logical incidences at a fixed $u \in \mathcal{E}$ are

$$(A, u) \longleftrightarrow (F, u), \qquad (B, u) \longleftrightarrow (G, u). \tag{978}$$

There are no additional incidences at odd middle parity. The four cross-KL equations obtained by dressing X_1 with I, Z_3, Z_4, Z_5, followed by inversion of Eq. (977), therefore give

$$\overline{a_u} f_u + \overline{b_u} g_u = 0 \qquad (u \in \mathcal{E}). \tag{979}$$

The corresponding incidence pairs for X_2 are $(A, u) \leftrightarrow (E, u)$ and $(C, u) \leftrightarrow (G, u)$, while those for X_6 are $(B, u) \leftrightarrow (E, u)$ and $(C, u) \leftrightarrow (F, u)$. The same four-point Walsh inversion gives

$$\overline{a_u} e_u + \overline{c_u} g_u = 0, \tag{980}$$

$$\overline{b_u} e_u + \overline{c_u} f_u = 0, \tag{981}$$

for every $u \in \mathcal{E}$. All operators used here have Pauli weight at most two.

Taking absolute values in Eqs. (979)–(981) gives

$$A_u F_u = B_u G_u, \qquad A_u E_u = C_u G_u, \qquad B_u E_u = C_u F_u. \tag{982}$$

Fix u and set $\Sigma_u = E_u + F_u + G_u$. Substitution of the moment reflection (974) into (982) factors the three equations exactly as

$$A_u F_u - B_u G_u = \frac{2}{3}(F_u - G_u)\Sigma_u = 0, \tag{983}$$

$$A_u E_u - C_u G_u = \frac{2}{3}(E_u - G_u)\Sigma_u = 0, \tag{984}$$

$$B_u E_u - C_u F_u = \frac{2}{3}(E_u - F_u)\Sigma_u = 0. \tag{985}$$

By Eq. (975), there is at least one $u \in \mathcal{E}$ with $E_u > 0$. For that u, one has $\Sigma_u > 0$, so Eqs. (983)–(985) force

$$E_u = F_u = G_u =: t_u > 0. \tag{986}$$

Equation (974) then gives $A_u = B_u = C_u = t_u$. Hence all six amplitudes in Eqs. (979)–(981) are nonzero. Nevertheless, the following division-free linear combination of those three equations gives

$$\begin{aligned} \overline{c_u}\big(\overline{a_u} f_u + \overline{b_u} g_u\big) - \overline{b_u}\big(\overline{a_u} e_u + \overline{c_u} g_u\big) + \overline{a_u}\big(\overline{b_u} e_u + \overline{c_u} f_u\big) \\ = 2\,\overline{a_u}\,\overline{c_u}\, f_u. \end{aligned} \tag{987}$$

The left-hand side vanishes by the three cross-KL equations, whereas the right-hand side is nonzero. This contradiction proves the no-go theorem.

Theorem 22.18 (Orbit 32 no-go)**.** *No exact* $((6, 2, d \geq 3))$ *qubit code can arise from the projective order-four moment orbit represented by*

$$(q, m, a) = \big(4, 4, (1, 1, 2, 2, 2, 3)\big). \tag{988}$$

Remark 22.19 (Moment-reflection obstruction)**.** The exact moment reflection and three Walsh-inverted weight-one flip systems force any locally positive triple into a six-amplitude odd signed triangle, while the row sums guarantee that such a positive triple exists. Thus the contradiction follows from the moment equations and these weight-one flip systems.

22.9 Orbit 62: an odd triangle of reciprocal flip relations

Here

$$(q, m, a) = \big(8, 4, (1, 1, 1, 5, 5, 5)\big). \tag{989}$$

The residue-zero support is

$$\{000000, 001111, 010111, 100111, 111001, 111010, 111100\}, \tag{990}$$

and the residue-two support is its bitwise complement. The moment solution is unique. With $x_0 = 000000$, its probabilities are

$$p_0(x_0) = p_1(\bar{x}_0) = \frac{1}{4}, \qquad p_0(x) = p_1(\bar{x}) = \frac{1}{8} \quad \big(x \in S_0 \setminus \{x_0\}\big). \tag{991}$$

Hence every amplitude on the two displayed supports is nonzero.

For complementary indexing define

$$\rho_x := \frac{d_{\bar{x}}}{\overline{c_x}}. \tag{992}$$

A weight-two flip with exactly two reciprocal incidences gives $\rho_y = -\rho_x$. Choose

$$x_0 = 000000, \qquad x_1 = 111100, \qquad x_2 = 010111. \tag{993}$$

The three masks

$$X_5X_6, \qquad X_2X_4, \qquad X_1X_3 \tag{994}$$

have, respectively, only the reciprocal index pairs (x_0, x_1), (x_1, x_2), and (x_2, x_0). Therefore

$$\rho_{x_1} = -\rho_{x_0}, \qquad \rho_{x_2} = -\rho_{x_1}, \qquad \rho_{x_0} = -\rho_{x_2}. \tag{995}$$

The first two imply $\rho_{x_2} = \rho_{x_0}$, contradicting the third.

Proposition 22.20 (Orbit 62 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code arises from orbit 62.*

22.10 Orbit 63: a fixed-probability two-edge obstruction

Finally, take

$$(q, m, a) = \big(8, 4, (1, 1, 2, 4, 5, 5)\big). \tag{996}$$

The moment family is one-dimensional, but four probabilities are fixed:

$$p_0(000000) = p_0(110111) = \frac{1}{4}, \qquad p_1(001000) = p_1(111111) = \frac{1}{4}. \tag{997}$$

The X_3-incidence set consists exactly of

$$001000 \longmapsto 000000, \qquad 111111 \longmapsto 110111. \tag{998}$$

The two sources have opposite first bits. Thus the equations for X_3 and X_3Z_1 give

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \overline{c_{000000}} d_{001000} \\ \overline{c_{110111}} d_{111111} \end{pmatrix} = 0. \tag{999}$$

Both products have magnitude $1/4$, contradicting invertibility of the Walsh matrix.

Proposition 22.21 (Orbit 63 is impossible)**.** *No exact* $((6, 2, d \geq 3))$ *code arises from orbit 63.*

22.11 Complete projective-order-four theorem

The ten exact arguments above exclude the moment orbits

$$25, 26, 27, 28, 29, 30, 31, 32, 62, 63. \tag{1000}$$

They are exhaustive because Eq. (758) leaves only $q = 4$ and $q = 8$, and the diagonal enumeration gives exactly the ten orbits in Table 4.

Theorem 22.22 (Six-qubit projective-order-four no-go)**.** *No exact* $((6, 2, d \geq 3))$ *qubit code admits a projective diagonal transversal logical element of order* 4*.*

Proof. Every controlled primitive lift of a projective order-four element has $q \in \{4, 8\}$. The exact diagonal enumeration gives precisely the ten moment orbits in Eq. (1000). The preceding subsections exclude each of them, including all boundary faces of their moment polytopes. Hence no order-four code exists. □

Corollary 22.23 (Complete six-qubit projective-order classification)**.** *For exact six-qubit codes encoding one qubit and having distance at least three, the possible nontrivial projective orders of diagonal transversal logical elements are exactly*

$$\boxed{\{2, 3, 5\}}. \tag{1001}$$

Including the identity, the order set is

$$\boxed{\{1, 2, 3, 5\}}, \tag{1002}$$

and $M_3(6) = 5$.

Proof. The controlled-lift theorem gives $m \le q \le 9$. Exact constructions realize $m = 2, 3, 5$. Theorem 22.22 excludes $m = 4$, while Theorems 20.5, 19.6, 18.3, and 17.2 exclude $m = 6, 7, 8, 9$. These exhaust all nontrivial orders permitted by the lift bound. □

At this stage of the argument, the high-order theorems eliminate twelve orbits and the order-four theorem eliminates ten more. Thus twenty-two of the sixty-six moment orbits have been excluded, leaving forty-four at the realized target orders $m = 2, 3, 5$. Section 21 will eliminate seven further order-five orbits, giving twenty-nine no-go orbits and thirty-seven nonexcluded orbits at that stage. Section 23 then excludes order-three orbits 5, 6, and 47–52, giving thirty-seven no-go orbits and twenty-nine nonexcluded orbits. Finally, Section 24 excludes fourteen order-two orbits, for the final totals of fifty-one no-go orbits and fifteen nonexcluded orbits.

23 Exact no-go theorems for order-three orbits 5, 6, and 47–52

The rank-compatible projective-order-three sector consists of orbits 5–9 with $(q, m) = (3, 3)$ and orbits 47–52 with $(q, m) = (6, 3)$ in Table 1. Orbits 7, 8, and 9 are realized; their complete weight-enumerator images are determined in Sections 7–9. This section proves that every other order-three moment orbit is impossible. The two minimal $q = 3$ cases are most naturally handled by grouping their zero-angle qubits into amplitude vectors. The six lifted $q = 6$ cases are handled by exact moment normal forms and finite Walsh systems. Every argument uses only Knill–Laflamme equations for physical errors of weight at most two and therefore proves a no-go for every exact $((6, 2, d \ge 3))$ code.

23.1 Orbit 5: a two-qubit-vector obstruction

Proposition 23.1 (Orbit 5 no-go)**.** *Orbit 5 admits no exact* $((6, 2, d \ge 3))$ *completion.*

Proof. Orbit 5 is represented by

$$(q, m, a) = \big(3, 3, (0, 0, 1, 1, 1, 1)\big). \tag{1003}$$

Let A denote the first two zero-angle qubits, and relabel the final four active coordinates by $V = \{1, 2, 3, 4\}$. Write

$$\Omega = 1111, \qquad \widehat{e}_i = \Omega \oplus e_i. \tag{1004}$$

The active-coordinate parts of the two fibers are

$$T_0 = \{0000, \widehat{e}_1, \widehat{e}_2, \widehat{e}_3, \widehat{e}_4\}, \qquad T_1 = \{e_1, e_2, e_3, e_4, \Omega\}. \tag{1005}$$

The two bits on A are arbitrary, so each active tail carries a vector in $\mathcal{H}_A \cong \mathbb{C}^4$. Every candidate pair can therefore be written as

$$|0_L\rangle = u \otimes |0000\rangle + \sum_{i=1}^{4} a_i \otimes |\widehat{e}_i\rangle, \tag{1006}$$

$$|1_L\rangle = \sum_{i=1}^{4} b_i \otimes |e_i\rangle + v \otimes |\Omega\rangle, \tag{1007}$$

with $u, v, a_i, b_i \in \mathbb{C}^4$.

Put

$$U = \|u\|^2, \qquad A_i = \|a_i\|^2, \qquad B_i = \|b_i\|^2, \qquad V_0 = \|v\|^2. \tag{1008}$$

For distinct active coordinates j, k, equality of the occupation moment $\langle n_j n_k \rangle$ gives

$$\sum_{i \notin \{j,k\}} A_i = V_0. \tag{1009}$$

The left side is a sum of two of the four A_i's. Varying j, k therefore forces

$$A_1 = A_2 = A_3 = A_4 =: \alpha, \qquad V_0 = 2\alpha. \tag{1010}$$

The one-coordinate moment at j is

$$\sum_{i \neq j} A_i = B_j + V_0, \tag{1011}$$

so $B_j = \alpha$. Normalization of $|1_L\rangle$ gives $4\alpha + 2\alpha = 1$. Consequently

$$\boxed{\|u\|^2 = \|v\|^2 = \frac{1}{3}, \qquad \|a_i\|^2 = \|b_i\|^2 = \frac{1}{6} \quad (1 \le i \le 4).} \tag{1012}$$

In particular, all ten vectors in Eqs. (1006)–(1007) are nonzero.

Fix $i \in V$. The active flip X_i has exactly two aggregated incidence classes,

$$0000 \longleftrightarrow e_i, \qquad \widehat{e}_i \longleftrightarrow \Omega. \tag{1013}$$

The off-diagonal equations for X_i and for $Z_j X_i$, with any active $j \neq i$, are respectively

$$u^\dagger b_i + a_i^\dagger v = 0, \qquad u^\dagger b_i - a_i^\dagger v = 0. \tag{1014}$$

Thus

$$u^\dagger b_i = 0, \qquad a_i^\dagger v = 0 \quad (1 \le i \le 4). \tag{1015}$$

Every operator on the two-qubit subsystem A has physical weight at most two. The diagonal KL equality for all such operators is therefore equality of the complete reduced density matrices on A:

$$uu^\dagger + \sum_{i=1}^{4} a_i a_i^\dagger = \sum_{i=1}^{4} b_i b_i^\dagger + vv^\dagger. \tag{1016}$$

Taking the expectation in u and using Eq. (1015) gives

$$\|u\|^4 + \sum_i |u^\dagger a_i|^2 = |u^\dagger v|^2. \tag{1017}$$

By Eq. (1012) and Cauchy–Schwarz,

$$|u^\dagger v|^2 \le \|u\|^2\|v\|^2 = \frac{1}{9} = \|u\|^4. \tag{1018}$$

Hence equality holds throughout: $u^\dagger a_i = 0$ for every i, and u and v are parallel. Rephasing $|1_L\rangle$, we may assume

$$v = u. \tag{1019}$$

Then

$$u^\dagger a_i = u^\dagger b_i = 0, \qquad \sum_i a_i a_i^\dagger = \sum_i b_i b_i^\dagger. \tag{1020}$$

We next use the matrix-valued form of the same one-flip equations. Identify a two-qubit vector $x = \sum_{r,s} X_{rs}|rs\rangle$ with its 2×2 coefficient matrix $\mathsf{X} = (X_{rs})$. Then

$$\mathrm{Tr}_2(|x\rangle\langle y|) = \mathsf{X}\mathsf{Y}^\dagger, \qquad \mathrm{Tr}_1(|x\rangle\langle y|) = \mathsf{X}^T\overline{\mathsf{Y}}. \tag{1021}$$

For an arbitrary one-qubit operator F on either factor of A, the error FX_i has weight at most two, so

$$u^\dagger F b_i + a_i^\dagger F u = 0. \tag{1022}$$

Taking adjoints and using the fact that $F^\dagger$ ranges over the same one-qubit operator space, this is equivalent to the vanishing of both one-qubit partial traces of $|u\rangle\langle b_i| + |a_i\rangle\langle u|$. Local unitaries on the two zero-angle qubits commute with the physical diagonal gate, so put u in Schmidt form

$$\mathsf{U} = \frac{1}{\sqrt{3}}\begin{pmatrix} s & 0 \\ 0 & t \end{pmatrix}, \qquad s,t \ge 0, \qquad s^2 + t^2 = 1. \tag{1023}$$

Writing $\mathsf{A}_i, \mathsf{B}_i$ for a_i, b_i, the two partial-trace equations are

$$\mathsf{U}\mathsf{B}_i^\dagger + \mathsf{A}_i\mathsf{U}^\dagger = 0, \qquad \mathsf{U}^T\overline{\mathsf{B}_i} + \mathsf{A}_i^T\overline{\mathsf{U}} = 0. \tag{1024}$$

Two-qubit flip-kernel dichotomy. Under Eqs. (1020) and (1024), exactly one of the following two possibilities holds.

(i) If $s \ne t$, including the product-state endpoints, all a_i and b_i lie on one common one-dimensional line in $u^\perp$.

(ii) If $s = t = 1/\sqrt{2}$, then

$$\mathrm{Tr}\,\mathsf{A}_i = 0, \qquad \mathsf{B}_i = -\mathsf{A}_i^\dagger \quad (1 \le i \le 4). \tag{1025}$$

To prove the dichotomy, suppress the index i and write

$$\mathsf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \mathsf{B} = \begin{pmatrix} p & q \\ r & w \end{pmatrix}. \tag{1026}$$

The off-diagonal entries of Eq. (1024) give

$$s\overline{r} + tb = 0, \qquad t\overline{r} + sb = 0, \qquad t\overline{q} + sc = 0, \qquad s\overline{q} + tc = 0. \tag{1027}$$

If $s \neq t$, the two 2×2 coefficient matrices in Eq. (1027) have determinant $s^2 - t^2 \neq 0$; hence $b = c = q = r = 0$. The diagonal equations and orthogonality to u then confine both A and B to the line generated by $\mathrm{diag}(t, -s)$. This conclusion also covers $s = 0$ or $t = 0$. If $s = t$, then $s = t = 1/\sqrt{2}$ and the same equations give $p = -\overline{a}$, $q = -\overline{c}$, $r = -\overline{b}$, $w = -\overline{d}$, while $u^\dagger a_i = 0$ gives $a + d = 0$. This is precisely Eq. (1025).

Suppose first that $s \neq t$. By the dichotomy, there is a fixed nonzero vector $w \in u^\perp$ and nonzero scalars λ_i, μ_i such that

$$a_i = \lambda_i w, \qquad b_i = \mu_i w. \tag{1028}$$

For distinct i, j, let $\{k, \ell\} = V \setminus \{i, j\}$. The weight-two flip $X_k X_\ell$ has exactly the two aggregated incidences

$$\widehat{e_i} \longleftrightarrow e_j, \qquad \widehat{e_j} \longleftrightarrow e_i. \tag{1029}$$

Its off-diagonal equation is

$$a_i^\dagger b_j + a_j^\dagger b_i = 0. \tag{1030}$$

Substituting Eq. (1028) and defining the nonzero numbers $\theta_i = \mu_i / \overline{\lambda_i}$ gives

$$\theta_i + \theta_j = 0 \qquad (i \neq j). \tag{1031}$$

The three equations for the pairs $(1,2), (1,3), (2,3)$ form an odd triangle and force $\theta_1 = \theta_2 = \theta_3 = 0$, a contradiction.

It remains to exclude the Bell branch $s = t = 1/\sqrt{2}$. Use the orthonormal traceless Pauli basis

$$\tau_r = \frac{\sigma_r}{\sqrt{2}}, \qquad r = 1, 2, 3, \tag{1032}$$

and write

$$\mathsf{A}_i = \sum_{r=1}^{3} z_{ri} \tau_r, \qquad z_i = (z_{1i}, z_{2i}, z_{3i})^T \in \mathbb{C}^3. \tag{1033}$$

Equation (1012) gives

$$z_i^\dagger z_i = \frac{1}{6}. \tag{1034}$$

Using $\mathsf{B}_i = -\mathsf{A}_i^\dagger$ in Eq. (1030) yields

$$\mathrm{Tr}(\mathsf{A}_i \mathsf{A}_j) = 0, \qquad \text{equivalently} \qquad z_i^T z_j = 0 \quad (i \neq j). \tag{1035}$$

The frame equality in Eq. (1020) becomes

$$S := \sum_{i=1}^{4} z_i z_i^\dagger = \overline{S}. \tag{1036}$$

Thus S is a real positive semidefinite 3×3 matrix. Using Eqs. (1036) and (1035),

$$\mathrm{Tr}(S^2) = \mathrm{Tr}(S\overline{S}) = \sum_{i,j} |z_i^T z_j|^2 = \sum_i |z_i^T z_i|^2 \leq 4 \left(\frac{1}{6}\right)^2 = \frac{1}{9}. \tag{1037}$$

On the other hand,

$$\mathrm{Tr}\, S = \sum_i \|z_i\|^2 = \frac{2}{3}, \tag{1038}$$

so positivity in dimension three gives

$$\mathrm{Tr}(S^2) \geq \frac{(\mathrm{Tr}\, S)^2}{3} = \frac{4}{27} > \frac{1}{9}. \tag{1039}$$

This contradiction excludes the Bell branch and completes the proof. □

23.2 Orbit 6: a one-qubit-spinor obstruction

Proposition 23.2 (Orbit 6 no-go)**.** *Orbit 6 admits no exact* $((6, 2, d \geq 3))$ *completion.*

Proof. Orbit 6 is represented by

$$(q, m, a) = \big(3, 3, (0, 1, 1, 1, 1, 1)\big). \tag{1040}$$

Let A be the first zero-angle qubit and identify a computational string on the five active qubits with a subset of $V = \{1, 2, 3, 4, 5\}$. The active parts of the two fibers are

$$T_0 = \{S \subseteq V : |S| \in \{0, 3\}\}, \qquad T_1 = \{S \subseteq V : |S| \in \{1, 4\}\}. \tag{1041}$$

Grouping the amplitudes on A into spinors in $\mathbb{C}^2$, write

$$|0_L\rangle = u \otimes |\varnothing\rangle + \sum_{i<j} a_{ij} \otimes |V \setminus \{i, j\}\rangle, \tag{1042}$$

$$|1_L\rangle = \sum_i b_i \otimes |\{i\}\rangle + \sum_i c_i \otimes |V \setminus \{i\}\rangle, \tag{1043}$$

where $u, a_{ij}, b_i, c_i \in \mathbb{C}^2$.

Set

$$r = \|u\|^2, \qquad x_{ij} = \|a_{ij}\|^2, \qquad y_i = \|b_i\|^2, \qquad t_i = \|c_i\|^2, \tag{1044}$$

and put

$$X = \sum_{i<j} x_{ij}, \qquad Y = \sum_i y_i, \qquad T = \sum_i t_i. \tag{1045}$$

Normalization gives

$$r + X = 1, \qquad Y + T = 1. \tag{1046}$$

For distinct active coordinates k, ℓ, the two-coordinate occupation moment is

$$\sum_{\{i,j\} \subseteq V \setminus \{k,\ell\}} x_{ij} = T - t_k - t_\ell. \tag{1047}$$

Summing over all ten pairs $\{k, \ell\}$ counts every x_{ij} three times and gives

$$X = 2T. \tag{1048}$$

The one-coordinate moment at k is

$$\sum_{\substack{i<j \\ k \notin \{i,j\}}} x_{ij} = y_k + T - t_k. \tag{1049}$$

Summing over k yields $3X = Y + 4T$. Together with Eqs. (1046) and (1048), this gives

$$T = \frac{1}{3}, \qquad X = \frac{2}{3}, \qquad Y = \frac{2}{3}, \qquad r = \frac{1}{3}. \tag{1050}$$

Let $d_k = \sum_{j \neq k} x_{kj}$ be the weighted edge degree at k. Summing Eq. (1047) over $\ell \neq k$ gives

$$2(X - d_k) = 3T - 3t_k, \qquad \text{hence} \qquad d_k = \frac{1}{6} + \frac{3}{2} t_k. \tag{1051}$$

Equation (1049) then gives

$$y_k = \frac{1}{6} - \frac{1}{2}t_k. \tag{1052}$$

Finally, rewriting Eq. (1047) as $X - d_i - d_j + x_{ij} = T - t_i - t_j$ gives

$$x_{ij} = \frac{t_i + t_j}{2}. \tag{1053}$$

Thus the complete aggregate-norm solution is the four-simplex

$$\boxed{\begin{aligned} &\|u\|^2 = \frac{1}{3}, \\ &\|c_i\|^2 = t_i, \\ &\|b_i\|^2 = \frac{1}{6} - \frac{t_i}{2}, \qquad t_i \geq 0, \quad \sum_{i=1}^{5} t_i = \frac{1}{3}. \\ &\|a_{ij}\|^2 = \frac{t_i + t_j}{2}, \end{aligned}} \tag{1054}$$

The seventeen aggregate scalar equations have rank seventeen; taking the pivot columns in lexicographic order gives a 17×17 minor of determinant 144. Hence Eq. (1054) is the full aggregate moment solution. In particular,

$$\sum_{i<j} \|a_{ij}\|^2 = \frac{2}{3}, \qquad \sum_{i} \|b_i\|^2 = \frac{2}{3}. \tag{1055}$$

Fix $k \in V$. The complete active-label incidences of X_k are

$$\varnothing \longleftrightarrow \{k\}, \qquad V \setminus \{k, j\} \longleftrightarrow V \setminus \{j\} \quad (j \neq k). \tag{1056}$$

Define

$$p_k = u^\dagger b_k, \qquad r_{kj} = a_{kj}^\dagger c_j \quad (j \neq k). \tag{1057}$$

The equation for X_k is

$$p_k + \sum_{j \neq k} r_{kj} = 0. \tag{1058}$$

For $\ell \neq k$, the equation for $Z_\ell X_k$ is

$$p_k + r_{k\ell} - \sum_{\substack{j \neq k \\ j \neq \ell}} r_{kj} = 0. \tag{1059}$$

Ordering the four indices $j \neq k$, the five equations have coefficient matrix

$$W_5 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 & 1 \end{pmatrix}, \qquad \det W_5 = -48. \tag{1060}$$

Therefore

$$u^\dagger b_k = 0, \qquad a_{kj}^\dagger c_j = 0 \quad (k \neq j). \tag{1061}$$

Interchanging i and j gives, for every edge,

$$a_{ij}^\dagger c_i = a_{ij}^\dagger c_j = 0. \tag{1062}$$

Since $\sum_i t_i = 1/3$, at least one c_s is nonzero. Let $\mathcal{C} = \mathrm{span}\{c_s\}$. If $j \neq s$ and $t_j > 0$, then

$$\|a_{sj}\|^2 = \frac{t_s + t_j}{2} > 0, \tag{1063}$$

and Eq. (1062) places both c_s and c_j in the one-dimensional space $a_{sj}^\perp$. Hence all nonzero c_i are parallel and belong to $\mathcal{C}$. Whenever $a_{ij} \neq 0$, at least one of t_i, t_j is positive, so

$$a_{ij} \in \mathcal{C}^\perp. \tag{1064}$$

Let $\mathcal{U} = \mathrm{span}\{u\}$. Equation (1061) likewise gives

$$b_i \in \mathcal{U}^\perp. \tag{1065}$$

The diagonal KL equations for all one-qubit operators on the zero-angle qubit require equality of its two reduced density matrices. By Eqs. (1050), (1055), (1064), and (1065), they are

$$\rho_A^{(0)} = \frac{1}{3} P_{\mathcal{U}} + \frac{2}{3} P_{\mathcal{C}^\perp}, \tag{1066}$$

$$\rho_A^{(1)} = \frac{2}{3} P_{\mathcal{U}^\perp} + \frac{1}{3} P_{\mathcal{C}}. \tag{1067}$$

Equality and $P_{\mathcal{L}^\perp} = I - P_{\mathcal{L}}$ give

$$P_{\mathcal{U}} = P_{\mathcal{C}}. \tag{1068}$$

Thus

$$u, c_i \in \mathcal{U}, \qquad a_{ij}, b_i \in \mathcal{U}^\perp. \tag{1069}$$

Finally let F be an arbitrary operator on the zero-angle qubit. The error FX_k has weight at most two, and its off-diagonal equation is

$$u^\dagger F b_k + \sum_{j \neq k} a_{kj}^\dagger F c_j = 0 \qquad \text{for every } F \in M_2(\mathbb{C}). \tag{1070}$$

Equivalently,

$$b_k u^\dagger + \sum_{j \neq k} c_j a_{kj}^\dagger = 0. \tag{1071}$$

Projecting Eq. (1071) on the left by $P_{\mathcal{U}^\perp}$ and on the right by $P_{\mathcal{U}}$ kills every term in the sum and leaves

$$b_k u^\dagger = 0. \tag{1072}$$

Since $u \neq 0$, this forces $b_k = 0$ for every k, contradicting the exact shell mass $\sum_k \|b_k\|^2 = 2/3$ in Eq. (1055). □

Notation for the lifted cases. The remaining six cases in this section have $(q, m) = (6, 3)$ and $\delta = q/m = 2$. Their physical diagonal lifts have order six, while the projective logical eigenvalue ratio is $\omega_6^2 = \omega_3$. Write

$$|0_L\rangle = \sum_{x \in S_0} c_x |x\rangle, \qquad |1_L\rangle = \sum_{y \in S_2} d_y |y\rangle, \tag{1073}$$

and put

$$p_x = |c_x|^2, \qquad q_y = |d_y|^2. \tag{1074}$$

All strings below are ordered lexicographically. Exact row reduction is performed on the full diagonal system (5)–(6), with the two normalization rows first, then the six one-coordinate rows, and then the fifteen two-coordinate rows in lexicographic order.

For a coordinate i, define the X_i incidence set

$$\mathcal{I}_i := \{(x, y) \in S_0 \times S_2 : x = y \oplus e_i\}. \tag{1075}$$

For every $(x, y) \in \mathcal{I}_i$, set

$$z_{x,y} := \overline{c_x} d_y. \tag{1076}$$

The off-diagonal KL equations for X_i and $X_i Z_j$, $j \neq i$, are

$$\sum_{(x,y) \in \mathcal{I}_i} z_{x,y} = 0, \tag{1077}$$

$$\sum_{(x,y) \in \mathcal{I}_i} (-1)^{x_j} z_{x,y} = 0. \tag{1078}$$

Here $x_j = y_j$ because $j \neq i$, so the sign may be evaluated at either endpoint.

Lemma 23.3 (Full-rank flip-character obstruction)**.** *Fix a coordinate i, enumerate $\mathcal{I}_i = \{(x^{(r)}, y^{(r)}) : 1 \le r \le N\}$, and choose coordinates $j_1, \dots, j_{N-1} \neq i$. If the character matrix*

$$W = \begin{pmatrix} 1 & \cdots & 1 \\ (-1)^{x^{(1)}_{j_1}} & \cdots & (-1)^{x^{(N)}_{j_1}} \\ \vdots & & \vdots \\ (-1)^{x^{(1)}_{j_{N-1}}} & \cdots & (-1)^{x^{(N)}_{j_{N-1}}} \end{pmatrix} \tag{1079}$$

is nonsingular, then every incidence monomial $\overline{c_{x^{(r)}}} d_{y^{(r)}}$ vanishes. Consequently, if the diagonal moments force one incidence to have both endpoint probabilities positive, no exact distance-three completion exists.

Proof. Equations (1077) and (1078) for the selected dressings are $Wz = 0$, where $z_r = \overline{c_{x^{(r)}}} d_{y^{(r)}}$. Nonsingularity gives $z = 0$. If $p_{x^{(r)}} q_{y^{(r)}} > 0$, however, then $|z_r|^2 = p_{x^{(r)}} q_{y^{(r)}} > 0$, a contradiction. □

23.3 Orbit 47: a two-incidence Walsh obstruction

Proposition 23.4 (Orbit 47 no-go)**.** *Orbit 47 admits no exact $((6, 2, d \ge 3))$ completion.*

Proof. Orbit 47 is represented by

$$(q, m, a) = \big(6, 3, (1, 1, 1, 1, 2, 2)\big). \tag{1080}$$

Its two fibers are

$$S_0 = \{000000, 001111, 010111, 011011, 100111, \\ 101011, 110011, 111101, 111110\}, \tag{1081}$$

$$S_2 = \{000001, 000010, 001100, 010100, 011000, \\ 100100, 101000, 110000, 111111\}. \tag{1082}$$

Order the probabilities according to these displays. Exact row reduction of the diagonal moment equations gives the complete affine solution

$$(p_x)_{x\in S_0} = \left(\frac{1}{3}, \beta, \alpha, \gamma, \gamma, \alpha, \beta, \frac{1}{6}, \frac{1}{6}\right), \tag{1083}$$

$$(q_y)_{y\in S_2} = \left(\frac{1}{6}, \frac{1}{6}, \beta, \alpha, \gamma, \gamma, \alpha, \beta, \frac{1}{3}\right), \tag{1084}$$

where

$$\gamma = \frac{1}{6} - \alpha - \beta, \qquad \alpha, \beta, \gamma \geq 0. \tag{1085}$$

The moment matrix has eighteen columns and rank sixteen; one maximal minor has determinant 48. Hence the displayed two-parameter family is the complete moment polytope.

The complete X_5 incidence set consists of exactly two pairs:

$$000000 \longleftrightarrow 000010, \qquad 111101 \longleftrightarrow 111111. \tag{1086}$$

Put

$$A = \overline{c_{000000}} d_{000010}, \qquad B = \overline{c_{111101}} d_{111111}. \tag{1087}$$

The equations for X_5 and $X_5 Z_1$ are

$$A + B = 0, \qquad A - B = 0. \tag{1088}$$

Thus $A = B = 0$. But

$$|A|^2 = p_{000000} q_{000010} = \frac{1}{3} \cdot \frac{1}{6} = \frac{1}{18} > 0, \tag{1089}$$

a contradiction. □

23.4 Orbit 48: a four-incidence Hadamard obstruction

Proposition 23.5 (Orbit 48 no-go)**.** *Orbit 48 admits no exact* $((6, 2, d \geq 3))$ *completion.*

Proof. Orbit 48 is represented by

$$(q, m, a) = \big(6, 3, (1, 1, 1, 2, 2, 2)\big). \tag{1090}$$

Its fibers are

$$S_0 = \{000000, 000111, 011011, 011101, 011110, \\ 101011, 101101, 101110, 110011, 110101, 110110\}, \tag{1091}$$

$$S_2 = \{000001, 000010, 000100, 011000, 011111, \\ 101000, 101111, 110000, 110111\}. \tag{1092}$$

Exact row reduction gives

$$(p_x)_{x\in S_0} = \left(\frac{1}{3},\frac{1}{6},h,h,h,\frac{\alpha}{2},\frac{\alpha}{2},\frac{\alpha}{2},\frac{\beta}{2},\frac{\beta}{2},\frac{\beta}{2}\right), \tag{1093}$$

$$(q_y)_{y\in S_2} = \left(\frac{1}{6},\frac{1}{6},\frac{1}{6},h,k,\frac{\alpha}{2},\alpha,\frac{\beta}{2},\beta\right), \tag{1094}$$

where

$$h=\frac{1}{6}-\frac{\alpha+\beta}{2},\qquad k=\frac{1}{3}-\alpha-\beta,\qquad \alpha,\beta\ge 0,\quad \alpha+\beta\le\frac{1}{3}. \tag{1095}$$

The moment matrix has twenty columns and rank eighteen; one maximal minor has determinant -384. Hence the displayed two-parameter family is complete.

The complete X_4 incidence set is

$$\begin{aligned} &000000 \longleftrightarrow 000100,\\ &011011 \longleftrightarrow 011111,\\ &101011 \longleftrightarrow 101111,\\ &110011 \longleftrightarrow 110111. \end{aligned} \tag{1096}$$

Define the four incidence monomials in the displayed order by

$$\begin{aligned} A &= \overline{c_{000000}}\,d_{000100}, \quad B = \overline{c_{011011}}\,d_{011111},\\ C &= \overline{c_{101011}}\,d_{101111}, \quad D = \overline{c_{110011}}\,d_{110111}. \end{aligned} \tag{1097}$$

The first three bits of the four bottom endpoints are $000, 011, 101, 110$. Therefore the equations for $X_4, X_4Z_1, X_4Z_2, X_4Z_3$ are

$$\begin{pmatrix} 1 & 1 & 1 & 1\\ 1 & 1 & -1 & -1\\ 1 & -1 & 1 & -1\\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} A\\ B\\ C\\ D \end{pmatrix} = 0. \tag{1098}$$

The determinant is -16, so $A=B=C=D=0$. Yet

$$|A|^2 = p_{000000}q_{000100} = \frac{1}{3}\cdot\frac{1}{6} = \frac{1}{18} > 0, \tag{1099}$$

a contradiction. □

For orbits 49–51 the sum of the six angle residues is $14 \equiv 2 \pmod 6$. Consequently bitwise complementation interchanges the fibers:

$$S_2 = \{\overline{x} : x \in S_0\}. \tag{1100}$$

Order $S_0 = \{x_0, \dots, x_{N-1}\}$ and use complementary indexing on the top fiber:

$$|0_L\rangle = \sum_i c_i |x_i\rangle, \qquad |1_L\rangle = \sum_i d_i |\overline{x_i}\rangle. \tag{1101}$$

In each of these three orbits the complete moment solution satisfies $|c_i|^2 = |d_i|^2 =: p_i$. Whenever $p_i > 0$, define

$$\rho_i = \frac{d_i}{\overline{c_i}}, \qquad |\rho_i| = 1, \tag{1102}$$

and put $z_{ij} = \overline{c_i} d_j$. Then

$$z_{ij} = \rho_j\,\overline{c_i c_j}. \tag{1103}$$

Thus, whenever $p_i p_j > 0$,

$$z_{ij} + z_{ji} = 0 \quad\Longrightarrow\quad \rho_i = -\rho_j, \tag{1104}$$

$$z_{ij} - z_{ji} = 0 \quad\Longrightarrow\quad \rho_i = \rho_j. \tag{1105}$$

23.5 Orbit 49: moment pinning and an odd triangle

Proposition 23.6 (Orbit 49 no-go)**.** *Orbit 49 admits no exact* $((6,2,d\geq 3))$ *completion.*

Proof. Orbit 49 is represented by

$$(q,m,a)=\big(6,3,(1,1,1,3,3,5)\big). \tag{1106}$$

Order the bottom fiber as

$$\begin{aligned} &x_0=000000,\quad x_1=000110,\quad x_2=001001,\quad x_3=001111,\\ &x_4=010001,\quad x_5=010111,\quad x_6=100001,\quad x_7=100111,\\ &x_8=111010,\quad x_9=111100. \end{aligned} \tag{1107}$$

The top fiber is $\{\overline{x_i}:0\leq i\leq 9\}$. Exact row reduction gives the complete probability vector

$$\begin{aligned} (p_0,\dots,p_9)=\Big(&\gamma,\tfrac{1}{6}-\gamma,\beta,\tfrac{1}{6}-\beta,\alpha,\tfrac{1}{6}-\alpha,\\ &\tfrac{1}{3}-\alpha-\beta-\gamma,\alpha+\beta+\gamma-\tfrac{1}{6},\tfrac{1}{6},\tfrac{1}{6}\Big), \end{aligned} \tag{1108}$$

with exact domain

$$0\leq\alpha,\beta,\gamma\leq\frac{1}{6},\qquad \frac{1}{6}\leq\alpha+\beta+\gamma\leq\frac{1}{3}. \tag{1109}$$

The 23×20 moment matrix has rank seventeen; one 17×17 minor has determinant -48. Hence Eq. (1108) is the full three-dimensional moment polytope. In particular,

$$p_{2r}+p_{2r+1}=\frac{1}{6}\quad(0\leq r\leq 3),\qquad p_8=p_9=\frac{1}{6}. \tag{1110}$$

The four Pauli rows supported on each displayed weight-two flip mask have the following row-reduced consequences. Here e_r denotes the rth physical coordinate and $z_{ij}=\overline{c_i}d_j$.

index pair	flip mask	isolated reciprocal equations
$\{0,1\}$	e_4+e_6	$z_{08}+z_{80}=0,\quad z_{19}+z_{91}=0$
	e_5+e_6	$z_{09}+z_{90}=0,\quad z_{18}+z_{81}=0$
$\{2,3\}$	e_3+e_4	$z_{28}+z_{82}=0,\quad z_{39}+z_{93}=0$
	e_3+e_5	$z_{29}+z_{92}=0,\quad z_{38}+z_{83}=0$
$\{4,5\}$	e_2+e_4	$z_{48}+z_{84}=0,\quad z_{59}+z_{95}=0$
	e_2+e_5	$z_{49}+z_{94}=0,\quad z_{58}+z_{85}=0$
$\{6,7\}$	e_1+e_4	$z_{68}+z_{86}=0,\quad z_{79}+z_{97}=0$
	e_1+e_5	$z_{69}+z_{96}=0,\quad z_{78}+z_{87}=0.$

(1111)

For example, the first mask has incidence order $(0,8),(1,9),(8,0),(9,1)$, and its exact Walsh row reduction is

$$z_{08}+z_{80}=0,\qquad z_{19}+z_{91}=0. \tag{1112}$$

The other seven rows are identical up to index permutation.

Since $p_0+p_1=1/6$, at least one of indices $0,1$ is active. Its two edges in Eq. (1111) connect it to both 8 and 9, so Eqs. (1104) imply

$$\rho_8=\rho_9=:\rho. \tag{1113}$$

The remaining rows of the table show that every active index $i \le 7$ obeys $\rho_i = -\rho$. If $p_i = 0$, then $c_i = d_i = 0$. Hence the uniform identity

$$d_i = -\rho\,\overline{c_i} \qquad (0 \le i \le 7) \tag{1114}$$

holds on the entire closed moment polytope.

Four further weight-two masks have only one independent cross equation:

mask	incidences	reduced equation
$e_1 + e_2$	$(0,3),(1,2),(2,1),(3,0)$	$z_{03} + z_{12} + z_{21} + z_{30} = 0$
$e_1 + e_3$	$(0,5),(1,4),(4,1),(5,0)$	$z_{05} + z_{14} + z_{41} + z_{50} = 0$
$e_1 + e_6$	$(2,5),(3,4),(4,3),(5,2)$	$z_{25} + z_{34} + z_{43} + z_{52} = 0$
$e_2 + e_3$	$(0,7),(1,6),(6,1),(7,0)$	$z_{07} + z_{16} + z_{61} + z_{70} = 0.$

(1115)

Using Eq. (1114), these become

$$c_0 c_3 + c_1 c_2 = 0, \tag{1116}$$

$$c_0 c_5 + c_1 c_4 = 0, \tag{1117}$$

$$c_2 c_5 + c_3 c_4 = 0, \tag{1118}$$

$$c_0 c_7 + c_1 c_6 = 0. \tag{1119}$$

Taking absolute values in Eq. (1116) gives

$$\gamma\left(\frac{1}{6} - \beta\right) = \left(\frac{1}{6} - \gamma\right)\beta, \tag{1120}$$

so $\gamma = \beta$. Equation (1117) similarly gives $\gamma = \alpha$. Put $\alpha = \beta = \gamma = t$. Taking absolute values in Eq. (1119) then gives

$$t\left(3t - \frac{1}{6}\right) = \left(\frac{1}{6} - t\right)\left(\frac{1}{3} - 3t\right), \tag{1121}$$

which reduces to $12t - 1 = 0$. Thus

$$p_0 = \cdots = p_7 = \frac{1}{12}, \qquad p_8 = p_9 = \frac{1}{6}. \tag{1122}$$

All amplitudes in Eqs. (1116)–(1118) are now nonzero. Define

$$r_A = \frac{c_0}{c_1}, \qquad r_B = \frac{c_2}{c_3}, \qquad r_C = \frac{c_4}{c_5}. \tag{1123}$$

Dividing Eqs. (1116)–(1118) by the corresponding nonzero products gives

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} r_A \\ r_B \\ r_C \end{pmatrix} = 0. \tag{1124}$$

The determinant is -2, forcing all three ratios to vanish, contrary to Eq. (1122). □

23.6 Orbit 50: a boundary-safe active odd triangle

Proposition 23.7 (Orbit 50 no-go)**.** *Orbit 50 admits no exact* $((6,2,d\ge 3))$ *completion.*

Proof. Orbit 50 is represented by

$$(q,m,a)=\big(6,3,(1,1,1,3,4,4)\big). \tag{1125}$$

Order the bottom fiber as

$$\begin{aligned} x_0&=000000, \quad x_1=001111, \quad x_2=010111,\\ x_3&=011001, \quad x_4=011010, \quad x_5=100111,\\ x_6&=101001, \quad x_7=101010, \quad x_8=110001,\\ x_9&=110010, \quad x_{10}=111100. \end{aligned} \tag{1126}$$

The top fiber is the complementary set. Exact row reduction gives the full four-parameter moment normal form

$$\begin{aligned} (p_0,\dots,p_{10})=\Big(&\frac16,\alpha,\beta,\gamma,\frac13-\alpha-\beta-\gamma,\frac13-\alpha-\beta,\\ &\tau,\beta-\tau,\frac16-\tau-\gamma,\tau+\gamma+\alpha-\frac16,\frac16\Big). \end{aligned} \tag{1127}$$

Its exact nonnegativity region is

$$\begin{aligned} \alpha,\beta,\gamma,\tau\ge 0, \qquad \alpha+\beta+\gamma\le\frac13, \qquad \tau\le\beta,\\ \tau+\gamma\le\frac16, \qquad \tau+\gamma+\alpha\ge\frac16. \end{aligned} \tag{1128}$$

The 23×22 moment matrix has rank eighteen, and one 18×18 minor has determinant -48. Two consequences of Eq. (1127) are

$$p_0=p_{10}=\frac16, \qquad p_1+p_2+p_5=\frac13. \tag{1129}$$

Thus indices 0 and 10 are always active, and at least one $k\in\{1,2,5\}$ is active.

The weight-two flip e_5+e_6 has only the reciprocal incidences $(0,10),(10,0)$, so

$$z_{0,10}+z_{10,0}=0, \qquad \rho_0+\rho_{10}=0. \tag{1130}$$

The three possible edges from 0 are

k	flip mask	complete incidence set
1	e_1+e_2	$(0,1),(1,0)$
2	e_1+e_3	$(0,2),(2,0)$
5	e_2+e_3	$(0,5),(5,0)$.

(1131)

Hence, for every active $k\in\{1,2,5\}$,

$$\rho_0+\rho_k=0. \tag{1132}$$

The corresponding k–10 flips have six incidences:

k	flip mask	incidence order
1	e_3+e_4	$(1,10),(3,7),(4,6),(6,4),(7,3),(10,1)$
2	e_2+e_4	$(2,10),(3,9),(4,8),(8,4),(9,3),(10,2)$
5	e_1+e_4	$(5,10),(6,9),(7,8),(8,7),(9,6),(10,5)$.

(1133)

For each row, the undressed equation is the sum of all six monomials. The Z_4-dressed row has sign -1 on the two outer incidences and $+1$ on the four middle incidences. Adding and subtracting the two rows therefore isolates

$$z_{k,10} + z_{10,k} = 0. \tag{1134}$$

Equivalently, the exact row-reduced Walsh system contains

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}. \tag{1135}$$

Thus every active $k \in \{1, 2, 5\}$ also satisfies

$$\rho_k + \rho_{10} = 0. \tag{1136}$$

Choose an active k, whose existence is guaranteed by Eq. (1129). The three unit-modulus numbers $\rho_0, \rho_{10}, \rho_k$ obey

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \rho_0 \\ \rho_{10} \\ \rho_k \end{pmatrix} = 0. \tag{1137}$$

The determinant is -2, so all three would vanish, contradicting their unit modulus. The argument is boundary-safe because it requires only one active index among $1, 2, 5$, not full support. □

23.7 Orbit 51: kernel pinning and a second odd triangle

Proposition 23.8 (Orbit 51 no-go)**.** *Orbit 51 admits no exact* $((6, 2, d \geq 3))$ *completion.*

Proof. Orbit 51 is represented by

$$(q, m, a) = \big(6, 3, (1, 1, 2, 3, 3, 4)\big). \tag{1138}$$

Order the bottom fiber as

$$\begin{aligned} &x_0 = 000000, \quad x_1 = 000110, \quad x_2 = 001001, \quad x_3 = 001111, \\ &x_4 = 011010, \quad x_5 = 011100, \quad x_6 = 101010, \quad x_7 = 101100, \\ &x_8 = 110001, \quad x_9 = 110111. \end{aligned} \tag{1139}$$

The top fiber is the complementary set. The complete moment normal form is

$$\begin{aligned} (p_0, \ldots, p_9) = \Big(&\gamma, \frac{1}{6} - \gamma, \beta, \frac{1}{6} - \beta, \alpha, \frac{1}{6} - \alpha, \\ &\frac{1}{6} - \alpha, \alpha, \frac{1}{3} - \beta - \gamma, \beta + \gamma\Big), \end{aligned} \tag{1140}$$

where

$$0 \leq \alpha, \beta, \gamma \leq \frac{1}{6}. \tag{1141}$$

The moment matrix has rank seventeen; one maximal minor has determinant 48. Hence Eq. (1140) is the full three-dimensional moment cube.

For the weight-one flip X_3, the complete incidence order is

$$(z_{09}, z_{18}, z_{81}, z_{90}). \tag{1142}$$

The rows supplied by X_3, X_3Z_1, X_3Z_4 have coefficient matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \tag{1143}$$

whose kernel is spanned by $(1,-1,-1,1)$. Therefore

$$(z_{09}, z_{18}, z_{81}, z_{90}) = \lambda(1,-1,-1,1). \tag{1144}$$

Similarly, X_6 has incidence order $(z_{29}, z_{38}, z_{83}, z_{92})$ and gives

$$(z_{29}, z_{38}, z_{83}, z_{92}) = \mu(1,-1,-1,1). \tag{1145}$$

Taking absolute values in Eqs. (1144) and (1145) gives

$$\gamma(\beta+\gamma) = \left(\frac{1}{6}-\gamma\right)\left(\frac{1}{3}-\beta-\gamma\right), \tag{1146}$$

$$\beta(\beta+\gamma) = \left(\frac{1}{6}-\beta\right)\left(\frac{1}{3}-\beta-\gamma\right). \tag{1147}$$

These reduce to

$$3\beta+9\gamma = 1, \qquad 9\beta+3\gamma = 1, \tag{1148}$$

so

$$\beta = \gamma = \frac{1}{12}. \tag{1149}$$

Consequently

$$p_0 = p_1 = p_2 = p_3 = \frac{1}{12}, \qquad p_8 = p_9 = \frac{1}{6}, \tag{1150}$$

and all six corresponding amplitudes are nonzero.

The reciprocal equalities inside the two kernel vectors give

$$\rho_0 = \rho_9, \qquad \rho_1 = \rho_8, \qquad \rho_2 = \rho_9, \qquad \rho_3 = \rho_8. \tag{1151}$$

It remains to synchronize ρ_8 and ρ_9. The masks e_1+e_4 and e_1+e_5 have the row-reduced reciprocal equations

$$z_{68} + z_{86} = 0, \qquad z_{79} + z_{97} = 0, \tag{1152}$$

$$z_{69} + z_{96} = 0, \qquad z_{78} + z_{87} = 0. \tag{1153}$$

Since

$$p_6 + p_7 = \left(\frac{1}{6}-\alpha\right) + \alpha = \frac{1}{6}, \tag{1154}$$

at least one of indices $6, 7$ is active. If 6 is active, the first equations in Eqs. (1152)–(1153) give $\rho_6 = -\rho_8 = -\rho_9$; if 7 is active, the second equations give the same conclusion. Thus, including $\alpha = 0$ and $\alpha = 1/6$,

$$\rho_8 = \rho_9 =: \rho. \tag{1155}$$

Combining with Eq. (1151),

$$\rho_0 = \rho_1 = \rho_2 = \rho_3 = \rho_8 = \rho_9 = \rho. \tag{1156}$$

Equation (1144) contains $z_{09}+z_{18}=0$, and Eq. (1145) contains $z_{29}+z_{38}=0$. Using Eq. (1156), these become

$$c_0c_9+c_1c_8=0, \qquad c_2c_9+c_3c_8=0. \tag{1157}$$

The weight-two mask e_1+e_2 has incidence order $(z_{03},z_{12},z_{21},z_{30})$ and one independent equation, which reduces to

$$c_0c_3+c_1c_2=0. \tag{1158}$$

All amplitudes in these equations are nonzero by Eq. (1150). Define

$$r_A=\frac{c_0}{c_1}, \qquad r_B=\frac{c_2}{c_3}, \qquad r_C=\frac{c_8}{c_9}. \tag{1159}$$

Then Eqs. (1157)–(1158) give

$$r_A+r_C=0, \qquad r_B+r_C=0, \qquad r_A+r_B=0. \tag{1160}$$

This is the same determinant--2 odd triangle as Eq. (1124), so it has only the zero solution, contradicting nonzero r_A,r_B,r_C. □

23.8 Orbit 52: a boundary-safe four-incidence obstruction

Proposition 23.9 (Orbit 52 no-go)**.** *Orbit 52 admits no exact* $((6,2,d\geq 3))$ *completion.*

Proof. Orbit 52 is represented by

$$(q,m,a)=\big(6,3,(1,2,3,3,4,4)\big). \tag{1161}$$

Its fibers are

$$\begin{aligned} S_0=\{&000000,001100,010001,010010,011101,\\ &011110,100111,101011,110100,111000\}, \end{aligned} \tag{1162}$$

$$\begin{aligned} S_2=\{&000011,001111,010000,011100,100101,\\ &100110,101001,101010,110111,111011\}. \end{aligned} \tag{1163}$$

The complete moment normal form is

$$(p_x)_{x\in S_0}=\left(\frac16-\frac{\alpha}{2},\frac{\alpha}{2},\frac16-\frac{\alpha}{2},\frac16-\frac{\alpha}{2},\frac{\alpha}{2},\frac{\alpha}{2},\frac13-2\beta,2\beta,\frac16-\beta,\beta\right), \tag{1164}$$

$$(q_y)_{y\in S_2}=\left(\frac16-\frac{\alpha}{2},\frac{\alpha}{2},\frac13-\alpha,\alpha,\frac16-\beta,\frac16-\beta,\beta,\beta,\frac16-\beta,\beta\right), \tag{1165}$$

with

$$0\leq\alpha\leq\frac13, \qquad 0\leq\beta\leq\frac16. \tag{1166}$$

The moment matrix has twenty columns and rank eighteen, and one maximal minor has determinant 192.

The complete X_2 incidence set is

$$\begin{aligned} &000000\longleftrightarrow 010000,\\ &001100\longleftrightarrow 011100,\\ &100111\longleftrightarrow 110111,\\ &101011\longleftrightarrow 111011. \end{aligned} \tag{1167}$$

Let the corresponding monomials be

$$\begin{aligned} A &= \overline{c_{000000}} d_{010000}, \quad B = \overline{c_{001100}} d_{011100}, \\ C &= \overline{c_{100111}} d_{110111}, \quad D = \overline{c_{101011}} d_{111011}. \end{aligned} \tag{1168}$$

On coordinates $1, 3, 4$, the four bottom endpoints have the patterns $000, 011, 101, 110$. Thus the equations for $X_2, X_2Z_1, X_2Z_3, X_2Z_4$ give

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = 0. \tag{1169}$$

Hence $A = B = C = D = 0$. But the exact moments satisfy

$$p_{000000} = \frac{1}{2} q_{010000}, \qquad p_{001100} = \frac{1}{2} q_{011100}, \tag{1170}$$

$$q_{010000} + q_{011100} = \frac{1}{3}. \tag{1171}$$

Therefore

$$|A|^2 = \frac{1}{2} q_{010000}^2, \qquad |B|^2 = \frac{1}{2} q_{011100}^2. \tag{1172}$$

The equations $A = B = 0$ force both probabilities in Eq. (1171) to vanish, a contradiction. This includes both boundary points and the full interior of the moment rectangle. □

Theorem 23.10 (Complete order-three no-go classification)**.** *None of the eight rank-compatible moment orbits*

$$5,\ 6,\ 47,\ 48,\ 49,\ 50,\ 51,\ 52 \tag{1173}$$

admits an exact six-qubit distance-three completion.

Proof. Propositions 23.1, 23.2, 23.4, 23.5, 23.6, 23.7, 23.8, and 23.9 exclude the eight orbits separately. The proofs are exact and cover every boundary face of the corresponding moment polytopes. □

Corollary 23.11 (Complete existence and enumerator-image status at order three)**.** *The only rank-compatible projective-order-three moment orbits admitting exact* $((6, 2, 3))$ *completions are*

$$\boxed{7, 8, 9.} \tag{1174}$$

Their complete Shor–Laflamme weight-enumerator images are those proved in Sections 7–9. Thus the $m = 3$ *sector is complete at the existence and weight-enumerator-image level. The finer classification of all exact projectors in orbits 7, 8, and 9 up to local-unitary or local-monomial equivalence remains separate.*

24 Exact no-go theorems for fourteen projective-order-two orbits

The rank-compatible projective-order-two sector contains twenty-five moment orbits. This section gives exact no-go certificates for the fourteen orbits

$$12, 13, 14, 15, 17, 18, 19, 20, 24, 41, 42, 43, 44, 46. \tag{1175}$$

The arguments fall into several small structural classes. Orbits 18 and 41 have restricted Walsh projectors that isolate forced positive anchor incidences. Orbits 42, 43, 44, and 46 have nonsingular ten-incidence half-angle character systems. Orbit 19 uses both half-angle coordinates to annihilate two complete anchor blocks. Orbit 17 reduces to a scaled fractional perfect matching of K_6 whose phases would have to satisfy all four-vertex hafnian equations. Orbit 20 reduces to uniform four-point parity blocks and an odd phase triangle. The zero-angle structures of orbits 12 and 13 lead to matrix-valued spinor contradictions. Finally, orbits 14, 15, and 24 are all consequences of one parity-four moment, support-connectivity, and frame-orthogonality lemma.

Every proof is exact and uses only physical Pauli operators of weight at most two. Hence all conclusions hold for $((6, 2, d \geq 3))$ codes and remain valid on every boundary face of the relevant moment polytope.

24.1 The common half-angle one-flip system

Let q be even, let $\delta = q/2$, and suppose that a representative $a \in \mathbb{Z}_q^6$ has $a_i = \delta$. For every $x \in S_0(a; q)$,

$$a \cdot (x \oplus e_i) \equiv a \cdot x + \delta(1 - 2x_i) \equiv \delta \pmod{q}. \tag{1176}$$

Since the same calculation in the reverse direction holds on S_δ, the coordinate flip is a bijection

$$S_\delta = S_0 \oplus e_i. \tag{1177}$$

Write

$$|0_L\rangle = \sum_{x \in S_0} c_x |x\rangle, \qquad |1_L\rangle = \sum_{y \in S_\delta} d_y |y\rangle, \tag{1178}$$

and, for $y_x = x \oplus e_i$, define the matched cross monomial

$$z_x = \overline{c_x}\, d_{y_x}. \tag{1179}$$

For $T \subseteq \{1, \ldots, 6\}$ put

$$\chi_T(y) = (-1)^{\sum_{j \in T} y_j}, \qquad R_T = \sum_{x \in S_0} \chi_T(y_x) z_x. \tag{1180}$$

Up to irrelevant nonzero global phases, the off-diagonal KL equations for

$$X_i, \qquad Y_i, \qquad X_i Z_j, \qquad Y_i Z_j \quad (j \neq i) \tag{1181}$$

are, respectively,

$$R_\varnothing = 0, \qquad R_{\{i\}} = 0, \qquad R_{\{j\}} = 0, \qquad R_{\{i,j\}} = 0. \tag{1182}$$

Every error in Eq. (1181) has weight at most two. Thus any restricted Walsh identity on the top fiber that expresses a point indicator as a linear combination of the characters in Eq. (1182) isolates the corresponding monomial z_x.

24.2 Orbit 18: an isolated fixed anchor

Proposition 24.1 (Orbit 18 no-go)**.** *Orbit 18 admits no exact* $((6, 2, d \geq 3))$ *completion.*

Proof. Orbit 18 is represented by

$$(q, m, a) = \big(4, 2, (1, 1, 1, 1, 1, 2)\big), \qquad \delta = 2. \tag{1183}$$

Use the half-angle coordinate $i = 6$, and set

$$w = x_1 + \cdots + x_5, \qquad t = x_6. \tag{1184}$$

The residue condition depends only on (w, t). The bottom and top fibers split into the shell types

	shell types (w,t)		
S_0	$(0,0)$	$(4,0)$	$(2,1)$
S_2	$(2,0)$	$(0,1)$	$(4,1)$.

(1185)

Let A, B, C be the total bottom probabilities on the three displayed S_0 shells and let D, E, F be the corresponding total top probabilities. In particular,

$$A = p_{000000}, \qquad E = q_{000001}. \tag{1186}$$

The two normalizations and equality of the diagonal moments of t, wt, w, and $\binom{w}{2}$ give

$$\begin{aligned} A + B + C &= 1, & D + E + F &= 1, \\ C &= E + F, & 2C &= 4F, \\ 4B + 2C &= 2D + 4F, & 6B + C &= D + 6F. \end{aligned} \tag{1187}$$

These are sums of the ordinary weight-one and weight-two diagonal KL relations. The system is nonsingular and has the unique solution

$$(A, B, C, D, E, F) = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right). \tag{1188}$$

Hence

$$p_{000000} = q_{000001} = \frac{1}{4}. \tag{1189}$$

The full diagonal moment matrix has thirty-two columns and rank twenty-three; one maximal minor has determinant -1024, providing an independent exact rank certificate for the nine-dimensional moment polytope.

Now restrict the characters in Eq. (1180) to the top fiber S_2. Since

$$\sum_{j=1}^{5} \chi_{\{j\}}(y) = 5 - 2w, \tag{1190}$$

the three top shell types in Eq. (1185) give the exact restricted identity

$$\frac{1}{8}\left(1 - 2\chi_{\{6\}}(y) + \sum_{j=1}^{5} \chi_{\{j\}}(y)\right) = \mathbf{1}_{\{000001\}}(y) \qquad (y \in S_2). \tag{1191}$$

Indeed, the left side takes the values $0, 1, 0$ on the shell types $(2,0), (0,1), (4,1)$, respectively. Taking the same linear combination of the KL rows in Eq. (1182) yields

$$z_{000000} = \frac{1}{8}\left(R_{\varnothing} - 2R_{\{6\}} + \sum_{j=1}^{5} R_{\{j\}}\right) = 0. \tag{1192}$$

But Eq. (1189) gives

$$|z_{000000}|^2 = p_{000000} q_{000001} = \frac{1}{16} > 0, \tag{1193}$$

which is impossible. □

24.3 Orbit 41: a four-character anchor projector

Proposition 24.2 (Orbit 41 no-go)**.** *Orbit 41 admits no exact* $((6,2,d\geq 3))$ *completion.*

Proof. Orbit 41 is represented by

$$(q,m,a)=\big(6,2,(1,1,1,1,2,3)\big),\qquad \delta=3. \tag{1194}$$

Use $i=6$, and write

$$w=x_1+x_2+x_3+x_4,\qquad s=x_5,\qquad t=x_6. \tag{1195}$$

The bottom and top shell types are

$$\begin{array}{c|cccc} S_0 & (0,0,0) & (1,1,1) & (3,0,1) & (4,1,0)\\ S_3 & (0,0,1) & (1,1,0) & (3,0,0) & (4,1,1). \end{array} \tag{1196}$$

Let their total probabilities be A,B,C,D and E,F,G,H, respectively. The first shell in each row is a singleton, so

$$A=p_{000000},\qquad E=q_{000001}. \tag{1197}$$

Normalization and equality of the diagonal moments of st, wt, $\binom{w}{2}$, t, s, and w give

$$\begin{aligned} A+B+C+D&=1, & E+F+G+H&=1,\\ B&=H, & B+3C&=4H,\\ 3C+6D&=3G+6H, & B+C&=E+H,\\ B+D&=F+H, & B+3C+4D&=F+3G+4H. \end{aligned} \tag{1198}$$

The equations successively imply

$$C=B,\qquad E=B,\qquad F=D,\qquad G=D,\qquad D=B. \tag{1199}$$

The normalizations then give

$$A=B=C=D=E=F=G=H=\frac{1}{4}. \tag{1200}$$

In particular,

$$p_{000000}=q_{000001}=\frac{1}{4}. \tag{1201}$$

The complete twenty-variable diagonal moment matrix has rank seventeen; a maximal minor has determinant 108.

On the top fiber, the exact character identity

$$\frac{1}{4}\big(1-\chi_{\{6\}}(y)+\chi_{\{1\}}(y)-\chi_{\{1,6\}}(y)\big)=\frac{1}{4}(1+\chi_{\{1\}})(1-\chi_{\{6\}})=\mathbf{1}_{\{000001\}}(y) \tag{1202}$$

holds. The factor $1-\chi_{\{6\}}$ selects $t=1$; among the two top shells with $t=1$, the factor $1+\chi_{\{1\}}$ selects 000001 rather than 111111. Hence the four KL rows for X_6, Y_6, X_6Z_1, Y_6Z_1 imply

$$z_{000000}=\frac{1}{4}\left(R_{\varnothing}-R_{\{6\}}+R_{\{1\}}-R_{\{1,6\}}\right)=0. \tag{1203}$$

Equation (1201) instead gives

$$|z_{000000}|^2=\frac{1}{16}, \tag{1204}$$

and the contradiction proves the claim. □

24.4 A full-rank ten-incidence criterion

The remaining four orbits have ten states in each fiber. The half-angle flip is a complete matching, and the diagonal moments identify the two probability vectors under that matching.

Lemma 24.3 (Full-rank matched-flip obstruction)**.** *Let*

$$S_0 = \{x_0, \dots, x_9\}, \qquad y_r = x_r \oplus e_i \in S_\delta, \qquad z_r = \overline{c_{x_r}} d_{y_r}. \tag{1205}$$

Choose four distinct coordinates $j_1, \dots, j_4 \neq i$, and let W be the 10×10 matrix whose rows, evaluated on $y_0, \dots, y_9$, are

$$1,\ \chi_{\{i\}},\ \chi_{\{j_1\}},\ \chi_{\{i,j_1\}}, \dots, \chi_{\{j_4\}},\ \chi_{\{i,j_4\}}. \tag{1206}$$

If $\det W \neq 0$, the weight-at-most-two cross-KL equations force $z_0 = \cdots = z_9 = 0$. If, in addition, the diagonal moments give

$$|c_{x_r}|^2 = |d_{y_r}|^2 =: p_r \qquad (0 \le r \le 9), \tag{1207}$$

then no normalized exact code exists.

Proof. The ten equations in Eq. (1206) are precisely those for $X_i, Y_i, X_i Z_{j_k}, Y_i Z_{j_k}$, up to nonzero row phases. Thus

$$W(z_0, \dots, z_9)^T = 0. \tag{1208}$$

Nonsingularity gives $z_r = 0$ for every r. But Eq. (1207) gives

$$|z_r|^2 = p_r^2, \tag{1209}$$

so every p_r vanishes, contradicting $\sum_r p_r = 1$. □

24.5 Orbits 42 and 43: nonsingular ten-incidence systems

Proposition 24.4 (Orbits 42 and 43 no-go)**.** *Neither orbit 42 nor orbit 43 admits an exact* $((6, 2, d \ge 3))$ *completion.*

Proof. The representatives are

$$a_{42} = (1, 1, 1, 1, 3, 3), \qquad a_{43} = (1, 1, 1, 2, 3, 3), \qquad (q, m, \delta) = (6, 2, 3). \tag{1210}$$

Use the half-angle coordinate $i = 6$. In the matching order $y_r = x_r \oplus e_6$, the fibers are as follows:

$$S_0^{(42)} = \{000000, 000011, 011101, 011110, 101101, 101110, 110101, 110110, 111001, 111010\}, \tag{1211}$$

$$S_3^{(42)} = \{000001, 000010, 011100, 011111, 101100, 101111, 110100, 110111, 111000, 111011\}, \tag{1212}$$

$$S_0^{(43)} = \{000000, 000011, 001101, 001110, 010101, 010110, 100101, 100110, 111001, 111010\}, \tag{1213}$$

$$S_3^{(43)} = \{000001, 000010, 001100, 001111, 010100, 010111, 100100, 100111, 111000, 111011\}. \tag{1214}$$

Exact row reduction of the diagonal moment equations gives the same complete three-parameter normal form in both cases:

$$\begin{aligned}(p_0, \dots, p_9) &= (q_0, \dots, q_9) \\ &= \left(\frac{1}{4}, \frac{1}{4}, s, s, \alpha, \alpha, \beta, \beta, \gamma, \gamma\right), \qquad s = \frac{1}{4} - \alpha - \beta - \gamma,\end{aligned} \tag{1215}$$

with

$$\alpha, \beta, \gamma, s \ge 0. \tag{1216}$$

Here q_r is ordered by the matched top fiber in Eqs. (1212) and (1214). Each complete moment matrix has rank seventeen. Maximal minors have determinants 108 and -108 for orbits 42 and 43, respectively, so Eq. (1215) is the complete moment polytope in each case.

Take $j_1, j_2, j_3, j_4 = 1, 2, 3, 4$ in Lemma 24.3. Exact evaluation of the ten character rows gives

$$\det W_{42} = 73728, \qquad \det W_{43} = 73728. \tag{1217}$$

Both matrices are nonsingular. Since the matched probability vectors are identical, Lemma 24.3 gives a contradiction in both cases. In fact, the first matched monomial alone would have $|z_0|^2 = (1/4)^2 = 1/16$ after full rank forces $z_0 = 0$. □

24.6 Orbit 44: paired moment masses and full Walsh rank

Proposition 24.5 (Orbit 44 no-go)**.** *Orbit 44 admits no exact* $((6, 2, d \geq 3))$ *completion.*

Proof. Orbit 44 is represented by

$$(q, m, a) = \big(6, 2, (1, 1, 2, 3, 3, 4)\big), \qquad \delta = 3. \tag{1218}$$

Use the half-angle coordinate $i = 5$. In the flip-matching order,

$$S_0^{(44)} = \{000000, 000110, 001001, 001111, 011010, 011100, 101010, 101100, 110001, 110111\}, \tag{1219}$$

$$S_3^{(44)} = \{000010, 000100, 001011, 001101, 011000, 011110, 101000, 101110, 110011, 110101\}. \tag{1220}$$

The complete three-parameter moment normal form is

$$\begin{aligned}(p_0, \ldots, p_9) &= (q_0, \ldots, q_9) \\ &= (u, u, \alpha, \alpha, v, v, \beta, \beta, \gamma, \gamma),\end{aligned} \tag{1221}$$

where

$$u = \frac{1}{4} - \alpha - \gamma, \qquad v = \frac{1}{4} - \beta, \tag{1222}$$

and

$$\alpha, \beta, \gamma \geq 0, \qquad \alpha + \gamma \leq \frac{1}{4}, \qquad \beta \leq \frac{1}{4}. \tag{1223}$$

The moment matrix has rank seventeen and a maximal minor of determinant -108.

Use the rows for

$$X_5,\ Y_5,\ X_5Z_j,\ Y_5Z_j, \qquad j = 1, 2, 3, 4. \tag{1224}$$

On the matched fibers above, their character determinant is

$$\det W_{44} = -32768 \neq 0. \tag{1225}$$

Lemma 24.3 forces all ten cross monomials to vanish. Equation (1221) would then force $u = \alpha = v = \beta = \gamma = 0$, whereas normalization gives

$$2(u + \alpha + v + \beta + \gamma) = 1. \tag{1226}$$

This contradiction excludes the orbit. □

24.7 Orbit 46: a second paired-mass full-rank obstruction

Proposition 24.6 (Orbit 46 no-go)**.** *Orbit 46 admits no exact* $((6,2,d\geq 3))$ *completion.*

Proof. Orbit 46 is represented by

$$(q,m,a)=\big(6,2,(1,2,2,2,3,3)\big),\qquad \delta=3. \tag{1227}$$

Use $i=6$. The flip-matched fibers are

$$S_0^{(46)}=\{000000,000011,011100,011111,100101,100110,101001,101010,110001,110010\}, \tag{1228}$$

$$S_3^{(46)}=\{000001,000010,011101,011110,100100,100111,101000,101011,110000,110011\}. \tag{1229}$$

Exact diagonal reduction gives

$$\begin{aligned}(p_0,\dots,p_9)&=(q_0,\dots,q_9)\\&=(u,u,\alpha,\alpha,v,v,\beta,\beta,\gamma,\gamma),\end{aligned} \tag{1230}$$

where

$$u=\frac{1}{4}-\alpha,\qquad v=\frac{1}{4}-\beta-\gamma, \tag{1231}$$

and

$$0\leq\alpha\leq\frac{1}{4},\qquad \beta,\gamma\geq 0,\qquad \beta+\gamma\leq\frac{1}{4}. \tag{1232}$$

The moment matrix has rank seventeen and a maximal minor of determinant -108.

The ten rows for

$$X_6,\ Y_6,\ X_6Z_j,\ Y_6Z_j,\qquad j=1,2,3,4, \tag{1233}$$

have exact character determinant

$$\det W_{46}=-73728\neq 0. \tag{1234}$$

The full-rank matched-flip lemma again forces every p_r to vanish, contradicting normalization. Thus orbit 46 is impossible. □

24.8 Orbit 19: two half-angle flips annihilate the anchor blocks

Proposition 24.7 (Orbit 19 no-go)**.** *Orbit 19 admits no exact* $((6,2,d\geq 3))$ *completion.*

Proof. Orbit 19 is represented by

$$(q,m,a)=\big(4,2,(1,1,1,1,2,2)\big),\qquad \delta=2. \tag{1235}$$

Write

$$w=x_1+x_2+x_3+x_4,\qquad s=x_5,\qquad t=x_6. \tag{1236}$$

The two residue fibers are

$$S_0=\{x: w+2(s+t)\equiv 0 \pmod 4\},\qquad S_2=\{y: w+2(s+t)\equiv 2 \pmod 4\}. \tag{1237}$$

Only $w=0,2,4$ occurs, and the fiber condition is equivalently

	$w\in\{0,4\}$	$w=2$
S_0	$s=t$	$s\neq t$
S_2	$s\neq t$	$s=t.$

(1238)

The four distinguished bottom states are

$$000000, \quad 000011, \quad 111100, \quad 111111, \tag{1239}$$

and the four distinguished top states are

$$000001, \quad 000010, \quad 111101, \quad 111110. \tag{1240}$$

For

$$|0_L\rangle = \sum_{x \in S_0} c_x |x\rangle, \qquad |1_L\rangle = \sum_{y \in S_2} d_y |y\rangle, \tag{1241}$$

put $p_x = |c_x|^2$, $q_y = |d_y|^2$, and define the anchor masses

$$P_0 = p_{000000} + p_{000011}, \qquad P_4 = p_{111100} + p_{111111}, \tag{1242}$$

$$Q_0 = q_{000001} + q_{000010}, \qquad Q_4 = q_{111101} + q_{111110}. \tag{1243}$$

On the three-point set $w \in \{0, 2, 4\}$, the degree-two polynomials

$$g_0(w) = \frac{(w-2)(w-4)}{8}, \qquad g_4(w) = \frac{w(w-2)}{8} \tag{1244}$$

are the indicators of $w = 0$ and $w = 4$, respectively. Since

$$w^2 = w + 2 \sum_{1 \le i < j \le 4} x_i x_j, \tag{1245}$$

both are linear combinations of diagonal Pauli operators of physical weight at most two. Their diagonal KL equalities give

$$P_0 = Q_0, \qquad P_4 = Q_4. \tag{1246}$$

Let

$$h(w) = g_0(w) + g_4(w), \qquad u(s,t) = s + t - 2st = (s-t)^2. \tag{1247}$$

Here h is the indicator of $w \in \{0, 4\}$ and u is the indicator of $s \neq t$. By Eq. (1238),

$$u = 1 - h \quad \text{on } S_0, \qquad u = h \quad \text{on } S_2. \tag{1248}$$

The observable $u = n_5 + n_6 - 2n_5 n_6$ is diagonal of degree at most two. If

$$H = P_0 + P_4 = Q_0 + Q_4, \tag{1249}$$

then its diagonal KL equality reads $1 - H = H$. Therefore

$$\boxed{P_0 + P_4 = Q_0 + Q_4 = \frac{1}{2}.} \tag{1250}$$

Both coordinates 5 and 6 are half-angle coordinates, so each coordinate flip is a bijection from S_0 to S_2. For $i = 5, 6$ define

$$z_x^{(i)} = \overline{c_x}\, d_{x \oplus e_i}. \tag{1251}$$

On the top fiber put

$$\chi_j(y) = (-1)^{y_j}, \qquad \Sigma(y) = \chi_1(y) + \chi_2(y) + \chi_3(y) + \chi_4(y). \tag{1252}$$

For $\alpha, \beta \in \{\pm 1\}$ define

$$\Phi^{(5)}_{\alpha,\beta}(y) = \frac{1+\beta\chi_5(y)}{16}\left(2+\alpha\Sigma(y)-2\beta\chi_6(y)\right). \tag{1253}$$

Restricted to S_2, this is the indicator of the unique anchor with

$$\chi_1 = \chi_2 = \chi_3 = \chi_4 = \alpha, \qquad \chi_5 = \beta, \qquad \chi_6 = -\beta. \tag{1254}$$

Indeed, on the $w = 2$ part of S_2 one has $\Sigma = 0$ and $\chi_5 = \chi_6$, so Eq. (1253) vanishes. On the $w = 0, 4$ part, the two factors select one of the four strings in Eq. (1240). Expanding Eq. (1253) uses only

$$1, \quad \chi_5, \quad \chi_j, \quad \chi_5\chi_j \qquad (j \neq 5), \tag{1255}$$

which are exactly the KL rows supplied by X_5, Y_5, Z_jX_5, Z_jY_5. Consequently the four X_5-matched anchor incidences vanish individually:

$$\overline{c_{000000}}d_{000010} = 0, \qquad \overline{c_{000011}}d_{000001} = 0, \tag{1256}$$

$$\overline{c_{111100}}d_{111110} = 0, \qquad \overline{c_{111111}}d_{111101} = 0. \tag{1257}$$

Interchanging coordinates 5 and 6, define

$$\Phi^{(6)}_{\alpha,\beta}(y) = \frac{1+\beta\chi_6(y)}{16}\left(2+\alpha\Sigma(y)-2\beta\chi_5(y)\right). \tag{1258}$$

The corresponding X_6, Y_6, Z_jX_6, Z_jY_6 equations isolate the other four anchor incidences:

$$\overline{c_{000000}}d_{000001} = 0, \qquad \overline{c_{000011}}d_{000010} = 0, \tag{1259}$$

$$\overline{c_{111100}}d_{111101} = 0, \qquad \overline{c_{111111}}d_{111110} = 0. \tag{1260}$$

Thus all four products between the two bottom $w = 0$ anchors and the two top $w = 0$ anchors vanish. Taking squared moduli and summing gives

$$P_0Q_0 = 0. \tag{1261}$$

By Eq. (1246), $P_0 = Q_0$, so $P_0 = Q_0 = 0$. The same argument at $w = 4$ gives $P_4 = Q_4 = 0$. This contradicts Eq. (1250), proving the no-go theorem.

The full moment matrix has size 23×32 and rank twenty, with a maximal minor of determinant -128. Each one-flip character matrix has size 12×16 and rank twelve; selected maximal minors have determinant -2^{20}. □

24.9 Orbit 17: a fractional perfect matching and vanishing hafnians

Proposition 24.8 (Orbit 17 no-go)**.** *Orbit 17 admits no exact* $((6, 2, d \geq 3))$ *completion.*

Proof. Orbit 17 is represented by

$$(q, m, a) = \big(4, 2, (1, 1, 1, 1, 1, 1)\big). \tag{1262}$$

Let $V = \{1, \ldots, 6\}$. Its two fibers are the Hamming-weight sectors

$$S_0 = \{\mathrm{wt} = 0, 4\}, \qquad S_2 = \{\mathrm{wt} = 2, 6\}. \tag{1263}$$

Index weight-two strings by edges $e \in \binom{V}{2}$ and write $V \setminus e$ for the complementary weight-four string. An arbitrary candidate is

$$|0_L\rangle = a|\varnothing\rangle + \sum_{e\in\binom{V}{2}} c_e|V \setminus e\rangle, \tag{1264}$$

$$|1_L\rangle = \sum_{e\in\binom{V}{2}} d_e|e\rangle + b|V\rangle. \tag{1265}$$

Put

$$x_e = |c_e|^2, \qquad y_e = |d_e|^2, \qquad X = \sum_e x_e, \qquad Y = \sum_e y_e. \tag{1266}$$

Normalization, equality of the sum of the six one-coordinate occupation moments, and equality of the sum of the fifteen two-coordinate moments give

$$|a|^2 + X = 1, \qquad Y + |b|^2 = 1, \tag{1267}$$

$$4X = 2Y + 6|b|^2, \qquad 6X = Y + 15|b|^2. \tag{1268}$$

Solving,

$$|a|^2 = |b|^2 = \frac{1}{4}, \qquad X = Y = \frac{3}{4}. \tag{1269}$$

For $i \in V$ define the weighted edge degrees

$$d_i^x = \sum_{e\ni i} x_e, \qquad d_i^y = \sum_{e\ni i} y_e. \tag{1270}$$

The one-coordinate equality at i gives

$$d_i^x + d_i^y = \frac{1}{2}. \tag{1271}$$

Summing the two-coordinate equalities involving i gives

$$3d_i^x + d_i^y = 1. \tag{1272}$$

Hence

$$d_i^x = d_i^y = \frac{1}{4} \qquad (i \in V). \tag{1273}$$

Substitution into the individual pair-moment equality for an edge e yields

$$x_e = y_e \qquad (e \in \tbinom{V}{2}). \tag{1274}$$

Thus the complete diagonal moment normal form is

$$\boxed{|a|^2 = |b|^2 = \frac{1}{4}, \qquad |c_e|^2 = |d_e|^2 =: x_e, \qquad \sum_{e\ni i} x_e = \frac{1}{4}.} \tag{1275}$$

The edge magnitudes are a scaled fractional perfect matching of K_6; in particular their support graph has no isolated vertex. The exact 23×32 moment matrix has rank twenty-three and a maximal minor of determinant 1024.

Fix an edge $f = \{i, j\}$. Since the cross reduced operator on the two qubits in f vanishes, its local $00 \leftrightarrow 11$ entry gives

$$\overline{a}\, d_f + \overline{c_f}\, b = 0. \tag{1276}$$

Because $a \neq 0$, define

$$\lambda = -\frac{b}{\overline{a}}. \tag{1277}$$

Equation (1269) gives $|\lambda| = 1$, and Eq. (1276) gives, including inactive edges,

$$d_f = \lambda\, \overline{c_f} \qquad (f \in \tbinom{V}{2}). \tag{1278}$$

The opposite local entry on the same two qubits contains the six incidences obtained by partitioning the complementary four-set $Q = V \setminus f$ into two edges. Therefore

$$\sum_{\{e,g\} \in \mathcal{M}(Q)} (\overline{c_e} d_g + \overline{c_g} d_e) = 0, \tag{1279}$$

where $\mathcal{M}(Q)$ is the set of the three perfect matchings of Q. Using Eq. (1278), this becomes

$$\boxed{c_{ij}c_{k\ell} + c_{ik}c_{j\ell} + c_{i\ell}c_{jk} = 0} \tag{1280}$$

for every four-set $\{i, j, k, \ell\} \subseteq V$. These are the fifteen four-vertex hafnian equations.

It remains to show that a graph with no isolated vertex cannot support nonzero edge labels satisfying all equations (1280). Let $G \subseteq K_6$ be the support graph $E(G) = \{e : c_e \neq 0\}$.

If $\Delta(G) \leq 2$ and G is disconnected, choose one edge from each of two components. Their four endpoints induce exactly one active perfect matching, so the corresponding hafnian has one nonzero monomial. If G is connected, it is a path or cycle on six vertices, and four consecutive vertices induce a P_4, again with exactly one active perfect matching. Both alternatives contradict Eq. (1280).

Suppose next that $\Delta(G) = 3$. Let v have neighbors a, b, c and nonneighbors u, w. The hafnian equations are invariant under nonzero vertex scalings

$$c_{ij} \longmapsto s_i s_j c_{ij}, \qquad s_i \neq 0. \tag{1281}$$

Normalize $c_{va} = c_{vb} = c_{vc} = 1$. For distinct $r, s \in \{a, b, c\}$,

$$0 = H_{\{v,u,r,s\}} = c_{ur} + c_{us}. \tag{1282}$$

The three pair-sum equations force $c_{ua} = c_{ub} = c_{uc} = 0$, and the same argument applies to w. Since neither u nor w may be isolated, $c_{uw} \neq 0$. But then

$$H_{\{v,a,u,w\}} = c_{va}c_{uw} = c_{uw} \neq 0, \tag{1283}$$

a contradiction.

If $\Delta(G) = 4$, let v have four neighbors R and one nonneighbor u, and normalize $c_{vr} = 1$ for $r \in R$. For every distinct $r, s \in R$,

$$0 = H_{\{v,u,r,s\}} = c_{ur} + c_{us}. \tag{1284}$$

The six pair-sum equations have full column rank four, so $c_{ur} = 0$ for all $r \in R$. Together with $c_{uv} = 0$, this makes u isolated.

Finally, suppose $\Delta(G) = 5$. Let v be universal and normalize $c_{vi} = 1$ for $i \neq v$. For every triple $i, j, k \neq v$,

$$0 = H_{\{v,i,j,k\}} = c_{ij} + c_{ik} + c_{jk}. \tag{1285}$$

The 10×10 triangle-versus-edge incidence matrix of K_5 has determinant -48, so all edges among the five neighbors vanish. Returning to the original, unscaled magnitudes, the support is a star. Every leaf has weighted degree $1/4$, hence every star edge has squared magnitude $1/4$; the center then has weighted degree $5/4$, contradicting Eq. (1273).

All possible maximum degrees have been excluded, so orbit 17 is impossible. □

24.10 Orbit 20: block uniformity and an odd phase triangle

Proposition 24.9 (Orbit 20 no-go)**.** *Orbit 20 admits no exact* $((6,2,d\geq 3))$ *completion.*

Proof. Orbit 20 is represented by

$$(q,m,a)=\big(4,2,(1,1,1,2,2,2)\big). \tag{1286}$$

Write a bit string as $(r,y)\in\mathbb{F}_2^3\times\mathbb{F}_2^3$, with r the first three bits. Only the even-parity patterns

$$\mathcal{R}=\{000,011,101,110\} \tag{1287}$$

occur. Put

$$\epsilon(000)=0,\qquad \epsilon(r)=1\quad (r\neq 000), \tag{1288}$$

and

$$P_r=\{y\in\mathbb{F}_2^3:\mathrm{wt}(y)\equiv\epsilon(r)\pmod 2\}. \tag{1289}$$

For fixed r, the bottom last-three-bit patterns are P_r and the top patterns have the opposite parity. Since three-bit complementation reverses parity, index the top block by $\bar{y}=y\oplus 111$ and write

$$|0_L\rangle=\sum_{r\in\mathcal{R}}\sum_{y\in P_r}c_{r,y}|r,y\rangle, \tag{1290}$$

$$|1_L\rangle=\sum_{r\in\mathcal{R}}\sum_{y\in P_r}\widetilde{d}_{r,y}|r,\bar{y}\rangle. \tag{1291}$$

On the four-point set $\mathcal{R}$, the functions $1,\chi_1,\chi_2,\chi_3$, where $\chi_j(r)=(-1)^{r_j}$, form a basis. The restricted point projector is

$$\Pi_r(s)=\frac{1}{4}\left(1+\sum_{j=1}^{3}(-1)^{r_j}\chi_j(s)\right). \tag{1292}$$

Normalization and the three diagonal Z_j equalities therefore separate the four blocks and give

$$\sum_{y\in P_r}|c_{r,y}|^2=\sum_{y\in P_r}|\widetilde{d}_{r,y}|^2=:\mu_r. \tag{1293}$$

Each of qubits $4,5,6$ is a half-angle coordinate. Fix one such coordinate k. The KL rows from

$$X_k,\quad Y_k,\quad X_kZ_j,\quad Y_kZ_j,\qquad j=1,2,3, \tag{1294}$$

span the restricted characters

$$\Pi_r(s)\frac{1\pm\chi_k}{2}. \tag{1295}$$

For fixed r and a fixed value of the kth top bit, this projector selects exactly two flip incidences. Their bottom strings $y,y'\in P_r$ differ in the two coordinates other than k, or equivalently

$$y'=y\oplus(111\oplus e_k). \tag{1296}$$

The isolated KL equation is therefore

$$\boxed{\overline{c_{r,y}}\,\widetilde{d}_{r,y'}+\overline{c_{r,y'}}\,\widetilde{d}_{r,y}=0.} \tag{1297}$$

Any two distinct strings in a fixed three-bit parity class differ in exactly two coordinates, so Eq. (1297) holds for every unordered pair $y\neq y'$ in P_r.

Put

$$p_{r,y} = |c_{r,y}|^2, \qquad \widetilde{q}_{r,y} = |\widetilde{d}_{r,y}|^2. \tag{1298}$$

Taking squared moduli in Eq. (1297) gives

$$p_{r,y}\widetilde{q}_{r,y'} = p_{r,y'}\widetilde{q}_{r,y} \qquad (y \neq y'). \tag{1299}$$

Thus all 2×2 minors of the two-row nonnegative matrix formed by $p_{r,\bullet}$ and $\widetilde{q}_{r,\bullet}$ vanish. The two rows have the same sum by Eq. (1293), so they coincide, including on zero-mass boundary blocks:

$$|c_{r,y}|^2 = |\widetilde{d}_{r,y}|^2 =: w_{r,y}. \tag{1300}$$

Under this matching the actual top point is $\bar{y}$, so $\chi_k(\bar{y}) = -\chi_k(y)$ for $k = 4, 5, 6$. The diagonal equalities for Z_k and $Z_i Z_k$, $i = 1, 2, 3$, give

$$\sum_{r,y} w_{r,y}\chi_k(y) = 0, \tag{1301}$$

$$\sum_{r,y} w_{r,y}\chi_i(r)\chi_k(y) = 0. \tag{1302}$$

Since $1, \chi_1, \chi_2, \chi_3$ span all functions on $\mathcal{R}$, these relations separate the blocks:

$$\sum_{y \in P_r} w_{r,y}\chi_k(y) = 0 \qquad (r \in \mathcal{R},\ k = 4, 5, 6). \tag{1303}$$

On each four-point parity class P_r, the functions $1, \chi_4, \chi_5, \chi_6$ form a Hadamard basis. Hence every block is uniform:

$$\boxed{w_{r,y} = \alpha_r \quad (y \in P_r),} \qquad 4\sum_{r \in \mathcal{R}} \alpha_r = 1, \qquad \alpha_r \geq 0. \tag{1304}$$

At least one block has $\alpha_r > 0$; all eight amplitudes in its matched bottom and top blocks are then nonzero.

Fix such a block and define

$$\rho_y = \frac{\widetilde{d}_{r,y}}{c_{r,y}}, \qquad |\rho_y| = 1. \tag{1305}$$

Equation (1297) becomes

$$\rho_y + \rho_{y'} = 0 \qquad (y \neq y'). \tag{1306}$$

For any three distinct $y_1, y_2, y_3 \in P_r$,

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \rho_{y_1} \\ \rho_{y_2} \\ \rho_{y_3} \end{pmatrix} = 0. \tag{1307}$$

The determinant is -2, forcing all three phase ratios to vanish, contrary to unit modulus. This proves the no-go theorem.

The exact original moment matrix has rank twenty and a maximal minor -512. After adjoining Eq. (1300), the augmented system has rank twenty-nine and a maximal minor -4096; its complete three-dimensional solution is Eq. (1304). □

24.11 Orbit 12: a zero-angle spinor and complementary-edge contradiction

Proposition 24.10 (Orbit 12 no-go)**.** *Orbit 12 admits no exact* $((6,2,d\geq 3))$ *completion.*

Proof. Orbit 12 is represented by

$$(q,m,a)=\big(4,2,(0,1,1,1,1,2)\big). \tag{1308}$$

Let A be the zero-angle first qubit, let $V=\{2,3,4,5\}$ be the four coefficient-one qubits, and let $H=6$ be the half-angle qubit. Identifying a pattern on V with a subset of a four-element set, the bottom active fiber is

$$(\varnothing,0),\qquad (V,0),\qquad (e,1)\quad (e\in\tbinom{V}{2}), \tag{1309}$$

and the top active fiber is

$$(\varnothing,1),\qquad (V,1),\qquad (e,0)\quad (e\in\tbinom{V}{2}). \tag{1310}$$

Collect the amplitudes over A into spinors in $\mathbb{C}^2$:

$$|0_L\rangle=u\otimes|\varnothing,0\rangle+v\otimes|V,0\rangle+\sum_{e\in\binom{V}{2}}a_e\otimes|e,1\rangle, \tag{1311}$$

$$|1_L\rangle=\widetilde{u}\otimes|\varnothing,1\rangle+\widetilde{v}\otimes|V,1\rangle+\sum_{e\in\binom{V}{2}}b_e\otimes|e,0\rangle. \tag{1312}$$

Let w be the Hamming weight on V. The two degree-two shell projectors in Eq. (1244), now with $w\in\{0,2,4\}$, give

$$\|u\|^2=\|\widetilde{u}\|^2,\qquad \|v\|^2=\|\widetilde{v}\|^2. \tag{1313}$$

Put

$$X=\sum_e\|a_e\|^2,\qquad Y=\sum_e\|b_e\|^2. \tag{1314}$$

The n_H equality gives

$$X=\|\widetilde{u}\|^2+\|\widetilde{v}\|^2. \tag{1315}$$

Together with normalization and Eq. (1313), this implies

$$X=Y=\frac{1}{2},\qquad \|u\|^2+\|v\|^2=\|\widetilde{u}\|^2+\|\widetilde{v}\|^2=\frac{1}{2}. \tag{1316}$$

The diagonal observable $wn_H=\sum_{i\in V}n_in_H$ has expectation $2X$ on the bottom word and $4\|\widetilde{v}\|^2$ on the top word. Hence

$$\boxed{\|u\|^2=\|v\|^2=\|\widetilde{u}\|^2=\|\widetilde{v}\|^2=\frac{1}{4}.} \tag{1317}$$

Write $x_e=\|a_e\|^2$ and $y_e=\|b_e\|^2$. The n_in_j equalities give $x_e=y_e$, while the n_Hn_i equalities give

$$\sum_{e\ni i}x_e=\frac{1}{4}\qquad (i\in V). \tag{1318}$$

For complementary edges $e,\bar{e}=V\setminus e$, equality of all four weighted degrees implies $x_e=x_{\bar{e}}$. Labeling $V=\{1,2,3,4\}$, the complete aggregate norm polytope is therefore

$$\begin{aligned}&x_{12}=x_{34}=\alpha,\\ &x_{13}=x_{24}=\beta,\qquad \alpha,\beta,\gamma\geq 0,\qquad \alpha+\beta+\gamma=\frac{1}{4}.\\ &x_{14}=x_{23}=\gamma,\end{aligned} \tag{1319}$$

In particular, at least one complementary edge pair has positive mass.

Apply the cross-KL equation to the two-qubit subsystem AH. Its $|1\rangle_H\langle 0|_H$ block is

$$\widetilde{u}\,u^\dagger + \widetilde{v}\,v^\dagger = 0. \tag{1320}$$

All four vectors are nonzero by Eq. (1317). Two nonzero rank-one matrices can cancel only when their column spaces and row spaces coincide. Hence

$$\operatorname{span}\{u, v\} =: \mathcal{L}, \qquad \operatorname{span}\{\widetilde{u}, \widetilde{v}\} =: \mathcal{M} \tag{1321}$$

are one-dimensional. The diagonal $H = 0$ block of the equality of the two AH reduced states is

$$uu^\dagger + vv^\dagger = \sum_e b_e b_e^\dagger. \tag{1322}$$

Positivity forces every b_e into $\mathcal{L}$. The $H = 1$ block similarly forces every a_e into $\mathcal{M}$. Choose unit vectors $\ell \in \mathcal{L}$, $m \in \mathcal{M}$, and write

$$u = \alpha_0 \ell, \qquad v = \alpha_4 \ell, \qquad \widetilde{u} = \beta_0 m, \qquad \widetilde{v} = \beta_4 m, \tag{1323}$$

$$a_e = A_e m, \qquad b_e = B_e \ell. \tag{1324}$$

The four anchor scalars are nonzero, and Eq. (1320) becomes

$$\beta_0\overline{\alpha_0} + \beta_4\overline{\alpha_4} = 0. \tag{1325}$$

Choose an edge e with $x_e > 0$. Then $x_{\bar{e}} = x_e > 0$, so A_e and $B_{\bar{e}}$ are nonzero. The cross reduced operator on the physical pair e has a local entry with exactly the two incidences $(e, 1) \to (\varnothing, 1)$ and $(V, 0) \to (\bar{e}, 0)$, giving

$$a_e^\dagger \widetilde{u} + v^\dagger b_{\bar{e}} = 0. \tag{1326}$$

The complementary physical pair $\bar{e}$ gives

$$u^\dagger b_{\bar{e}} + a_e^\dagger \widetilde{v} = 0. \tag{1327}$$

Substitution of Eqs. (1323)– (1324) yields

$$\overline{A_e}\beta_0 + \overline{\alpha_4} B_{\bar{e}} = 0, \tag{1328}$$

$$\overline{\alpha_0} B_{\bar{e}} + \overline{A_e}\beta_4 = 0. \tag{1329}$$

Eliminating the nonzero $B_{\bar{e}}$ gives

$$\beta_0\overline{\alpha_0} = \beta_4\overline{\alpha_4}, \tag{1330}$$

in direct contradiction with Eq. (1325) and the nonzero anchor norms. Thus orbit 12 is impossible. □

24.12 Orbit 13: blockwise equality versus complete orthogonality

Proposition 24.11 (Orbit 13 no-go)**.** *Orbit 13 admits no exact* $((6, 2, d \geq 3))$ *completion.*

Proof. Orbit 13 is represented by

$$(q, m, a) = \big(4, 2, (0, 1, 1, 1, 2, 2)\big). \tag{1331}$$

Let A be the zero-angle first qubit, write $r = (x_2, x_3, x_4)$ and $h = (x_5, x_6)$, and let

$$\mathcal{R} = \{000, 011, 101, 110\}. \tag{1332}$$

Put $\epsilon(000) = 0$, $\epsilon(r) = 1$ for $r \neq 000$, and

$$P_r = \{h \in \mathbb{F}_2^2 : h_5 \oplus h_6 = \epsilon(r)\}. \tag{1333}$$

The bottom fiber has $h \in P_r$, while the top fiber has the opposite h-parity. Index the top amplitudes through the X_6 matching:

$$|0_L\rangle = \sum_{r\in\mathcal{R}} \sum_{h\in P_r} c_{r,h} \otimes |r, h\rangle, \tag{1334}$$

$$|1_L\rangle = \sum_{r\in\mathcal{R}} \sum_{h\in P_r} d_{r,h} \otimes |r, h \oplus e_6\rangle, \tag{1335}$$

with $c_{r,h}, d_{r,h} \in \mathbb{C}^2$. Write $\bar{h} = h \oplus (1, 1)$ for the other element of P_r.

On $\mathcal{R}$, the restricted point projector is

$$\Pi_r(s) = \frac{1}{4}\left(1 + \sum_{j=2}^{4} (-1)^{r_j} \chi_j(s)\right), \qquad \chi_j(s) = (-1)^{s_j}. \tag{1336}$$

The underlying 4×4 Hadamard matrix has determinant -16. The weight-at-most-two errors

$$X_6, \quad Y_6, \quad Z_j X_6, \quad Z_j Y_6, \qquad j = 2, 3, 4, \tag{1337}$$

produce the restricted operators $\Pi_r |1 - h_6\rangle_6\langle h_6|_6$. For fixed r and h_6, exactly one $h \in P_r$ remains, so every X_6 incidence is isolated:

$$c_{r,h}^\dagger d_{r,h} = 0. \tag{1338}$$

The corresponding 8×8 restricted character determinant has absolute value 4096.

Flipping qubit 5 sends the bottom pattern h to the top pattern $h \oplus e_5 = \bar{h} \oplus e_6$. The analogous rows for $X_5, Y_5, Z_j X_5, Z_j Y_5$ therefore give

$$c_{r,h}^\dagger d_{r,\bar{h}} = 0. \tag{1339}$$

This second restricted 8×8 determinant also has absolute value 4096. Since P_r has two elements, the two systems together give the complete block orthogonality

$$\boxed{c_{r,h}^\dagger d_{r,h'} = 0 \quad \text{for all } h, h' \in P_r.} \tag{1340}$$

Define the positive semidefinite block frames

$$C_r = \sum_{h\in P_r} c_{r,h} c_{r,h}^\dagger, \qquad D_r = \sum_{h\in P_r} d_{r,h} d_{r,h}^\dagger. \tag{1341}$$

For an arbitrary operator F on the zero-angle qubit, the restricted operator $F_A \Pi_r$ is a linear combination of F_A and $F_A Z_j$, $j = 2, 3, 4$, all of physical weight at most two. Its diagonal KL equality implies

$$\mathrm{Tr}(F C_r) = \mathrm{Tr}(F D_r) \qquad \text{for every } F \in M_2(\mathbb{C}), \tag{1342}$$

so

$$C_r = D_r. \tag{1343}$$

On the other hand, Eq. (1340) gives

$$C_r D_r = \sum_{h,h' \in P_r} c_{r,h}\big(c^\dagger_{r,h} d_{r,h'}\big) d^\dagger_{r,h'} = 0. \tag{1344}$$

Thus $C_r^2 = 0$. Positivity forces $C_r = 0$ for every r, contradicting normalization of $|0_L\rangle$. Therefore orbit 13 is impossible. □

24.13 A parity-four moment and support-connectivity lemma

The last three order-two no-go proofs share one exact four-bit lemma. Let

$$\mathcal{E} = \{x \in \mathbb{F}_2^4 : \mathrm{wt}(x) \equiv 0 \pmod 2\}, \qquad \mathcal{O} = \{y \in \mathbb{F}_2^4 : \mathrm{wt}(y) \equiv 1 \pmod 2\}. \tag{1345}$$

For $S \subseteq \{1,2,3,4\}$ put $\chi_S(z) = (-1)^{\sum_{i\in S} z_i}$.

Lemma 24.12 (Parity-four moment cross-polytope)**.** *Suppose normalized nonnegative distributions $p = (p_x)_{x\in\mathcal{E}}$ and $q = (q_y)_{y\in\mathcal{O}}$ satisfy*

$$\sum_{x\in\mathcal{E}} p_x \chi_S(x) = \sum_{y\in\mathcal{O}} q_y \chi_S(y) \qquad (|S| \le 2). \tag{1346}$$

Then there are real numbers $\alpha_1, \dots, \alpha_4$ such that

$$p_x = \frac{1}{8}\left(1 + \sum_{i=1}^4 \alpha_i \chi_i(x)\right), \qquad x \in \mathcal{E}, \tag{1347}$$

$$q_y = \frac{1}{8}\left(1 + \sum_{i=1}^4 \alpha_i \chi_i(y)\right), \qquad y \in \mathcal{O}, \tag{1348}$$

and nonnegativity is equivalent to

$$|\alpha_1| + |\alpha_2| + |\alpha_3| + |\alpha_4| \le 1. \tag{1349}$$

Hence the exact moment polytope is the four-dimensional cross-polytope.

Proof. For a two-set S, let S^c be the complementary two-set. On $\mathcal{E}$, $\chi_S = \chi_{S^c}$, whereas on $\mathcal{O}$, $\chi_S = -\chi_{S^c}$. Applying Eq. (1346) to both S and S^c shows that every two-body Fourier coefficient vanishes on both fibers. Let α_i be the common one-body coefficient. Fourier inversion on the two parity classes gives Eqs. (1347)–(1348). Across the union $\mathcal{E} \cup \mathcal{O}$, the sign vector $(\chi_1, \dots, \chi_4)$ ranges over all of $\{\pm 1\}^4$. Therefore all sixteen probabilities are nonnegative exactly when $1 + \alpha \cdot s \ge 0$ for every sign vector s, which is equivalent to Eq. (1349). □

For a point α in this cross-polytope, form the bipartite support graph G_α: its left vertices are the even strings with $p_x > 0$, its right vertices are the odd strings with $q_y > 0$, and edges join strings at Hamming distance one.

Lemma 24.13 (Connectivity of the positive parity support)**.** *For every α satisfying Eq. (1349), the graph G_α is connected. Moreover, after fixing any coordinate $z_i = b$ and retaining only edges in the other three coordinate directions, the resulting positive-support graph is either empty or connected.*

Proof. If $\|\alpha\|_1 < 1$, every one of the sixteen probabilities is positive, so the graph is the full four-cube. Suppose $\|\alpha\|_1 = 1$ and put $s_i = \operatorname{sgn}(\alpha_i)$ for nonzero coordinates. Uniformly on both parity classes,

$$8\pi_z = 1 + \sum_i \alpha_i \chi_i(z) = \sum_{i:\alpha_i \neq 0} |\alpha_i| \big(1 + s_i \chi_i(z)\big), \tag{1350}$$

where π_z denotes p_z or q_z. Hence $\pi_z > 0$ exactly when $\chi_i(z) = s_i$ for at least one nonzero α_i. The positive support is a union of coordinate three-cubes. Each is connected and any two such cubes intersect, proving the first claim. Intersecting this union with a fixed coordinate slice gives either the whole three-cube or a union of pairwise-intersecting coordinate two-cubes, proving the second claim. □

Lemma 24.14 (Isolation of every parity-changing edge)**.** *Let $c_x, d_y \in \mathbb{C}^2$ be vector amplitudes on the even and odd fibers. If the off-diagonal KL equations hold for all physical errors of weight at most two, then*

$$c_x^\dagger d_y = 0 \qquad \text{whenever } d_H(x,y) = 1, \tag{1351}$$

provided the four effective parity coordinates are represented by physical single-qubit Z characters and the flipped coordinate is a physical single-qubit flip.

Proof. Fix a coordinate k. Flipping it bijects $\mathcal{E}$ and $\mathcal{O}$. On the eight incidences $x \mapsto x \oplus e_k$, the KL rows from

$$X_k, \quad Y_k, \quad Z_j X_k, \quad Z_j Y_k \qquad (j \neq k) \tag{1352}$$

are, up to row phases,

$$1, \quad \chi_k, \quad \chi_j, \quad \chi_k \chi_j \qquad (j \neq k). \tag{1353}$$

These are the complete character table of a three-dimensional Boolean group. The resulting 8×8 matrix W_k satisfies $W_k W_k^T = 8 I_8$, so

$$|\det W_k| = 4096. \tag{1354}$$

Every incidence overlap is therefore zero. Repeating over the four coordinates gives Eq. (1351). □

Lemma 24.15 (Connected orthogonality versus equal frames)**.** *Let a connected bipartite graph carry nonzero vectors $c_x, d_y \in \mathbb{C}^2$ on its two vertex classes and suppose $c_x^\dagger d_y = 0$ on every edge. Then there is a line $\mathcal{L} \subset \mathbb{C}^2$ such that all $c_x \in \mathcal{L}$ and all $d_y \in \mathcal{L}^\perp$. Consequently, if the two sides have the same positive total mass, their frame operators cannot be equal.*

Proof. Choose one edge. In $\mathbb{C}^2$, the orthogonal complement of a nonzero vector is one-dimensional. Propagating across a connected graph places all left vectors on one line and all right vectors on its orthogonal line. If the common total mass is $\mu > 0$, the two frame operators are $\mu P_{\mathcal{L}}$ and $\mu P_{\mathcal{L}^\perp}$, which are unequal. □

24.14 Orbit 15: a fixed qubit and the parity-four obstruction

Proposition 24.16 (Orbit 15 no-go)**.** *Orbit 15 admits no exact $((6, 2, d \geq 3))$ completion.*

Proof. Orbit 15 is represented by

$$(q, m, a) = \big(4, 2, (0, 1, 2, 2, 2, 2)\big). \tag{1355}$$

Write the physical bits as $(x_1, x_2, y_1, y_2, y_3, y_4)$. The residue equation is

$$x_2 + 2\,\mathrm{wt}(y) \equiv 0 \text{ or } 2 \pmod 4. \tag{1356}$$

It follows that $x_2 = 0$ on both fibers, and then

$$y \in \mathcal{E} \quad \text{on } S_0, \qquad y \in \mathcal{O} \quad \text{on } S_2. \tag{1357}$$

Collect the amplitudes on the zero-angle qubit 1 into spinors:

$$|0_L\rangle = |0\rangle_2 \sum_{x\in\mathcal{E}} c_x \otimes |x\rangle_{3,4,5,6}, \tag{1358}$$

$$|1_L\rangle = |0\rangle_2 \sum_{y\in\mathcal{O}} d_y \otimes |y\rangle_{3,4,5,6}, \tag{1359}$$

with $c_x, d_y \in \mathbb{C}^2$. The diagonal moments on qubits $3,4,5,6$ put the norm distributions $p_x = \|c_x\|^2$, $q_y = \|d_y\|^2$ in the cross-polytope of Lemma 24.12. Every operator F on qubit 1 has physical weight at most one, so its diagonal KL equality gives

$$\sum_{x\in\mathcal{E}} c_x c_x^\dagger = \sum_{y\in\mathcal{O}} d_y d_y^\dagger. \tag{1360}$$

For each of qubits $3,4,5,6$, Lemma 24.14 gives edgewise orthogonality on the positive-support graph G_α. This graph is connected by Lemma 24.13. Lemma 24.15 now contradicts Eq. (1360). Hence orbit 15 is impossible. □

24.15 Orbit 24: an effective repetition qubit and parity four

Proposition 24.17 (Orbit 24 no-go)**.** *Orbit 24 admits no exact* $((6,2,d\geq 3))$ *completion.*

Proof. Orbit 24 is represented by

$$(q,m,a) = \big(4,2,(1,2,2,2,2,3)\big). \tag{1361}$$

Write $y = (x_2,x_3,x_4,x_5)$. The residue is

$$x_1 + 3x_6 + 2\,\mathrm{wt}(y) \equiv x_1 - x_6 + 2\,\mathrm{wt}(y) \pmod 4. \tag{1362}$$

For this to be 0 or 2, one must have $x_1 = x_6 =: r$. The contribution $x_1 + 3x_6 = 4r$ then vanishes modulo four, and

$$y \in \mathcal{E} \quad \text{on } S_0, \qquad y \in \mathcal{O} \quad \text{on } S_2. \tag{1363}$$

Let $V_R|r\rangle = |rr\rangle_{1,6}$ be the repetition isometry, and collect the amplitudes over the effective repetition qubit into $c_x, d_y \in \mathbb{C}^2_R$:

$$|0_L\rangle = \sum_{x\in\mathcal{E}} V_R c_x \otimes |x\rangle_{2,3,4,5}, \tag{1364}$$

$$|1_L\rangle = \sum_{y\in\mathcal{O}} V_R d_y \otimes |y\rangle_{2,3,4,5}. \tag{1365}$$

The full one-qubit operator algebra on the repetition qubit is represented by physical operators of weight at most two:

$$V_R^\dagger Z_1 V_R = Z_R, \qquad V_R^\dagger X_1 X_6 V_R = X_R, \qquad V_R^\dagger Y_1 X_6 V_R = Y_R. \tag{1366}$$

Hence the diagonal KL equations imply equality of the complete repetition frames,

$$\sum_{x\in\mathcal{E}} c_x c_x^\dagger = \sum_{y\in\mathcal{O}} d_y d_y^\dagger. \tag{1367}$$

The degree-at-most-two diagonal moments on qubits $2,3,4,5$ give the cross-polytope of Lemma 24.12. For each middle qubit, the errors in Eq. (1352), with the other three middle qubits supplying the Z dressings, isolate every parity-changing incidence. Thus the vector families are edgewise orthogonal on the connected support graph G_α. Lemma 24.15 contradicts Eq. (1367), proving the no-go theorem. □

24.16 Orbit 14: a repetition slice and blockwise frame equality

Proposition 24.18 (Orbit 14 no-go)**.** *Orbit 14 admits no exact* $((6,2,d\geq 3))$ *completion.*

Proof. Orbit 14 is represented by

$$(q,m,a)=\big(4,2,(0,1,1,2,2,2)\big). \tag{1368}$$

Let qubit 1 be the zero-angle qubit and put $y=(x_4,x_5,x_6)$. The residue condition

$$x_2+x_3+2\,\mathrm{wt}(y)\equiv 0 \text{ or } 2 \pmod 4 \tag{1369}$$

forces $x_2=x_3=:r$. With the four effective parity bits $z=(r,y_1,y_2,y_3)$,

$$z\in\mathcal{E} \quad \text{on } S_0, \qquad z\in\mathcal{O} \quad \text{on } S_2. \tag{1370}$$

Collecting amplitudes on qubit 1, write

$$|0_L\rangle=\sum_{z=(r,y)\in\mathcal{E}} c_z\otimes|rr\rangle_{2,3}|y\rangle_{4,5,6}, \tag{1371}$$

$$|1_L\rangle=\sum_{w=(r,y)\in\mathcal{O}} d_w\otimes|rr\rangle_{2,3}|y\rangle_{4,5,6}, \tag{1372}$$

with $c_z,d_w\in\mathbb{C}^2$. The diagonal moments of the four effective parity bits again give Lemma 24.12; the character of r is represented physically by Z_2.

For $r=0,1$, define

$$C_r=\sum_{\substack{z\in\mathcal{E}\\ z_1=r}} c_z c_z^\dagger, \qquad D_r=\sum_{\substack{w\in\mathcal{O}\\ w_1=r}} d_w d_w^\dagger. \tag{1373}$$

For an arbitrary operator F on zero-angle qubit 1, the diagonal KL equalities for F and FZ_2 give, respectively, equality of the sums and differences of the two blocks. Therefore

$$C_0=D_0, \qquad C_1=D_1. \tag{1374}$$

Fix one of the three half-angle qubits $k\in\{4,5,6\}$. In the effective four-bit description, the other three characters are the repetition character, supplied by Z_2, and the two remaining half-angle characters. Thus the eight rows in Eq. (1353) are all available through errors of physical weight at most two. They isolate

$$c_z^\dagger d_{z\oplus e_k}=0 \tag{1375}$$

for every incidence in a y direction. By the sliced statement in Lemma 24.13, after fixing r the positive support using these three directions is either empty or connected.

Normalization guarantees a block r with positive mass. In that block, connected edgewise orthogonality puts all bottom vectors on a line $\mathcal{L}_r$ and all top vectors on $\mathcal{L}_r^\perp$. If the common block mass is $\mu_r>0$, then

$$C_r=\mu_r P_{\mathcal{L}_r}, \qquad D_r=\mu_r P_{\mathcal{L}_r^\perp}, \tag{1376}$$

contradicting Eq. (1374). Therefore orbit 14 is impossible. □

Theorem 24.19 (Fourteen exact order-two no-go orbits)**.** *None of the fourteen rank-compatible moment orbits*

$$\boxed{12, 13, 14, 15, 17, 18, 19, 20, 24, 41, 42, 43, 44, 46} \tag{1377}$$

admits an exact $((6, 2, d \geq 3))$ *completion.*

Proof. The first six exclusions are Propositions 24.1, 24.2, 24.4, 24.5, and 24.6. The remaining eight are Propositions 24.7, 24.8, 24.9, 24.10, 24.11, 24.18, 24.16, and 24.17. Every proof uses only physical errors of weight at most two and is valid on all boundary faces of the relevant moment polytope. □

Corollary 24.20 (Exact reduction of the order-two sector)**.** *Among the twenty-five rank-compatible projective-order-two moment orbits, the only orbits not excluded by an exact no-go theorem are*

$$\boxed{1, 2, 3, 4, 10, 11, 16, 21, 22, 23, 45.} \tag{1378}$$

Among these eleven, Sections 13 and 13.2 determine the exact weight-enumerator images of orbits 10, 11, 16, 21, 22, 23, and 45. Sections 11 and 12 also determine orbits 2 and 3. Consequently only orbits 1 and 4 remain unresolved at the weight-enumerator-image level.

Part IV
Enumeration tables and availability of code and data

25 Complete count tables

Table 7 records angle-vector counts. The columns after "nonempty" are still angle records, so multiple vectors producing the same fiber pair are counted separately.

q	m	δ	raw	primitive	nonempty	coarse rank	moment LP	rank compatible
2	2	1	7	6	6	6	4	4
3	3	1	28	27	26	26	15	15
4	2	2	84	77	74	57	40	40
4	4	1	84	77	75	57	23	23
5	5	1	210	209	200	132	30	30
6	2	3	462	428	416	52	52	30
6	3	2	462	428	408	36	33	21
6	6	1	462	428	412	36	13	13
7	7	1	924	923	879	34	18	18
8	2	4	1716	1632	1556	38	38	0
8	4	2	1716	1632	1544	13	8	8
8	8	1	1716	1632	1560	14	9	9
9	3	3	3003	2975	2815	4	4	0
9	9	1	3003	2975	2797	5	5	5
Total			**13877**	**13449**	**12768**	**510**	**292**	**216**

Table 7: Exact angle-record counts through the diagonal SSLP pipeline.

Table 8 gives the corresponding labeled fiber-pair counts after hashing and the final orbit counts.

q	m	nonempty pairs	coarse-rank pairs	moment-LP pairs	rank-compatible pairs	orbits
2	2	6	6	4	4	4
3	3	26	26	15	15	5
4	2	69	55	39	39	15
4	4	75	57	23	23	8
5	5	197	132	30	30	8
6	2	372	52	52	30	6
6	3	396	36	33	21	6
6	6	397	36	13	13	4
7	7	845	34	18	18	5
8	2	1260	38	38	0	0
8	4	1431	13	8	8	2
8	8	1460	14	9	9	2
9	3	2346	4	4	0	0
9	9	2542	5	5	5	1
Total		**11422**	**508**	**291**	**215**	**66**

Table 8: Exact labeled fiber-pair and orbit counts. The labels (q, m) are retained throughout.

26 Code and data availability

The code, supporting data, exact certificates, and Lean proof sources will be made publicly available at https://github.com/xh-xihe/sslp-classification, together with instructions for reproducing the results.

References

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